CSE Study Guide

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1. Common Conversion Factors / Equations

1.1 Conversion Factors:

1.1.1 Common Factors:

Unit	=	Unit
Gallon	8.34	Lbs Water @ 60°F
Density of Water	**62.4 Lbs/Ft^3**	
Density of Air	**0.07649 Lbs/Ft^3**	
SG Water @ 60°F	1	
MW of Air	29	
SG of Liquid	MW of Liquid / 18.02	
SG of Gas	MW of Gas / 29	

1.1.2 Distance Factors:

Multiply	By	To Obtain
Inch	2.54	Centimeters
Centimeter	0.3937	Inch
Foot	0.3048	Meter
Meter	3.28083	Foot

1.1.3 Volume Factors:

Multiply	By	To Obtain
Gallon	0.13368	FT^3
Gallon	0.003754	M^3
Gallon	3.7853	Liter
Liter	0.2642	Gallon
Liter	0.03531	FT^3
Liter	0.001	M^3
FT^3	7.481	Gallon
FT^3	28.3205	Liter
FT^3	0.028317	M^3
M^3	35.3147	FT^3
M^3	3.28083	Gallon
M^3	1000	Liter

1.1.4 Mass Factors:

Multiply	By	To Obtain
Pound	0.4536	Kilogram
Kilogram	2.2046	Pound

1.1.5 Force Factors:

Multiply	By	To Obtain
Newton	0.22481	Pound-Force
Pound-Force	4.4482	Newton

1.1.6 Energy Factors:

Multiply	By	To Obtain
BTU	778.17	Ft-Lbf
BTU	1.055	KJoules
BTU/Hr	0.293	Watt
HP	0.7457	Kilowatt
HP	2545	BTU/Hr

1.1.7 Temperature Factors:

Unit	Use Equation	To Obtain Unit
°F	(°F – 32)*1.8	°C
°F	(°F + 459.67) / 1.8	°K
°F	(°F + 459.67)	°R
°C	(°C × 1.8) + 32	°F
°C	°C + 273.15	°K
°C	(°C × 1.8) + 32 + 459.67	°R
°K	(°K × 1.8) – 459.67	°F
°K	°K - 273.15	°C
°K	°K × 1.8	°R
°R	°R – 459.67	°F
°R	(°R – 32 – 459.67) / 1.8	°C
°R	°R / 1.8	°K

1.1.8 Pressure Factors:

Multiply	By	To Obtain
Atm	1.01295	Bar
Atm	29.9213	“Hg
Atm	760	mm Hg
Atm	406.86	“WC
Atm	14.696	PSI
Atm	1.01295×10^5	N/M^2 or Pa
Bar	0.9872	Atm
Bar	29.54	“ Hg
Bar	750.2838	mm Hg
Bar	401.65	“WC
“WC	0.03612	PSI
“WC	0.07354	“Hg
“WC	1868.1	mm Hg
“WC	248.9	N/M^2 or Pa
“WC	0.001868	Micron or mtorr
PSI	27.68	“WC
PSI	2.036	“Hg
PSI	51.71	mm Hg
PSI	0.068046	Atm
PSI	0.068948	Bar
PSI	6892.7	N/M^2 or Pa
Micron or mtorr	0.0005353	“WC
N/M^2 or Pa	0.004018	“WC
N/M^2 or Pa	0.00014508	PSI

1.1.9 Viscosity

Multiply	By	To Obtain
cs	$0.999 g/cm^3$	cp
cp	$1 / 0.999 g/ cm^3$	cs

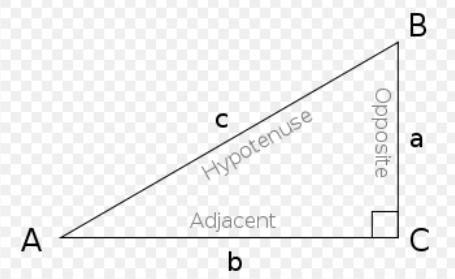
Kinematic viscosity (stoke) = Absolute viscosity (poise) / S.G.

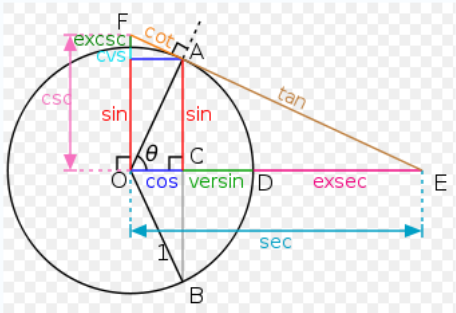
1.2 Equations:

1.2.1 General

1.2.1.1 Angles

$$\frac{r}{2\pi r} x 360° = \frac{180}{\pi} \text{deg}rees \qquad 1 \cdot radian = \frac{180}{\pi}° = 57.3°$$





$$\sin A = \frac{opposite}{hypotenuse} = \frac{a}{c} \qquad \csc A = \frac{hypotenuse}{opposite} = \frac{c}{a} = \frac{1}{\sin A}$$

$$\cos A = \frac{adjacent}{hypotenuse} = \frac{b}{c} \qquad \sec A = \frac{hypotenuse}{adjacent} = \frac{c}{b} = \frac{1}{\cos A}$$

$$\tan A = \frac{opposite}{adjacent} = \frac{a}{b} = \frac{\sin A}{\cos A} \qquad \cot A = \frac{adjacent}{opposite} = \frac{b}{a} = \frac{\cos A}{\sin A}$$

1.2.2 Pressure:

$$P = \frac{F}{A}$$

F = Force applied
A = Area

1.2.3 Boyle's Law

$$P_1 V_1 = P_2 V_2$$

Boyle's law states that *at constant temperature*, the absolute pressure and the volume of a gas are inversely proportional. The law can also be stated in a slightly different manner, that the product of absolute pressure and volume is always constant

P = Pressure in PSIA
V = Volume in FT^3

1.2.4 Charles's Law

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad \textbf{OR} \quad V_1 T_2 = V_2 T_1$$

Charles' law states that *at constant pressure*, the volume of a given mass of an ideal gas increases or decreases by the same factor as its temperature on the absolute temperature scale (i.e. the gas expands as the temperature increases).

T = Temperature in °R
V = Volume in FT^3

1.2.5 Gay-Lussac's Law

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \quad \textbf{OR} \quad P_1T_2 = P_2T_1$$

The pressure of a fixed mass and fixed volume of a gas is directly proportional to the gas's temperature.

T = Temperature in °R
P = Pressure in PSIA

1.2.6 Ideal Gas Law

(for compressibles):

$$PV = RT$$

R = Gas Constant (Value = 1544 / MW)
P = Pressure in PSIA
V = Volume in FT^3
T = Temperature in °R

1.2.7 Pascal's Law

(a change in the pressure of an enclosed incompressible fluid is conveyed undiminished to every part of the fluid and to the surfaces of its container)

$$\Delta P = \rho g(\Delta h)$$

ΔP = Hydrostatic pressure
ρ = Mass Density
g= Gravitation constant
Δh = Difference in elevation between the two points within the fluid column

1.2.8 Bernoulli's

(states that as the speed of a moving fluid increases, the pressure within the fluid decreases):

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$P + \frac{1}{2} \rho v^2 + \rho gh = \text{Constant}$
P = Pressure in PSIA
ρ = Mass Density
g = Gravitation constant
h = Height above reference level
v = Velocity

1.2.9 Flow:

$$Q = AV \qquad Q_{(gpm)} = 3.12\ A_{(sq\ in)} \times V_{(ft/sec)}$$ *Make sure units match*

Q = Volumetric Flow Rate
A = Cross Sectional Area of the Pipe
V = Velocity of the Fluid

1.2.10 Darcy's Formula (general formula for pressure drop):

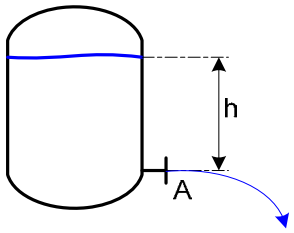
$$h = \frac{fLV^2}{2Dg}$$

h = Pressure drop in feet of fluid
L = Length of pipe
V = Velocity of the fluid
g = acceleration of gravity (32.2 ft/sec^2)
D = Pipe ID
f = The fanning friction factor f = 16 ÷ Re

1.2.11 Velocity of Exiting Fluid:

$$V = \sqrt{2gh} \qquad Q = A\sqrt{2gh}$$

V = Velocity of the Fluid
g = Gravitation constant
h = Height above reference level (in feet)
A = Area of opening (in sq ft)



1.2.12 Convert ACFM to SCFM:

$$ACFM = SCFM\left[\frac{14.7}{P_a} \times \frac{T_a}{520}\right] \qquad \text{equivalent to } \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

P_a = Actual pressure (PSIA)
T_s = Standard temperature (520°R) NOTE: °R =60°F+460
T_a = Actual temperature (°R)

1.2.13 Joule–Thomson (Kelvin) coefficient:

The rate of change of temperature *T* with respect to pressure *P* in a Joule–Thomson process (that is, at constant enthalpy *H*) is the *Joule–Thomson (Kelvin) coefficient* μ_{JT}. This coefficient can be expressed in terms of the gas's volume *V*, its heat capacity at constant pressure C_p, and its coefficient of thermal expansion α as:

$$\mu_{JT} \equiv \left(\frac{\partial T}{\partial P}\right)_H = \frac{V}{C_P}(\alpha T - 1)$$

The value of μJT is typically expressed in °C/bar (SI units: K/Pa)

In practice, the Joule–Thomson effect is achieved by allowing the gas to expand through a throttling device (usually a valve) which must be very well insulated to prevent any heat transfer to or from the gas. No external work is extracted from the gas during the expansion (the gas must not be expanded through a turbine, for example)
In a gas expansion the pressure decreases, so the sign δP of is always negative. With that in mind, the following table explains when the Joule–Thomson effect cools or warms a real gas:

If the gas temperature is	then μ_{JT} is	since ∂P is	thus ∂T must be	so the gas
below the inversion temperature	positive	always negative	negative	cools
above the inversion temperature	negative	always negative	positive	warms

1.2.14 Differentiation:

Method to compute the rate at which a dependent output y changes with respect to the change in the independent input x. This rate of change is called the **derivative** of y with respect to x. In more precise language, the dependence of y upon x means that y is a function of x. If x and y are real numbers, and if the graph of y is plotted against x, the derivative measures the slope of this graph at each point. This functional relationship is often denoted y = f(x), where f denotes the function.
The simplest case is when y is a linear function of x, meaning that the graph of y against x is a straight line. In this case, y = f(x) = m x + c, for real numbers m and c, and the slope m is given by:

$$m = \frac{\textit{change in } y}{\textit{change in } x} = \frac{\Delta y}{\Delta x}$$

The idea is to compute the rate of change as the limiting value of the ratio of the differences Δy / Δx as Δx becomes infinitely small. In Leibniz's notation, such an infinitesimal change in x is denoted by dx, and the derivative of y with respect to x is written:

$$\frac{dy}{dx}$$

Differentiation Rules:

- Constant rule: if $f(x)$ is constant, then
  f' = 0
- Sum rule: for all functions f and g and all real numbers a and b.
  (af + bg)' = af' +bg'
- Product rule: for all functions f and g.
  (fg)' = f'g + fg'
- Quotient rule: for all functions f and g where g ≠ 0.
  $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
- Chain rule: If f(x) = $h(g(x))$, then
  F'(x) = h'(g(x)) * g'(x)

For Differential tables Reference Section 25.10

Example computation

The derivative of

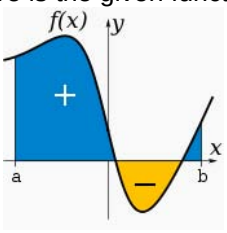
$f(x) = x^4 + \sin(x^2) - \ln(x)e^x + 7$ is:

$f'(x) = 4x^{(4-1)} + \frac{d(x^2)}{dx}\cos(x^2) - \frac{d(\ln x)}{dx}e^x - \ln x \frac{d(e^x)}{dx} + 0$ simplified is:

$f'(x) = 4x^3 + 2x\cos(x^2) - \frac{1}{x}e^x - \ln(x)e^x$

1.2.15 Integration:

Defined informally to be the net signed area of the region in the *xy*-plane bounded by the graph of f, the *x*-axis, and the vertical lines $x = a$ and $x = b$.
The term *integral* may also refer to the notion of antiderivative, a function *F* whose derivative is the given function f.



$\int_a^b f(x)dx = F(b) - F(a)$ For Integral tables Reference Section 25.11

1.2.16 Logarithms:

The logarithm of *x* to the base *b* is written $\log_b(x)$ or, if the base is implicit, as log(*x*). So, for a number *x*, a base *b* and an exponent *y*,

If $x = b^y$, then $y = \log_b(x)$

An important feature of logarithms is that they reduce multiplication to addition, by the formula:

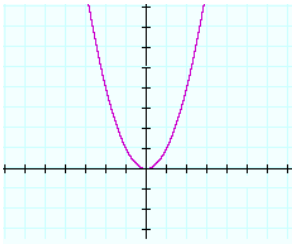
Log(xy) = log x + log y That is, the logarithm of the product of two numbers

- The exponential equation $4^3 = 64$ could be written in terms of a logarithmic equation as $\log_4(64) = 3$.
- The exponential equation 5-2 = 1 / 25 can be written as the logarithmic equation $\log_5(1/25) = -2$.

The antilogarithm function $\text{antilog}_b(y)$ is the inverse function of the logarithm function $\log_b(x)$; it can be written in closed form as b^y

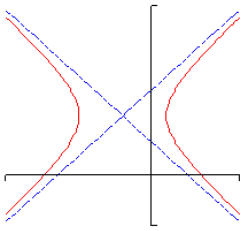
1.2.17 Parabola Equation:

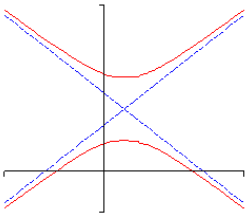
$$(y-k)^2 = 4a(x-h))$$



1.2.18 Hyperbola Equation:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$





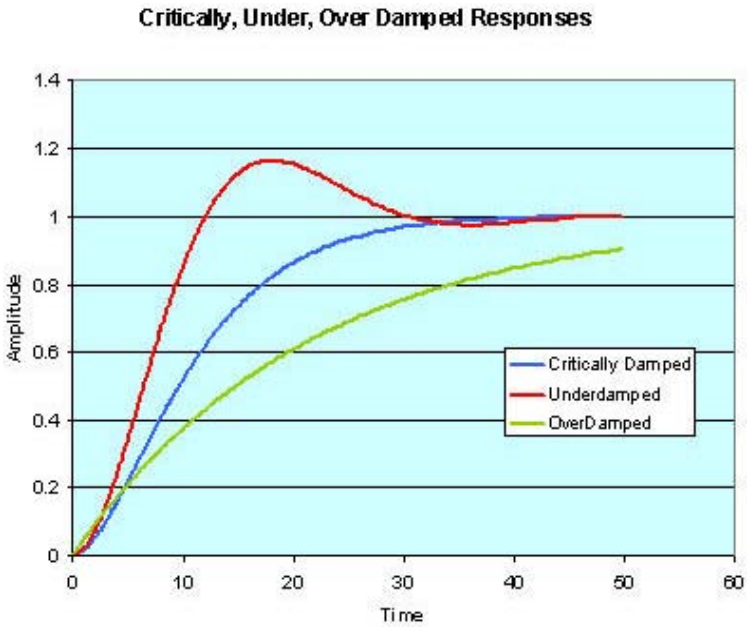
1.2.19 Laplace Transforms:

LaPlace Transforms:

The Laplace transform is very useful in the area of circuit analysis. It is often easier to analyze the circuit in its Laplace form, than to form differential equations.

The techniques of Laplace transform are not only used in circuit analysis, but also in

- Proportional-Integral-Derivative (PID) controllers
- DC motor speed control systems
- DC motor position control systems
- Second order systems of differential equations (underdamped, overdamped and critically damped)



Inverse of Laplace Transforms:

If $G(s) = \mathcal{L}\{g(t)\}$, then the **inverse transform** of $G(s)$ is defined as: $\mathcal{L}^{-1}G(s) = g(t)$

Some Properties of the Inverse Laplace Transform
Property 1: Linearity Property

$\mathcal{L}^{-1}\{a\,G_1(s) + b\,G_2(s)\} = a\,g_1(t) + b\,g_2(t)$

Property 2: Shifting Property

If $\mathcal{L}^{-1}G(s) = g(t)$, then $\mathcal{L}^{-1}G(s - a) = e^{at}g(t)$

Property 3

If $\mathcal{L}^{-1}G(s) = g(t)$, then $\mathcal{L}^{-1}\left\{\frac{G(s)}{s}\right\} = \int_0^t g(t)dt$

Property 4

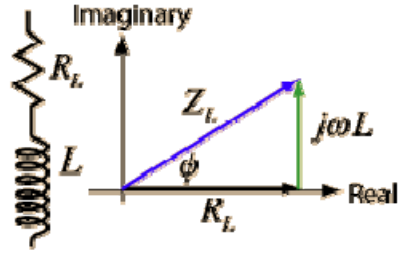
If $\mathcal{L}^{-1}G(s) = g(t)$, then $\mathcal{L}^{-1}\{e^{-as}G(s)\} = u(t - a) \bullet g(t - a)$

For Laplace tables Reference Section 25.12

1.2.20 Electrical Equations:

- Ohm's Law (DC): $E = I \times R$
  - Resistors in parallel: $R_T = \frac{1}{R_1^{-1} + R_2^{-1} + R_3^{-1} + R_N^{-1}}$
- Ohm's Law (AC): $E_{RMS} = I_{RMS} \times Z$
  - Inductive Reactance: $X_L = \omega L = 2\pi fL$
  - Inductive Capacitance: $X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$

This depicts the phasor diagrams and complex impedance expressions for RL and RC circuits in polar form. They can also be expressed in cartesian form.

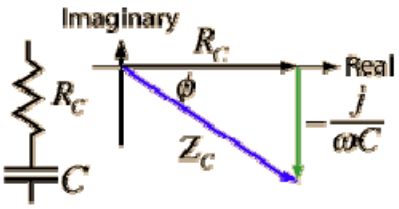


Cartesian form: $Z_L = R_L + j\omega L$

Polar form: $Z_L = |Z_L|e^{j\phi}$

where $|Z_L| = \sqrt{R_L^2 + \omega^2 L^2}$

$\phi = \tan^{-1}\frac{\omega L}{R_L}$



Cartesian form: $Z_C = R_C - \frac{j}{\omega C}$

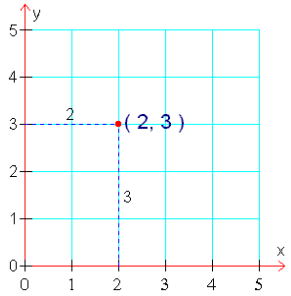
Polar form: $Z_C = |Z_C|e^{j\phi}$

where $|Z_C| = \sqrt{R_C^2 + \frac{-1}{\omega C R_C}}$

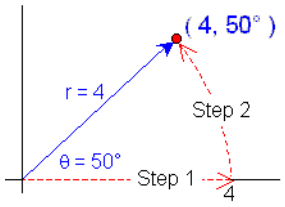
$\phi = \tan^{-1}\frac{-1}{\omega C R_C}$

o Polar to Rectangular Conversion:

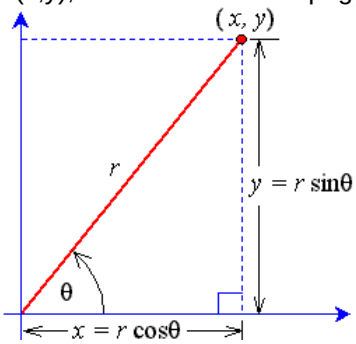
Rectangular coordinates are in the form (x,y), where 'x' and 'y' are the horizontal and vertical distances from the origin:



Polar coordinates are in the form: (r,q), where 'r' is the distance from the origin to the point, and 'q' is the angle measured from the positive 'x' axis to the point:



To convert between polar and rectangular coordinates, make a right triangle to the point (x,y), like shown on next page:



Polar to Rectangular:

From the diagram above, these formulas convert polar coordinates to rectangular coordinates:

x = r cosθ, y = r sinθ

So the polar point: (r,q) can be converted to rectangular coordinates like this:

(r cosθ, r sinθ) ⇨ (x, y)

***Example**: A point has polar coordinates: (5, 30º). Convert to rectangular coordinates.*
***Solution**: (x,y) = (5cos30º, 5sin30º) = (4.3301, 2.5)*

Rectangular to Polar:

Again, from the diagram above, these formulas convert **rectangular** coordinates **to polar** coordinates:

By the rule of Pythagoras: $r = \sqrt{x^2 + y^2}$ and $\tan\theta = \frac{y}{x}$ so $\therefore\ q = \tan^{-1}\left(\frac{y}{x}\right)$

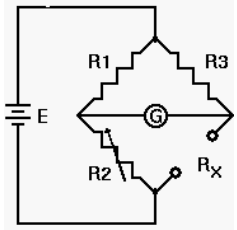
So the **rectangular** point: (x,y) can be converted **to polar** coordinates like shown on the next page:

$$\left[\sqrt{x^2+y^2}\ ,\ \tan^{-1}\left(\frac{y}{x}\right)\right] \quad \Rightarrow \quad (r,\ \theta)$$

Example: *A point has rectangular coordinates: (3, 4). Convert to polar coordinates.*
Solution: *r = square root of (3² + 4²) = 5, q = tan-1(4/3) = 53.13º*
so (r,q) = (5, 53.13º)

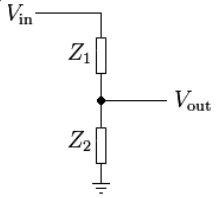
1.2.21 Wheatstone Bridge:

The wheatstone bridge is an instrument used to measure electrical resistance by means of balancing a bridge circuit. The bridge circuit contains two legs, one of which contains the unknown resistance. Variations in wheatstone bridge can be employed to measure inductance, capacitance, and impedance also



In its basic application, a dc voltage (E) is applied to the Wheatstone Bridge, and a galvanometer (G) is used to monitor the balance condition. The values of R1 and R3 are precisely known, but do not have to be identical. R2 is a calibrated variable resistance, whose current value may be read from a dial or scale.
An unknown resistor, R_X, is connected as the fourth side of the circuit, and power is applied. R2 is adjusted until the galvanometer, G, reads zero current. At this point, $R_X = R2 \times R3/R1$.
This circuit is most sensitive when all four resistors have similar resistance values. However, the circuit works quite well in any event. If R2 can be varied over a 10:1 resistance range and R1 is of a similar value, we can switch decade values of R3 into and out of the circuit according to the range of value we expect from R_X. Using this method, we can accurately measure any value of R_X by moving one multiple-position switch and adjusting one precision potentiometer.

Voltage Divider Rule:
Simple linear circuit that produces an output voltage (V_{out}) that is a fraction of its input voltage (V_{in}). **Voltage division** refers to the partitioning of a voltage among the components of the divider.
A simple example of a voltage divider consists of two resistors in series or a potentiometer. It is commonly used to create a reference voltage, and may also be used as a signal attenuator at low frequencies.

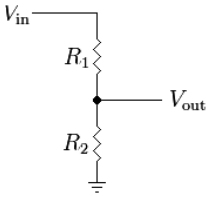


Voltage Divider

$$V_{OUT} = \frac{Z_2}{Z_1 + Z_2} \times V_{IN}$$

Proof (Ohm's Law)

$$V_{IN} = I \times (Z_1 + Z_2)$$
$$V_{OUT} = I \times Z_2$$
$$I = \frac{V_{IN}}{Z_1 + Z_2} \quad \therefore \quad V_{OUT} = V_{IN} \times \frac{Z_2}{Z_1 + Z_{2\,IN}}$$

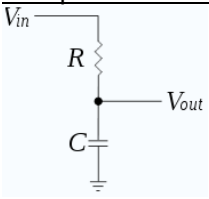


Resistive Voltage Divider

$$V_{OUT} = \frac{R_2}{R_1 + R_2} \times V_{IN}$$

A resistive divider is a special case where both impedances, Z_1 and Z_2, are purely resistive
Substitute $Z_1 = R_1$ and $Z_2 = R_2$ into the previous expression:

Low-pass RC filter:



Consider a divider consisting of a resistor and capacitor as shown above. Comparing with the general case, we see Z_1 = R and Z_2 is the impedance of the capacitor, given by:

$$Z_2 = jX_C = \frac{1}{j\omega C} = \frac{1}{j2\pi fC}$$

X_C = Capacitive Reactance
C = is the capacitance of the capacitor
j = the imaginary unit
ω = (omega) is the radian frequency of the input voltage.

This divider will then have the voltage ratio:

$$\frac{V_{OUT}}{V_{IN}}\frac{Z_2}{Z_1+Z_2} = \frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C}+R} = \frac{1}{1+j\omega RC}$$

The product of τ (tau) = RC is called the *time constant* of the circuit. The ratio then depends on frequency, in this case decreasing as frequency increases. *This circuit is, in fact, a basic (first-order) lowpass filter*. The ratio contains an imaginary number, and actually contains both the amplitude and phase shift information of the filter. To extract just the amplitude ratio, calculate the magnitude of the ratio, that is:

$$\left|\frac{V_{OUT}}{V_{IN}}\right| = \frac{1}{\sqrt{1+(\omega RC)^2}}$$

Inductive divider:

Inductive dividers split DC input according to resistive divider rules above. Inductive dividers split AC input according to inductance:

$$V_{OUT} = V_{IN} \times \frac{L_2}{L_1+L_2}$$

The above equation is for ideal conditions. In the real world the amount of mutual inductance will alter the results.

Capacitive divider:
Capacitive dividers do not pass DC input.
For an AC input a simple capacitive equation is:

$$V_{OUT} = V_{IN} \times \frac{C_2}{C_1+C_2}$$

Capacitive dividers are limited in current by the capacitance of the elements used. This effect is opposite to resistive division and inductive division.

1.2.22 Mass Flow – Gas Equations:

Substitute Q for V/t: *Substitute for Q:*

$$w = \frac{m}{t} = \frac{M}{10^3 R}\left(\frac{V}{t}\right)\left(\frac{p}{T}\right) \qquad w = \frac{MQ}{10^3 R}\left(\frac{p}{T}\right) \qquad Q = k\sqrt{D}\,;\, k = \frac{Mk_f}{10^3 R}$$

Simplified: $w = k\sqrt{D\left(\frac{p}{T}\right)}$

w = Mass flow rate (kg/sec)
Q = Volume flow rate (m^3/sec)
p = Absolute pressure (pascal)
T = Absolute temperature (Kelvin)
M = MW (g/mol)
R = Universal gas constant = 8.314 J ÷ (K x mol)
D = Flowmeter D/P (pascal)
k = Mass flow proportionality constant
k_f = Flowmeter proportionality constant

$$M = \rho AV$$

M = Mass flow rate (lbs/sec)
A = Cross sectional area (ft^2)
ρ = Fluid density (lbs/ft^3)
V = Velocity (ft/sec)

Density will vary in reverse proportion to temperature, and in direct proportion to pressure.

1.2.23 Volume Formulas:

- Sphere: $\frac{4}{3}\pi r^3$
- Right Circular Cone: $\frac{1}{3}\pi r^2 h$
- Right Circular Cylinder: $\pi r^2 h$
- Pyramid: $\frac{1}{3}A \bullet h$ (A = Area of base)

1.2.24 Surface Area Formulas:

- Sphere: $4\pi r^2$
- Right Circular Cone: $\pi r^2 + \pi rs$
- Right Circular Cylinder: $2\pi rh + 2\pi r^2$
- Pyramid: Area of Base + Area of the (4) Triangular Sides

2. Sizing Calculations

2.1 Orifice Plate Sizing:

Beta Ratio (β): d / D

- **Liquid Orifice** (LK Spink)

$$S = \frac{Q_M * G_b}{ND^2\sqrt{G_F\sqrt{h_M}}}$$

Ratios

$$\boxed{\frac{F_2}{F_1} = \sqrt{\frac{\Delta P_2}{\Delta P_1}}} \qquad \boxed{\frac{P_2}{P_1} = \left(\frac{F_1}{F_2}\right)^2} \qquad \boxed{A_1V_1 = A_2V_2}$$

Basic Equation: $Q_M = 5.667SD^2\sqrt{\frac{h_M}{G_F}}$

Q_M = Maximum flow in **GPM**
G_b = Base S.G. [(S.G. of liquid @ 60°F (Water @ 60°F = 1)]
N = 5.667 for GPM
D = Pipe ID in inches
G_F = Flowing SG of liquid @ flowing temperature (see Crane A-6)
h_M = Meter differential in "WC
S = Orifice ratio (reference Spink pg. 167 Table 12 for corresponding β)

- **Liquid Orifice** (Cameron Hydraulic Book)

$$Q = 19.636Cd_1^2\sqrt{h}\sqrt{\frac{1}{1-\left(\frac{d_1}{d_2}\right)^4}}$$ Where d1 ÷ d2 > 0.3

$$Q = 19.636Cd_1^2\sqrt{h}$$ Where d1 ÷ d2 < 0.3

Q = Flow (in GPM)
d1 = Diameter of orifice or nozzle opening (in inches)
d2 = Diameter of pipe in which orifice is placed (in inches)
h = Differential head at orifice (in FEET of liquid)
C = Discharge coefficient (typical values below for water)(Ref. Cameron Book pg 2-8): Sharp Edge: C = 0.61
Square Edge: C = 0.61
Well Rounded: C = 0.98

- **Steam or Gas Orifice** (LK Spink)

$$S = \frac{W}{359 * D^2\sqrt{S_W\sqrt{h_M}}}$$

Basic Equation Steam ↓

$$W_{lbs/hr} = 359SD^2\sqrt{h_mS_W}$$

Basic Equation Gas

$$Q_{scfh} = 218.4SD^2\frac{T_{abs}}{P_{abs}}\sqrt{\frac{h_mP_f}{T_fG_f}} \qquad T_f = T_{abs}\ in\ °R$$

$P_{abs} = 14.7$

SG_{gas}=MW÷29
W = Flow in lbs / hr
S_W = Specific Weight of vapor in lbs/ft^3 = 1 ÷ Specific Volume
For Steam, reference Crane A12 thru A18 (use 1/specific volume)
For Gas, reference Crane A-8, column rho 'ρ')
h_M = Meter differential in "WC
D = Pipe ID in inches
S = Orifice ratio (reference Spink pg. 167 Table 12 for corresponding β)

A rule of thumb to use in gas flow is that critical flow is reached when the downstream pipe tap registers an absolute pressure to approximately 50% or less than the upstream pipe tap.

2.2 Venturi Sizing (liquid):

$$Q_m = \frac{CA_{throat}\sqrt{2\rho\Delta P}}{\sqrt{1-\mathrm{B}^4}} \qquad Q_v = \frac{Q_m}{\rho}$$

A = Area of Throat
C = Coefficient of Discharge
ΔP = Differential Pressure
Qm = Mass Flow Rate
Qv = Volumetric Flow Rate
P = Density

(From Cameron Hydraulic Book):

$$Q = 19.05\, d_1^{\,2}\sqrt{H}\sqrt{\frac{1}{1-\left(\frac{d_1}{d_2}\right)^4}}$$ for any Venturi Tube

$Q = 19.17\, d_1^{\,2}\sqrt{H}$ for Venturi Tube in which d1 = 0.33d2

Q = Flow (in GPM)
d_1 = Diameter of Venturi Throat (in inches)
d_2 = Diameter of Main Pipe (in inches)
H = Diff. in head between upstream end and throat (in feet)

2.3 V-Cone Sizing:

$$\mathrm{B} = \frac{\sqrt{D^2-d^2}}{D} \qquad k_1 = \frac{\pi}{576}\sqrt{2G_C}\,\frac{D^2\mathrm{B}^2}{\sqrt{1-\mathrm{B}^4}}C_F \qquad ACFS = k_1\sqrt{\frac{5.197\Delta P}{\rho}}$$

B = V-Cone Beta Ratio
K_1 = Flow Constant
C_G = Gravitational Constant
D = Pipe ID
d = Cone Diameter
C_F = Flowmeter Coefficient (use 1 if unknown)

2.4 Elbow Flowmeter Sizing:

$$S = 0.68\sqrt{\frac{r_b}{D}} \qquad Q_n = SND^2F_a\frac{\sqrt{G_f}}{G_l}\sqrt{h_w} \text{ **OR** } h_w = \left[\frac{Q_nG_l}{SNDF_a\sqrt{G_f}}\right]^2$$

S = Elbow ratio (reference Spink pg. 180 Table 14 for corresponding S)
r_b = Radius to the center of mass of the fluid flowing in the elbow from the center of curvature of the bend.
D = Elbow ID
N = Constant (reference Spink pg. 154 Table 4 for corresponding N)
F_a = Ratio to correct for thermal expansion of elbow (reference Spink pg. 156 Table 7)
G_f = S.G. at flowing temperature
C_l = S.G. at base temperature
H_w = Operating D/P in "WC
Q_n = Operating Flow Rate

2.5 Pitot / Annubar Sizing:

Liquid:

$$\Delta P = \frac{Q^2 S_f}{K^2 D^4 32.14}$$

ΔP = D/P in "WC
Q = Flowrate in GPM.
S_f = S.G. at flowing conditions
K = Flow Coefficient (use 1 if unknown)
D = Pipe ID

Steam or Gas:

$$\Delta P = \frac{Q^2 (lb/hr)}{K^2 D^4 \rho 128900} \quad \text{or} \quad \Delta P = \frac{Q^2 (scfm) S_s T_R}{K^2 D^4 P 16590}$$

ΔP = D/P in "WC
S_s = S.G. at 60°F
K = Flow Coefficient (use 1 if unknown)
D = Pipe ID
ρ = Density (in lb/ft^3)
P = Static Line Pressure (in PSIA)
T_R = Temperature in °R

2.6 Magmeter Sizing:

$$Q_V = v \times A = \frac{U_e}{B \times L} \times A$$

Q_V = Flowrate in GPM.
v = Flow velocity
U_e = Induced Measuring Voltage
A = Pipe Cross-sectional Area
B = Magnetic Field Strength
L = Distance Between Electodes

2.7 Weir Sizing:

(From Cameron Hydraulic Book):

Weir (Rectangular Notch):

$$Q = 13.33(L - 0.2H)H^{1.5}$$ Francis Formula (Ref Cameron Book pg 2-10)

Q = FT^3 of water flowing per second
L = Length of weir opening in feet (should be 4 to 8 times H)
H = Head on weir in feet (to be measured 6ft back of weir opening)

Weir (V - Notch):

$$Q = C \times 0.2667 \times L \times H\sqrt{2gH}$$ Thompson Formula (Ref Cameron Book pg 2-11)

Q = Flow of water in FT^3/second
L = Width of notch in feet at H distance above apex
H = Head of water above apex of notch (in feet)
C = Constant varying with conditions, 0.57 for the table in Cameron Book

2.8 Control Valve Sizing:

2.8.1 Liquid (Turbulent Flow):

Volumetric Flow Rate: (From Fisher Control Valve handbook)

$$C_V = \frac{Q}{N_1 F_P}\sqrt{\frac{G_f}{P_1 - P_2}} \quad \textbf{OR} \quad Q = N_1 F_P C_V \sqrt{\frac{P_1 - P_2}{G_f}}$$

Mass Flow Rate:

$$C_v = \frac{w}{N_6 F_P \sqrt{(P_1 - P_2)\gamma_1}} \quad \textbf{OR} \quad w = N_6 F_P C_V \sqrt{(P_1 - P_2)\gamma_1}$$

General Equation: $C_V = Q\sqrt{\frac{G}{\Delta P}}$ Q in GPM; G = SG

Q = Volumetric Flow Rate
w = Weight or Mass Flow Rate
G_f = Liquid Specific Gravity
P_1 = Inlet Pressure in PSIA
P_2 = Outlet Pressure in PSIA
N = Numerical Constants of Units of Measure Used (Ref. Table below)
γ_1 = Specific Weight (upstream conditions)
d = Nominal Valve Size
D = Pipe ID

F_P = Piping Geometry Factor $F_P = \sqrt{\left(\frac{\Sigma K \times C_V{}^2}{N_2 d^4} + 1\right)}$

Inlet Reducer Only: $K_1 = 0.5\left(1 - \frac{d^2}{D^2}\right)^2$

Outlet Reducer Only: $K_2 = 1.0\left(1 - \frac{d^2}{D^2}\right)^2$

When Inlet & Outlet Reducers are same size: $K_1 + K_2 = 1.5\left(1 - \frac{d^2}{D^2}\right)^2$

Numerical Constants N for Liquid Flow:

Constant		Units Used in Equations					
N		w	Q	$P_1\Delta P$	d,D	γ_1	v
N_1	0.0865		m^3/h	kPa			
	0.865		m^3/h	Bar			
	1.00		gpm	psia			
N_2	0.00214				mm		
	890				in		
N_4	76000		m^3/h		mm		Centistokes*
	17300		gpm		in		Centistokes*
N_6	2.73	kg/h		kPa		kg/m^3	
	2.73	kg/h		Bar		kg/m^3	
	63.3	lb/h		psia		lb/ft^3	

* To convert m^2/s to centistokes multiply by 10^6
To convert centipoise to centistokes, divide by G_f

Chocked Flow & Noise:

- Valves in flashing service can be recognized using the comparison below:
  When $P_2 < P_V$ and $\Delta P_{(choked)} < \Delta P_{(actual)}$ = Flashing Service
- Valves in cavitation service can be recognized using the comparison below:
  When $P_2 > P_V$ and $\Delta P_{(choked)} < \Delta P_{(actual)}$ = Cavitation Service

Check for critical flow by calculating the allowable ΔP

$$\Delta P_{allow} = F_L{}^2 \left(P_1 - F_F P_V\right)$$

F_L = Pressure Recovery Coefficient (globe ~ 0.85; ball ~ 0.6)
P_1 = Inlet Pressure in PSIA
P_V = Liquid Vapor Pressure in PSIA
P_C = Pressure at Thermodynamic Critical Point (in PSIA)(eg Wtr = 3206)
F_F = Liquid Critical Pressure Ratio Factor

$$F_F = 0.96 - 0.28\sqrt{\frac{P_V}{P_C}}$$

If $\Delta P > \Delta P_{allow}$ then use this equation: $C_V = \frac{Q}{F_L}\sqrt{\frac{G_F}{P_1 - F_F}P_V}$

2.8.2 Steam:

2.8.2.1 Saturated Steam:

Basic equation

$$C_V = \frac{W}{63.3 \times Y\sqrt{XP_1S_W}} \qquad W = N_1N_6F_PC_VY\sqrt{xP_1S_W} \qquad C_V = \frac{W}{N_1N_6F_PY\sqrt{xP_1S_W}}$$

N_1 = Always = 1 for PSIA
N_6 = 63.3
W = Flow Rate in lbs/hr
P_1 = Inlet Pressure in PSIA
S_w = Specific Weight in lbs/ft^3 (1/specific volume) (See Crane A12 thru 15 and use the inverse of specific volume)
Y = Expansion Factor

$$Y = 1 - \frac{x}{3X_T}$$

x = Pressure Drop Ratio

$$x = \frac{\Delta P}{P_1}$$

$X_T = 0.85F_L{}^2$ (F_L depends on valve style: globe = 0.85; ball = 0.060)

If $\Delta P/P_1 < 0.1$ the equation above can be simplified to:

$$C_V = \frac{W}{2.1\sqrt{\Delta P\left(P_1 + P_2\right)}}$$

The flow coefficient must be corrected for superheated steam flow:

$$C_V = \frac{W\left(1 + 0.0007T_{SH}\right)}{2.1\sqrt{\Delta P\left(P_1 + P_2\right)}}$$

T_{SH} = Steam superheat in °F above saturation temp.

2.8.3 Gas (Compressible Fluid):

For Volumetric Flow Rate Units:

S.G. of Gas Known:

$$C_V = \frac{Q}{N_7 F_P P_1 Y \sqrt{\frac{x}{C_g T_1 Z}}}$$

MW of Gas Known:

$$C_V = \frac{Q}{N_9 F_P P_1 Y \sqrt{\frac{x}{M T_1 Z}}}$$

For Mass Flow Rate Units:

Specific Weight of Gas Known:

$$C_v = \frac{w}{N_6 F_P Y \sqrt{x P_1 \gamma_1}}$$

MW of Gas Known:

$$C_V = \frac{w}{N_8 F_P P_1 Y \sqrt{\frac{xM}{T_1 Z}}}$$

Aerodynamic Noise Prediction:

$$C_g = 40 \times C_V \sqrt{X_T}$$

Q = Volumetric Flow Rate
w = Weight or Mass Flow Rate
M = Molecular Weight (MW of air =
C_g = SG of Gas *Cg = MW ÷ 29*
P_1 = Inlet Pressure in PSIA
T_1 = Inlet Temperature in °R
N = Numerical Constants of Units of Measure Used (Ref. Table on next page)
γ_1 = Specific Weight (upstream conditions)
F_K = Ratio of Specific Heats (use 1 if unknown)
Z = Compressibility Factor (1.0 for pressures less than 100 psia – ideal gas)
d = Nominal Valve Size
D = Pipe ID
Y = Expansion Factor
X = Pressure Drop Ratio

$$X = \frac{\Delta P}{P_1}$$

$X_T = 0.85 F_L^2$ (F_L depends on valve style: globe = 0.85; ball = 0.060)

F_P = Piping Geometry Factor $F_P = \sqrt{\left(\frac{\Sigma K \times C_V{}^2}{N_2 d^4} + 1\right)}$

Inlet Reducer Only: $K_1 = 0.5\left(1 - \frac{d^2}{D^2}\right)^2$

Outlet Reducer Only: $K_2 = 1.0\left(1 - \frac{d^2}{D^2}\right)^2$

When Inlet & Outlet Reducers are same size: $K_1 + K_2 = 1.5\left(1 - \frac{d^2}{D^2}\right)^2$

Numerical Constants N for Gas Flow:

Constant		Units Used in Equations					
N		w	Q	$P_1\Delta P$	d,D	γ_1	T_1
N_5	0.00241				mm		
	1000				in		
N_2	2.73	kg/h		kPa		kg/m^3	
	27.3	kg/h		Bar		kg/m^3	
	63.3	lb/h		psia		lb/ft^3	
N_7	4.17		m^3/h	kPa			°K
	417		m^3/h	Bar			°K
	1360		scfh	psia			°R
N_8	0.948	kg/h		kPa			°K
	94.8	kg/h		Bar			°K
	19.3	lb/h		psia			°R
N_9	22.5		m^3/h	kPa			°K
	2250		m^3/h	Bar			°K
	7320		scfh	psia			°R

2.7 Pressure Relief Valve Sizing:

2.7.1 Gas & Vapor Service:

10% Over-Pressure (lb/hr) *ASME VIII Code Equation*

$$A = \frac{W_{lb/hr}\sqrt{TZ}}{CKP_1K_b\sqrt{M}\,(CCF)} \qquad W = K_bCKAP_1\sqrt{\frac{M}{TZ}} \qquad C = 520\sqrt{k\left(\frac{2}{k+1}\right)^{\frac{k+1}{k-1}}}$$

Combination derating factor when used in conjunction with rupture disk = 0.9

A = Minimum required orifice area, in^2
W = Required relieving rate, lb/hr
T = Relieving temperature, °R
Z = Compressibility factor (use 1 if unknown)
M = Molecular weight
C = Gas constant = a function of (Cp / Cv) (use 315 if unknown)*(see equation above)*
Cp = specific heat at constant pressure (consistent units)
Cv = specific heat at constant volume (consistent units)
k = Specific heats ratio
K = Coefficient of discharge, (0.975)
K_b = Backpressure correction factor, (use 1.0 for atmospheric)
P_1 = Relief pressure + allowable accumulation, psia
CCF = Combination De-Rating Factor (1 if not combination, otherwise 0.9)

10% Over-Pressure (scfm)

$$A = \frac{W_{scfm}\sqrt{TGZ}}{1.175CKP_1K_b(CCF)}$$

G = S.G. of the gas or vapor

2.7.2 Steam Service:

10% Over-Pressure (lb/hr)

$$A = \frac{W_{lb/hr}}{51.5KP_1K_{SH}K_nK_b}$$

K_n = Correction factor for dry saturated steam
= 1.0 where P_1 < 1515 psia

$$\frac{0.1906P_1 - 1000}{0.2292P_1 - 1061}$$ Where P1 > 1515 psia and ≤ 3215 psia

K = Coefficient of Discharge (0.975)
K_b = Vapor gas correction for constant backpressure above critical pressure
K_{SH} = Superheat correction factor (for saturated use 1.0)(reference table on next page)

Superheat Correct Factor (K_{SH}) Table:

Set pressure (psi g)	Temperature (°F)									
	300	400	500	600	700	800	900	1 000	1 100	1 200
15	1.00	0.98	0.93	0.88	0.84	0.80	0.77	0.74	0.72	0.70
20	1.00	0.98	0.93	0.88	0.84	0.80	0.77	0.74	0.72	0.70
40	1.00	0.99	0.93	0.88	0.84	0.81	0.77	0.74	0.72	0.70
60	1.00	0.99	0.93	0.88	0.84	0.81	0.77	0.75	0.72	0.70
80	1.00	0.99	0.93	0.88	0.84	0.81	0.77	0.75	0.72	0.70
100	1.00	0.99	0.94	0.89	0.84	0.81	0.77	0.75	0.72	0.70
120	1.00	0.99	0.94	0.89	0.84	0.81	0.78	0.75	0.72	0.70
140	1.00	0.99	0.94	0.89	0.85	0.81	0.78	0.75	0.72	0.70
160	1.00	0.99	0.94	0.89	0.85	0.81	0.78	0.75	0.72	0.70
180	1.00	0.99	0.94	0.89	0.85	0.81	0.78	0.75	0.72	0.70
200	1.00	0.99	0.95	0.89	0.85	0.81	0.78	0.75	0.72	0.70
220	1.00	0.99	0.95	0.89	0.85	0.81	0.78	0.75	0.72	0.70
240		1.00	0.95	0.90	0.85	0.81	0.78	0.75	0.72	0.70
260		1.00	0.95	0.90	0.85	0.81	0.78	0.75	0.72	0.70
280		1.00	0.96	0.90	0.85	0.81	0.78	0.75	0.72	0.70
300		1.00	0.96	0.90	0.85	0.81	0.78	0.75	0.72	0.70
350		1.00	0.96	0.90	0.86	0.82	0.78	0.75	0.72	0.70
400		1.00	0.96	0.91	0.86	0.82	0.78	0.75	0.72	0.70
500		1.00	0.96	0.92	0.86	0.82	0.78	0.75	0.73	0.70
600		1.00	0.97	0.92	0.87	0.82	0.79	0.75	0.73	0.70
800			1.00	0.95	0.88	0.83	0.79	0.76	0.73	0.70
1 000			1.00	0.96	0.89	0.84	0.78	0.76	0.73	0.71
1 250			1.00	0.97	0.91	0.85	0.80	0.77	0.74	0.71
1 500			1.00	1.00	0.93	0.86	0.81	0.77	0.74	0.71

2.7.3 Liquid Service:

Spring Loaded:

$$A = \frac{Q_g \sqrt{G}}{28.14 K_U K_W \sqrt{(\Delta P)}}$$

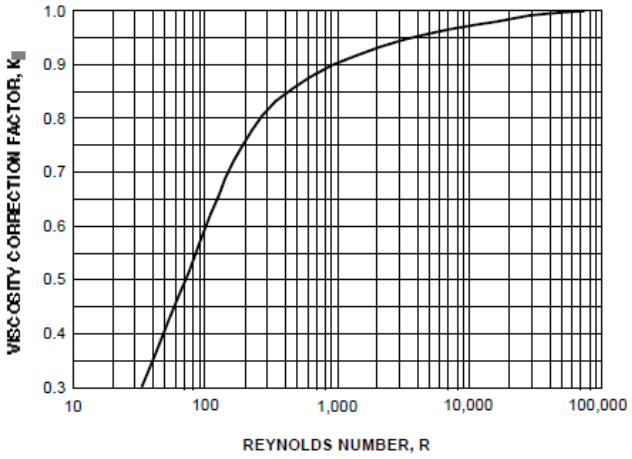
Pilot Operated:

$$A = \frac{Q_g \sqrt{G}}{36.81 K_U \sqrt{(\Delta P)}}$$

Basic Equation

$$A = \frac{Q_g}{27.2 \sqrt{\frac{\Delta P}{G}} (K_P K_U K_W)}$$

Q_g = Relieving rate in GPM
G = S.G. of liquid at flowing conditions
ΔP = Set pressure (psig) + over pressure – back pressure (PSID)
K_p = Overpressure correction for liquid = 0.60
K_w = (Bellows Seal Valves Only) Variable or constant backpressure factor
K_U = Correction factor due to viscosity at flowing conditions



2.8 Rupture Disk Sizing:

Vapor: *Sonic Flow* Subsonic Flow

$$A = 9.02\frac{V_A}{C}\sqrt{\frac{M_W}{ZT}}$$ *Volume-actual*

$$C = 520\sqrt{k\left[\frac{2}{k+1}\right]^{\frac{k+1}{k-1}}}$$

$$C = 735\sqrt{\frac{k}{k-1}\left[\left(\frac{P_2}{P_1}\right)^{\frac{2}{k}} - \left(\frac{P_2}{P_1}\right)^{\frac{k+1}{k}}\right]}$$

$$A = \frac{V_S}{3.92CP_1}\sqrt{ZM_W T}$$ *Volume-standard*

Liquid:

$$A = \frac{Q}{186}\sqrt{\frac{\rho}{P}}$$ *Volume* $$A = \frac{W}{1492\sqrt{P\rho}}$$ *Mass*

Steam:

$$A = \frac{W}{51.5KP_1}$$ $$A = \frac{W}{51.5\left(\frac{0.1906P_1 - 1000}{0.2292P_1 - 1061}\right)}$$ $$A = \frac{W(1 - 0.012 \times \Delta X)}{KP_1}$$

Dry Sat ≤1500psig *1500 < Dry Sat < 3200psig* *Wet Steam*

ρ = density in lbs/ft^3 *(to use SG instead of ρ: SG x 62.37)*
C = Gas Constant (function of ratio of specific heat)
Z = Compressibility Factor
A = Area in square inches
W = Lbs/hour
M_W = Molecular Weight
P_1 = Inlet Pressure PSIA
Q = Relieving Rate (in GPM)
SG = Liquid SG, where water = 1.0
T = Relieving Temperature (in °R)
K = 0.62 per ASME code
k = Ratio of Specific Heats
ΔX = (100 - % steam quality)

2.9 Pressure Regulator Sizing:

2.9.1 Steam or Gas:

2.9.1.1 Steam flows when P_1 is < 1000 psig:

$$Q_{lb/hr} = \left[\left(\frac{C_S P_1}{1 + 0.00065T_{SH}}\right)\right] SIN\left[\left(\frac{3417}{C_1}\right)\sqrt{\frac{\Delta P}{P_1}}\right] Deg$$

$C_1 = C_G / C_V$
C_S = Steam sizing coefficient C_G / 20
C_G = Gas sizing coefficient
T_H = Degrees of superheat, °F
P_1 = Inlet Pressure
$Q_{lb/hr}$ = Steam or vapor flow rate, pounds per hour

2.9.1.2 Predict flow for perfect or non-perfect gas sizing applications

For any vapor including steam, at any service condition when fluid density is known:

$$Q_{lb/hr} = 1.06\sqrt{d_1 P_1} \times C_G SIN\left(\frac{3417}{C_1}\right)\sqrt{\frac{\Delta P}{P_1}} Deg$$

d_1 = Density of steam or vapor at inlet, lb/ft^3

2.9.1.3 Predict flow for either high or low recovery valves:

for any gas adhering to the perfect gas lows, and under any service conditions:
Universal Gas Sizing Equations

$$Q_{SCFH} = \sqrt{\frac{520}{GT}} C_G P_1 SIN\left[\left(\frac{59.64}{C_1}\right)\sqrt{\frac{\Delta P}{P_1}}\right] Rad$$

OR

$$Q_{SCFH} = \sqrt{\frac{520}{GT}} C_G P_1 SIN\left[\left(\frac{3417}{C_1}\right)\sqrt{\frac{\Delta P}{P_1}}\right] Deg$$

$C_1 = C_G / C_V$
C_G = Gas sizing coefficient
T = Absolute temperature of gas at inlet, °R
P_1 = Inlet Pressure
G = S.G. at flowing conditions
Q_{SCFH} = Gas flow rate, SCFH

2.9.1.4 Very low pressure drop:

(ΔP/P1) ratios of 0.02 or less:

$$Q_{SCFH} = 59.64 \times C_V P_1 \sqrt{\frac{\Delta P}{P_1}} \sqrt{\frac{520}{GT}}$$

P_1 = Inlet Pressure
C_V = Liquid sizing coefficient
G = S.G. at flowing conditions
T = Temperature in °R

2.9.1.5 Determine critical flow capacity:

at a given inlet pressure

$$Q_{CRIT} = C_G P_1 \sqrt{\frac{520}{GT}}$$

C_G = Gas sizing coefficient
T = Absolute temperature of gas at inlet, °R
P_1 = Inlet Pressure
G = S.G. at flowing conditions
Q_{CRIT} = Critical flow rate, SCFH

2.9.2 Liquid:

2.9.2.1 Basic liquid sizing equation:

$$Q = C_V \sqrt{\frac{\Delta P}{G}} \quad \textbf{OR} \quad C_V = Q\sqrt{\frac{G}{\Delta P}}$$

C_V = Valve sizing coefficient
P_1 = Inlet Pressure
P_2 = Outlet Pressure
$\Delta P = P_1 - P_2$
G = S.G. at flowing conditions
Q = Liquid flow rate, GPM

2.10 Voltage Drop:

2.10.1 DC

$$V_d = \left(\frac{2 \times L}{1000}\right) \times I \times R$$

2.10.2 AC

Single Phase:

$$V_d = \left(\frac{2 \times L}{1000}\right) \times I \times Z_e$$ *(Note: Ze with PF = 100% is equal to dc Resistance)*

Three Phase:

$$V_d = \left(\frac{\sqrt{3} \times L}{1000}\right) \times I \times Z_e$$ *(Note: Ze with PF = 100% is equal to dc Resistance)*

Cable Sizing:

Single Phase:

$$cm = \left(\frac{2 \times L \times I \times k}{V_d}\right)$$

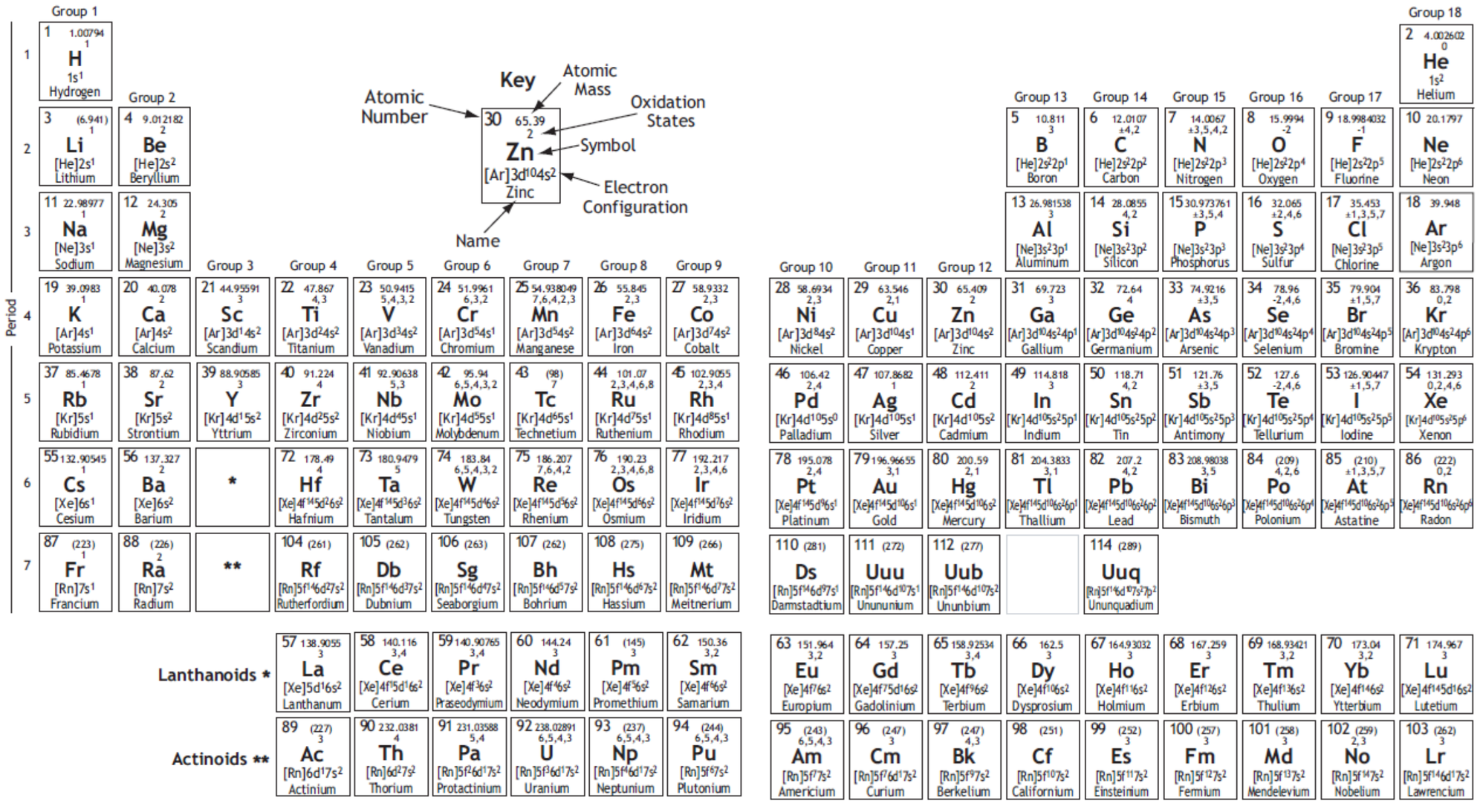
Three Phase:

$$cm = \left(\frac{\sqrt{3} \times L \times I \times k}{V_d}\right)$$

k = specific resistance of copper = 12 (for 75°C)

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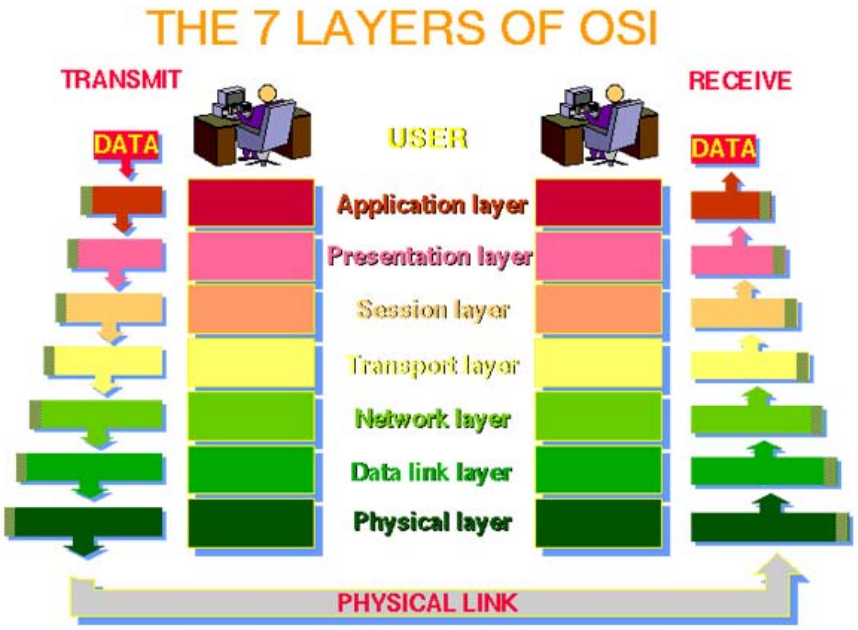
3 Periodic Table of Elements:



4 Networks

4.1 OSI Model:

- **O**pen **S**ystem **I**nterconnection
- In its most basic form, it divides network architecture into seven layers



- **Layer 7 (Application Layer):** This layer supports application and end-user processes. Communication partners are identified, quality of service is identified, user authentication and privacy are considered, and any constraints on data syntax are identified. Everything at this layer is application-specific. This layer provides application services for file transfers, e-mail, and other network software services. Telnet and FTP are applications that exist entirely in the application level. Tiered application architectures are part of this layer.
- **Layer 6 (Presentation Layer):** This layer provides independence from differences in data representation (e.g., encryption) by translating from application to network format, and vice versa. The presentation layer works to transform data into the form that the application layer can accept. This layer formats and encrypts data to be sent across a network, providing freedom from compatibility problems. It is sometimes called the *syntax layer*.
- **Layer 5 (Session Layer):** This layer establishes, manages and terminates connections between applications. The session layer sets up, coordinates, and terminates conversations, exchanges, and dialogues between the applications at each end. It deals with session and connection coordination.
- **Layer 4 (Transport Layer):** This layer provides transparent transfer of data between end systems, or hosts, and is responsible for end-to-end error recovery and flow control. It ensures complete data transfer. Layer 4 data units are also called packets, but when you're talking about specific protocols, like TCP, they're "segments" or "datagrams" in UDP (User Datagram Protocol). This layer is responsible for getting the entire message, so it must keep track of fragmentation, out-of-order packets, and other perils. Another way to think of layer 4 is that it provides end-to-end management of communication. Some protocols, like TCP, do a very good job of making sure the communication is reliable. Some don't really care if a few packets are lost--UDP is the prime example.
- **Layer 3 (Network Layer):** This layer provides switching and routing technologies, creating logical paths, known as *virtual circuits*, for transmitting data from node to node. Routing and forwarding are functions of this layer, as well as addressing, internetworking, error handling, congestion control and packet sequencing. If you're talking about an IP address, you're dealing with layer 3 and "packets" instead of layer 2's "frames." IP is part

of layer 3, along with some routing protocols, and ARP (Address Resolution Protocol). Everything about routing is handled in layer 3. Addressing and routing is the main goal of this layer.

- **Layer 2 (Data Link Layer):** At this layer, data packets are encoded and decoded into bits. It furnishes transmission protocol knowledge and management and handles errors in the physical layer, flow control and frame synchronization. The data link layer is divided into two sub layers: The *Media Access Control* (MAC) layer and the *Logical Link Control* (LLC) layer. The MAC sub layer controls how a computer on the network gains access to the data and permission to transmit it. The LLC layer controls frame synchronization, flow control and error checking. Layer two is Ethernet, among other protocols. Switches, as they're called nowadays, are bridges. They all operate at layer 2, paying attention only to MAC addresses on Ethernet networks. If you're talking about MAC address, switches, or network cards and drivers, you're in the land of layer 2. Hubs live in layer 1 land, since they are simply electronic devices with zero layer 2 knowledge.
- **Layer 1 (Physical Layer):** This layer conveys the bit stream - electrical impulse, light or radio signal -- through the network at the electrical and mechanical level. It provides the hardware means of sending and receiving data on a carrier, including defining cables, cards and physical aspects. Fast Ethernet, RS232, and ATM are protocols with physical layer components. Layer one is simply wiring, fiber, network cards, and anything else that is used to make two network devices communicate.

4.1.1 Acronyms / Definitions

Acronym	Definition
ATM	Asynchronous Transfer Mode In electronic digital data transmission systems, the network protocol Asynchronous Transfer Mode (ATM) encodes data traffic into small fixed-sized cells. The standards for ATM were first developed in the mid 1980s. The goal was to design a single networking strategy that could transport real-time video and audio as well as image files, text and email. Two groups, the International Telecommunications Union and the ATM Forum were involved in the creation of the standards. ATM, as a connection-oriented technology, establishes a virtual circuit between the two endpoints before the actual data exchange begins. ATM is a cell relay, packet switching protocol which provides data link layer services that run over Layer 1 links. This differs from other technologies based on packet-switched networks (such as the Internet Protocol or Ethernet), in which variable sized packets (known as frames when referencing Layer 2) are used. ATM exposes properties from both circuit- and packet switched networking, making it suitable for wide area data networking as well as real-time media transport. It is a core protocol used in the SONET/SDH backbone of the public switched telephone network.
FTP	File Transfer Protocol File Transfer Protocol (FTP) is a network protocol used to transfer data from one computer to another through a network such as the Internet. FTP is a file transfer protocol for exchanging and manipulating files over a TCP computer network. An FTP client may connect to an FTP server to manipulate files on that server.

Acronym	Definition
Telnet	Telecommunication Network Telnet is a client-server protocol, based on a reliable connection-oriented transport. Typically this protocol is used to establish a connection to TCP port 23, where a getty-equivalent program (telnetd) is listening, although Telnet predates TCP/IP and was originally run on NCP. The protocol has many extensions, some of which have been adopted as Internet standards. IETF standards STD 27 through STD 32 define various extensions, most of which are extremely common. Other extensions are on the IETF standards track as proposed standards.

4.1.1.1 OSI Example:

Pretend you're an operating system on a network. Your network card, operating at layers 1 and 2, will notify you when there's data available. The driver handles the shedding of the layer 2 frame, which reveals a bright, shiny layer 3 packet inside (hopefully). You, as the operating system, will then call your routines for handling layer 3 data. If the data has been passed to you from below, you know that it's a packet destined for yourself, or it's a broadcast packet (unless you're also a router). If you decide to keep the packet, you will unwrap it, and reveal a layer 4 packet. If it's TCP, the TCP subsystem will be called to unwrap and pass the layer 7 data onto the application that's listening on the port it's destined for.
When it's time to respond to the other computer on the network, everything happens in reverse. The layer 7 application will ship its data onto the TCP people, who will stick additional headers onto the chunk of data. In this direction, the data gets larger with each progressive step. TCP hands a valid TCP segment onto IP, who give its packet to the Ethernet people, who will hand it off to the driver as a valid Ethernet frame. And then off it goes, across the network. Routers along the way will partially disassemble the packet to get at the layer 3 headers in order to determine where the packet should be shipped. If the destination is on the local Ethernet subnet, the OS will simply ARP for the computer instead of the router, and send it directly to the host.

4.2 Network Hardware:

4.2.1 Switches:

- **Network switch** is a small hardware device that joins multiple computers together within one local area network (LAN). Technically, network switches operate at layer two (Data Link Layer) of the OSI model using MAC addresses.
- Network switches appear nearly identical to network hubs, but a switch generally contains more "intelligence" (and a slightly higher price tag) than a hub. Unlike hubs, network switches are capable of inspecting data packets as they are received, determining the source and destination device of that packet, and forwarding it appropriately. By delivering each message only to the connected device it was intended for, a network switch conserves network bandwidth and offers generally better performance than a hub. One of the main benefits of using a switch over a hub is micro-segmentation. It allows you to have dedicated bandwidth on point to point connections with every computer and to therefore run in full duplex with no collisions. contrarily, a hub can only run in half duplex and there would be collisions and retransmissions.
- **Role of switches in networks**
  - Network switch is a marketing term rather than a technical one. Switches may operate at one or more OSI layers, including physical, data link, network, or transport (i.e., end-to-end). A device that operates simultaneously at more than one of these layers is called a multilayer switch, although use of the term is diminishing.
  - In switches intended for commercial use, built-in or modular interfaces make it possible to connect different types of networks, for example Ethernet, Fiber Channel, ATM, and 802.11. This connectivity can be at any of the layers mentioned. While Layer 2 functionality is adequate for speed-shifting within one technology, interconnecting technologies such as Ethernet and token ring are easier at Layer 3.

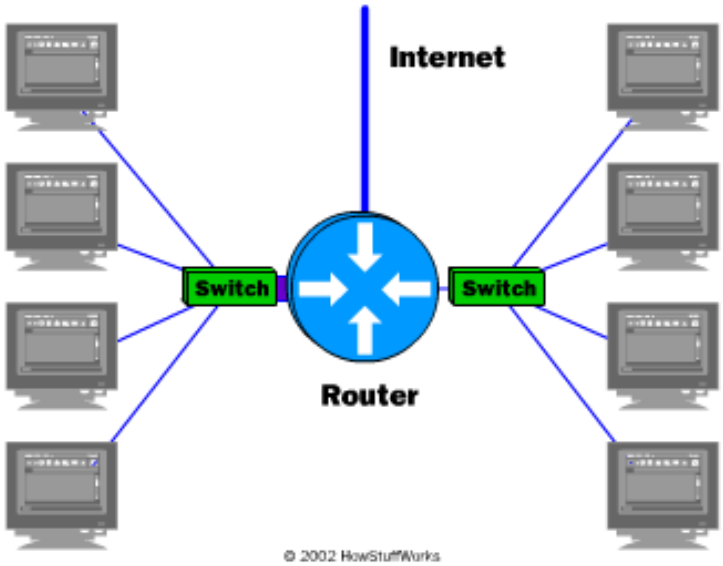
- Interconnection of different Layer 3 networks is done by routers. If there are any features that characterize "Layer-3 switches" as opposed to general-purpose routers, it tends to be that they are optimized, in larger switches, for high-density Ethernet connectivity.
- In some service provider and other environments where there is a need for much analysis of network performance and security, switches may be connected between WAN routers as places for analytic modules. Some vendors provide firewall, network intrusion detection, and performance analysis modules that can plug into switch ports. Some of these functions may be on combined modules.
- In other cases, the switch is used to create a "mirror" image of data that can go to an external device. Since most switch port mirroring provides only one mirrored stream, network hubs can be useful for fanning out data to several read-only analyzers, such as intrusion detection systems and packet sniffers.

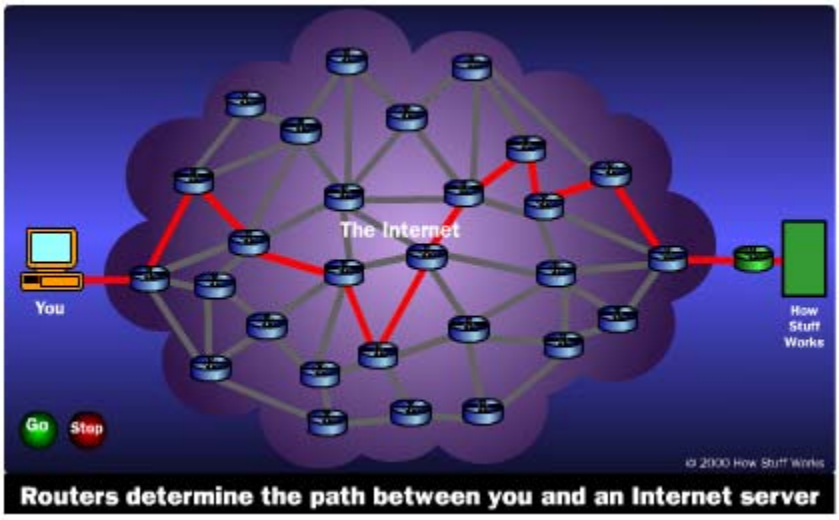
4.2.2 Router:

- **Network router** is a network device that forwards packets from one network to another. Based on internal routing tables, routers read each incoming packet and decide how to forward it. The destination address in the packets determines which line (interface) outgoing packets are directed to. In large-scale enterprise routers, the current traffic load, congestion, line costs and other factors determine which line to forward to.
- **Routers** work at Layer 3 (Network) with Layer 3 addresses (IP, IPX or Appletalk, depending on which Layer 3 protocols are being used).
- A router, then, has two separate but related jobs:

    1. The router ensures that information doesn't go where it's not needed. This is crucial for keeping large volumes of data from clogging the connections of "innocent bystanders."
    2. The router makes sure that information does make it to the intended destination.

    In performing these two jobs, a router is extremely useful in dealing with two separate computer networks. It joins the two networks, passing information from one to the other and, in some cases, performing translations of various protocols between the two networks. It also protects the networks from one another, preventing the traffic on one from unnecessarily spilling over to the other. As the number of networks attached to one another grows, the configuration table for handling traffic among them grows, and the processing power of the router is increased. Regardless of how many networks are attached, though, the basic operation and function of the router remains the same. Since the Internet is one huge network made up of tens of thousands of smaller networks, its use of routers is an absolute necessity.





4.2.3 Hub:

- A network hub is a fairly un-sophisticated broadcast device. Hubs do not manage any of the traffic that comes through them, and any packet entering any port is broadcast out on every other port (other than the port of entry). Since every packet is being sent out through every other port, packet collisions result--which greatly impedes the smooth flow of traffic.
- The need for hosts to be able to detect collisions limits the number of hubs and the total size of the network. For 10 Mbit/s networks, up to 5 segments (4 hubs) are allowed between any two end stations. For 100 Mbit/s networks, the limit is reduced to 3 segments (2 hubs) between any two end stations, and even that is only allowed if the hubs are of the low delay variety. Some hubs have special (and generally manufacturer specific) stack ports allowing them to be combined in a way that allows more hubs than simple chaining through Ethernet cables, but even so, a large Fast Ethernet network is likely to require switches to avoid the chaining limits of hubs.
- Most hubs detect typical problems, such as excessive collisions on individual ports, and partition the port, disconnecting it from the shared medium. Thus, hub-based Ethernet is generally more robust than coaxial cable-based Ethernet, where a misbehaving device can disable the entire segment. Even if not partitioned automatically, a hub makes troubleshooting easier because status lights can indicate the possible problem source or, as a last resort, devices can be disconnected from a hub one at a time much more easily than a coaxial cable. They also remove the need to troubleshoot faults on a huge cable with multiple taps.
- Hubs classify as Layer 1 devices in the OSI model. At the physical layer, hubs can support little in the way of sophisticated networking. Hubs do not read any of the data passing through them and are not aware of their source or destination. Essentially, a hub simply receives incoming packets, possibly amplifies the electrical signal, and broadcasts these packets out to all devices on the network - including the one that originally sent the packet!
- Technically speaking, three different types of hubs exist:
  1. Passive
  2. Active
  3. Intelligent

  Passive hubs do not amplify the electrical signal of incoming packets before broadcasting them out to the network. Active hubs, on the other hand, do perform this amplification, as does a different type of dedicated network device called a repeater. Some people use the terms concentrator when referring to a passive hub and multiport repeater when referring to an active hub.

  Intelligent hubs add extra features to an active hub that are of particular importance to businesses. An intelligent hub typically is stackable (built in such a way that multiple units can be placed one on top of the other to conserve space). It also typically includes remote management capabilities via SNMP and virtual LAN (VLAN) support.
- Hubs remain a very popular device for small networks because of their low cost.

4.2.4 Server:

- Hardware requirements for servers vary, depending on the server application. Absolute CPU speed is not as critical to a server as it is to a desktop. Servers' duties to provide service to many users over a network lead to different requirements like fast network connections and high I/O throughput. Since servers are typically accessed over a network, servers emphasize function over form, without regard to aesthetics like appearance and noise level, because users may never lay eyes on the machine itself. Servers may accordingly run in headless mode without a monitor in order to free up processing cycles for other tasks. In general, a server becomes more specialized and therefore more efficient as unnecessary and unused services are eliminated. For this reason, many servers lack a graphical user interface, or GUI, because it consumes resources that could be allocated elsewhere. Similarly, servers often lack audio and USB interfaces.
- By definition, servers provide services, but it is not always possible to predict when users will need those services. For this reason, servers are often online for weeks or months without interruption, making hardware durability extremely important. Although servers can be built from commodity computer parts, mission-critical servers use specialized hardware with low failure rates in order to maximize uptime. For example, servers may incorporate faster, higher-capacity hard drives, larger computer fans or water cooling to help remove heat, and uninterruptible power supplies that ensure the servers continue to function in the event of a power failure. These components offer higher performance and reliability at a correspondingly higher price. The dominant paradigm in servers is parallel computing, and thus high-performance servers are often placed in rack-mounted configurations to save space inside server rooms or "closets." These special rooms help mute the large amount of noise produced and also restrict physical access to the system administrators for security purposes.
- Servers have a unique property in that, the more powerful and complex the system, the longer it takes for the hardware to turn on and begin loading the operating system. Servers often do extensive preboot memory testing and verification and start up of remote management services. The hard drive controllers then start up banks of drives sequentially, rather than all at once, so as not to overload the power supply, and afterwards they initiate RAID system prechecks for correct operation of redundancy. It is not uncommon for a machine to take several minutes to turn on and yet not require a restart for the next calendar year.

4.2.5 RAID (Redundant Array of Independent Disks):

RAID 2, 3, and 4 are rarely used, many hardware controllers don't support these modes anymore

4.2.5.1 Data Striping (for improved performance)

Data striping transparently distributes data over multiple disks to make them appear as a single fast, large disk. Striping improves aggregate I/O performance by allowing multiple I/Os to be serviced in parallel. There are 2 aspects to this parallelism.

- Multiple, independent requests can be serviced in parallel by separate disks. This decreases the queuing time seen by I/O requests.
- Single, multiple block requests can be serviced by multiple disks acting in co-ordination. This increases the effective transfer rate seen by a single request. The performance benefits increase with the number of disks in the array. Unfortunately, a large number of disks lowers the overall reliability of the disk array.

Most of the redundant disk array organizations can be distinguished based on 2 features:
The granularity of data interleaving and
The way in which the redundant data is computed and stored across the disk array.

- Data interleaving can be either fine grained or coarse grained.
- Fine grained disk arrays conceptually interleave data in relatively small units so that all I/O requests, regardless of their size, access all of the disks in the disk array. This results in very high data transfer rate for all I/O requests but has the disadvantages that only one logical I/O request can be in service at any given time and all disks must waste time positioning for every request.

Coarse grained disk arrays interleave data in relatively large units so that small I/O requests need access only a small number of disks while large requests can access all

the disks in the disk array. This allows multiple small requests to be serviced simultaneously while still allowing large requests to see the higher transfer rates afforded by using multiple disks.

4.2.5.2 Redundancy

Since larger number of disks lower the overall reliability of the array of disks, it is important to incorporate redundancy in the array of disks to tolerate disk failures and allow for the continuous operation of the system without any loss of data.

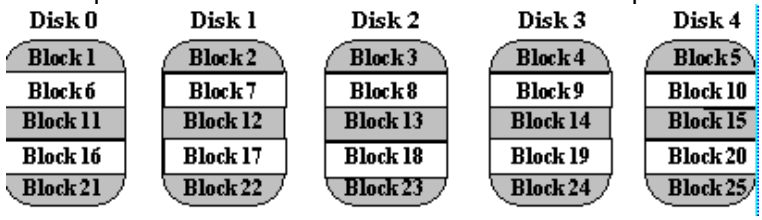
The incorporation of redundancy in disk arrays brings up two problems:

1. Selecting the method for computing the redundant information. Most redundant disks arrays today use parity, though some use Hamming or Reed-Solomon codes.
2. Selecting a method for distribution of the redundant information across the disk array. The distribution method can be classified into 2 different schemes:
   - 2a. Schemes that concentrate redundant information on a small number of disks.
   - 2b. Schemes that distribute redundant information uniformly across all of the disks.

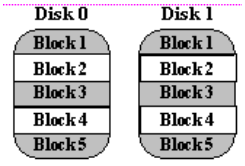
Such schemes are generally more desirable because they avoid hot spots and other load balancing problems suffered by schemes that do not uniformly distribute redundant information.

- This array distributes data across several disks, but the array is seen by the computer user and operating system as one single disk. RAID can be set up to serve several different purposes.
- Redundancy is a way that extra data is written across the array, which are organized so that the failure of one (sometimes more) disks in the array will not result in loss of data. A failed disk may be replaced by a new one, and the data on it reconstructed from the remaining data and the extra data. A redundant array allows less data to be stored. For instance, a 2-disk RAID 1 array loses half of the total capacity that would have otherwise been available using both disks independently, and a RAID 5 array with several disks loses the capacity of one disk. Other RAID level arrays are arranged so that they are faster to write to and read from than a single disk.
- There are various combinations of these approaches giving different trade-offs of protection against data loss, capacity, and speed. RAID levels 0, 1, and 5 are the most commonly found, and cover most requirements.

- **RAID 0** (striped disks) distributes data across several disks in a way that gives improved speed and full capacity, but all data on all disks will be lost if any one disk fails. Sequential blocks of data are written across multiple disks in stripes, as follows:



- **RAID 1** (mirrored settings/disks) could be described as a real-time backup solution. Two (or more) disks each store exactly the same data, at the same time, and at all times. Data is not lost as long as one disk survives. Total capacity of the array is simply the capacity of one disk. At any given instant, each disk in the array is simply identical to every other disk in the array.



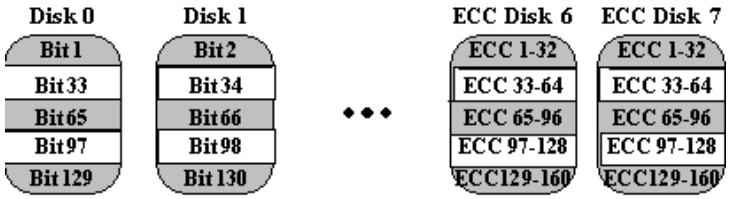
- **RAID 2** (Memory Style) Memory systems have provided recovery from failed components with much less cost than mirroring by using Hamming codes. Hamming codes contain parity for distinct overlapping subsets of components. In one version of this scheme, four disks require three redundant disks, one less than mirroring. Since the number of redundant disks is proportional to the log of the total number of the disks on the system, storage efficiency increases as the number of data disks increases.

  If a single component fails, several of the parity components will have inconsistent values, and the failed component is the one held in common by each incorrect subset. The lost information is recovered by reading the other components in a subset, including the parity component, and setting the missing bit to 0 or 1 to create proper parity value for that subset. Thus, multiple redundant disks are needed to identify the failed disk, but only one is needed to recover the lost information.

  In you are unaware of parity, you can think of the redundant disk as having the sum of all data in the other disks. When a disk fails, you can subtract all the data on the good disks form the parity disk; the remaining information must be the missing information. Parity is simply this sum modulo 2.

  A RAID 2 system would normally have as many data disks as the word size of the computer, typically 32. In addition, RAID 2 requires the use of extra disks to store an error-correcting code for redundancy. With 32 data disks, a RAID 2 system would require 7 additional disks for a Hamming-code ECC. Such an array of 39 disks was the subject of a U.S. patent granted to Unisys Corporation in 1988, but no commercial product was ever released.

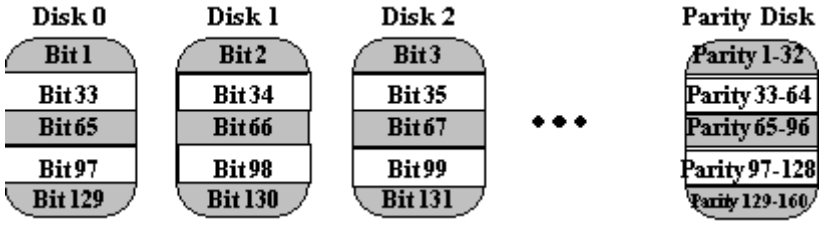
  For a number of reasons, including the fact that modern disk drives contain their own internal ECC, RAID 2 is not a practical disk array scheme.



- RAID 3 (Bit-Interleaved Parity) One can improve upon memory-style ECC disk arrays by noting that, unlike memory component failures, disk controllers can easily identify which disk has failed. Thus, one can use a single parity rather than a set of parity disks to recover lost information.

  In a bit-interleaved, parity disk array, data is conceptually interleaved bit-wise over the data disks, and a single parity disk is added to tolerate any single disk failure. Each read request accesses all data disks and each write request accesses all data disks and the parity disk. Thus, only one request can be serviced at a time. Because the parity disk contains only parity and no data, the parity disk cannot participate on reads, resulting in slightly lower read performance than for redundancy schemes that distribute the parity and data over all disks. Bit-interleaved, parity disk arrays are frequently used in applications that require high bandwidth but not high I/O rates. They are also simpler to implement than RAID levels 4, 5, and 6.

  Here, the parity disk is written in the same way as the parity bit in normal Random Access Memory (RAM), where it is the Exclusive Or of the 8, 16 or 32 data bits. In RAM, parity is used to detect single-bit data errors, but it cannot correct them because there is no information available to determine which bit is incorrect. With disk drives, however, we rely on the disk controller to report a data read error. Knowing which disk's data is missing, we can reconstruct it as the Exclusive Or (XOR) of all remaining data disks plus the parity disk.

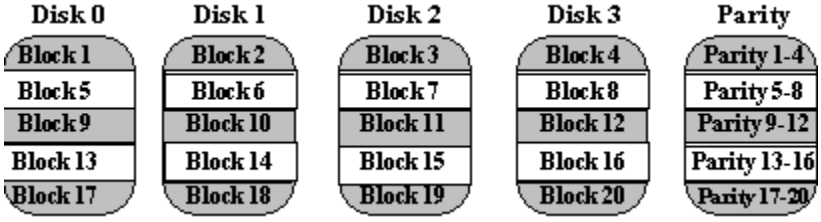


As a simple example, suppose we have 4 data disks and one parity disk. The sample bits are:

Disk 0	Disk 1	Disk 2	Disk 3	Parity
0	1	1	1	1

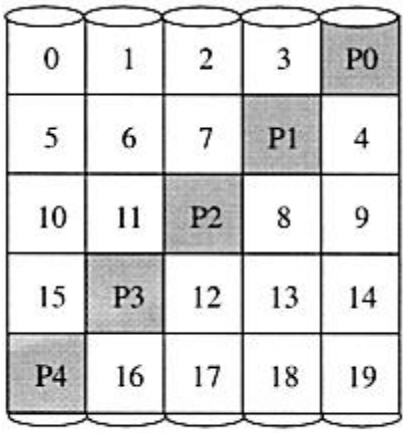
The parity bit is the XOR of these four data bits, which can be calculated by adding them up and writing a 0 if the sum is even and a 1 if it is odd. Here the sum of Disk 0 through Disk 3 is "3", so the parity is 1. Now if we attempt to read back this data, and find that Disk 2 gives a read error, we can reconstruct Disk 2 as the XOR of all the other disks, including the parity. In the example, the sum of Disk 0, 1, 3 and Parity is "3", so the data on Disk 2 must be 1.

- **RAID 4** (Block-Interleaved Parity) The block-interleaved, parity disk array is similar to the bit-interleaved, parity disk array except that data is interleaved across disks of arbitrary size rather than in bits. The size of these blocks is called the striping unit. Read requests smaller than the striping unit access only a single data disk. Write requests must update the requested data blocks and must also compute and update the parity block. For large writes that touch blocks on all disks, parity is easily computed by exclusive-or'ing the new data for each disk. For small write requests that update only one data disk, parity is computed by noting how the new data differs from the old data and applying those differences to the parity block. Small write requests thus require four disk I/Os: one to write the new data, two to read the old data and old parity for computing the new parity, and one to write the new parity. This is referred to as a read-modify-write procedure. Because a block-interleaved, parity disk array has only one parity disk, which must be updated on all write operations, the parity disk can easily become a bottleneck. Because of this limitation, the block-interleaved distributed parity disk array is universally preferred over the block-interleaved, parity disk array.



- **RAID 5** (striped disks with parity) The block-interleaved distributed-parity disk array eliminates the parity disk bottleneck present in the block-interleaved parity disk array by distributing the parity uniformly over all of the disks. An additional, frequently overlooked advantage to distributing the parity is that it also distributes data over all of the disks rather than over all but one. This allows all disks to participate in servicing read operations in contrast to redundancy schemes with dedicated parity disks in which the parity disk cannot participate in servicing read requests. Block-interleaved distributed-parity disk array have the best small read, large write performance of any redundancy disk array. Small write requests are somewhat inefficient compared with redundancy schemes such as mirroring however, due to the need to perform read-modify-write operations to update parity. This is the major performance weakness of RAID level 5 disk arrays.

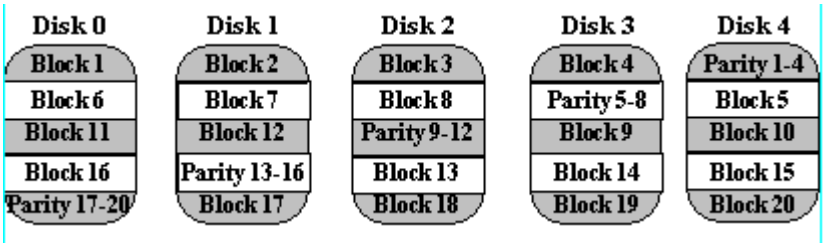
The exact method used to distribute parity in block-interleaved distributed-parity disk arrays can affect performance. Following figure illustrates left-symmetric parity distribution.



(Left-Symmetric)

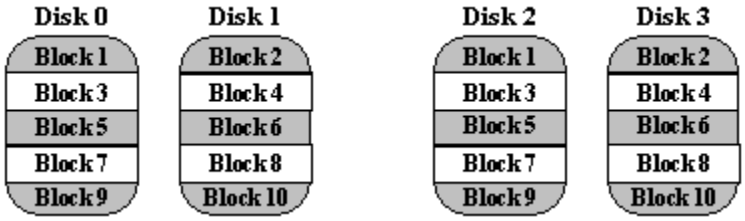
Each square corresponds to a stripe unit. Each column of squares corresponds to a disk. P0 computes the parity over stripe units 0, 1, 2 and 3; P1 computes parity over stripe units 4, 5, 6, and 7 etc.

A useful property of the left-symmetric parity distribution is that whenever you traverse the striping units sequentially, you will access each disk once before accessing any disk device. This property reduces disk conflicts when servicing large requests.



- **RAID 6** (striped disks with dual parity) (less common) can recover from the loss of two disks.

- **RAID 10** (or 1+0) uses both striping and mirroring. "01" or "0+1" is sometimes distinguished from "10" or "1+0": a striped set of mirrored subsets and a mirrored set of striped subsets are both valid, but distinct, configurations. Obviously, RAID 10 uses more disk space to provide redundant data than RAID 5. However, it also provides a performance advantage by reading from all disks in parallel while eliminating the write penalty of RAID 5. In addition, RAID 10 gives better performance than RAID 5 while a failed drive remains unreplaced. Under RAID 5, each attempted read of the failed drive can be performed only by reading all of the other disks. On RAID 10, a failed disk can be recovered by a single read of its mirrored pair.



- RAID can involve significant computation when reading and writing information. With traditional "real" RAID hardware, a separate controller does this computation. In other cases the operating system or simpler and less expensive controllers require the host computer's processor to do the computing, which reduces the computer's performance on processor-intensive tasks (see "Software RAID" and "Fake RAID" below). Simpler RAID controllers may provide only levels 0 and 1, which require less processing.

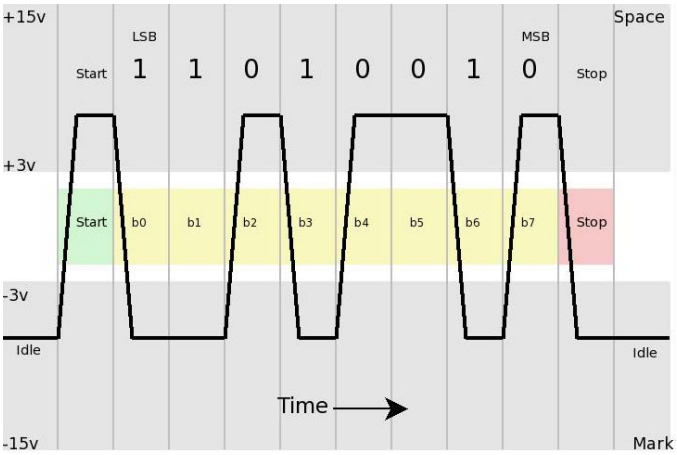
RAID systems with redundancy continue working without interruption when one, or sometimes more, disks of the array fail, although they are vulnerable to further failures. When the bad disk is replaced by a new one the array is rebuilt while the system continues to operate normally. Some systems have to be shut down when removing or adding a drive; others support hot swapping, allowing drives to be replaced without powering down. RAID with hot-swap drives is often used in high availability systems, where it is important that the system keeps running as much of the time as possible.

RAID is not a good alternative to backing up data. Data may become damaged or destroyed without harm to the drive(s) on which they are stored. For example, part of the data may be overwritten by a system malfunction; a file may be damaged or deleted by user error or malice and not noticed for days or weeks; and of course the entire array is at risk of catastrophes such as theft, flood, and fire.

4.3 Network Communications:

4.3.1 RS232

- **R**ecommended **S**tandard 232 (issued in 1969)
    - Now EIA-232
    - Current version EIA/TIA-232E (issued 1991)
    - Standard for serial binary data signals connecting between a *DTE* (Data Terminal Equipment, such as a computer or a printer)) and a *DCE* (Data Circuit-terminating Equipment, such as a modem). It is commonly used in computer serial ports
- User data is sent as a time-series of bits. Both synchronous and asynchronous transmissions are supported by the standard
- Widely-used rule-of-thumb indicates that cables more than 50 feet (15 meters) long will have too much capacitance, unless special cables are used
- The 20kbps rate is too slow for many applications. (originally developed for modems)
- The RS-232 standard defines the voltage levels that correspond to logical one and logical zero levels. Valid signals are plus or minus 3 to 15 volts. The range near zero volts is not a valid RS-232 level; logic one is defined as a negative voltage, the signal condition is called marking, and has the functional significance of OFF. Logic zero is positive, the signal condition is spacing, and has the function ON. The standard specifies a maximum open-circuit voltage of 25 volts; signal levels of ±5 V, ±10V, ±12 V, and ±15 V are all commonly seen depending on the power supplies available within a device. RS-232 drivers and receivers must be able to withstand indefinite short circuit to ground or to any voltage level up to ±25 volts. The slew rate, or how fast the signal changes between levels, is also controlled.



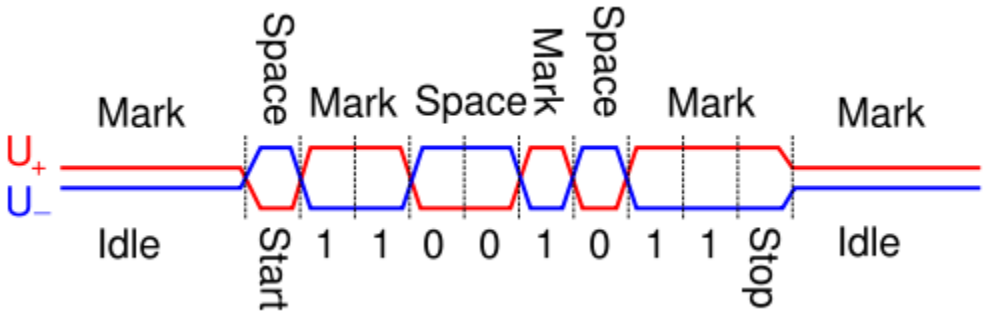
Oscilloscope Trace

Pin# DB9 DTE	Pin Assignment		Pin# DB25 DTE	
Pin 1	Protective Ground		Pin 1	Protective Ground
Pin 2	Transmit Data		Pin 2	Transmit Data
Pin 3	Received Data		Pin 3	Received Data
Pin 4	Request To Send		Pin 4	Request To Send
Pin 5	Clear To Send		Pin 5	Clear To Send
Pin 6	Data Set Ready		Pin 6	Data Set Ready
Pin 7	Signal Ground		Pin 7	Signal Ground
Pin 8	Received Line Signal Detector (Data Carrier Detect)		Pin 8	Received Line Signal Detector (Data Carrier Detect)
Pin 9	Data Terminal Ready		Pin 20	Data Terminal Ready
NA			Pin 22	Ring Indicator

RS-232 Pinout

4.3.2 RS485

- **R**ecommended **S**tandard 485
  - Now EIA-485
  - Extension of RS-422 (increases the number of transmitters and receivers permitted on the line)
  - Electrical specification of a two-wire, half-duplex, multipoint serial communications channel. Since it uses a differential balanced line over twisted pair (like EIA-422), it can span relatively large distances (up to 4,000 feet)
- Offers high data transmission speeds
  - 1200m (~4000ft) 90kbps
  - 6m (~20ft) 10Mbps
- Full duplex operation can be made full-duplex by using four wires
  - Full duplex allows send and receive data at the same time
  - Half-duplex, meaning information can move in only one direction at a time
- Multi-point network topology (it is possible to connect 32 devices to the network)
- Used as the physical layer underlying many standard and proprietary automation protocols used to implement Industrial Control Systems, including the most common versions of Modbus and Profibus. These are used in programmable logic controllers and on factory floors. Since it is differential, it resists electromagnetic interference from motors and welding equipment.
- The EIA-485 differential line consists of two pins:
  - A aka '−' aka TxD-/RxD – aka inverting pin
  - B aka '+' aka TxD+/RxD+ aka non-inverting pin
  - The B line is positive (compared to A) when the line is idle (i.e., data is 1).



Waveform Example

4.3.3 RS422

- **R**ecommended **S**tandard 422
  - Now EIA-422
- RS422 allows a multi-drop network topology, rather than a multi-point network where all nodes are considered equal and every node has send and receive capabilities over the same line (This allows one central control unit to send commands in parallel to up to ten slave devices)

4.3.4 ModBus

Modbus:

- Serial communications protocol published by Modicon in 1979
- Serial Modbus connections can use two basic transmission modes, **ASCII** or **RTU**
  - *Modbus/ASCII*, the messages are in a readable ASCII format
  - *Modbus/RTU* format uses binary coding which makes the message unreadable when monitoring, but reduces the size of each message which allows for more data exchange in the same time span.
- Modbus Addressing:
  - The first information in each Modbus message is the address of the receiver. This parameter contains one byte of information. In Modbus/ASCII it is coded with two hexadecimal characters, in Modbus/RTU one byte is used. Valid addresses are in the range 0…247. The values 1…247 are assigned to individual Modbus devices and 0 is used as a broadcast address. Messages sent to the latter address will be accepted by all slaves. A slave always responds to a Modbus message. When responding it uses the same address as the master in the request. In this way the master can see that the device is actually responding to the request.
  - Within a Modbus device, the holding registers, inputs and outputs are assigned a number between 1 and 10000. One would expect, that the same addresses are used in the Modbus messages to read or set values. Unfortunately this is not the case. In the Modbus messages addresses are used with a value between 0 and 9999. If you want to read the value of output (coil) 18 for example, you have to specify the value 17 in the Modbus query message. More confusing is even, that for input and holding registers an offset must be subtracted from the device address to get the proper address to put in the Modbus message structure. This leads to common mistakes and should be taken care of when designing applications with Modbus. The following table shows the address ranges for coils, inputs and holding registers and the way the address in the Modbus message is calculated given the actual address of the item in the slave device.

Device and Modbus address ranges

Device address	Modbus address	Description
1...10000*	address - 1	Coils (outputs)
10001...20000*	address - 10001	Inputs
40001...50000*	address - 40001	Holding registers

* Maximum value is device dependent

- Modbus Function Codes:
  - The second parameter in each Modbus message is the function code. This defines the message type and the type of action required by the slave. The parameter contains one byte of information. In Modbus/ASCII this is coded with two hexadecimal characters, in Modbus/RTU one byte is used. Valid function codes are in the range 1…255. Not all Modbus devices recognize the same set of function codes. The most common codes are discussed here.
  - Normally, when a Modbus slave answers a response, it uses the same function code as in the request. However, when an error is detected, the highest bit of the function code is turned on. In that way the master can see the difference between success and failure responses.

Common Modbus function codes

Code	Description
01	Read coil status
02	Read input status
03	Read holding registers
04	Read input registers
05	Force single coil
06	Preset single register
07	Read exception status
15	Force multiple coils
16	Preset multiple registers
17	Report slave ID

- Function 01: Read coil status:
  - In Modbus language, a coil is a discrete output value. Modbus function **01** can be used to read the status of such an output. It is only possible to query one device at a time. Broadcast addressing is not supported with this Modbus function. The function can be used to request the status of various coils at once. This is done by defining an output range in the data field of the message.

Function 01 query structure

Byte	Value	Description
1	**1...247**	Slave device address
2	**1**	Function code
3	**0...255**	Starting address, high byte
4	**0...255**	Starting address, low byte
5	**0...255**	Number of coils, high byte
6	**0...255**	Number of coils, low byte
7(...8)	**LRC/CRC**	Error check value

  - When receiving a Modbus query message with function **01**, the slave collects the necessary output values and constructs an answer message. The length of this message is dependent on the number of values that have to be returned. In general, when **N** values are requested, a number of ((**N**+7) mod 8) bytes are necessary to store these values. The actual number of databytes in the datablock is put in the first

byte of the data field. Therefore the general structure of an answer to a Modbus function **01** query is:

Function 01 answer structure

Byte	Value	Description
1	**1...247**	Slave device address
2	**1**	Function code
3	**0...255**	Number of data bytes **N**
4...N+3	**0...255**	Bit pattern of coil values
N+4(...N+5)	**LRC/CRC**	Error check value

- Function 02: Read input status:
  - Reading input values with Modbus is done in the same way as reading the status of coils. The only difference is that for inputs Modbus function 02 is used. Broadcast addressing mode is not supported. You can only query the value of inputs of one device at a time. Like with coils, the address of the first input, and the number of inputs to read must be put in the data field of the query message. Inputs on devices start numbering at 10001. This address value is equivalent to address 0 in the Modbus message.

Function 02 query structure

Byte	Value	Description
1	**1...247**	Slave device address
2	**2**	Function code
3	**0...255**	Starting address, high byte
4	**0...255**	Starting address, low byte
5	**0...255**	Number of inputs, high byte
6	**0...255**	Number of inputs, low byte
7(...8)	**LRC/CRC**	Error check value

  - After receiving a query message with Modbus function **02**, the slave puts the requested input values in a message structure and sends this message back to the Modbus master. The length of the message depends on the number of input values returned. This causes the length of the output message to vary. The number of databytes in the data field that contain the input values is passed as the first byte in the data field. Each Modbus answering message has the following general structure.

Function 02 answer structure

Byte	Value	Description
1	**1...247**	Slave device address
2	**2**	Function code
3	**0...255**	Number of data bytes **N**
4...N+3	**0...255**	Bit pattern of input values
N+4(...N+5)	**LRC/CRC**	Error check value

- Function 03: Read holding registers:
  - Internal values in a Modbus device are stored in holding registers. These registers are two bytes wide and can be used for various purposes. Some registers contain configuration parameters where others are used to return measured values (temperatures etc.) to a host. Registers in a Modbus compatible device start counting

at 40001. They are addressed in the Modbus message structure with addresses starting at 0. Modbus function 03 is used to request one or more holding register values from a device. Only one slave device can be addressed in a single query. Broadcast queries with function 03 are not supported.

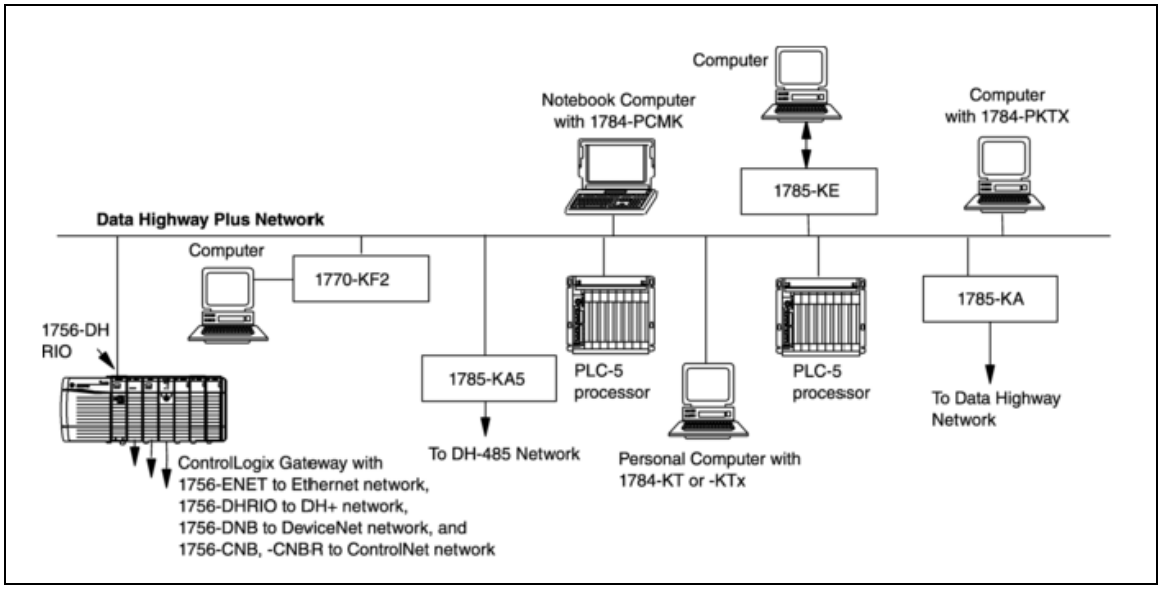
Function 03 query structure

Byte	Value	Description
1	**1...247**	Slave device address
2	**3**	Function code
3	**0...255**	Starting address, high byte
4	**0...255**	Starting address, low byte
5	**0...255**	Number of registers, high byte
6	**0...255**	Number of registers, low byte
7(...8)	**LRC/CRC**	Error check value

- After processing the query, the Modbus slave returns the 16 bit values of the requested holding registers. Because of the size of the holding registers, every register is coded with two bytes in the answering message. The first data byte contains the high byte, and the second the low byte of the register. The Modbus answer message starts with the slave device address and the function code **03**. The next byte is the number of data bytes that follow. This value is two times the number of registers returned. An error check is appended for the host to check if a communication error occurred.

4.3.5 DH+

- Data Highway/Data Highway (Plus) both developed by Allen Bradley as a [proprietary], industrial bus. The interconnecting cable for the Network Link is called a "Blue Hose" due to the use of Belden 9463 cable [78 ohms, Shielded Twin Axial Cable]. Interconnection between nodes is done over the [Differential LAN] DH, DH+, DH485, or ControlNet Link. DH and DH+ allows 64 nodes, DH485 allows 32, and ControlNet 99 nodes. The protocol used to interconnect the Network link and a PC is called DF1 and provides baseband link for a local area network over RS232 or RS422. Other derivations include; DH+E Link (Plus E), DHIIE Link (IIE), and ControlNet.
- **DH** uses a trunk cable of [up to] 3048 meters, and drop cables [to each node] of 30.48 meters. Uses peer-to-peer communication in which each node [PLC] bids on being the Master, called Floating Master.
- **DH+** is used with smaller networks. Uses peer-to-peer communication implementing Token passing. Nominal voltages on the bus are 8 to 12 volts p-p; how ever the bus is +/- 200mV sensitive over the 2 differential lines. Each node on the bus is transformer coupled onto the bus. The bus should be terminated to 150 ohms at each end of the bus. Data bits are Manchester encoded [Clock and data are synchronous], and run at 57.6Kb or 115.2Kb [half-duplex]. Each data packet is sent with a 16 bit CRC, or [may also use BCC ~Block check character]. All messages on the bus are either a command or reply.



DH+ Specifications

Cabling	(1770-CD shielded twinaxial): ▪ trunkline - 10,000 ft max. ▪ droplines - 100 ft max.
Transmission Characteristics	▪ token passing ▪ 57.6 kbit/s ▪ 230 kbit/s on selected products
Operating Temperature	▪ 1770-KF2: 0...50 °C (32...122 °F) ▪ all other Data Highway Plus backplane-resident modules: 0...60 °C (32...140 °F)

4.3.6 HART:

(Highway **A**ddressable **R**emote **T**ransducer Protocol)

- The protocol was developed by Rosemount Inc., built off the Bell 202 early communications standard, in the mid-1980s as proprietary digital communication protocol for their smart field instruments. Soon it evolved into HART. In 1986, it was made an open protocol. Since then, the capabilities of the protocol have been enhanced by successive revisions to the specification.

- There are two main operational modes of HART instruments: analog/digital mode, and multidrop mode.

  Peer-to-Peer mode (analog/digital) Here the digital signals are overlayed on the 4-20 mA loop current. Both the 4-20 mA current and the digital signal are valid output values from the instrument. The polling address of the instrument is set to "0". Only one instrument can be put on each instrument cable signal pair.
  Multi-drop mode (digital) In this mode only the digital signals are used. The analog loop current is fixed at 4 mA. In multidrop mode it is possible to have up to 15 instruments on one signal cable. The polling addresses of the instruments will be in the range 1-15. Each meter needs to have a unique address.

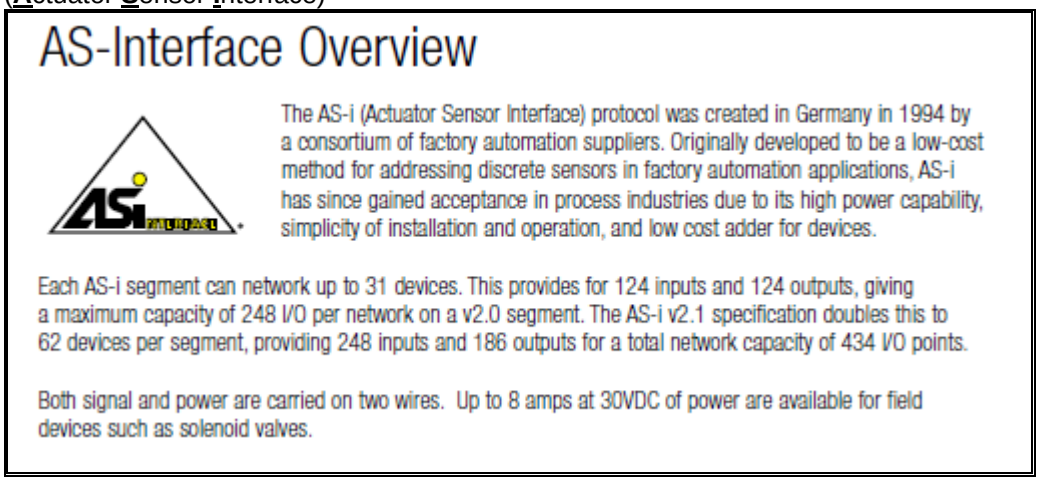
HART

Protocol Information

Type of Network	Device Bus (Process Automation)
Physical Media	Legacy 4-20 mA analog instrumentation wiring or 2.4GHz Wireless
Network Topology	One-on-One, Multidrop, Wireless Mesh
Maximum Devices	64 in multidrop
Maximum Speed	Depends on Physical Layer employed
Device Addressing	Hardware/Software
Governing Body	HART Communication Foundation

4.3.7 AS-I:

(**A**ctuator **S**ensor **I**nterface)



- Used to network sensors and actuators. ASi is a two wire interface; Power and Data. Based around ProfiSafe [developed from Profibus DP].ASI bus was developed by Siemens Automation. This is a Unshielded 2-wire [Yellow cable], Unterminated, Ungrounded Sensor Bus. The Topology may be either Bus, Ring, Tree, or Star at up to 100 meters. Power is provided by a 24V floating DC supply, which can supply at least 8 A over the network. The AS-Interface is an open standard based on IEC 62026-2 and EN 50295.

4.3.8 Profibus:

(PROcess **FI**eld **BUS)**

- Based on the EIA-485 bus and EN-50170, using a non-powered 2-wire bus. The connection is half-duplex over a shielded, twisted-pair cable. The bus will use either a 9 pin D (DIN 19245) or 12mm connector (EC50170). Data rates may be from 9600 to 12M baud, with message lengths of 244 bytes. At 12 Mbps the maximum distance is 100 meters. A maximum distance of 1200 meters may be achieved using a maximum data rate of 94kps. Up to 126 nodes may be connected in up two 5 segments, which are separated by repeaters. Each segment may contain up to 32 nodes which are laid out in a single node. Each node has one master and slave devices.

4.3.9 Foundation Fieldbus:

Foundation Fieldbus is an all-digital, serial, two-way communications system that serves as the base-level network in a plant or factory automation environment. It is an open architecture.Developed and administered by the Fieldbus Foundation. It's targeted for applications using basic and advanced regulatory control, and for much of the discrete control associated with those functions. Foundation fieldbus technology is mostly used in process industries, but nowadays it is being implemented in powerplants also.

Two related implementations of FOUNDATION fieldbus have been introduced to meet different needs within the process automation environment. These two implementations use different physical media and communication speeds.

- *H1* works at 31.25 kbit/s and generally connects to field devices. It provides communication and power over standard twisted-pair wiring. H1 is currently the most common implementation. Can support up to 32 devices on one segment, though in reality more like 4 – 15. Minimum power requirement of 8mA, Minimum device operating voltage of 9V, Maximum bus voltage of 32V

  Segment Calculations: When calculating how many devices can fit on a fieldbus segment, the primary factors to be taken into account are the maximum current requirement of each device and the resistance of the segment cable (because of voltage drops along the length). The calculation is a simple Ohm's law problem, with the aim of showing that at least 9V can be delivered at the farthest end of the segment, after taking into account all the voltage drops from the total segment current. - - For example, driving 16x20mA devices requires 320mA, so if the segment is based on cable with 50 Ohms/km/loop and a 25V power conditioner, the maximum cable length is 1000m to guarantee 9V at the end. Voltage available for cable = 25 – 9 = 16V

  Allowable resistance = 16V / 0.320A = 50 Ohms; equivalent to 1000m cable. Note that many users also specify a safety margin on top of the 9V minimum operating voltage to allow for unexpected current loads and for adding additional devices in the future.

  Recommended spur length:

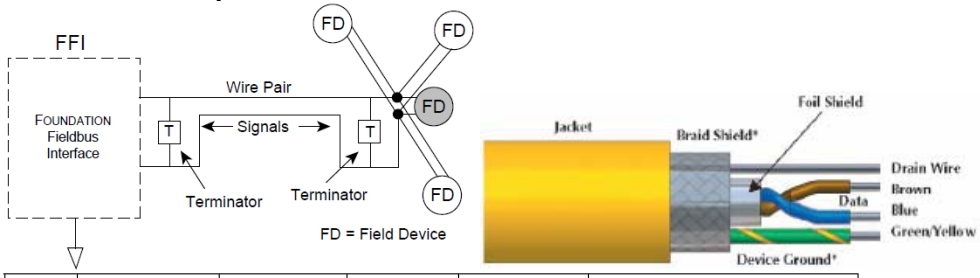
Total Devices	1 Device per Spur	2 Devices per Spur	3 Devices per Spur	4 Devices per Spur
25-32	1 (3)	1 (3)	1 (3)	1 (3)
19-24	30 (98)	1 (3)	1 (3)	1 (3)
15-18	60 (197)	30 (98)	1 (3)	1 (3)
13-14	90 (295)	60 (197)	30 (98)	1 (3)
1-12	120 (394)	90 (295)	60 (197)	30 (98)

  Recommended trunk length:

  The maximum allowed length of a fieldbus segment is 1900 meters (6232 ft) except where repeaters are installed. The total segment length is calculated by:

  Total Segment Length = Trunk + All Spurs

  Fieldbus Cable Specifications:



Type	Meters/Feet	Impedance Ohms	Resistivity	Attenuation (db/km)	Description
A	1900 / 6270	100	22	3	Individual shielded pairs
B	1200 / 3960	100	56	5	Multiple pairs with overall shield
C	400 /1320	Unknown	132	8	Multiple pairs with no shield
D	200 / 660	Unknown	20	8	Multiconductor, no pairing

- *HSE* (High-speed Ethernet) works at 100 Mbit/s and generally connects input/output subsystems, host systems, linking devices, gateways, and field devices using standard Ethernet cabling. It doesn't currently provide power over the cable, although work is under way to address this.

4.3.10 ARCNET:

(**A**ttached **R**esource **C**omputer **NET**work)

- **"**ARCNET" uses a token-passing protocol, with packet lengths of from 0 to 507 bytes at a data rate of 2.5 Mbps [10Mbps max]. ARCNET uses CRC-16. Depending on the topology the following cables may be used: Coax; RG-62/u, RG-59/u [BNC], or #24 or #22 AWG solid copper twisted-pair cable [RJ-11], or fiber optic cable [SMA or ST]. ARCNET is also used with [DC or AC coupled] RS-485.

4.3.11 BACnet:

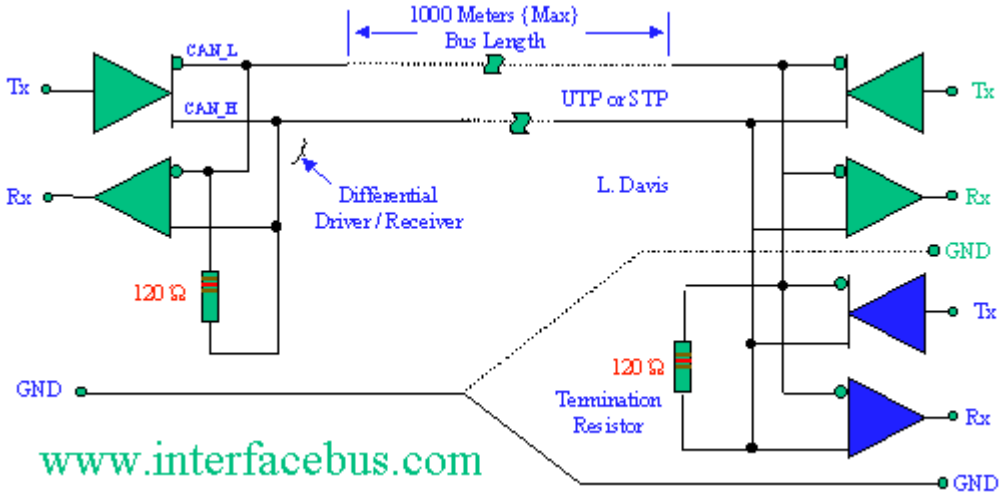
(**B**uilding **A**utomation and **C**ontrol **NET**work)

- BACnet is an ISO/ANSI/ASHRAE 135-1995 standard. Like the LonWorks protocol, it has many physical/data-link layers including RS-485, Ethernet, ARCNET, RS-232, IP, and LonTalk. ASHRAE: American Society of Heating, Refrigerating and Air-Conditioning Engineers

4.3.12 CAN Bus:

(**C**ontroller **A**rea **N**etwork)

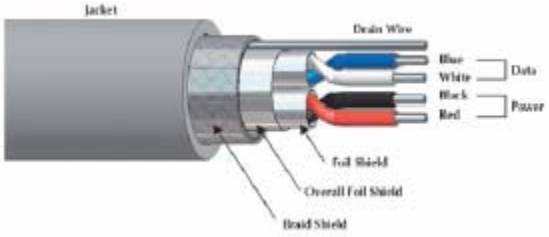
- The Controller Area Network (CAN) specification defines the Data Link Layer, ISO 11898 defines the Physical Layer. The CAN interface is an asynchronous transmission scheme controlled by start and stop bits at the beginning and end of each character. This interface is used, employing serial binary interchange. Information is passed from transmitters to receivers in a data frame. The data frame is composed of an Arbitration field, Control field, Data field, CRC field, ACK field. The frame begins with a 'Start of frame', and ends with an 'End of frame' space. The data field may be from 0 to 8 bits.
- The CAN bus [CANbus] is a Balanced (differential) 2-wire interface running over either a Shielded Twisted Pair (STP), Un-shielded Twisted Pair (UTP), or Ribbon cable. Each node uses a Male 9-pin D connector. The Bit Encoding used is: Non Return to Zero (NRZ) encoding (with bit-stuffing) for data communication on a differential two wire bus. The use of NRZ encoding ensures compact messages with a minimum number of transitions and high resilience to external disturbance.



CAN Bus Electrical Interface Circuit

4.3.13 DeviceNet:

- DeviceNet identifies the physical layer and is based on the CanBus protocol but does not use the same physical layer interface as ISO 11898. DeviceNet provides optical isolation for additional protection and does not use 9-pin subD connectors. DeviceNet only supports three baud rates: 125, 250 and 500 Kbaud (@ 500 meters) with up to 64 devices on the (differential) bus.. In addition the cable carries 24 volts which powers the devices. DeviceNet uses trunk and drop topology. The trunk is the main communication cable, and requires a 121Ω resistor at both ends. The maximum length of the trunk depends on the communication rate and the cable type. Drops are branches off the trunk, and may be from zero to 20ft in length. The cumulative drop lengths are dependent on the communication rate



Communication Rate	Thick Trunk Length (maximum)	Mid Trunk Length (maximum)	Thin Trunk Length (maximum)	Drop Length (maximum per drop)	Drop Length (cumulative)	Nodes (maximum)
125 kbps	500 m (1640 ft.)	300 m (984 ft.)	100 m (328 ft.)	6 m (20 ft.)	156 m (512 ft.)	64
250 kbps	250 m (820 ft.)	250 m (820 ft.)	100 m (328 ft.)	6 m (20 ft.)	78 m (256 ft.)	64
500 kbps	100 m (328 ft.)	100 m (328 ft.)	100 m (328 ft.)	6 m (20 ft.)	39 m (128 ft.)	64

4.3.14 OPC

A compound document standard developed by Microsoft (OPC Specification was based on the OLE, COM, and DCOM technologies). It enables creation of objects with one application and then link or embed them in a second application. Embedded objects retain their original format and the links to the application that created them. Essentially this reduces the number of interfaces required to one per application.

OPC = OLE for Process Control (OLE = Object Linking & Embedding)
COM = Component Object Model
DCOM = Distributed Component Object Model

- **OPC Data Access**: Used to move real-time data from PLCs, DCSs, and other control devices to HMIs and other display clients
- **OPC Alarms & Events**: Provides alarm and event notifications on demand (in contrast to the continuous data flow of Data Access). These include process alarms, operator actions, informational messages, and tracking/auditing messages
- **OPC Batch**: This spec carries the OPC philosophy to the specialized needs of batch processes. It provides interfaces for the exchange of equipment capabilities (corresponding to the S88.01 Physical Model) and current operating conditions
- **OPC Data Exchange**: This specification involves server-to-server with communication across Ethernet fieldbus networks
- **OPC Historical Data Access**: Where OPC Data Access provides access to real-time, continually changing data, OPC Historical Data Access provides access to data already stored. From a simple serial data logging system to a complex SCADA system, historical archives can be retrieved in a uniform manner.
- **OPC Security**: All the OPC servers provide information that is valuable to the enterprise and if improperly updated, could have significant consequences to plant processes. OPC Security specifies how to control client access to these servers in order to protect this sensitive information and to guard against unauthorized modification of process parameters
- **OPC XML** – DA: Provides flexible, consistent rules and formats for exposing plant floor data using XML, leveraging the work done by Microsoft and others on SOAP and Web Services (XML = Extensible Markup Language)

4.3.15 Common Ethernet Variations (e.g. 10Base5, etc)

Common Ethernet Variations							
	10Base5	**10Base2**	**10BaseT**	**10BaseFL**	**100BaseT**	**100BaseFX**	**1000Base-TX**
Common Name	Thicknet	Thinnet	10BaseT	10baseFL	Fast Ethernet	Fast Ethernet	Gigabit Ethernet
Media Access	CSMA/CD	CSMA/CD	CSMA/CD	CSMA/CD	CSMA/CD	CSMA/CD	CSMA/CD
Topology	Bus	Bus	Star	Star	Star	Star	Star
Cabling	RG8 Coax	RG58 Coax	UTP	Fiber	UTP	Fiber	UTP
Data Rate	10 Mbps	10 Mbps	10 Mbps	10 Mbps	100 Mbps	100 Mbps	1000 Mbps
Segment Length	500 m	185 m	100 m	2000 m	100 m	412 m	100 m

Thicknet: 50Ω "thick" (10mm) coaxial cable used with Ethernet 10Base5 networks.10Base5 is the original Ethernet system that supports a 10 Mb/s transmission rate over a 500 meter maximum supported segment length

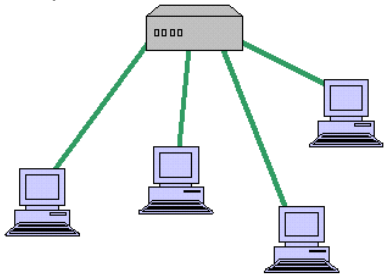
Thinnet: 10BASE2 is a variant of Ethernet that uses thin coaxial cable (RG-58A/U or similar terminated with BNC connectors.

CSMA/CD: Carrier **S**ense **M**ultiple **A**ccess with **C**ollision **D**etection

5. Bus Topology

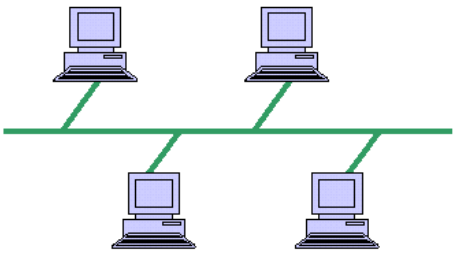
Star:

Many home networks use the star topology. A star network features a central connection point called a "hub" that may be a hub, switch or router. Devices typically connect to the hub with Unshielded Twisted Pair (UTP) Ethernet.
Compared to the bus topology, a star network generally requires more cable, but a failure in any star network cable will only take down one computer's network access and not the entire LAN. (If the hub fails, however, the entire network also fails.)



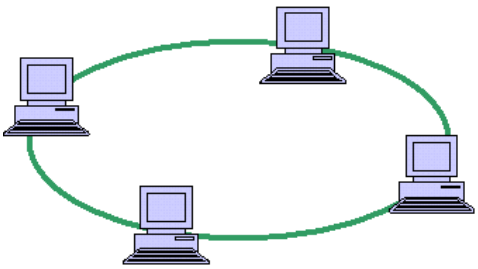
Bus:

Ethernet bus topologies are relatively easy to install and don't require much cabling compared to the alternatives. 10Base-2 ("ThinNet") and 10Base-5 ("ThickNet") both were popular Ethernet cabling options many years ago for bus topologies. However, bus networks work best with a limited number of devices. If more than a few dozen computers are added to a network bus, performance problems will likely result. In addition, if the backbone cable fails, the entire network effectively becomes unusable



Ring:

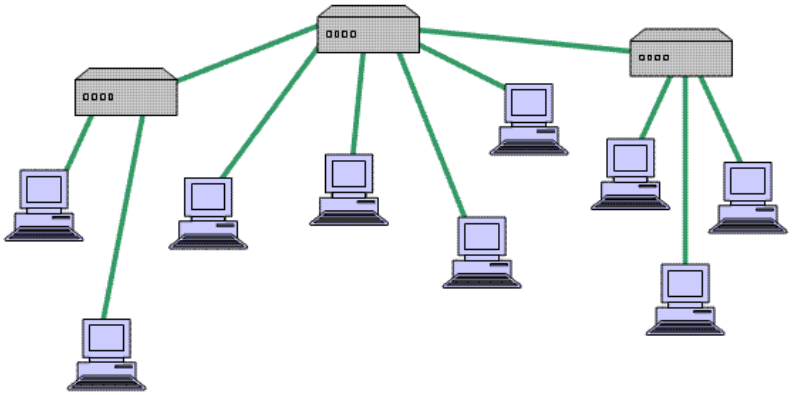
In a ring network, every device has exactly two neighbors for communication purposes. All messages travel through a ring in the same direction (either "clockwise" or "counterclockwise"). A failure in any cable or device breaks the loop and can take down the entire network.
To implement a ring network, one typically uses FDDI, SONET, or Token Ring technology. Ring topologies are found in some office buildings or school campuses.



Tree:

Tree topologies integrate multiple star topologies together onto a bus. In its simplest form, only hub devices connect directly to the tree bus, and each hub functions as the

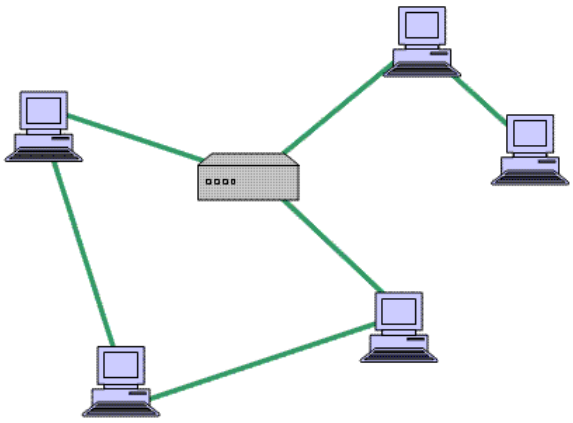
"root" of a tree of devices. This bus/star hybrid approach supports future expandability of the network much better than a bus (limited in the number of devices due to the broadcast traffic it generates) or a star (limited by the number of hub connection points) alone.



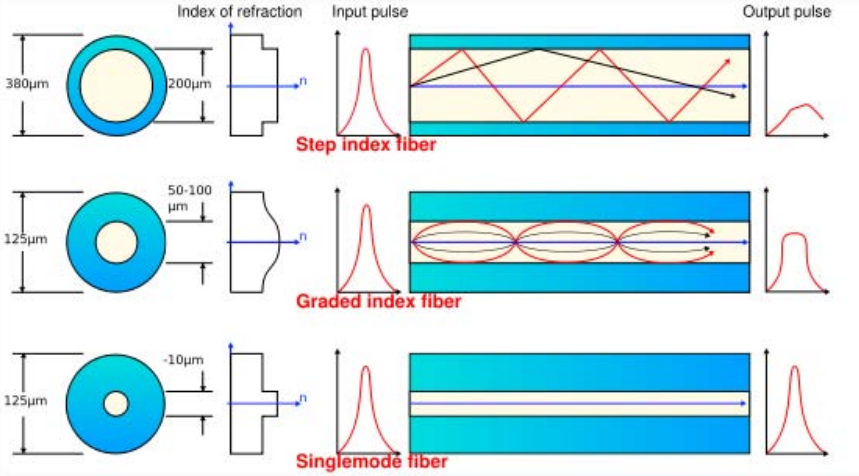
Mesh:

Mesh topologies involve the concept of routes. Unlike each of the previous topologies, messages sent on a mesh network can take any of several possible paths from source to destination. (Recall that even in a ring, although two cable paths exist, messages can only travel in one direction.) Some WANs, most notably the Internet, employ mesh routing.

A mesh network in which every device connects to every other is called a full mesh. As shown in the illustration below, partial mesh networks also exist in which some devices connect only indirectly to others.

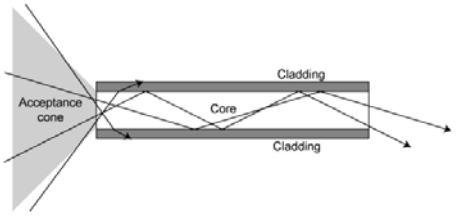


6. Fiber Optics



Multimode:

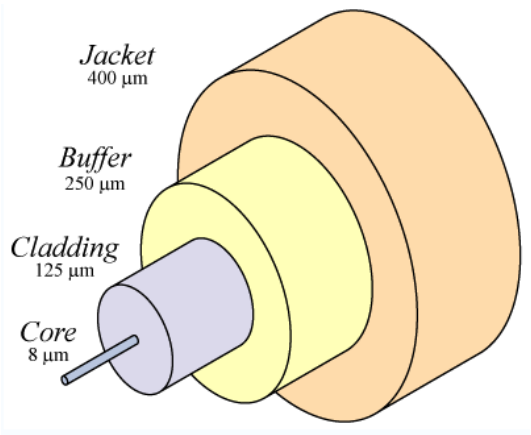
- Larger diameter core > 10µm (typically 50µm or 62.5 µm) this allows the rays of light to travel along several different angles between the core and cladding.
- Larger core size simplifies connections and also allows the use of lower-cost electronics such as light-emitting diodes (LEDs) and vertical-cavity surface-emitting lasers (VCSELs) which operate at the 850 nm wavelength
    - LEDs emit incoherent light: Light waves that lack a fixed-phase relationship are referred to as incoherent light.
    - VCSELs emit coherent light: Light waves with a fixed-phase relationship (both spatial and temporal) between points on the electromagnetic wave are referred to as coherent light
- Due to the modal dispersion in the fiber, multi-mode fiber has higher pulse spreading rates than single mode fiber, limiting multi-mode fiber's information transmission capacity.
- Shorter distance use for communication links (typically < 500m) such as within a building.
- Typical multimode links have data rates of 10 Mbit/s to 10 Gbit/s
- Used when higher power must be transmitted
- Typically less expensive than singlemode
- To distinguish multimode cables from singlemode, MM patch cables typically have orange jackets, while SM cable jackets are usually yellow.



Singlemode:

- Smaller diameter core 8 – 10µm (typically 9 µm), which allows only one path for the rays of light to travel thru the fiber.
- Light source is typically a singlemode laser
- Typically used for communication links > 200m
- Single-mode fibers are most often used in high-precision areas because the allowance of only one propagation mode of the light makes the light easier to focus properly
- Single mode fibers are therefore better at retaining the fidelity of each light pulse over long distances than are multi-mode fibers. For these reasons, single-mode fibers can have a higher bandwidth than multi-mode fibers
- Singlemode fiber have the broadest bandwidth.

- Designed for use in the NIR region.



Bandwidth:

- **Digital bandwidth** or just **bandwidth** is the capacity for a given system to transfer data over a connection. It is measured as a bit rate expressed in bits/s or multiples of it (kb/s Mb/s etc.). Digital bandwidth should not be confused with the network throughput, which is the average rate of *successful* data transfer through a connection. It should also be distinguished from "data transfer", which is the quantity of data transferred over a given period of time.

7. Copper Cabling

Twisted Pair

- Twisting wires together decreases interference because the loop area between the wires (which determines the magnetic coupling into the signal) is reduced.
- In balanced pair operation, the two wires typically carry equal and opposite signals (differential mode) which are combined by addition at the destination. The common-mode noise from the two wires (mostly) cancel each other in this addition because the two wires have similar amounts of EMI that are 180 degrees out of phase. This results in the same effect as subtraction. Differential mode also reduces electromagnetic radiation from the cable, along with the attenuation that it causes.
- The twist rate (also called pitch of the twist, usually defined in twists per meter) makes up part of the specification for a given type of cable. Where pairs are not twisted, one member of the pair may be closer to the source than the other, and thus exposed to slightly different induced electromotive force (EMF).
- Where twist rates are equal, the same conductors of different pairs may repeatedly lie next to each other, partially undoing the benefits of differential mode. For this reason it is commonly specified that, at least for cables containing small numbers of pairs, the twist rates must differ.
- In contrast to FTP (foiled twisted pair) and STP (shielded twisted pair) cabling, UTP (unshielded twisted pair) cable is not surrounded by any shielding. It is the primary wire type for telephone usage and is very common for computer networking, especially as patch cables or temporary network connections due to the high flexibility of the cables.

Cable Shielding

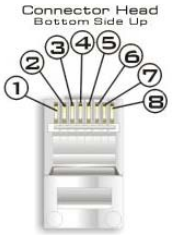
- Twisted pair cables are often shielded in attempt to prevent electromagnetic interference. Because the shielding is made of metal, it may also serve as a ground. However, usually a shielded or a screened twisted pair cable has a special grounding wire added called a drain wire. This shielding can be applied to individual pairs, or to the collection of pairs. When shielding is applied to the collection of pairs, this is referred to as screening. The shielding must be grounded for the shielding to work.

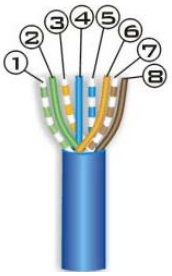
Characteristics of STP/UTP Cables		
Cable Type	**Data Rate**	**Common Usage**
Category 1	N/A	Voice Grade Analog
Category 2	4 Mbps	Digital Voice
Category 3	10 Mbps	10BaseT
Category 4	16 Mbps	Token Ring
Category 5	100 Mbps	100BaseT
Category 6 (5e)	1000 Mbps	1000BaseT

Cable Terminators

Copper Cable Connectors:

CAT 5/5e/6 RJ45 (RJ = Registered Jack)





1 White-Green
2 Green
3 White-Orange
4 Blue
5 White-Blue
6 Orange
7 White-Brown
8 Brown

CAT-5 Wire Order				
Pair #	Wire Color	Abbr.	Wire # (Pair)	Wire # (Cable)
1	White/Blue	W/B	1	1
	Blue	B	2	2
2	White/Orange	W/O	1	3
	Orange	O	2	4
3	White/Green	W/G	1	5
	Green	G	2	6
4	White/Brown	W/Br	1	7
	Brown	Br	2	8



Coaxial Cable Connectors:

BNC (Bayonet-Neill-Concelman) 50Ω & 75Ω: 50Ω impedance good for frequencies of up to 4 GHz and the 75Ω impedance for up to 2 GHz. Employs a bayonet mount mechanism for locking. (RG-8 and RG-58)



F allows for a solid 75Ω impedance with a match of up to 1 GHz.
RG-6/U: RG-6" is generally used to refer to coaxial cables with an 18 AWG center conductor and 75Ω characteristic impedance
RG-59/U: Often used for low-power video and RF signal connections



F.O. Cable Connectors:

SC (Subscriber Connector) is a connector with a push-pull latching mechanism which provides quick insertion and removal while also ensuring a positive connection. Typical matched SC connectors are rated for 1000 mating cycles and have an insertion loss of 0.25 dB



ST (Straight Tip) is a connector which uses a plug and socket which is locked in place with a half-twist bayonet lock. Typical match ST connectors are rated for 500 mating cycles and have an insertion loss of 0.25 dB.



MU is a small form factor SC. It has the same push/pull style, but can fit 2 channels in the same footprint of a single SC.



LC (Lucent Connector) is connector that uses a 1.25 mm ferrule, half the size of the ST.



SMA (Sub Miniature A) is a connectors that uses a threaded plug and socket.



MT-RJ (Mechanical Transfer Registered Jack) is a fiber-optic cable connector that is very popular for small form factor devices due to its small size. Housing two fibers and mating together with locating pins on the plug, the MT-RJ comes from the MT connector, which can contain up to 12 fibers.



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8. Cable Tray

(NEC Article 392) Cable Tray Types:

1. **Ladder Cable Tray** provides:
   Solid side rail protection and system strength with smooth radius fittings and a wide selection of materials and finishes.
   maximum strength for long span applications
   standard widths of 6,12,18, 24, 30, and 36 inches
   standard depths of 3, 4, 5, and 6 inches
   standard lengths of 10, 12, 20 and 24 feet
   rung spacing of 6, 9, 12, and 18 inches
   Ladder cable tray is generally used in applications with intermediate to long support spans, 12 feet to 30 feet.



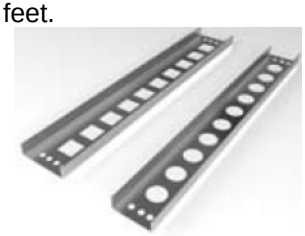
2. **Solid Bottom Cable Tray** provides:
   Nonventilated continuous support for delicate cables with added cable protection available in metallic and fiberglass.
   Solid bottom metallic with solid metal covers for nonplenum rated cable in environmental air areas
   standard widths of 6, 12, 18, 24, 30, and 36 inches
   standard depths of 3, 4, 5, and 6 inches
   standard lengths of 10, 12, 20 and 24 feet
   Solid Bottom cable tray is generally used for minimal heat generating electrical or telecommunication applications with short to intermediate support spans of 5 feet to 12 feet.



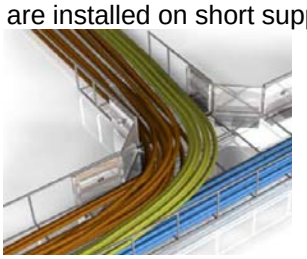
3. **Trough Cable Tray** provides:
   Moderate ventilation with added cable support frequency and with the bottom configuration providing cable support every 4 inches. Available in metal and nonmetallic materials.
   standard widths of 6, 12, 18, 24, 30, 36 inches
   standard depths of 3, 4, 5, and 6 inches
   standard lengths of 10, 12, 20 and 24 feet
   fixed rung spacing of 4 inch on center
   Trough cable tray is generally used for moderate heat generating applications with short to intermediate support spans of 5 feet to 12 feet.



4. **Channel Cable Tray** provides:
   an economical support for cable drops and branch cable runs from the backbone cable tray system.
   standard widths of 3, 4, and 6 inches in metal systems and up to 8 inches in nonmetallic systems.
   standard depths of 1¼-1¾ inches in metal systems and 1, 1 1/8, 1 5/" and 2 3/16 inches in nonmetallic systems
   standard length of 10, 12, 20 and 24 feet
   Channel cable tray is used for installations with limited numbers of tray cable when conduit is undesirable. Support frequency with short to medium support spans of 5 to 10 feet.



5. **Wire Mesh Cable Tray** provides:
   A job site, field adaptable support system primarily for low voltage, telecommunication and fiber optic cables. These systems are typically steel wire mesh, zinc plated.
   standard widths of 2, 4, 6, 8, 12, 16, 18, 20, and 24 inches
   standard depths of 1, 2, and 4 inches
   standard length of about 10 feet (118")
   Wire Mesh tray is generally used for telecommunication and fiber optic applications and are installed on short support spans, 4 to 8 feet.



6. **Single Rail Cable Tray** provides:
   These aluminum systems are the fastest systems to install and provide the maximum freedom fort cable to enter and exit the system.
   Single hung or wall mounted systems in single or multiple tiers.
   Standard widths are 6, 9, 12, 18, and 24 inches.
   Standard depths are 3, 4, and 6 inches.
   Standard lengths are 10 and 12 feet.
   Single Rail Cable Tray is generally used for low voltage and power cables installations where maximum cable freedom, side fill, and speed to install are factors.



9. Wireless

The IEEE standard 802.11 was approved in 1997. It specifies the technologies for wireless communications. It has been further revised into a family of substandards, each offering their own improvements and drawbacks. It specifies three different types of wireless communications: diffuse infrared, frequency-hopping spread spectrum, and direct-sequence spread spectrum. The most popular of these substandards is the 802.11b and 802.11g, which operate under the 2.GHz frequency range using direct-sequence spread spectrum. In the case of these two, the carrier frequency is divided into 13 channels of 22 MHz spaced 5 MHz apart in the range of 2.4000 GHz to 2.4835 GHz. In practice, only 3 channels can operate independently without overlap and subjecting to interference.

802.11a - 5 GHz, max speed 54 Mbit/s

Using the 5 GHz band allows 802.11a devices a significant advantage over 802.11b, both in terms of maximum speeds and because the 2.4 GHz band is heavily overused by many other devices. Because of operating at a higher frequency range, however, it does suffer from a range and penetration issues.

802.11b - 2.4 GHz, max speed 11 Mbit/s

This substandard grew quickly due to its longer range and cheaper price. With a high-gain antenna, access points based off this technology can penetrate walls, cover wide areas, and long distances, but are subject to co-channel interference from other devices operating in this frequency range, such as microwave ovens, cordless telephones, Bluetooth, etc.

802.11g - 2.4 GHz, max speed 54 Mbit/s

An improvement on 802.11b to increase the speed and further reduce manufacturing costs. It is backwards compatible with its predecessor, and mimics much of the 802.11a modulation techniques to achieve the speed increases. It suffers the same interference issues as 802.11b, and due to its wide acceptance, also suffers from usage density issues.

802.11n - 5 or 2.4 GHz, max speed 600 Mbit/s

A proposed substandard to improve all of the previous substandards. It operates in 5 GHz or 2.4 GHz bands, allowing it to bypass some of the interference issues with 802.11b/g. It utilizes 40 MHz channels as well as MIMO (multiple input multiple output) technology to use all channels of the frequency range. It also improves the range and coverage of the signals by allowing multiple antennas and splitting/combining signals between them. One of the main benefits of 802.11n is that it is backwards compatible with all previous substandards, and can modulate between different substandards simultaneously without significant impact.

802.11 Protocol	Release Date	Carrier Frequency (GHz)	Typical Speed (Mbit/s)	Max Speed (Mbit/s)	Modulation	Range (m) (inside/outside)
	1997	2.4	0.9	2	DSSS	20/100
a	1999	5	23	54	OFDM	35/120
b	1999	2.4	4.3	11	DSSS	38/140
g	2003	2.4	19	54	OFDM	38/140
n	2009	2.4 or 5	130	600	OFDM	70/250

DSSS = Direct Sequence Spread Spectrum
OFDM = Orthogonal Frequency Division Multiplexing

10. Flow Measurement

10.1 Flow Meter Evaluation Table

Non-Newtonian fluid is a fluid whose flow properties are not described by a single constant value of viscosity.

Flowmeter	Pipe Size, in. (mm)	Steam	Gases (Vapors)				Liquids								Reverse Flow	Pulsating Flow	High Temp.	Cryogenic	Semi-Filled Pipe	Non-Newtonian	Open Channel	Accuracy (Full Scale) unless noted as rate	Rangeability	Minimum Reyolds #	Viscosity Effect	Maximum Temperature degF (degC)	Maximum Pressure PSIG	Permanent Pressure Loss	Minimum Straight Piping Run Installation Requirements (Upstream)
					Pressure			Viscous					Slurries																	
			Clean	Dirty	High	Low	Clean	High	Low	Dirty	Corrosive	Very Corr.	Fibrous	Abrasive																
Square Root Scale: Maximum Single Range 4:1 (Typical)																													
Orifice																													
Sq-Edged	> 1.5" (40mm)	✔	✔	X	✔	✔	✔	X	?	X	?	X	X	X	SD	?	✔	✔	X	X	?	X	+/- 2 to 4%	4:1	> 10,000	High	1000F (540C)	4000	Medium	10
Integral	<= 0.5" (12mm)	?	✔	X	✔	✔	✔	X	?	X	?	X	X	X	SD	?	?	X	X	X	?	X	+/- 2 to 4%	4:1	> 10,000	High	1000F (540C)	4000	Medium	NA
Wedge	< 12" (300mm)	✔	✔	✔	✔	✔	✔	?	✔	?	?	X	?	?	SD	?	✔	✔	X	X	?	X	+/- 2 to 4%	4:1	> 10,000	High	1000F (540C)	4000	Medium	10
Eccentric	>= 2" (50mm)	?	?	✔	✔	✔	?	X	?	?	?	X	?	X	SD	?	✔	✔	X	X	?	X	+/- 2 to 4%	4:1	> 10,000	High	1000F (540C)	4000	Medium	10
Segmental	>= 4" (100mm)	?	?	✔	✔	✔	?	X	?	?	?	X	?	X	SD	?	✔	✔	X	X	?	X	+/- 2 to 4%	4:1	> 10,000	High	1000F (540C)	4000	Medium	10
V-Cone	0.5 - 72"	✔	✔	?	✔	✔	✔	?	✔	?	?	X	?	?	X	?	?	?	X	X	?	X	+/- 1%	10:1	> 8000	High	1000F (540C)	4000	Medium	3
Target	>= 0.5" (12mm)	?	✔	✔	✔	✔	✔	?	✔	✔	?	X	X	X	?	X	?	?	X	X	?	X	+/- 1 to 5%	10:1	> 2000	Medium	500F (260C)	15000	Medium	10
Venturi	>= 2" (50mm)	✔	✔	?	✔	✔	✔	?	✔	?	?	X	✔	?	X	?	?	?	X	X	?	X	+/- 1%	4:1	> 75,000	High	1000F (540C)	4000	Low	5
Flow Nozzle	>= 2" (50mm)	?	✔	?	✔	✔	✔	X	?	?	?	X	X	X	X	?	?	?	X	X	?	X	+/- 2%	4:1	> 50,000	High	1000F (540C)	4000	Medium	10
Pitot	>= 3" (75mm)	X	✔	X	✔	✔	✔	X	?	X	?	X	X	X	X	X	?	?	X	X	X	X	+/- 3 to 5%	3:1	> 100,000	Low	1000F (540C)	4000	Low	20
Avg Pitot	>= 1" (25mm)	X	✔	SD	✔	✔	✔	X	?	SD	?	X	X	X	SD	X	?	?	X	X	X	X	+/- 1 to 2%	3:1	> 15,000	Low	300F (149C)	275	Low	20
Elbow	>= 2" (50mm)	X	✔	?	✔	✔	✔	X	?	?	?	X	X	X	✔	X	?	?	X	?	X	X	+/- 5 to 10%	3:1	> 10,000	Low	1000F (540C)	4000	Very Low	30
Linear Scale Range 10:1 or Better																													
Magmeter	0.1 - 72"	X	X	X	X	X	✔	?	✔	✔	✔	✔	✔	✔	✔	✔	?	X	?	?	?	+/-0.5% of rate	40:1	> 4,500	None	360F (180C)	1500	None	5
PD Meter																													
Gas	< 12" (300mm)	X	✔	X	?	✔	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	+/- 1% of rate	10:1	NA	NA	250F (120C)	1400	High	None
Liquid	< 12" (300mm)	X	X	X	X	X	✔	✔	?	X	?	X	X	X	X	X	?	X	X	X	X	+/-0.5% of rate	10:1	NA	High	600F (315C)	1400	High	None
Turbine																													
Gas	0.25 - 24"	SD	✔	X	✔	✔	X	X	X	X	X	X	X	X	SD	SD	?	?	X	X	?	+/-0.5% of rate	10:1	NA	NA	500F (260C)	3000	High	5
Liquid	0.25 - 24"	X	X	X	X	X	✔	X	?	X	?	X	X	SD	SD	SD	?	?	X	X	?	+/-0.5% of rate	10:1	> 5,000	High	500F (260C)	3000	High	5
Ultrasonic																													
Time of Flight	>= 0.5" (12mm)	X	SD	SD	SD	SD	✔	?	?	X	✔	✔	?	?	✔	✔	X	?	X	X	?	+/- 1 to 5%	20:1	> 10,000	None	500F (260C)	Pipe rating	None	5
Doppler	>= 0.5" (12mm)	X	X	X	X	X	X	?	?	✔	✔	✔	✔	✔	✔	✔	X	X	X	?	X	+/- 5%	10:1	> 4,000	None	500F (260C)	Pipe rating	None	5
Rotameter	<= 3" (75mm)	?	✔	X	X	✔	✔	X	?	X	?	?	X	X	X	?	?	?	X	X	X	+/- 1 to 10%	10:1	NA	Medium	400F-Glass 1000F-Metal	350 - Glass 720 - Metal	Medium	None
Vortex	1.5 - 16"	✔	✔	?	✔	✔	✔	X	?	?	?	X	X	X	X	X	?	?	X	X	X	+/- 1% of rate	10:1	> 10,000	Medium	400F (200C)	1500	Medium	10
Mass																													
Coriolis	0.25 - 6"	?	?	?	✔	✔	✔	✔	✔	✔	?	?	?	✔	?	?	?	?	X	✔	X	+/-0.4% of rate	10:1	NA	None	800F (427C)	5700	Low	None
Thermal	>= 0.5" (12mm)	X	✔	?	✔	✔	✔	?	✔	✔	?	?	?	?	X	?	?	X	X	?	X	+/- 1%	10:1	NA	None	1500F (816C)	Pipe rating	Low	None
Open Channel																													
Weir		X	X	X	X	X	✔	X	✔	✔	?	?	?	?	X	SD	SD	X	✔	✔	✔	+/- 2 to 5%	100:1	NA	Very Low		Atmospheric	Very Low	None
Flume		X	X	X	X	X	✔	X	✔	✔	?	?	?	?	X	SD	SD	X	✔	✔	✔	+/- 2 to 5%	50:1	NA	Very Low		Atmospheric	Very Low	None

✔ **Designed for this application (generally suitable)**
? **May be applicable (worth consideration)**
X **Not Applicable**
SD Some designs may be applicable.

Flowmeter applications chart

Sensor	Rangeability	Accuracy	Advantages	Disadvantages
orifice	3.5:1	2-4% of full span	-low cost -extensive industrial practice	-high pressure loss -plugging with slurries
venturi	3.5:1	1% of full span	-lower pressure loss than orifice -slurries do not plug	-high cost -line under 15 cm
flow nozzle	3.5:1	2% full span	-good for slurry service -intermediate pressure loss	-higher cost than orifice plate -limited pipe sizes
elbow meter	3:1	5-10% of full span	-low pressure loss	-very poor accuracy
annubar	3:1	0.5-1.5% of full span	-low pressure loss -large pipe diameters	-poor performance with dirty or sticky fluids
turbine	20:1	0.25% of measurement	-wide rangeability -good accuracy	-high cost -strainer needed, especially for slurries
vortex shedding	10:1	1% of measurement	-wide rangeability -insensitive to variations in density, temperature, pressure, and viscosity	-expensive
positive displacement	10:1 or greater	0.5% of measurement	-high rangeability -good accuracy	-high pressure drop -damaged by flow surge or solids
Coriolis mass flow	100:1	0.05-0.15% of measurement	-good accuracy	-expensive

10.2 Reynolds Number

Reynolds number **Re** is a **dimensionless number** that gives a measure of the **ratio** of **inertial forces** (Vq) to **viscous** forces (**μ / *L***) and, consequently, it quantifies the relative importance of these two types of forces for given flow conditions.

The Reynolds Number is found from the equation:

$$\mathrm{Re} = \frac{3160 * Q * G}{D * \rho}$$ *For liquids*

$$\mathrm{Re} = \frac{6316 * Q}{D * \rho}$$ *For gases & steam*

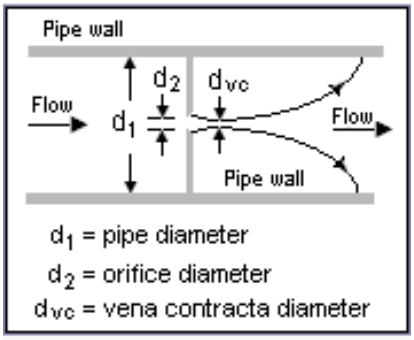
Q: Flow in GPM
G: Specific Gravity
D: Pipe ID
ρ: Viscosity (in cp)

Q: Flow in lb/hr

Flow type you can expect (**laminar**, **transitional**, or **turbulent**) based on the Reynolds Number equation result.
Laminar: Re < 2300
Transitional: 2300 < Re < 4000
Turbulent: Re > 4000

10.3 D/P Producers

10.3.1 Orifice Plate

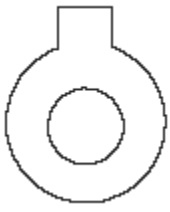


Bernoulli's principle which says that there is a relationship between the pressure of the fluid and the velocity of the fluid. When the velocity increases, the pressure decreases and vice versa.

- Beta Ratio shall be between 0.20 and 0.75
- Preferred D/P range for full scale flow: 50" or 100"

10.3.1.1 Orifice Plate Types

- **Concentric Square-Edged**

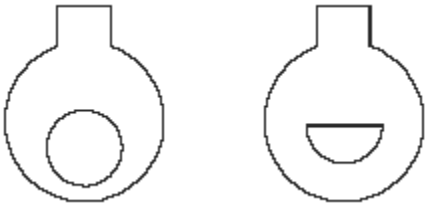


The concentric orifice plate is the most common of the 3 types. The orifice is equidistant (concentric) to the inside diameter of the pipe. Flow through a sharp-edged orifice plate is characterized by a change in velocity. As the fluid passes through the orifice, the fluid converges, and the velocity of the fluid increases to a maximum value. As the fluid diverges to fill the entire pipe, the velocity decreases back to the original value. The

pressure increases to about 60% to 80% of the original input value. The pressure loss is permanent, therefore the outlet pressure will always be less than the input pressure. The pressure on both sides of the orifice are measured (thru taps), resulting in a d/P which is proportional to the flow rate.

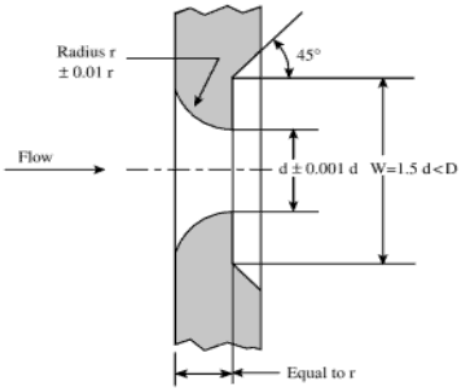
- **Vents / Drains** (flow through must be < 1% of total flow):
  - Vents: Hole located at the top of the Orifice Plate to allow entrained gases in a liquid flow to vent pass the Orifice Plate.
  - Drain: Prevent Build-up of condensate behind orifice.

- **Segmental and Eccentric (**usually used for sediment laden liquids or slurries)

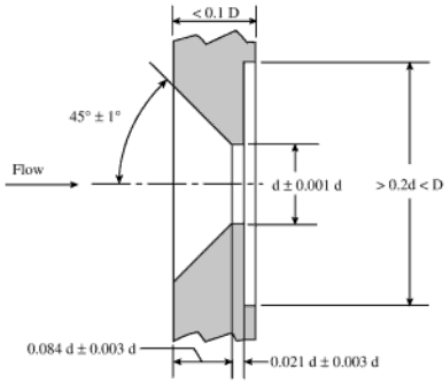


Segmental and Eccentric orifice plates are functionally identical to the concentric orifice plate. The circular section of the segmental orifice is concentric with the pipe. The segmental portion of the orifice eliminates damming of foreign materials on the upstream side of the orifice when mounted in a horizontal pipe. Depending on the type of fluid, the segmental section is placed either on top or on bottom of the horizontal pipe to increase the accuracy of the measurement.

- **Quadrant Edged Orifice** (not a sharp edge)
  Generally used for highly viscous applications Re < 10,000. Relatively immune to erosion and the deposits of solids at the surface of the orifice.

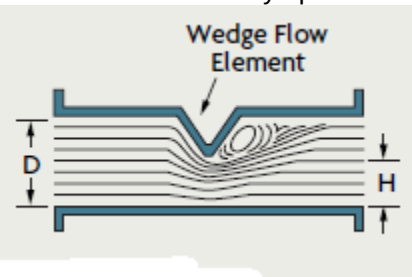


- **Conic Edged Orifice** (not a sharp edge): Generally used for even higher viscous applications than the quadrant edged. Has a 45° bevel edged facing upstream into the flowing stream. Typically utilizes corner taps.



- **Integral Orifice:** Similar to an orifice plate except typically used to measure very small flow rates. Upstream & downstream piping requirements are built into the meter body.

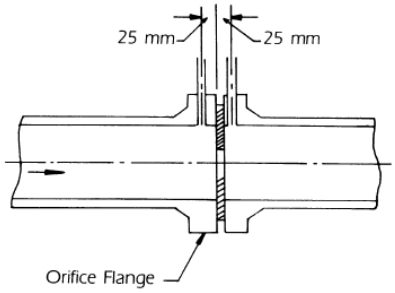
- **Segmental Wedge Orifice:** Proprietary device designed for use in slurry, corrosive, erosive, viscous or high temperature applications. Typically used in conjunction with chemical seals immediately upstream & downstream of the restriction.



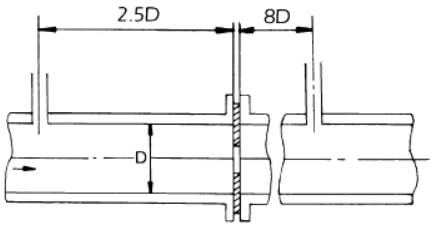
Two distinct disadvantages of orifices:
- High permanent pressure loss
- Erosion of the machined bore which will cause inaccuracies in the measured D/P.

10.3.1.2 Orifice Tap Types

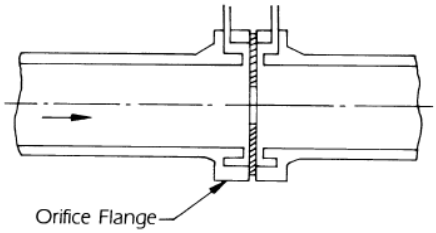
- **Flange Taps** (standard taps are ½" NPT) These taps are located 1" (centerline) from the upstream face of the orifice plate and 1" (centerline) from the downstream face with a ±1/64" to ±1/32" tolerance.



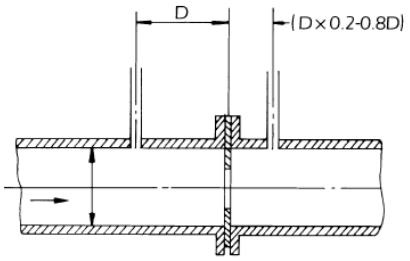
- **Pipe Taps** (standard taps are ½" NPT) (NOT very popular) The taps are located 2½ pipe diameters upstream and 8 pipe diameters downstream (point of maximum pressure recovery)



- **Corner Taps** (used more often in EU) The taps are located immediately adjacent to the plate faces, upstream and downstream. Usually pipe size is < 2".

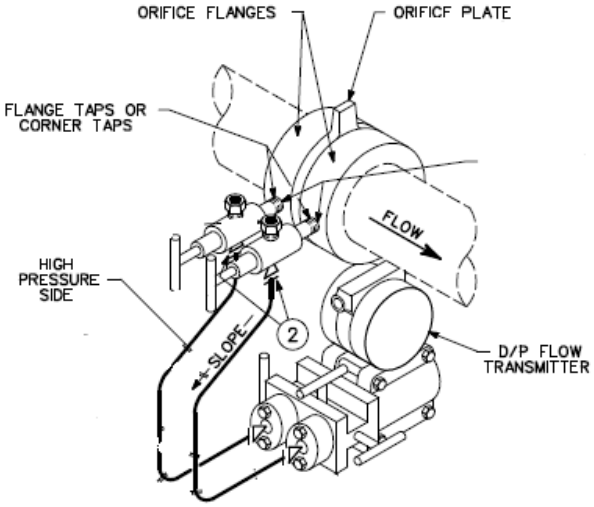


- **Vena Contracta Taps:** The taps are located one pipe diameter upstream and at the minimum pressure point downstream (vena-contracta). The minimum pressure point varies with the Beta ratio. Seldom used except where flows are very constant and plates are not changed. Pipe sizes > 6" lines.

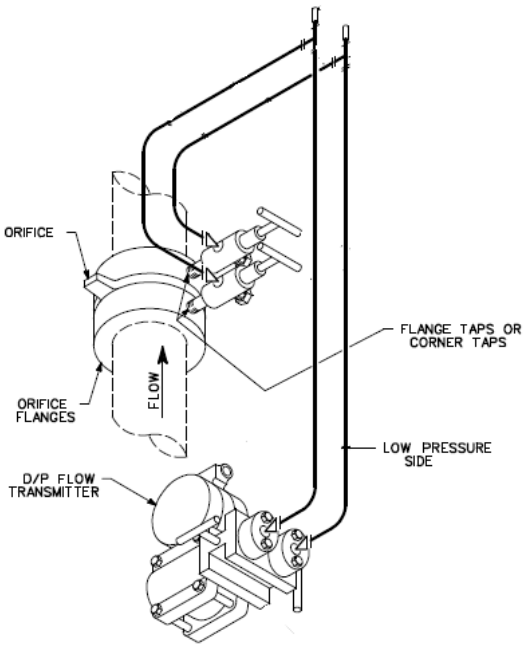


10.3.1.3 Installation Details

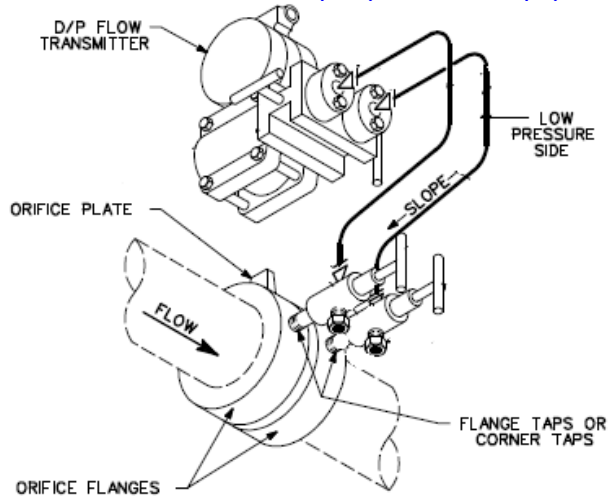
- Liquid or Steam Service (Horizontal Pipe)



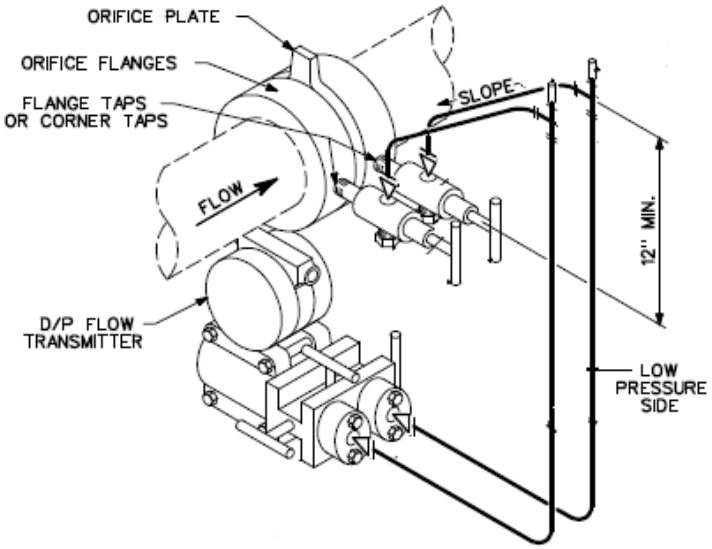
- Liquid or Steam Service (Vertical Pipe)



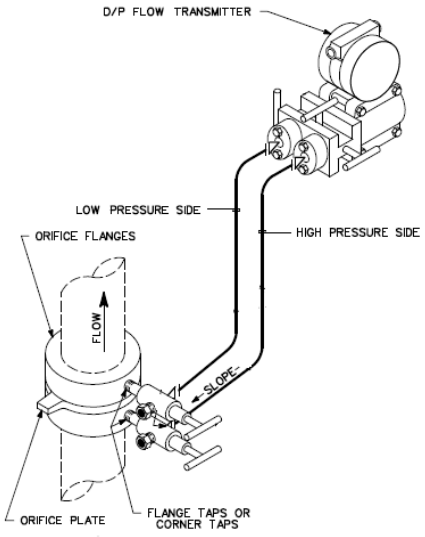
- Gas Service – Xmtr above Pipe (Horizontal Pipe)



- Gas Service – Xmtr below Pipe (Horizontal Pipe)

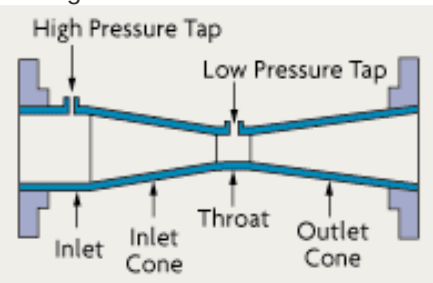


- Gas Service (Vertical Pipe)



10.3.2 Venturi Flowmeter

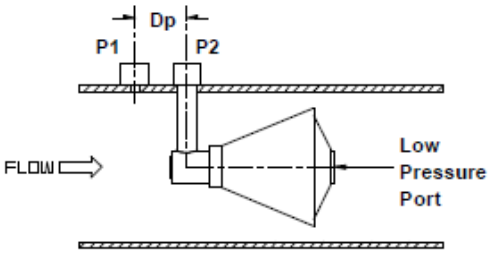
These meters create less of a permanent pressure loss than that of orifice plates, however, they have a higher degree of uncertainty (less accurate) due to the low d/p. In the venturi meter the fluid is accelerated through a converging cone of angle 15-20° and the pressure difference between the upstream side of the cone and the throat is measured and provides a signal for the rate of flow, initial cost is high, so primarily used on larger flows.



10.3.3 V-Cone Flowmeter:

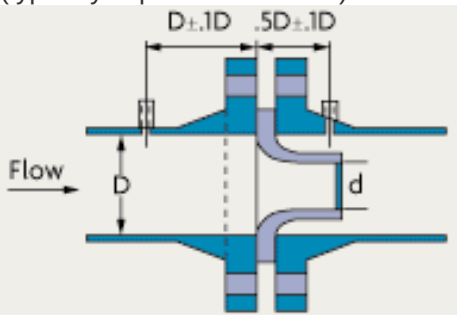
With its D/P built-in flow conditioning design, the V-Cone is especially useful in tight-fit and retrofit installations in which the long runs of straight pipe required by Orifice Plates, Venturi Tubes, etc.

- Differential pressure is created by a cone placed in the center of the pipe.
- The cone is shaped so that it "flattens" the fluid velocity profile in the pipe, creating a more stable signal across wide flow downturns.
- Flow rate is calculated by measuring the difference between the pressure upstream of the cone at the meter wall and the pressure downstream of the cone through its center.



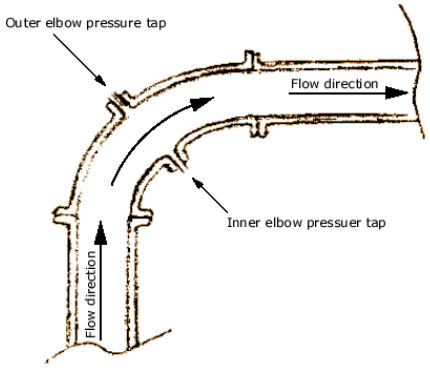
10.3.4 Flow Nozzle:

Permanent pressure loss of a flow nozzle is significantly higher than that of a venture (similar to an orifice). Use of this device is more common in EU than the USA. Due to its rigidity may be used in lieu of orifice plate when the pipeline velocities exceed 100 ft/sec (typically superheated steam).



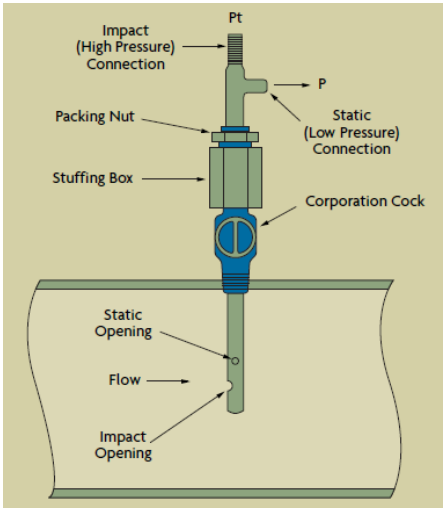
10.3.5 Elbow Flowmeter:

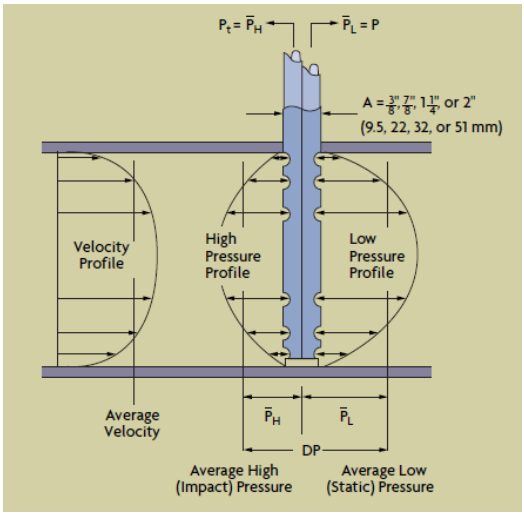
When a liquid flows through an elbow, the centrifugal forces cause a pressure difference between the outer and inner sides of the elbow. This difference in pressure is used to calculate the flow velocity. The pressure difference generated by an elbow flowmeter is smaller than that by other pressure differential flowmeters, but the upside is an elbow flowmeter has less obstruction to the flow and may be installed in an existing pipe elbow when an “order of magnitude” flow reading is desired.



10.3.6 Pitot Tube / Annubar:

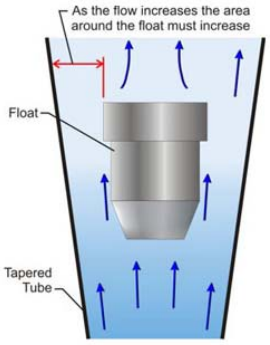
- The basic pitot tube consists of a tube pointing directly into the fluid flow. As this tube contains fluid, a pressure can be measured; the moving fluid is brought to rest (stagnates) as there is no outlet to allow flow to continue. This pressure is the stagnation pressure of the fluid, also known as the total pressure.
- The averaging pitot tube (Annubar) - The biggest difference between an annubar and a pitot tube is that an annubar takes multiple samples across a section of a pipe or duct. In this way, the annubar averages the differential pressures encountered accounting for variations in flow across the section. A pitot tube will give a similar reading if the tip is located at a point in the pipe cross section where the flowing velocity is close to the average velocity.





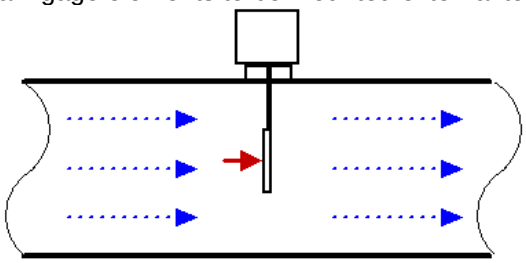
10.3.7 Variable Area / Rotameter:

The rotameter's operation is based on the variable area principle: fluid flow raises a float in a tapered tube, increasing the area for passage of the fluid. The greater the flow, the higher the float is raised. The height of the float is directly proportional to the flowrate. The float moves up or down in the tube in proportion to the fluid flowrate and the annular area between the float and the tube wall. The float reaches a stable position in the tube when the upward force exerted by the flowing fluid equals the downward gravitational force exerted by the weight of the float. MUST be oriented vertically.



10.3.8 Target Meter:

With the target meter a physical target is located directly in the fluid flow. The transducer is actually an electronic "inside-out" orifice plate. The flow produces a pressure differential between the front and the rear surfaces of target. The force of this pressure drag is transmitted via a cantilever arm to a flexure tube of unique design which permits the strain gage elements to be mounted external to the flowing medium.



The force on the target can be expressed as

$F = c_d \rho v^2 A_t / 2$

where

F = force on the target *(N)*

c_d = overall drag coefficient obtained from empirical data

ρ = density of fluid *(kg/m³)*

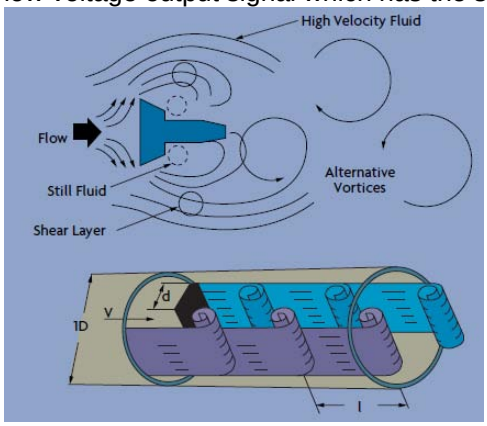
v = fluid velocity *(m/s)*

A_t = target area *(m²)*

10.4 Electronic Flowmeters:

10.4.1 Vortex Shedder:

The majority of vortex flowmeters use capacitance type sensors to detect the pressure oscillations around the bluff body. The sensors respond to the pressure oscillations with a low voltage output signal which has the same frequency as the oscillation.



f = SV/L

where,

f = the frequency of the vortices

L = the characteristic length of the bluff body

V = the velocity of the flow over the bluff body

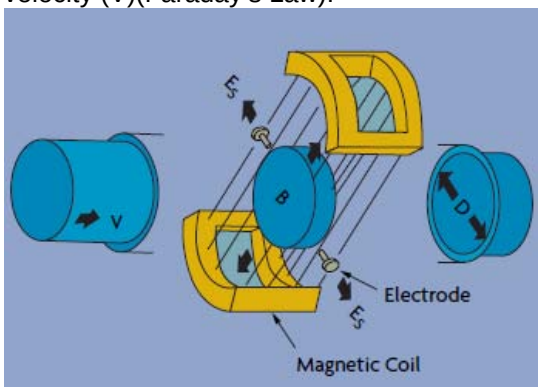
S = Strouhal Number and is a constant for a given body shape



Vortex Shedder Cross Sections

10.4.2 Magmeter:

The magmeter consists of a non-magnetic pipe lined with an insulating material and a pair of magnetic coils that penetrate the pipe and its lining. When a conductive fluid flows thru the pipe of diameter (D) through a magnetic field density (B) generated by the coils, the amount of voltage (E) developed across the electrodes will be proportional to the velocity (V)(Faraday's Law).



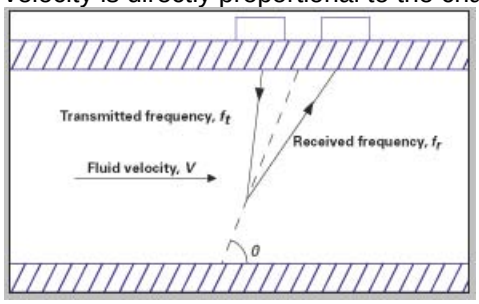
E = kBdV
where,
E = the induced voltage between the sensing electrodes
k = a constant
B = the magnetic flux density
d = the distance between electrodes (equivalent to the pipe diameter)
V = the velocity of the fluid

10.4.3 Ultrasonic Flowmeter:

The speed at which sound propagates through a fluid is dependent upon the fluid's density. If the density is constant, you can use the time of ultrasonic passage (reflection) to determine the velocity of a flowing fluid.

Doppler:

The shift in frequency is the basis upon which all Doppler ultrasonic flowmeters work. The transducer sends an ultrasonic pulse or beam into the flowing stream. The sound waves are reflected back by acoustical discontinuities such as particles or entrained gas. The meter detects the velocity of the discontinuities in calculating the flow rate. The flow velocity is directly proportional to the change in frequency.

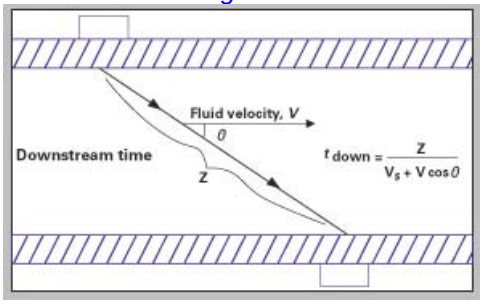


Doppler: $\Delta f = kV$
where $k = 2(f_t)\cos\theta / V_s$
f_t = Transmitted Frequency
V_s = Velocity of Sound in the Fluid

Time of Flight:

In this design, the time of flight of the ultrasonic signal is measured between two separate transducers, one upstream and one downstream. The difference in the elapsed time going with or against the flow determines the fluid velocity. When there is flow, the effect is to boost the speed of the ultrasonic signal vs. that of zero flow for in the downstream direction while decreasing it in the upstream direction.

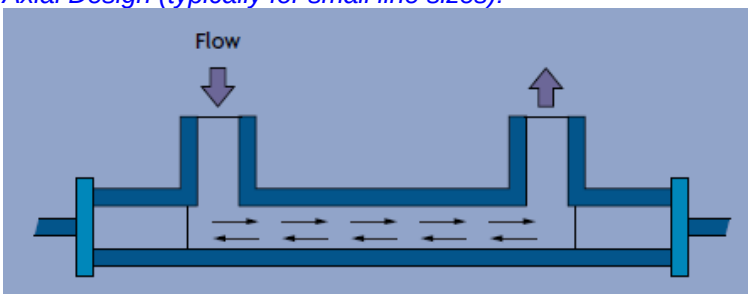
Radial Design:



Transit: $\Delta f = kV$
where $k = 2\cos\theta / Z$
Z = Distance from upstream transducer to downstream transducer

(For Axial Design Reference Next Page)

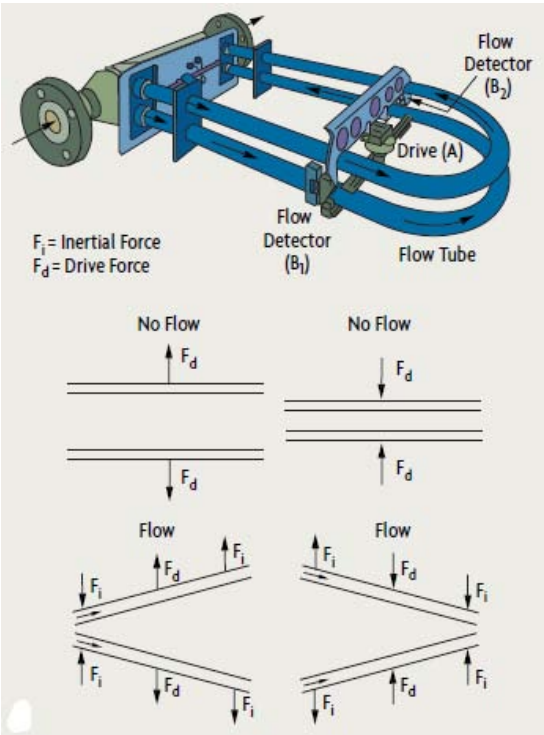
Axial Design (typically for small line sizes):



10.5 Mass Flowmeters:

10.5.1 Coriolis:

Tube design can be either straight or curved. When the design consist of two parallel tubes, the flow is divided equally between the two tubes and recombined at the end. Drivers (A) vibrate the tubes. These drivers consist of a coil connected to one tube and a magnet connected to the other. The transmitter applies an alternating current to the coil which causes the magnet to be attracted and repelled by turns, thereby forcing the tubes towards and away from one another. The sensor can detect the position, velocity or acceleration of the tubes. The magnet and the coil in the sensor change their relative positions as the tubes vibrate, causing a change in the magnetic field of the coil. Therefore the sinusoidal voltage output from the coil represents the motions of the tubes. When there is no flow in the tubes, the vibration caused by the coil and magnet result in identical displacements at the two sensing points (B1 and B2). When flow is present, coriolis forces act to produce a second twisting vibration, resulting in a small phase difference in the relative motions. This is detected at the sensing points. The deflection of the tubes only exists when both axial flow and tube vibration are present.



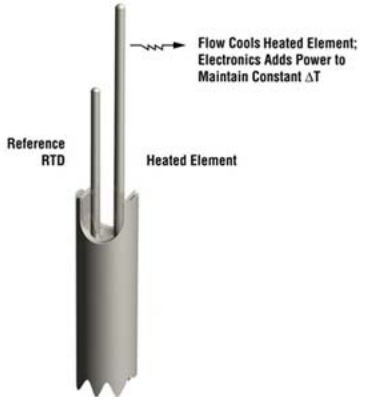
$F_{Cor} = -2m(w * v)$ **where**:
v = the velocity of the fluid in the measuring tube
w = the instantaneous angular speed of rotation
m = the mass of liquid in the tube section

10.5.2 Thermal Mass:

Based on an operational principle that states that the rate of heat absorbed by a flow stream is directly proportional to its mass flow. As molecules of a moving gas come into contact with a heat source, they absorb heat and thereby cool the source. At increased flow rates, more molecules come into contact with the heat source, absorbing even more heat. The amount of heat dissipated from the heat source in this manner is proportional to the number of molecules of a particular gas (its mass), the thermal characteristics of the gas, and its flow characteristics.

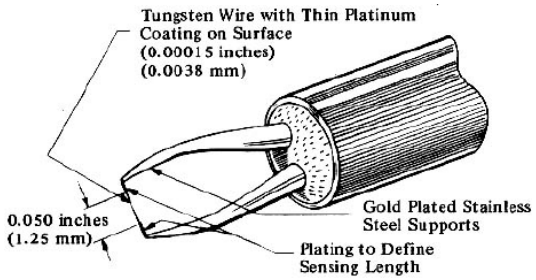
There are three basic operating methods which are commonly used to excite the sensor.

- **Constant temperature thermal mass flowmeters** require two active sensors (typically platinum RTDs) that are operated in a balanced state. One acts as a temperature sensor reference; the other is the active heated sensor. Heat loss produced by the flowing fluid tends to unbalance the heated flow sensor and it is forced back into balance by the electronics.
- **Constant power thermal mass flowmeters** are thermal (heat loss) mass flowmeters and require three active elements. A constant current heating element is coupled to an RTD. This heated RTD acts a heat loss flow sensor while a second RTD operates as an environmental temperature sensor. When the fluid is at rest the heat loss is at a minimum. Heat loss increases with increasing fluid velocity.
- **Calorimetric or energy balance thermal mass flowmeters** require one heating element and two temperature sensors. Typically the heater is attached to the middle of a flow tube with a constant heat input. Two matched RTDs or thermocouples are attached equidistant upstream and downstream of the heater. The temperature differential at flowing conditions is sensed, producing an output signal.



10.5.3 Hot-Wire Anemometer:

Consists of an electrically heated fine-wire element. Tungsten is traditionally used as the wire material because of its strength and high temperature coefficient of resistance. When placed in a moving stream of gas, the wire cools and the rate of cooling corresponds to the mass flow. As the fluid velocity increases, the rate of heat flow from the heated wire to the flow stream increases. Thus, a cooling effect on the wire electrode occurs. Causing the electrical resistance to change. In a constant-current anemometer, the fluid velocity is determined from a measurement of the resulting change in wire resistance. In a constant-resistance anemometer, fluid velocity is determined from the current needed to maintain a constant wire temperature, and thus, the resistance constant.

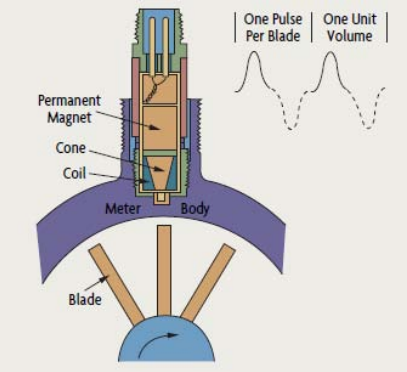


10.6 Mechanical Flowmeters:

10.6.1 Turbine Meter:

Consists of a multi-bladed rotor mounted at right angles to the flow and suspended in the fluid stream on a free running bearing. The speed of the rotor rotation is proportional to the volumetric flow. Turbine rotations can be detected by solid state devices (e.g. reluctance, inductance)

- **Reluctance** pickup, the coil is a permanent magnet and the turbine blades are made of material attracted to magnets. As each blade passes the coil, a voltage is generated in the coil. Each pulse represents a discrete volume of liquid. The number of pulses per unit volume is the meter's K-factor.



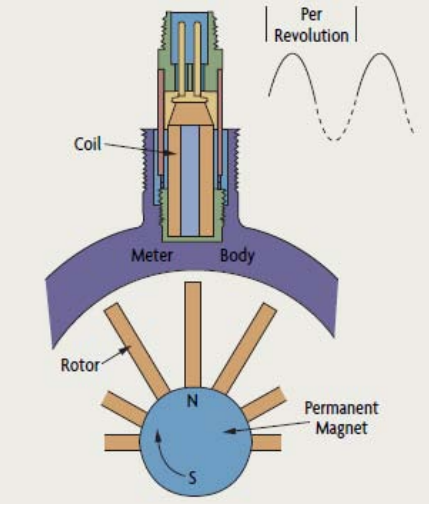
$V = KN$
V = Volume in cubic inches
K = Volume per pulse
N = Number of pulses

$$Q_{avg} = \frac{V}{\Delta t} = K\frac{N}{\Delta t}$$

But $f = \frac{N}{\Delta t}$ *is the number of pulses per unit time* $\therefore$

$$Q_{avg} = Kf$$

- **Inductance** pickup, the permanent magnet is embedded in the rotor, or the blades of the rotor are made of permanently magnetized material. As each blade passes the coil, it generates a voltage pulse.



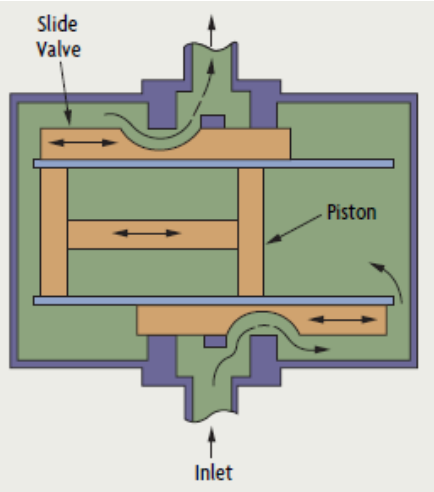
The outputs of reluctance and inductive pickup coils are continuous sine waves with pulse train's frequency proportional to the flow rate.

10.6.2 Positive-Displacement Meter:

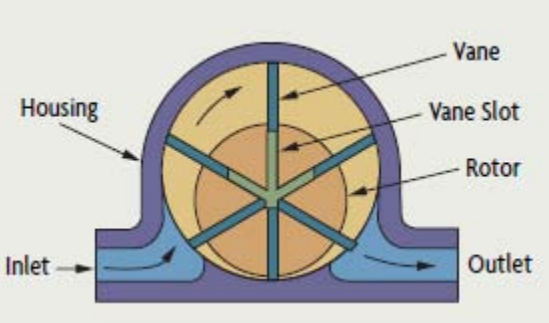
PD meters are operated by the kinetic energy of the flowing fluid. To avoid slippage of fluid between the mechanical components low viscosity applications should be avoided. Due to small clearances fluids must be clean for use with PD meters, slurries and abrasive fluids should be avoided. Most common use of PD meters is for custody transfer.

Common Types of PD Meters:

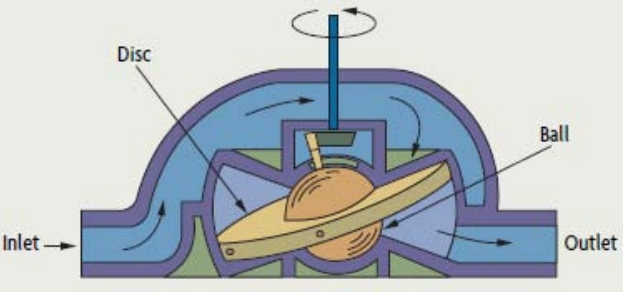
- **Piston** the piston pushes a known volume from the measuring chamber, a magnet is used to count the number of piston strokes.



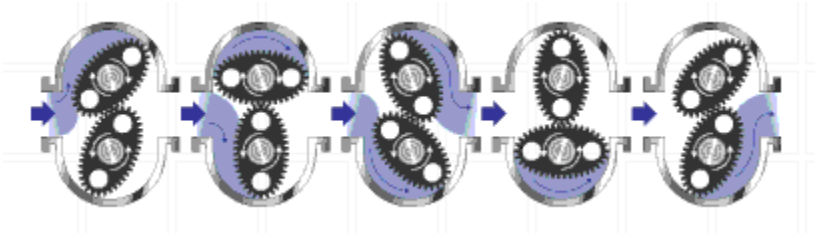
- **Rotary Vane** have spring loaded vanes that entrap increments of liquid between the eccentric mounted rotor and the casing. Rotation of the vanes moves the flow increment from inlet to outlet and discharge. The rotation is monitored by magnetic or photo-electric pickup, the frequency of the output being proportional to flow rate. Generally used for the petroleum industry.



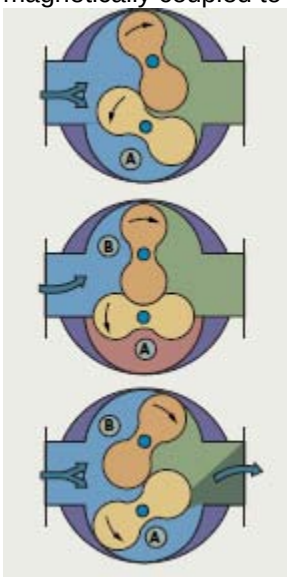
- **Nutating Disc** most common application is the water meter in a residential household. The movable element is a circular disk which is attached to a central ball. A shaft is attached to the ball and held in an inclined position. The disk is mounted in a chamber which has spherical side walls and conical top / bottom surfaces. The fluid enters an opening in the spherical wall on one side of the partition and leaves thru the other. As the fluid flows thru the chamber, the disk wobbles (nutating motion). Since the volume of fluid required to make the disk complete one revolution is known, the total flow can be calculated by counting the number of disk rotations by the known volume of fluid.



- **Oval Gear** consists of two fine toothed intermeshing gears, one mounted horizontal and the other vertically, which are rotated by the fluid passing thru the meter. This means that for every revolution of the pair of gears a specific quantity of liquid is carried thru the meter. A spindle extending from one of the gears is used to determine the number of revolutions which can then be converted to engineering units.



- **Rotating Lobe** is a variation of the oval gear meter that does not share its precise gearing. In the rotating lobe design, two impellers rotate in opposite directions within the housing. As they rotate, a fixed volume of liquid is entrapped and the transported to the outlet. Because the lobe gears remain in a fixed relative position it is only necessary to measure the rotational velocity of one of them. The gear is magnetically coupled to a register or transmitter.



10.6.3 Metering Pumps:

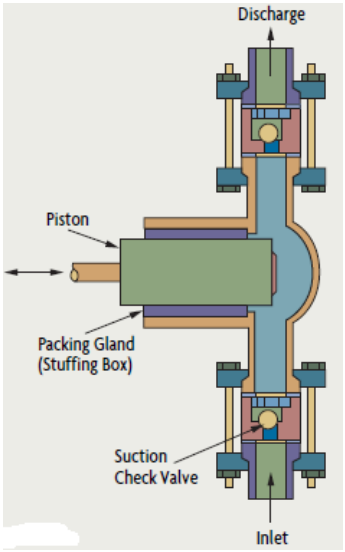
Metering pumps are PD meters that also impart kinetic energy to the process fluid. There are three basic designs of metering pumps:

Discharge line of a PD pump should never be blocked as dangerously high pressure could build up.

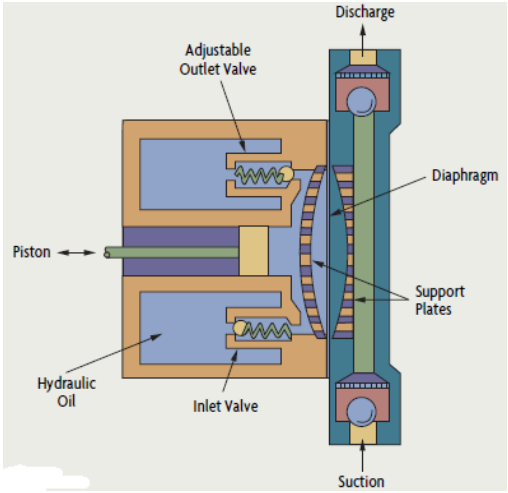
- **Peristaltic** operate by have fingers or a cam systematically squeeze a plastic tube against the housing. Each one of the fingers or cams produces a known volume. Typically used in labs, medical and pharma applications.



- **Piston** pumps deliver a fixed volume of liquid with each stroke. Piston pump generates a pulsating type flow. To minimize pulsation, dampening reservoirs may be installed.



- **Diaphragm** pumps are the most common industrial PD pump. Typical configuration consist of a single diaphragm, a chamber, and suction / discharge check valves to prevent backflow. The piston is what drives the diaphragm. Diaphragm pump generates a pulsating type flow. To minimize pulsation, dampening reservoirs may be installed.

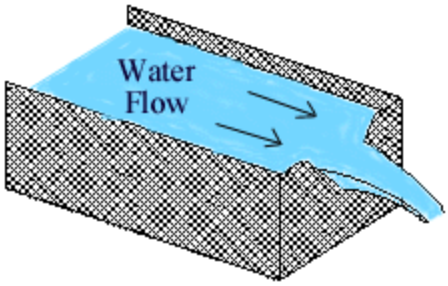


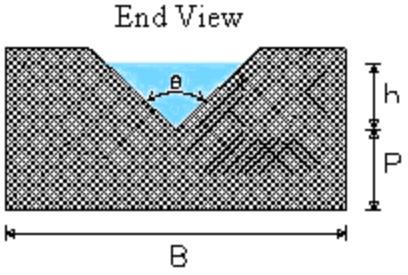
10.7 Open Channel Flow:

A common method of measuring flow through an open channel is to measure the height of the liquid as it passes over an obstruction as a flume or weir in the channel

10.7.1 Weir:

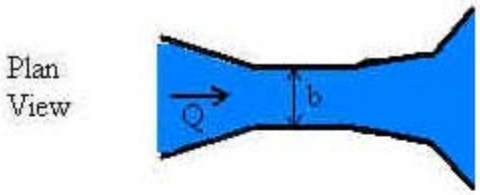
Weirs are typically installed in open channels such as streams to determine discharge (flowrate). The basic principle is that discharge is directly related to the water depth above the crotch (bottom) of the V; this distance is called head (h). The V-notch design causes small changes in discharge to have a large change in depth allowing more accurate head measurement than with a rectangular weir. Depth is usually measured with an ultrasonic level device.



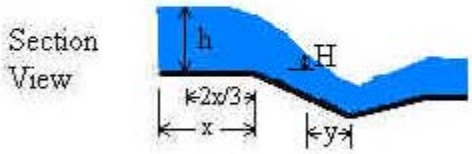


10.7.2 Flume:

A free flowing flume can be identified by the drop in water depth at the flume throat. In submerged flow, the downstream water backs up into the throat swallowing the drop making the drop difficult or impossible to identify. Analysis of submerged flow requires two head measurements - one in the approach channel and one in the throat. Whereas, free flow requires only the upstream head measurement.

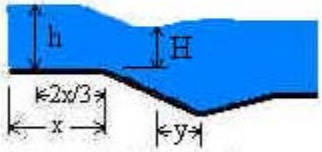


y=0.05 m for b<3.05 m
y=0.305 m for b≥3.05 m



Free flow

H/h≤0.6 for b<3.05 m
H/h≤0.8 for b≥3.05 m



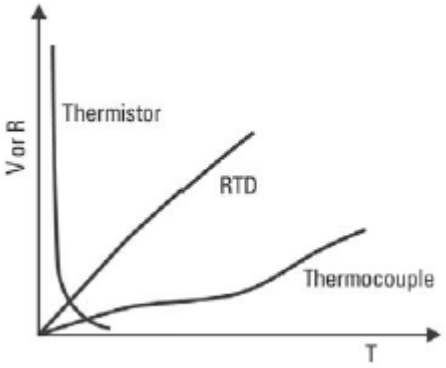
Submerged

H/h>0.6 for b<3.05 m
H/h>0.8 for b≥3.05 m

11 Temperature Measurement

11.1 Temperature Sensor Comparison:

- Linearity:



- Comparison Table:

Relative Advantage of Contact Temperature Sensors			
Quality	T/Cs	RTDs	Thermistors
Temp Range	**-400 to 4200°F**	**-200 to 1475°F**	**-176 to 392°**
Accuracy	**< RTD**	**> T/C**	**> T/C & RTD**
Ruggedness	**Highly Rugged**	**Sensitive to Shock**	**NOT Rugged**
Linearity	**Highly NON-Linear**	**Somewhat NON-Linear**	**Highly NON-Linear**
Drift	**Subject to Drift**	**< T/C**	**< T/C**
Cold Junction	**Required**	**None**	**None**
Compensation Response	**Fast**	**Relatively Slow**	**Faster than RTD**
Cost	**Low, except for noble metals**	**> RTDs**	**Low**

11.2 Thermocouple:

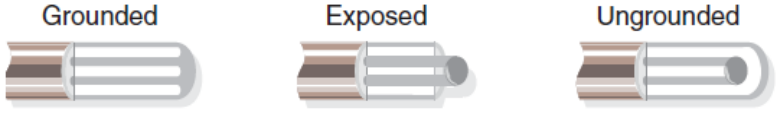
A thermocouple is a sensor for measuring temperature. It consists of two dissimilar metals, joined together at one end, which produce a small unique voltage at a given temperature. This voltage is measured and interpreted by a thermocouple thermometer.

The following criteria are used in selecting a thermocouple:

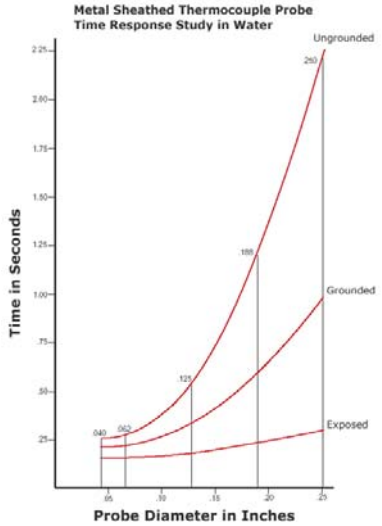
- Temperature range
- Chemical resistance of the thermocouple or sheath material
- Abrasion and vibration resistance
- Installation requirements (may need to be compatible with existing equipment; existing holes may determine probe diameter).

11.2.1 Thermocouple Junctions:

Sheathed thermocouple probes are available with one of three junction types: grounded, ungrounded or exposed. At the tip of a grounded junction probe, the thermocouple wires are physically attached to the inside of the probe wall. This results in good heat transfer from the outside, through the probe wall to the thermocouple junction. In an ungrounded probe, the thermocouple junction is detached from the probe wall. Response time is slowed down from the grounded style, but the ungrounded offers electrical isolation.



- The **grounded junction** is recommended for the measurement of static or flowing corrosive gas and liquid temperatures and for high pressure applications. The junction of a grounded thermocouple is welded to the protective sheath giving faster response than the ungrounded junction type.
- An **ungrounded junction** is recommended for measurements in corrosive environments where it is desirable to have the thermocouple electronically isolated from and shielded by the sheath. The welded wire thermocouple is physically insulated from the thermocouple sheath by MgO powder (soft).
- An **exposed junction** is recommended for the measurement of static or flowing non-corrosive gas temperatures where fast response time is required. The junction extends beyond the protective metallic sheath to give accurate fast response. The sheath insulation is sealed where the junction extends to prevent penetration of moisture or gas which could cause errors.



11.2.2 Thermocouple Types:

T/C Type	Names of Materials	Insulation Colors	Useful Range
B	**Platinum30%Rhodium (+) Platinum 6% Rhodium (-)**	**+ Grey - Red Overall: Grey**	**2500 -3100°F (1370-1700°C)**
C	**W5Re Tungsten 5% Rhenium (+) W26Re Tungsten 26% Rhenium (-)**	**+ White/Red (Glass Braid) - Red (Glass Braid) Overall: White/Red (Glass Braid)**	**3000-4200°F (1650-2315°C)**
E	**Chromel (+) Constantan (-)**	**+ Purple - Red Overall: Purple**	**200-1650°F (95-900°C)**
J	**Iron (+) Constantan (-)**	**+ White - Red Overall: Black**	**200-1400°F (95-760°C)**
K	**Chromel (+) Alumel (-)**	**+ Yellow - Red Overall: Red**	**200-2300°F (95-1260°C)**
N	**Nicrosil (+) Nisil (-)**	**+ Orange - Red Overall: Orange**	**1200-2300°F (650-1260°C)**
R	**Platinum 13% Rhodium (+) Platinum (-)**	**+ Black - Red Overall: Green**	**1600-2640°F (870-1450°C)**
S	**Platinum 10% Rhodium (+) Platinum (-)**	**+ Black - Red Overall: Green**	**1800-2640°F (980-1450°C)**
T	**Copper (+) Constantan (-)**	**+ Blue - Red Overall: Red**	**-330-660°F (-200-350°C)**

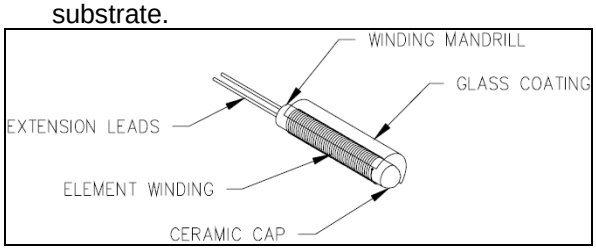
11.2.3 Thermocouple RASS Rule:

- For every T/C you have a cold junction effect when you connect the leads to a measurement device or a simulated voltage input device.
- T/C measurement devices will automatically add or subtract the cold junction effect by internal circuitry. Common VOMs and voltage sources do not have the built-in circuitry, therefore compensations need to be made:
  - **Receive** – **Add** the cold junction effect
  - **Send – Subtract** the cold junction effect

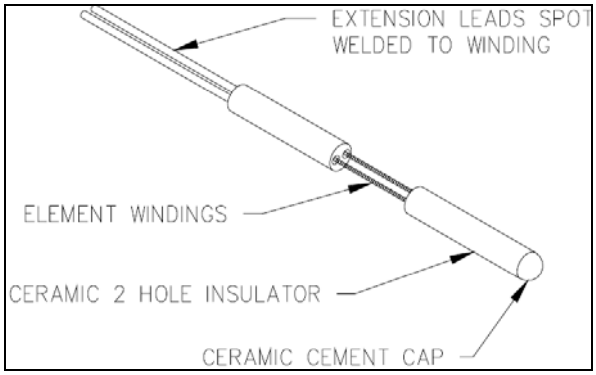
11.3 RTD:

Resistance Temperature Detectors (RTDs) are temperature sensors that contain a resistor changes resistance value as its temperature changes.
A typical RTD consists of a fine platinum wire wrapped around a mandrel and covered with a protective coating (basically three types of RTD construction:

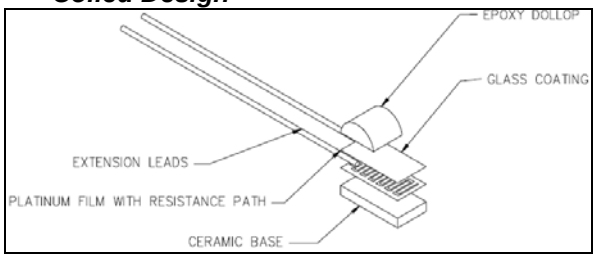
- **Wire Wound:** Simplest design, the sensor wire is wrapped around an insulating core or mandrel. Care must be taken during the manufacture to reduce mechanical strain on the winding and core materials.
- **Coiled:** "Strain Free", allows the sensing wire to expand and contract.
- **Thin Film:** Manufactured by depositing a very thin layer of platinum on a ceramic substrate.



Wire Wound



Coiled Design



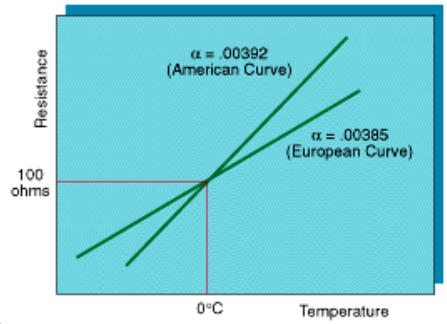
Thin Film

11.3.1 RTD Standards:

There are two standards for Pt RTDs: the European Standard (DIN/IEC) and the American Standard:

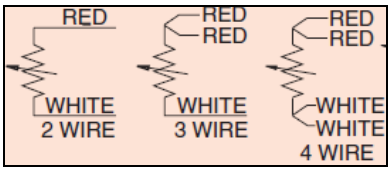
- **European** Standard (IEC751) is considered the world-wide standard for Pt RTDs. This standard requires the RTD to have an electrical resistance of 100.0Ω at 100°C and a temperature coefficient of resistance of 0.00385Ω/Ω/°C between 0 and 100°C

- **American** Standard used mostly in North America, has a resistance of 100.00 ±0.10Ω at 0°C and a temperature coefficient of 0.00392Ω/Ω/°C between -100 and 457°C



11.3.2 RTD Wiring Configuration:

- **2-Wire** loop construction, the sensor resistor measurement includes the lead wire resistance. The loop resistance is measured and subtracted from the sensor resistance. The 2-Wire construction is typically used only with high resistance sensors, when the lead length will be very short.
- **3-Wire** loop construction is the most common design, found in industrial process and monitoring applications. The lead wire resistance is factored out as long as all of the lead wires have the same resistance.
- **4-Wire** loop construction is typically found in laboratories and other applications where very precise measurement is needed. The fourth wired allows the measurement equipment to factor out all of the lead wire and other unwanted resistance from the measurement circuit.



11.3.3 RTD Accuracy:

- **Class A** RTD: Highest RTD Element tolerance and accuracy, Class A (IEC-751), Alpha = 0.00385 (accuracy of ±0.43Ω at 600°C)
- **Class B** RTD: Most Common RTD Element tolerance and accuracy, Class B (IEC-751), Alpha = 0.00385 (accuracy of ±1.06Ω at 600°C)

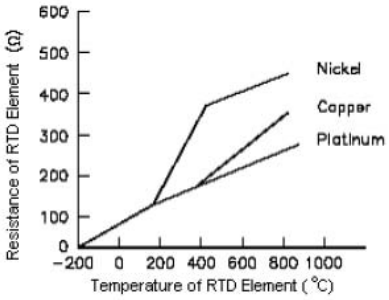
Permissible deviations from basic values

Class A		
Temperature °C	Deviation	
	ohms	°C
-200	±0.24	±0.55
-100	±0.14	±0.35
0	±0.06	±0.15
100	±0.13	±0.35
200	±0.20	±0.55
300	±0.27	±0.75
400	±0.33	±0.95
500	±0.38	±1.15
600	±0.43	±1.35
650	±0.46	±1.45

Class B		
Temperature °C	Deviation	
	ohms	°C
-200	±056	±1.3
-100	±0.32	±0.8
0	±0.12	±0.3
100	±0.30	±0.8
200	±0.48	±1.3
300	±0.64	±1.8
400	±0.79	±2.3
500	±0.93	±2.8
600	±1.06	±3.3
650	±1.13	±3.6
700	±1.17	±3.8
800	±1.28	±4.3
850	±1.34	±4.6

11.3.4 RTD Types:

- Copper is used occasionally as an RTD element. Its low resistivity forces the element to be longer than a platinum element, but its linearity and very low cost make it an economical alternative. Its upper temperature limit is only about 120ºC.
- The most common RTD's are made of either platinum, nickel, or nickel alloys (Balco). The economical nickel derivative wires are used over a limited temperature range. They are quite non-linear and tend to drift with time. For measurement integrity, platinum is the obvious choice.
  - **Balco**: Annealed resistance alloy with a nominal composition of 70% nickel and 30% iron.
  - **Platinum**: Exhibit linear response and stability over time. Most versatyile because of its wide temperature range (-200°C - 850°C)



11.4 Thermistor:

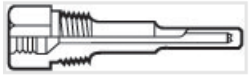
- A thermistor is a temperature-sensing element composed of sintered semiconductor material which exhibits a large change in resistance proportional to a small change in temperature. Thermistors usually have negative temperature coefficients which means the resistance of the thermistor decreases as the temperature increases.
- Thermistors are usually designated in accordance with their resistance at 25°C. The most common of these ratings is 2252 ohms.
- Limited Spans, also limited to low-medium temperatures (max is 100 to 200°C)

11.5 Thermowell:

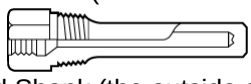
Thermowells are used in industrial temperature measurement to provide isolation between a temperature sensor and the environment whose temperature is to be measured. They are intrusive fittings and are subject to static and dynamic fluid forces. These forces govern their design. Vortex shedding is the dominant concern as it is capable of forcing the thermowell into flow-induced resonance and consequent fatigue failure. The latter is particularly significant at high fluid velocities.
The ASME Performance Test Code (PTC 19.3 – Temperature Measurement) is the most widely used basis for thermowell design

Traditional Shank Styles Available:

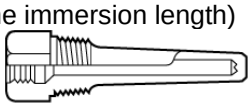
- Step Shank (has an outer diameter of ½" at the end of the thermowell immersion length to provide a quicker response time)



- Straight Shank (same size all along the immersion length)



- Tapered Shank (the outside diameter of the thermowell decreases gradually along the immersion length)



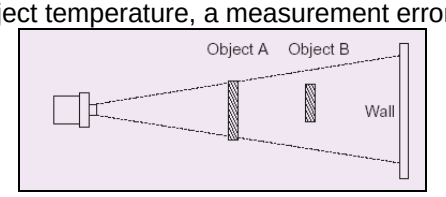
- Built-up Step Shank (typically use for high velocity application or long insertion lengths)

Lagging: The lagging extension of a thermowell is often referred to as the thermowell's "T" length. The lagging extension or T length is located on the cold side of the process connection and is usually an extension of the hex length of the thermowell. Typically, the T length enables the probe and thermowell to extend through insulation or walls.

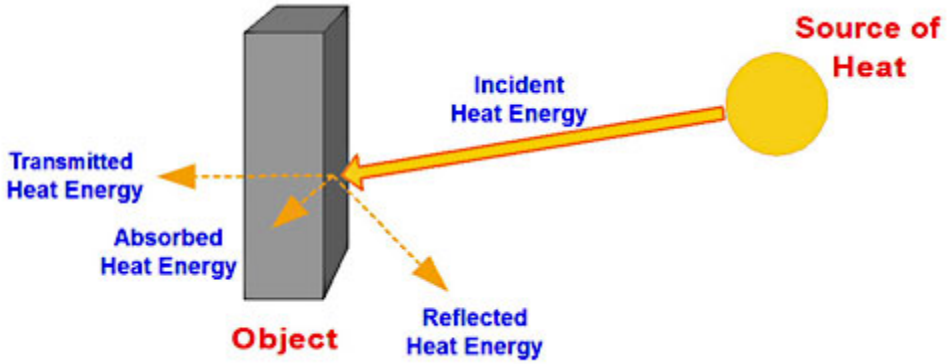
Thermowell Resonance: Thermowells have been known to fail due to induced vibrations from the fluid flowing past it. The problem is in general confined to the flow of gases as their high velocities lead to higher vortex shedding frequencies and the low mass and viscosities do not provide any damping to the thermowells. In general, if this phenomenon is likely to happen, you should design the thermowell in such a way that the maximum Strouhal frequency (fs) is not higher than 70% or 75% of the thermowell natural resonance frequency (ft). Options are to increase the diameter of the thermowell or shorten the length of the thermowell.

11.6 Infra-Red:

- These measure the amount of radiation emitted by a surface. Electromagnetic energy radiates from all matter regardless of its temperature. In many process situations, the energy is in the infrared region. As the temperature goes up, the amount of infrared radiation and its average frequency go up.
- Infrared pyrometers allow users to measure temperature in applications where conventional sensors cannot be employed. Specifically, in cases dealing with moving objects (i.e., rollers, moving machinery, or a conveyer belt), or where non-contact measurements are required because of contamination or hazardous reasons (such as high voltage), where distances are too great, or where the temperatures to be measured are too high for thermocouples or other contact sensors.
- The field of view is the angle of vision at which the instrument operates, and is determined by the optics of the unit. To obtain an accurate temperature reading, the target being measured should completely fill the field of view of the instrument. Since the infrared device determines the average temperature of all surfaces within the field of view, if the background temperature is different from the object temperature, a measurement error can occur.



WI = WR + WT + WA
Where:
WI = incident energy received by the object, W
WR = energy reflected off the object's surface, W
WT = energy transmitted by the object, W
WA = energy absorbed by the object, W



As the object absorbs energy and heats, it also emits energy. When an object is in a state of thermal equilibrium, the amount of energy it absorbs (WA) equals the amount of energy it emits (WE): WA = WE. When an object absorbs more energy and its temperature increases, the amount of radiation it emits also increases.
IR thermometry is based on the fact that any body (solid, liquid, or gaseous) that has a temperature above absolute zero (0°K or -273°C) emits radiant energy. This energy is proportional to the forth power of the body temperature, and the body's ability to absorb and emit IR energy is called emissivity. Energy radiated by a body can be expressed as follows:

$W = E \sigma T^4 A$

Where:

W = energy, W
E = emissivity
σ = Stefan-Boltzmann Constant = $5.6703\ 10^{-8}$, W/m^2K^4
T = absolute temperature, °K
A = emitting area, m^2

Practical considerations

- Avoid degrading measurement accuracy by environmental elements, such as dirt, dust, smoke, steam, other vapors, extremely high or low ambient temperatures, and electromagnetic interference from other devices.
- Select an IR thermometer with a wavelength band compatible with the measured object (especially high reflectivity objects) and with the media between the thermometer and measured object (especially glass, smoke, or steam).
- Select an instrument with a temperature range not much greater than the maximum application temperature. Wider than needed temperature ranges lead to lower accuracy or higher instrument cost.
- An IR thermometer averages the temperature of all objects within its field of view: Select the instrument with an appropriate FOV, and calculate the proper distance so that only the desired area is measured.
- Avoid hot objects near the measured object. They radiate energy that can be reflected or transmitted by the measured object into the thermometer FOV.

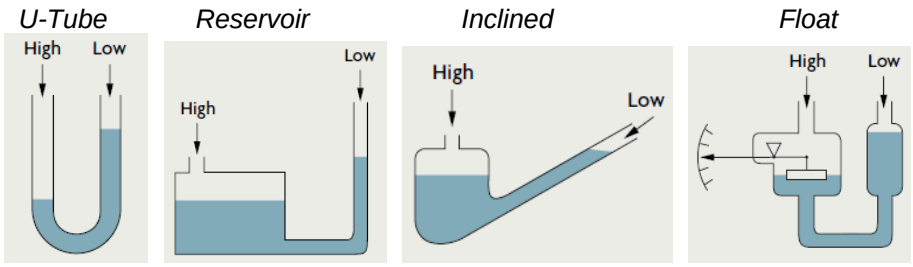
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12 Pressure Measurement

12.1 Sensing Elements:

12.1.1 Manometers:

A basic manometer consists of a reservoir filled with liquid and a vertical tube. The difference in the two column heights indicates the process vacuum. Capable of detecting vacuums down to 1 millitorr.



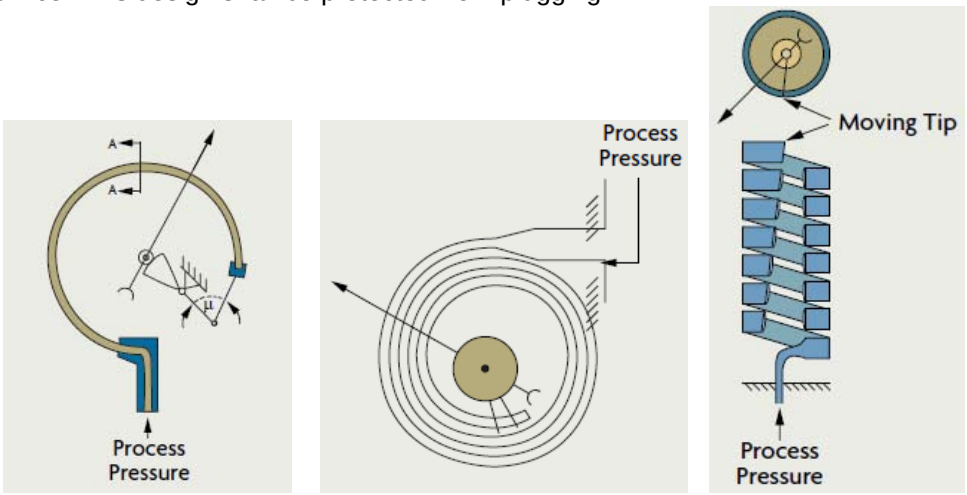
12.1.2 C / Spiral / Helical Bourdon Tube:

C / Spiral Bourdon Tube:

Consists of a thin-walled tube that is flattened diametrically on opposite sides to produce a cross-sectional area elliptical in shape, having two long flat sides and two short round sides. The tube is bent lengthwise into an arc of a circle (270 to 300°). Pressure applied to the inside of the tube causes dissention of the flat sections and tends to restore its original round cross-section. This change in cross-section causes the tube to straighten slightly. Since the tube is permanently attached at one end, the tip of the tube traces a curve that is the result of the change in angular position with respect to the center. (Should not be over-pressurized as this may stretch the bourdon tube)

Helical Bourdon Tube:

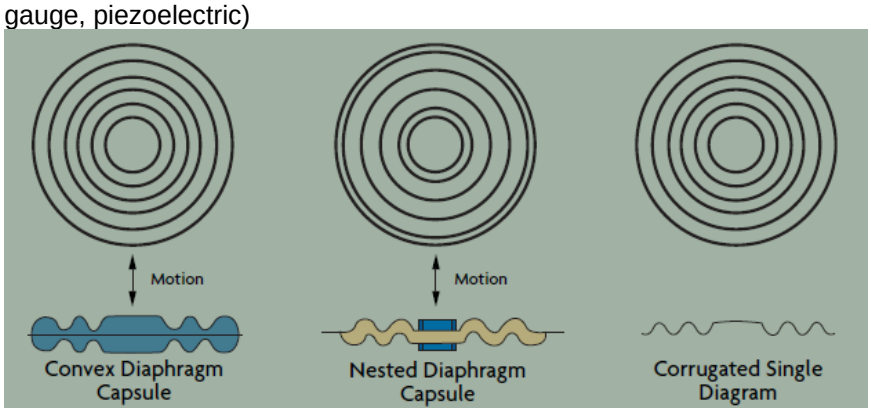
This device may include as many as 20 coils, and can measure pressures in excess of 10,000psig. Provides high over-range protection and is suitable for fluctuating pressure service. This design shall be protected from plugging.



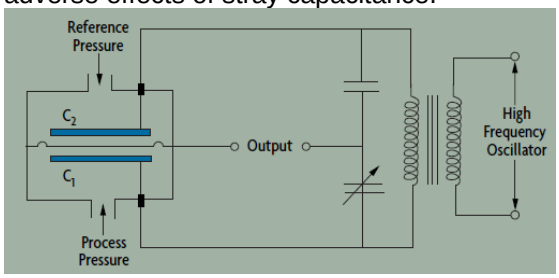
C Spiral Helical

12.1.3 Capsule / Diaphragm:

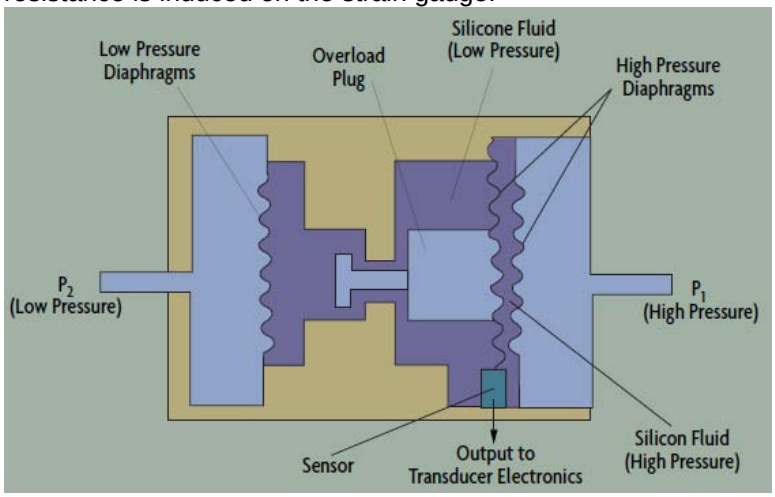
Diaphragms are popular because the require less space and because their motion (or force) they produce is sufficient for operating electronic transducers (e.g. capacitance, strain gauge, piezoelectric)



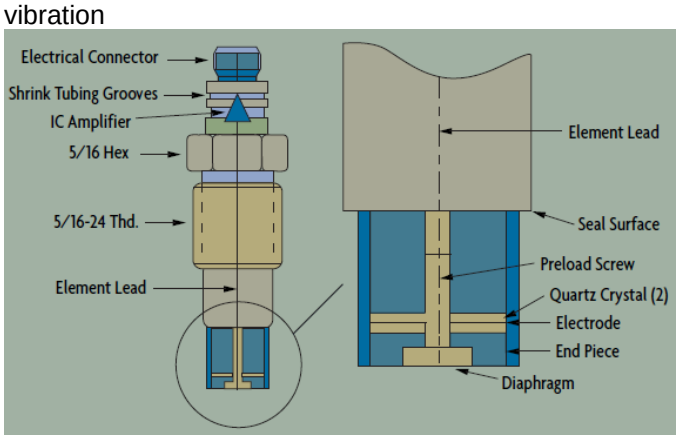
- **Capacitance** pressure sensors use a thin diaphragm, usually metal or metal-coated quartz, as one plate of a capacitor. The diaphragm is exposed to the process pressure on one side and to a reference pressure on the other. Changes in pressure cause it to deflect and change the capacitance. The change may or may not be linear with pressure and is typically a few percent of the total capacitance. The capacitance can be monitored by using it to control the frequency of an oscillator or to vary the coupling of an AC signal. It is good practice to keep the signal-conditioning electronics close to the sensor in order to mitigate the adverse effects of stray capacitance.



- **Strain Gauge** sensors make use of a strain gauge and a diaphragm. When a change in pressure causes the diaphragm to deflect, a corresponding change in resistance is induced on the strain gauge.

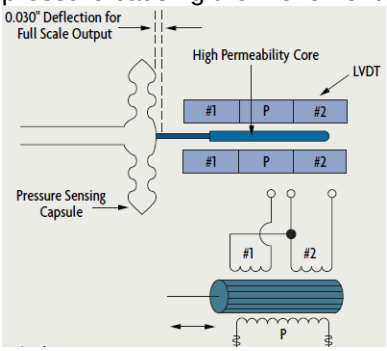


- **Piezoelectric** sensors take advantage of the electrical properties of naturally occurring crystals such as quartz. These crystals generate an electrical charge when they are strained. These sensors however, are very susceptible to shock and vibration



12.1.4 LVDT:

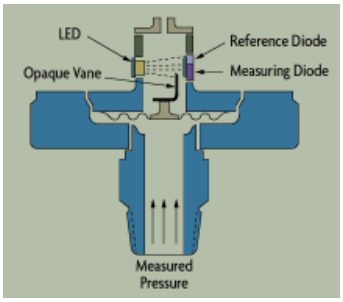
Linear **V**ariable **D**ifferential **T**ransformer sensor operates on the inductance ratio principle. In this design, three coils are wired onto an insulating tube containing an iron core, which is positioned within the tube by the pressure sensor. Alternating current is applied to the primary coil in the center, and if the core also is centered, equal voltages will be induced in the secondary coils (#1 and #2). Because the coils are wired in series, this condition will result in a zero output. As the process pressure changes and the core moves, the differential in the voltages induced in the secondary coils is proportional to the pressure causing the movement.



12.1.5 Optical:

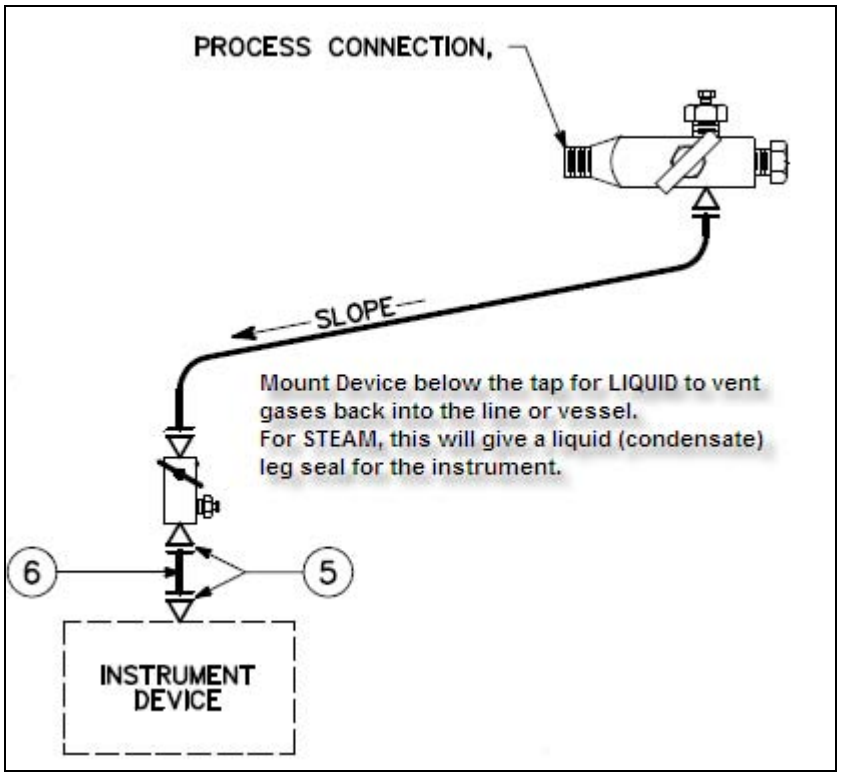
Optical pressure transducers detect the effects of minute motions due to changes in process pressure and generate a corresponding electronic output signal (Figure 3-11). A light emitting diode (LED) is used as the light source, and a vane blocks some of the light as it is moved by the diaphragm. As the process pressure moves the vane between the source diode and the measuring diode, the amount of infrared light received changes.

The optical transducer must compensate for aging of the LED light source by means of a reference diode, which is never blocked by the vane. This reference diode also compensates the signal for build-up of dirt or other coating materials on the optical surfaces.

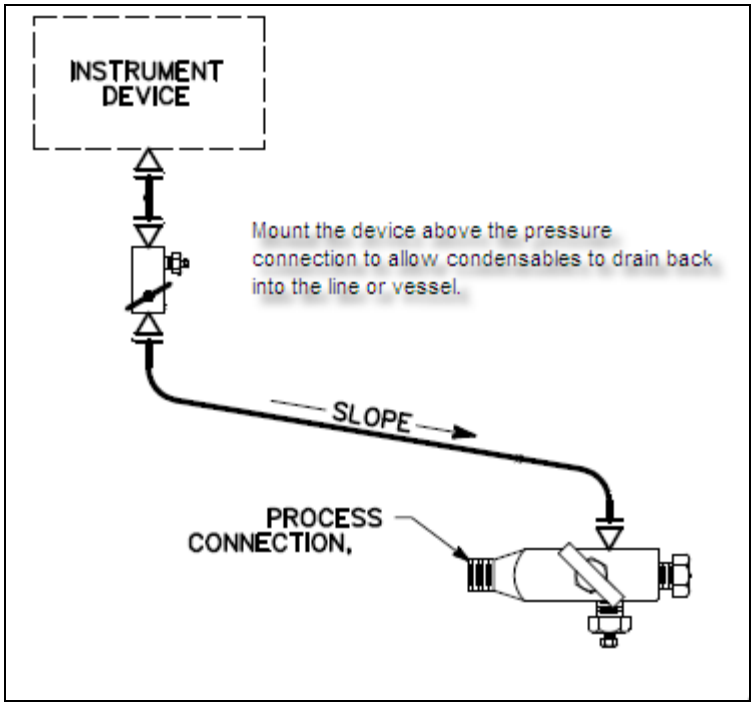


12.1.6 Pressure Installation Details:

12.1.6.1 Steam / Liquid Service



12.1.6.2 Gas Service



12.2 Pressure Regulators:

Within the broad categories of direct-operated and pilot-operated regulators fall virtually all of the general regulator designs, including:

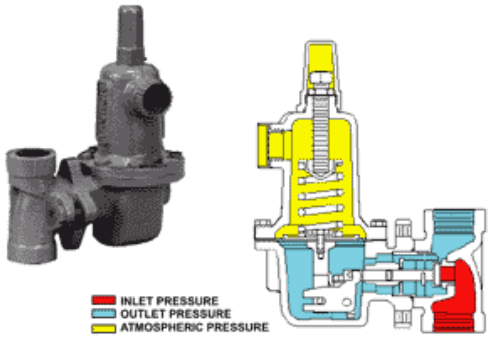
- Pressure reducing regulators
- Backpressure regulators
- Vacuum regulators and breakers

12.2.1 Pressure Reducing Regulator:

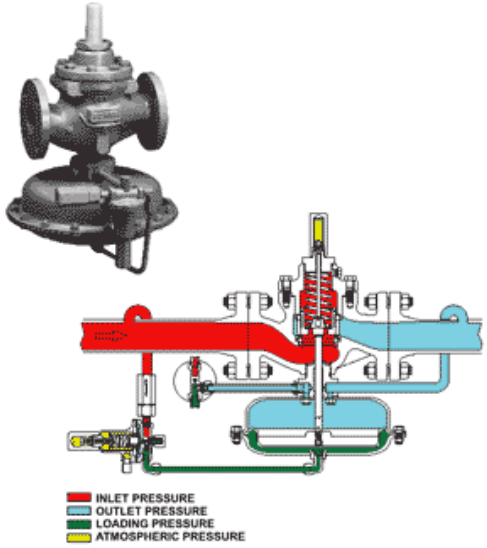
A pressure reducing regulator maintains a desired reduced outlet pressure while providing the required fluid flow to satisfy a downstream demand. The pressure which the regulator maintains is the outlet pressure setting (setpoint) of the regulator. Three-way switching valves direct inlet pressure from one outlet port to another whenever the sensed pressure exceeds or drops below a preset limit. All regulators fit into one of the following two categories:

Direct-operated regulators generally have faster response to quick flow changes than pilot-operated regulators

- **Direct-Operated** (also called Self-Operated): In operation, a direct-operated, pressure reducing regulator senses the downstream pressure through either internal pressure registration or an external control line. This downstream pressure opposes a spring which moves the diaphragm and valve plug to change the size of the flow path through the regulator.

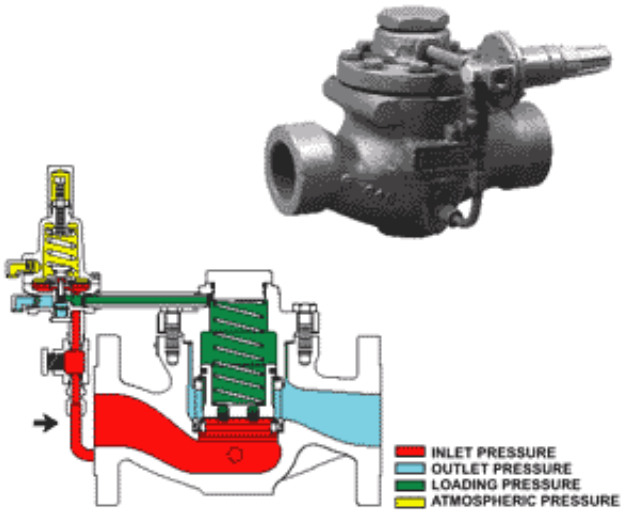


- **Pilot-Operated**: Preferred for high flow rates or where precise pressure control is required. A popular type of pilot-operated system uses two-path control. In two-path control, the main valve diaphragm responds quickly to downstream pressure changes, causing an immediate correction in the main valve plug position. At the same time, the pilot diaphragm diverts some of the reduced inlet pressure to the other side of the main valve diaphragm to control the final positioning of the main valve plug. Two-path control results in fast response.



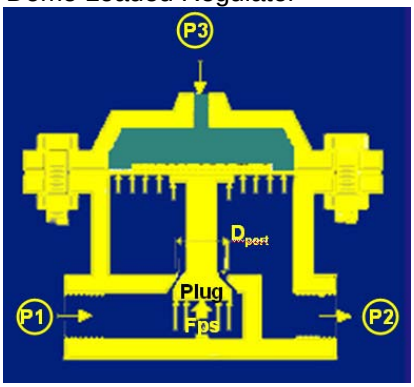
12.2.2 Back Pressure Regulator:

Backpressure regulator maintains a desired upstream pressure by varying the flow in response to changes in upstream pressure.



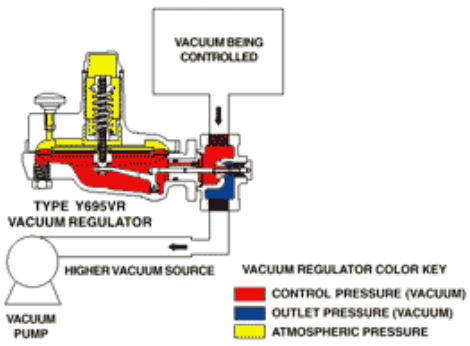
12.2.3 Pressure Loaded Regulator:

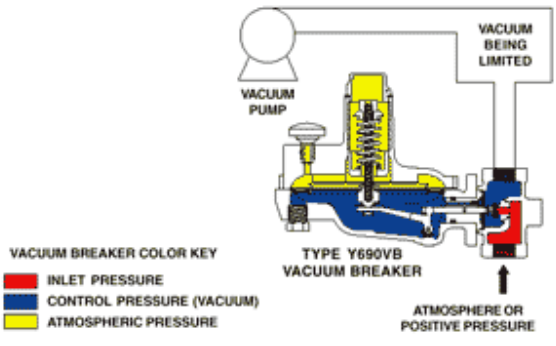
A regulator using a fixed volume and pressure of compressible fluid as a spring and set point reference to accomplish pressure reduction or back pressure regulation. sometimes called a Dome Loaded Regulator



12.2.4 Vacuum Regulators & Breakers:

Vacuum regulators and vacuum breakers are devices used to control vacuum. A vacuum regulator maintains a constant vacuum at the regulator inlet with a higher vacuum connected to the outlet. During operation, a vacuum regulator remains closed until a vacuum decrease (a rise in absolute pressure) exceeds the spring setting and opens the valve disk. A vacuum breaker prevents a vacuum from exceeding a specified value. During operation, a vacuum breaker remains closed until an increase in vacuum (a decrease in absolute pressure) exceeds the spring setting and opens the valve disk.





12.2.5 Applying Regulators:

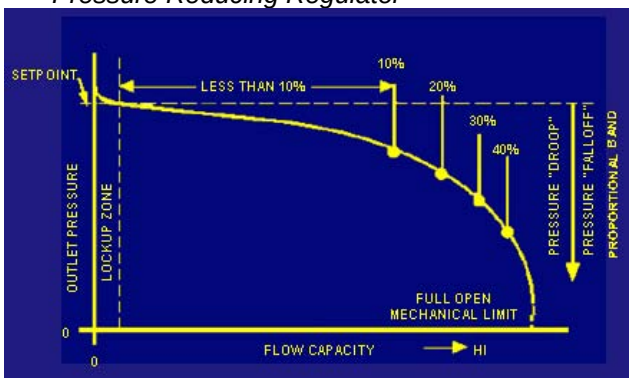
FACTORS CONSIDERED	BEST	NEXT	LAST
Line Size Cost– Thru 1-1/2”	REG	POR	CRV
– 2” thru 3”	POR	REG	CRV
– 4” and Up	CRV	POR	REG
Cost/CV – Thru 1-1/2”	POR	REG	CRV
– 2” and Up	CRV	POR	REG
Capacity	CRV	POR	REG
Outlet Pressure Level Capability	CRV	REG	POR
Output Pressure Level Maint of SP	CRV	POR	REG
Rangeability	FTO REG	CRV/POR	FTC REG
Requirements for External Power (Air or Electricity)	REG/POR		CRV
Stability	CRV	REG	POR
Speed of Response	REG	POR	CRV WOP CRV WP
Ability to Adapt to System Dynamics	CRV	POR	REG
Fail-Safe Action		CRV	POR/REG
Adaptability (Add accessories, modify action)	CRV	REG	PRO
Remote Set Point Capability	CRV	Regulator, Dome Loaded	POR/REG
Maintenance Cost/Spare Parts	REG	POR	CRV

Key:
REG = Self-contained Regulator (FTC – Flow to Close)(FTO – Flow to Open)
POR = Pilot Operated Regulator
CRV = Control Valve
CRV WP = Control Valve w/positioner
CRV WOP = Control Valve w/out positioner

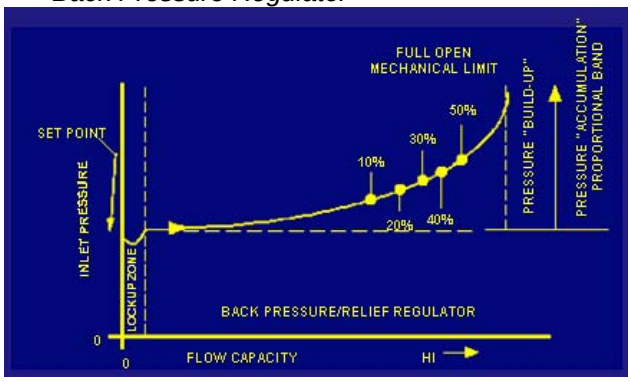
12.2.6 Regulator Droop:

Droop is the reduction of outlet pressure experienced by pressure-reducing regulators as the flow rate increases. It is stated as a percent, in inches of water column (mbar) or in pounds per square inch (bar) and indicates the difference between the outlet pressure setting made at low flow rates and the actual outlet pressure at the published maximum flow rate. Droop is also called offset or proportional band.

Pressure Reducing Regulator



Back Pressure Regulator



12.2.7 Regulator w/External Control Line:

Use control lines of equal or greater size than the control tap on the regulator. If a long control line is required, make it bigger. A rule of thumb is to use the next nominal pipe size for every 20 feet (6,1 meters) of control line. Small control lines cause a delayed response of the regulator, leading to increased chance of instability. 3/8-Inch OD tubing is the minimum recommended control line size.
Do not place control lines immediately downstream of rotary or turbine meters

12.2.8 Regulator Casing Vent:

Diaphragms leak a small amount due to migration of gas through the diaphragm material. To allow escape of this gas, be sure casing vents (where provided) remain open. Vents should be pointed down to help avoid the accumulation of water condensation or other materials in the spring case. Do not use small diameter, long vent lines. Use the rule of thumb of the next nominal pipe size every 10 feet (6,1 meters) of vent line and 3 feet (0,91 meters) of vent line for every elbow in the line

12.2.9 Regulator Hunting:

Do not oversize regulators. Pick the smallest orifice size or regulator that will work. Hunting is the rapid opening and closing of the regulator, this occurs when the regulator tries to respond to cyclic fluctuations caused by pulsations in the system (a very strong indication that the regulator is over-sized).

13 Level Measurement

13.1 Level Device Evaluation Table:

Level Device	Applications: Liquids: Clean	Viscous	Slurry/ Sludge	Interface	Foam	Solids: Powder	Chunky	Sticjy	Accuracy (Full Scale) unless noted as inches	Maximum Temp °F NonCoated (Coated)	Available as Non Contact	Limitations
D/P Type												
D/P	E	G-E	G	P	X	X	X	X	+/- 1 - 2%	1200		Not suitable for uses where SG changes
Bubbler	G	F	P	F	X	X	X	X	+/- 1 - 2%	NA		High maintenance, wastes air or N2
Capacitance	G	F-G	F	G-L	P	F	F	P	+/- 1 - 2%	1800 (500)		Excessive material buildup can prevent operation.
Conductivity	F	P	F	L	L	L	L	L	+/- 0.025"	1800		Process to be measured must be conductive
Displacer	E	P	P	F-G	X	X	X	X	0.50%	850		Not recommended for slurry/sludge service
Float	G	P	P	F	X	X	X	X	+/- 1%	500		Moving parts limits to clean service
Laser	L	G	G	X	F	F	F	F	+/- 0.4"	NA	✓	Typically used on solids
Level Gage	G	F	P	F	X	X	X	X	+/- 0.5"	700		Uses glass, Moving parts limits to clean service
Optical	G	F	E	F-G	F	F	P	F	+/- 0.25'	260	✓	Typically point switch use only
Magnetostrictive	E	P	P	F-G	X	X	X	X	0.50%	212		Not recommended for slurry/sludge service
Nuclear	G	E	E	G-L	F	G	E	E	+/- 0.25"	NA	✓	Requires NRC license
Radar	G	G	F	P	X	P	F	P	+/- 0.5"	450	✓	Interference from tank internals or excessive turbulence
Rotating Paddle	X	X	X	X	X	G	F	P	+/- 1"	500	✓	Limited to dry non-corrosive, low-pressure solids
Thermal	G	F	F	P	F	X	X	X	+/- 0.5"	850		Typically point switch use only
Ultrasonic	F-G	G	G	F-G	F	F	F	G	+/- 1 - 2%	300	✓	Does not work in vacuum.
Vibratory	F	G	G	F	F	F	F	G	+/- 0.25"	300		Excessive material buildup can prevent operation.
TDR/PDS	F	F	F	X	X	G	G	F	+/- 3"	220		Limited performance on sticky process materials

TDR = Time Domain Reflectometry
PDS = Phase Difference Sensors

E	EXCELLENT
G	GOOD
F	FAIR
P	POOR
L	LIMITED
X	DO NOT USE

13.2 D/P Level:

Liquid level can be measured (inferred) by measuring a D/P caused by the weight of the fluid column in a vessel balanced against a reference.

- For atmospheric vessels, the high side is connected to the bottom of the vessel, and the low side (reference is vented to atmosphere)
- For pressurized vessels, the high side is connected to the bottom of the vessel, and the low side is connected to the vapor space section (top) of the vessel.
- Extended diaphragms may be used on viscous, slurry or other plugging type applications. This design eliminates dead-ended cavities.
- Chemical seals may be used with D/P level devices where the process is corrosive or toxic. Whenever the length of the capillaries for the low and high side are of different lengths, this may introduce an error into the reading for small D/P readings as the thermal expansion will be different for both legs.

Calibration range = Maximum height of column liquid X S.G.
Whenever the transmitter is located at a different plane than the lower tap, zero adjustments must be made (reference 13.2.1)
Whenever chemical seals with capillaries are used, zero adjustments must be made to account for any head associated with the capillaries.

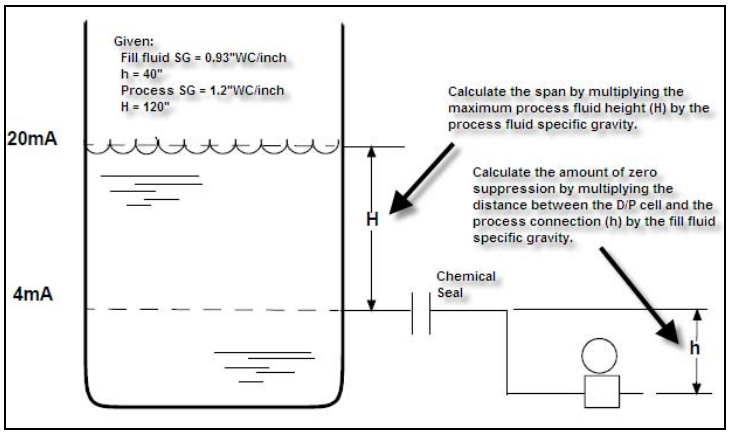
13.2.1 Zero Elevation / Suppression

Whenever the D/P cell is at an elevation other than the connecting nozzle. The zero of the D/P cell needs to be elevated or depressed. It is important to realize that two zero reference points exist. One is the level of the tank that is considered to be zero (lower range value). The other zero reference point is the point at which the D/P cell experiences zero differential.

Zero suppression: If you set the instrument zero to a positive value.

Zero elevation: If you set the instrument zero to a negative value.

Zero Suppression (1-seal system):



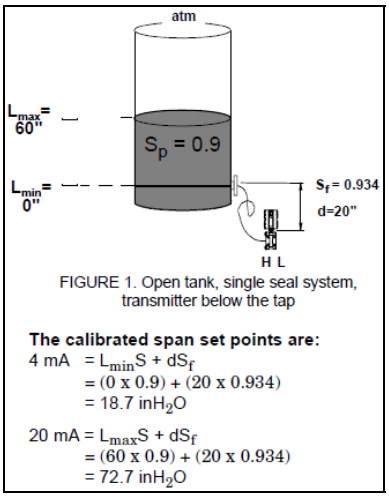


FIGURE 1. Open tank, single seal system, transmitter below the tap

The calibrated span set points are:

4 mA = $L_{min}S + dS_f$
= (0 x 0.9) + (20 x 0.934)
= 18.7 inH_2O

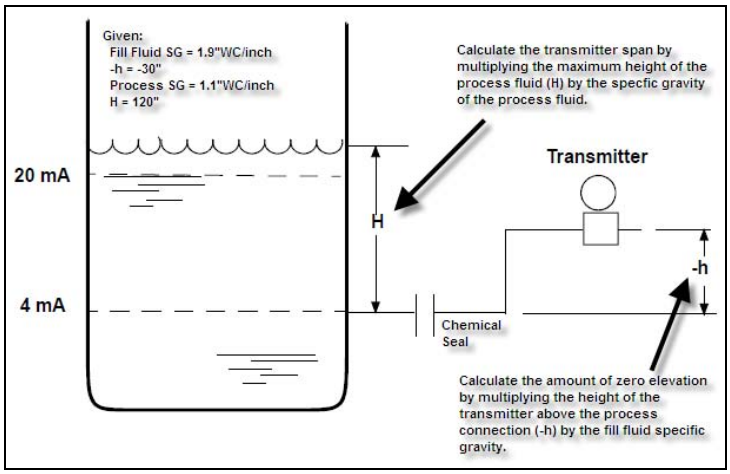
20 mA = $L_{max}S + dS_f$
= (60 x 0.9) + (20 x 0.934)
= 72.7 inH_2O

$$Zero_{\sup p} = h \times SG_F = 40" \times 0.93"WC/inch = 37.2"WC$$

$$Span = H \times SG_P = 120" \times 1.2"WC/inch = 144"WC$$

Calibration = $Zero_{supp}$ to ($Zero_{supp}$ + Span)
= 37.2"WC to (37.2"WC + 144"WC)
= 37.2"WC to 181.2"WC

Zero Elevation (1-seal system):



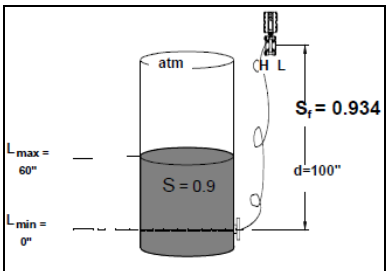


FIGURE 2. Open tank, single seal system, transmitter above tap

The calibrated span set points are:

4 mA = $L_{min}S - dS_f$
= (0 x 0.9) – (100 x 0.934)
= -93.4 inH_2O

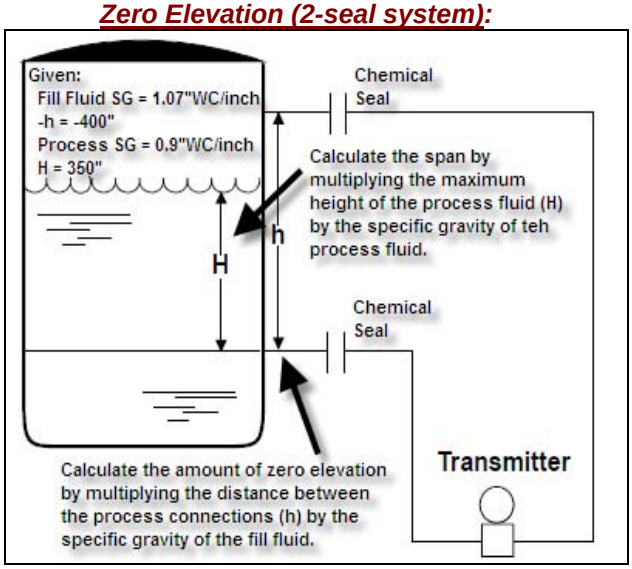
20 mA= $L_{max}S - dS_f$
= (60 x 0.9) – (100 x 0.934)
= –39.4 inH_2O

$$Zero_{elev} = -h \times SG_F = -30" \times 1.9"WC/inch = -57"WC$$

$$Span = H \times SG_P = 120" \times 1.1"WC/inch = 132"WC$$

Calibration = $Zero_{elev}$ to ($Zero_{elev}$ + Span)
= -57"WC to (-57"WC + 132"WC)
= -57"WC to 75"WC

Zero Elevation (2-seal system):



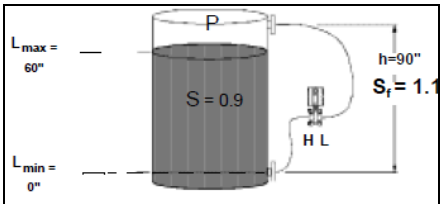


FIGURE 3. Closed tank, two seal system

The calibrated span set points are:

4 mA $= L_{min}S - hS_f$
$= (0 \times 0.9) - (90 \times 1.1)$
$= -99\ inH_2O$

20 mA $= L_{max}S - hS_f$
$= (60 \times 0.9) - (90 \times 1.1)$
$= -45\ inH_2O$

$$Zero_{elev} = -h \times SG_F = -400" \times 1.07"WC/inch = -428"WC$$

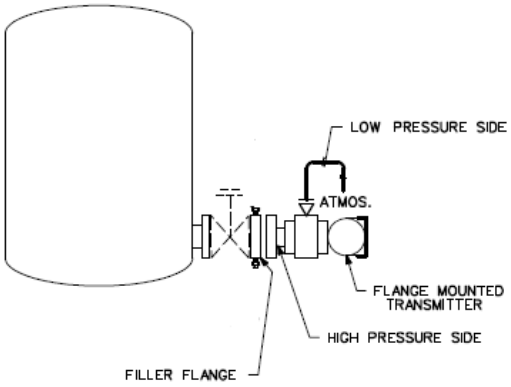
$$Span = H \times SG_P = 350" \times 0.9"WC/inch = 315"WC$$

Calibration = $Zero_{elev}$ to ($Zero_{elev}$ + Span)
= -428"WC to (-428"WC + 315"WC)
= -428"WC to -113"WC

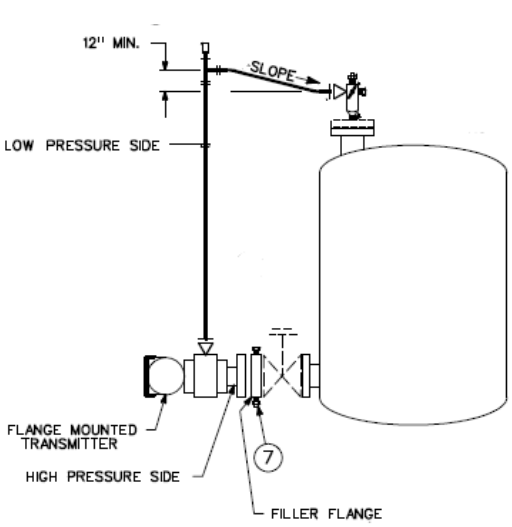
13.2.2 Installation Details:

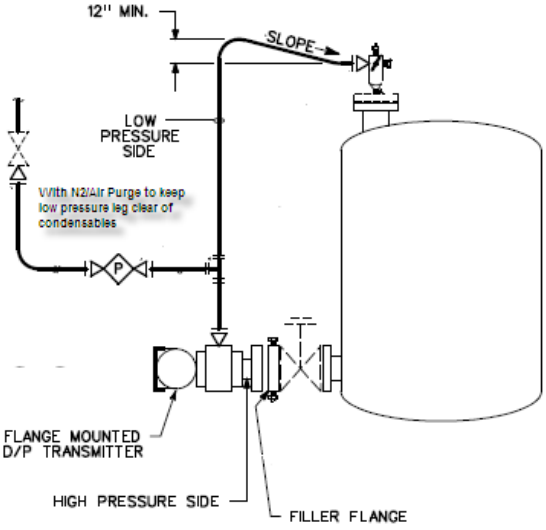
13.2.2.1 Close Coupled:

Atmospheric Vessel:



Pressurized Vessel:





13.3 Bubbler Level:

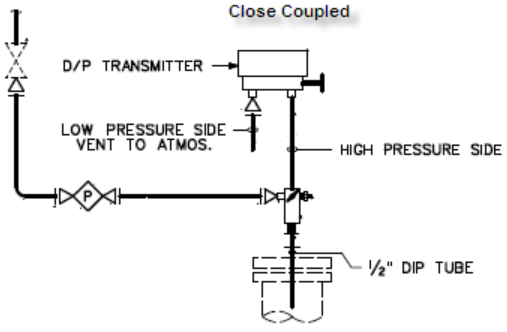
The operation of an air bubbler is similar to blowing air into a glass of water with a straw. The more water there is in the glass, the harder one needs to blow. If the air pressure entering the dip pipe is greater than the hydrostatic head of the process fluid in the tank, the air will bubble out of the bottom of the tank. As liquid level changes, the air pressure in the dip pipe also changes.

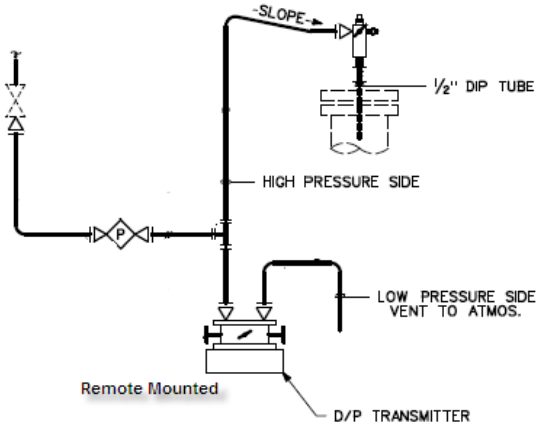
$$h = L_{TS} G_f$$

H = Head pressure in "WC
L_{TS} = Length of tube submerged in process fluid
G_f = SG of the process fluid

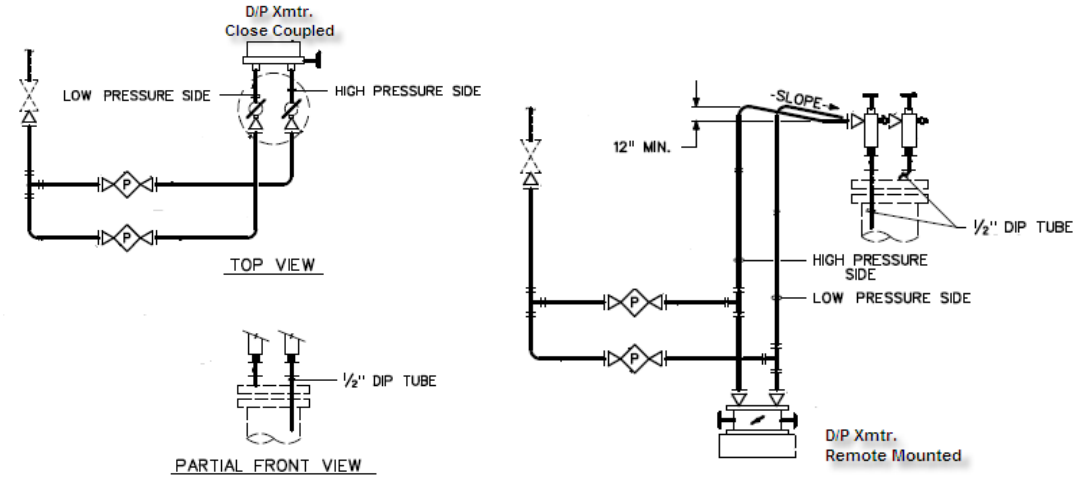
13.3.1 Installation Details:

Atmospheric Vessel:





Pressurized Vessel:

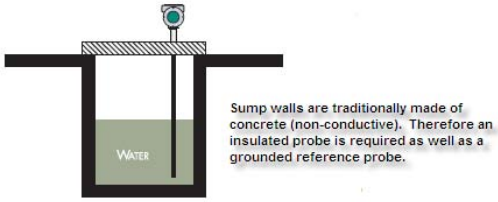


13.4 Capacitance Level:

Low voltage frequency is applied to the level probe, a very small current flow is caused from the probe to the ground. As changes in level cause a change in capacitance between the probe and ground, this affects the very small currents that are detected by the bridge circuits. Should not be used in applications where they are susceptible to coating. There are special design probes that can be used in a coating application which involves a secondary probe. However, if you have an application where buildup may occur, use a different technology other than capacitance.
Traditionally used more in point level applications than in continuous level applications.

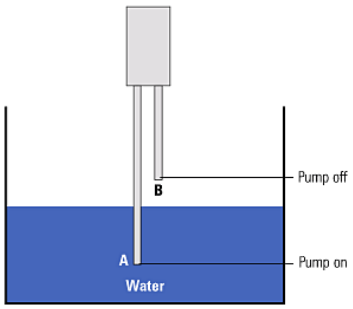
13.4.1 Installation Details:

Concrete Sump:



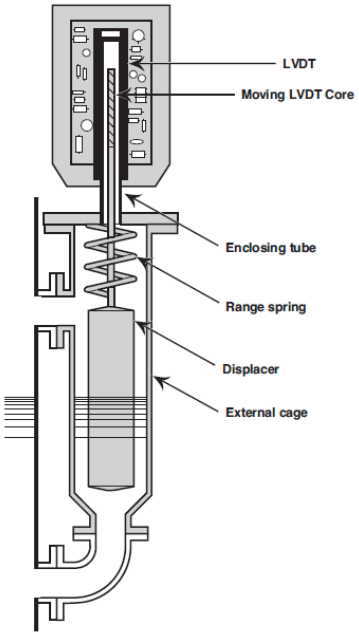
13.5 Conductivity Level:

Typically used for point level applications, not for continuous level applications.
The process itself is used to close an electrical circuit when the level rises to contact the probe.



13.6 Displacer Level:

Archimedes' principle states that a body wholly or partially immersed in a fluid is buoyed up by a force equal to the weight of the fluid displaced. The simplest form of this device involves a displacer weight that is heavier than the process fluid and is suspended from a transducer. When the liquid level is below the displacer, the full weight of the displacer is measured by the transducer. As the level rises, the apparent weight of the displacer decreases, thereby yielding a linear and proportional relationship between transducer tension and the level. LVDTs are typically used to convert the spring tension to an electrical signal.



Displaced Volume

$$V_{df} = (\pi r^2) * L_s$$

Displacement Force Upward

$$f = \frac{V_{df}}{231}(8.33)G_f$$

F = Bouyancy force in lbf
V_{df} = Total volume of displaced process fluid in in^3
L_s = The submerged length of the displacer in process fluid
231 = in3 in one gallon of water
8.33 = Weight of one gallon of water in pounds
Gf = SG of displaced process fluid

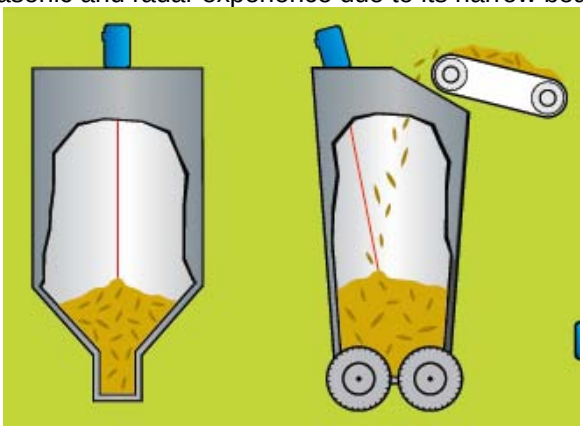
13.7 Float Level:

Typically used for point level applications, not for continuous level applications. Design uses a float which follows the liquid level.
Types of float designs included:

- Spring loaded: Magnet and switch are assembled on a swinging relay arm which operates on pivots. As the float rises it also carries the attractor with it. When the attractor reaches its preset position, it pulls the magnet on the swing arm which in turns actuates a mercury switch.
- Cam: Float actuates a cam which moves a permanent magnet into close proximity to a reed switch which in turn changes state.
- Float & Tube Guide: Float contains a permanent magnet which will change the state of a series of reed switches as it passes over them. Float is external to the tube guide, and the reed switches are contained within the tube guide.
- Tape Gauge: Float moves an indicator along a tape gauge that is external to a tank)

13.8 Laser Level:

Laser type level devices measure level by detecting how long it takes for a light pulse (infrared) o the process surface and back again (similar to ultrasonic and radar). Used primarily with solids level measurement. Does not have tank internal refection problems that ultrasonic and radar experience due to its narrow beam.

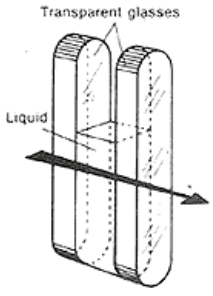


13.9 Level Gauge / Magnetic Flag Indicator:

- Tubular Glass Level Gauge is unsafe and no longer recommended for use in industrial areas (glass breakage).

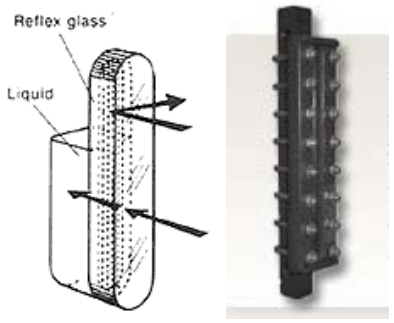


- Flat Glass Gauges come in two designs
  - Transparent: The transparent glass gauge has glass on opposite sides of the chamber allowing the liquid level to be viewed through the gauge.





- Reflex: The reflex gauge has a single glass with prisms cut in the glass on the process side. Light striking the glass in the vapor phase is reflected back appearing silvery white. Light striking the glass covered with liquid is reflected back appearing black.



o Magnetic Flag Indicator: In certain corrosive / toxic services, gasketed glass designs are not acceptable. In this design a non-magnetic metal cage contains an internal float that rides on the liquid level. A permanent magnet in the float is coupled to an external indicator. The external indicator typically consists of a series of wafers that have one color on the front and a contrasting color on the back. As the magnet in the float passes by, the wafers are flipped over indicating the level.

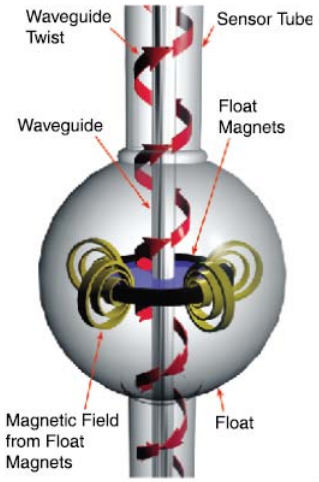


13.10 Optical Level:

Typically used for point level applications, not for continuous level applications. Sensors incorporate a phototransisitor that provides a digital output indicating the presence or absence of liquid. The sensor tip houses the transistor. The refractive index is changed when covered by liquid, thus indicating a level.

13.11 Magnetostrictive Level:

Device consists of a magnetostrictive wire in the stem and a permanent magnet inside the float. The float is the only moving part that travels vertically on the stem. Once a pulse current is induced from the end of the magnetostrictive wire, a tubular magnetic field emanates. As the float travels, torsional vibration is launched by the interaction between the float magnetic field and the magnetostrictive wire. The float position is determined by measuring the lapse of time from the launching of the torsional vibration to the return of the signal.

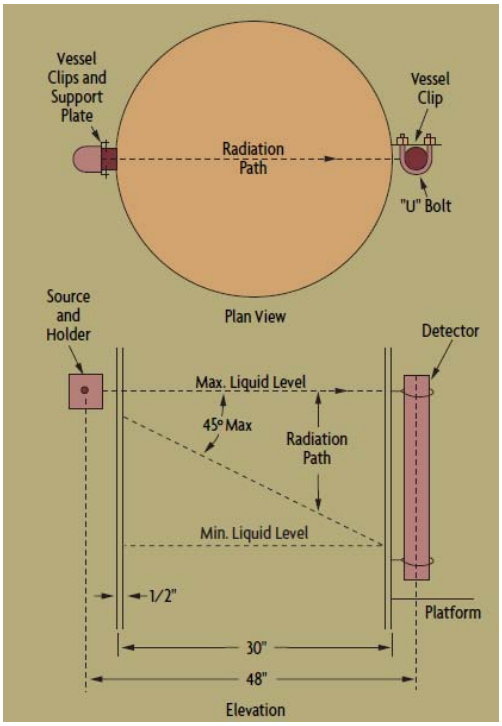


13.12 Nuclear Level:

- Gamma radiation sources (Cesium 137) are used because they have sufficient penetrating power to pass through metal tank. Detectors are typically the scintillation type detector.

    The scintillation counter has a layer of phosphor cemented in one of the ends of the photomultiplier. Its inner surface is coated with a photo-emitter with less work potential. This photoelectric emitter is called as photocathode and is connected to the negative terminal of a high tension battery. A number of anodes called dynodes are arranged in the tube at increasing positive potential. When a charged particle strikes the phosphor, a photon is emitted. This photon strikes the photocathode in the photomultipier, releasing an electron. This electron accelerates towards the first dynode and hits it. Multiple secondary electrons are emitted, which accelerate towards the second dynode. More electrons are emitted and the chain continues, multiplying the effect of the first charged particle. By the time the electrons reach the last dynode, enough have been released to send a voltage pulse across the external resistors. This voltage pulse is amplified and recorded by the electronic counter

- Used whenever tank penetrations are not permitted (e.g. present risk to human life).
- Nuclear level transmitters can be susceptible to x-rays (i.e. if located outdoors, shield transmitter with sufficient material whenever pipe x-ray activities are occurring).
- The sources are regulated by the NRC, any site that uses nuclear level detectors must have a named Nuclear Radiation Officer on their site who is the point of contact with the NRC, they also require a NRC license.



13.13 Rotating Paddle:

Typically used for point level applications, not for continuous level applications. Rotating type paddle switch is used to detect the presence or absence of solids in a silo. A small geared synchronous motor keeps the paddle in motion at very slow speed. When the level rises to the paddle wheel, it is stopped and torque applied to the drive assembly. This in turn actuates a switch.



13.14 Thermal Level Switch:

Operation of the thermal level switch is similar to that of a thermal flow switch. Thermal level sensors detect the difference in the thermal conductivity of the process. The probe contains a resistive heater that has a current flowing through it. If a probe is submerged in a fluid, heat generated by the resistive element will be carried into the fluid. Once the fluid falls below the sensor the probe temperature will begin to rise, the switch detects this rise.

13.15 Ultrasonic:

Ultrasonic level transmitter operates by generating an ultrasonic pulse and measuring the time it takes for an echo to return. The ultrasonic pulse needs a media to transmit the pulse, therefore this technology will not work in vacuum service. Ultrasonic can be used for both point and continuous level. Ultrasonic level measurement is also used in conjunction with open channel flow measurement.

13.16 Vibratory:

Typically used for point level applications, not for continuous level applications. In this design the probe is kept in oscillation or in natural frequency vibration, a relay is triggered when material in the tank reaches the probe and dampens out the vibration. When installing, the forks shall be in the vertical position, not in the horizontal to prevent material buildup on the fork. The switch shall be installed horizontal or at angle pointing down, it should never be installed at an angle pointing up.

13.17 TDR/PDS:

- The principle behind Time Domain Reflectometry is that a portion of the electrical signal will be reflected back towards its source by a discontinuity in the cable that is carrying the signal. By measuring the time that it takes for the signal to reach discontinuity plus time for that it takes for the reflected return, the discontinuity point can be located. This application works well in the power and communication industries to locate problems on transmission lines. Efforts to use this on level detection have not yet been successful.
- The principle behind Phase Difference Sensor is that a high frequency signal travels through parallel conductors at a fixed velocity until it is partially reflected by the stored material interface where the sensor impedance changes abruptly. Due to the travel distance of the two parallel wires there will be a phase difference between the input and reflected signals. Typically used in narrow grain silos.

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14 Analytical Measurement

14.1 Analyzer Selection for Specific Substances

Reference Page 848 in Lipták (3rd edition)

Code Numbers Used in Table starting on Next Page:

1. Electroconductivity, electrochemical, polarographic or fuel cell
2. Infrared
3. Selective ion or acid analyzer
4. Colorimeter, autoanalyzer or autotitrator
5. Electrolytic hygrometer
6. Capacitance
7. Polarographic
8. UV and visible photometers
9. Refractometers
10. Thermal conductivity
11. Phototape
12. Zirconium oxide
13. Mass spectrometer
14. Chromatography
15. Paramagnetic
16. Flame ionization
17. Diffusion elements
18. Amperometric (galvanic)
19. Catalytic combustion
20. Atomic absorption

Acetaldehyde—8
Acetic anhydride—8
Acetone—8, 9
Acidity—4
Acids in water—9
Acrylonitrile—8
Air humidity—5, 6
Alcohol in water—9
Alcyl chloride—1
Aldehydes—8
Alkalinity—4
Aluminum—4
Amines, ppm—16
Ammonia—1, 2, 3, 4, 10
Ammonia, ppb—16
Ammonium ions—3
Ammonium sulfate—9
Aniline—8
Argon—5, 10
Aromatics in vapors or in water—8
Benzene moisture content—6
Benzene in ethanol or cyclohexane—9
Benzene in raffinate or in ethyl alcohol—8
Benzotriazole in water, ppm—8
Boron—4, 9
Brine concentration—9
Bromide ions—3
Bromine—8

Butane—2, 5, 6
Butadiene—8, 9
Butadiene in butanes and butylenes—8
Butadiene in styrene—9
Cadmium ions—3
Caffeine—8
Calcium ions—3
Carbon bisulfide—8
Carbon disulfide—1
Carbon dioxide in carbonated beverages—2
Carbon dioxide in gases—1, 2, 4, 19
Carbon dioxide moisture content—5
Carbon monoxide in gases—1, 2
Carbon monoxide moisture content—5
Carbon tetrachloride in air—1
Carbon tetrachloride, ppm—16
Catsup and tomato paste—9
Caustic concentration—2, 9
Chloride—3, 4
Chlorine—1, 8, 10, 11
Chlorine in air—10
Chlorine in ethylene dichloride—8
Chlorine, ppm, in off-gas or in phosgene—8
Chlorine, ppm—16
Chlorine residual—4, 7, 18
Chloride—3, 4
Chlorobenzene—1
Chloroform—1
Chromium in water (hexavalent or total)—4
Citrus juice—9
COD—4
Color—4
Combustibles—16, 17, 19
Copper in water—3, 4
Cyanide in water—3, 4
Cyclohexane—4, 9
Cyclohexane in h-hexane and methyl cyclopentane—9
Diolefin vapors—8
Divalent ions—3
Divinyl acetylene in acrylonitrile—8
Ethane—2
Ethane moisture content—5
Ethanol—8
Ethanol in benzene or in water—9
Ethyl bromide—1
Ethyl chloride—1
Ethyl chloride moisture content—5
Ethylene in %—2
Ethylene in Cl–C6, H_2, CO_2—14
Ethylene bromide, ppm—16
Ethylene chloride, ppm—16
Ethylene glycol—9
Ethylene oxide—2, 10
Ethylene oxyde in methane, ethane, propane—13
Fluoride in water—3, 4
Fluorine, ppm—16
Freon—10
Freon moisture content—5
Furfural—8
Glycerine and salts in water—9
Green liquor or white liquor—9
Hardness (total) in water—4
Helium in oxygen, nitrogen—10
Helium moisture content—5
Hazardous gases—17
Hexane—2
Hexane moisture content—6
Hexavalent chromium—4
Hydrazine in water—4
Hydrazine, ppb—16
Hydrocarbon in H_2S, CO_2, air—13
Hydrocarbon vapors in ambient air—16, 17, 19
Hydrogen in chlorine—8
Hydrogen in nitrogen, oxygen, inert gases—10
Hydrogen chloride—1, 4
Hydrogen chloride, ppb—16
Hydrogen cyanide—1
Hydrogen fluoride—1
Hydrogen fluoride, ppm—16
Hydrogen impurities (in O_2, N_2, CO, H_2O)—13
Hydrogen sulfide in air—1, 8, 11, 17
Hydrogen sulfide in hydrocarbon liquids—4
Hydrogen sulfide in natural gas or in stack gas—1, 8, 17
Iodide ions—3
Iron (total) in water—4
Isobutane in n-butane (liquid)—9
Isobutane in C3, nC4, iC5—14
Isoprene in solvents—8
Jams and jellies—9
Kerosene moisture content—6
Ketones—8
Lead ions—3
Mercury in air—4, 8, 20
Mercury in water—4, 14, 20
Methane—2, 10
Methane moisture content—5

Methanol—8
Methanol in water—9
Methyl bromide—1
Methyl chloride—1
Methylene chloride—1, 2
Naphtha—6
Natural gas moisture content—5, 6
Neon moisture content—5
Nickel carbonyl, ppb—16
Nitrate and nitrite—4
Nitric acid in water—9
Nitric oxide in air—1
Nitrobenzene—8
Nitrogen—10
Nitrogen (ammonia, Kjeldahl, total, nitrite, nitrate, organic)—4
Nitrogen in argon—13
Nitrogen dioxide, ppm—16
Nitrous fumes—1
Nitrous oxide—2
Nitrogen moisture content—5
Nitrogen peroxide—8
Octane rating of gasoline—9
Oil in liquid freon—2
Oil in wax—9
Ortho- and meta-xylene in para-xylene—9
Ortho-phosphate—4
Oxygen—4, 7, 10, 12
Oxygen in argon, hydrogen—10
Oxygen in ethylene, argon—13
Oxygen in stack gases—12, 15
Oxygen in water—1
Oxygen moisture content—5
Ozone in air—8
Phenol in water—4, 8
Phosgene—8
Phosgene in air—1, 8, 11
Phosgene, ppm—16
Phosphoric acid concentration—9
Phosphorus (total or O-phosphate) in water—4
Polymer-solvent mixtures—9
Potassium dichromate—8
Potassium ions—3
Propane—2
Propane moisture content—5, 6
Propylene dichloride—1
Propylene glycol in water—9
Propylene moisture content—6
Proteins—8
Pyridine—8
Residual chlorine—4, 7, 18
Silicates in water and in seawater—4
Sodium carbonate in water—2
Sodium chloride in water—9
Sodium hydroxide in water—2
Sodium ions—3
Starch concentration—9
Steam in air—2
Styrene in ethyl benzene—9
Sucrose in water—4
Sugars in water, syrups—9
Sulfate and sulfite—4
Sulfur dioxide—1, 2, 4, 8, 10
Sulfur dioxide moisture content—5
Sulfur dioxide in stack gas—2, 8
Sulfur dioxide ppm—16
Sulfuric acid—3
Sulfuric acid in water—9
Tetrachloroethylene, ppm—16
Tetraethyl lead, ppb—16
Tetranitromethane in air—2
Toluene in hydrocarbons—8
Toxic gases—17
Trans-unsaturation of vegetable oils, detergents—2
Trichloroethylene—1
Trichloroethylene, ppm—16
Vinyl acetate in polymer vapor—2
Vinyl chloride—1
Water hardness—3
Water in liquid SO_2—2
Water in organic liquids—2
Water in methylene chloride liquid—2
Water vapor in air—2
Wax in oil—9
White liquor—9
Wine—9
Xylenes in hydrocarbon—8
Xylene in isomers—9

14.2 Analyzer Technologies

14.2.1 Combustible Gas Analyzers:

Bead Type: Typically utilizes two beads in a sensing head. One serves as an active sensing element and the other as a reference. The active bead is composed of a small coil of platinum wire embedded in a ceramic bead that is coated in a catalyst. The catalyst coating encourages oxidation of combustible gases. The reference bead is usually the same construction as the active bead except that it has in inert coating. The coils are then heated to achieve a temperature at which the active bead is considered to be highly effective at oxidizing combustible gases. If any gases are oxidized at the active bead, the heat of reaction should raise the active bead's temperature relative to the reference bead. This type of technology is subject to catalyst poisoning because of the direct contact of the gas with the catalytic surface it may be deactivated in some circumstances

Infrared Type: Hydrocarbons and many other gases absorb infrared radiation at specific wavelengths. The analyzer exposes gas samples to IR radiation and monitors the intensity of that radiation at wavelengths that will provide both good reference and good measure radiation intensity readings for the gas components that are anticipated to appear in the beam and be measured. Available as traditional point sensor and an open path version.

14.2.2 Moisture / Dew Point Analyzers:

Dew Cup / Chilled Mirror: In the chilled-mirror technique, a mirror is constructed from a material with good thermal conductivity such as silver or copper, and properly plated with an inert metal such as iridium, rubidium, nickel, or gold to prevent tarnishing and oxidation. The mirror is chilled using a thermoelectric cooler until dew just begins to form. A beam of light, typically from a solid-state broadband light emitting diode, is aimed at the mirror surface and a photodetector monitors reflected light. As the gas sample flows over the chilled mirror, dew droplets form on the mirror surface, and the reflected light is scattered. As the amount of reflected light decreases, the photodetector output also decreases. This in turn controls the thermoelectric heat pump that maintains the mirror temperature at the dew point. A precision miniature platinum resistance thermometer properly embedded in the mirror monitors the mirror temperature at the established dew point.

Electrolytic Hygrometer: The principle of measurement utilized involves the electrolysis of water into oxygen and hydrogen. Since two electrons are required for electrolysis of each water molecule, the electrolysis current is a measure of the water present in the sample. If the volumetric flow rate of the sample gas into the electrolysis cell is controlled at a fixed value, then the electrolysis current is a function of water concentration in the sample

Piezoelectric Hygrometer: In moisture measurement, advantage is taken of the oscillating crystal's sensitivity to deposits of foreign material on its surface. Commercially available crystals will show a frequency change of 2000 cycles per second (cps) per microgram of material deposited. For moisture measurement, the quartz crystals are coated with a hygroscopic material and exposed to the sample; water from the sample is absorbed by the crystal coating, thus increasing the total mass and decreasing the oscillating frequency of the crystal. In order to measure changes of decreasing moisture concentration and to simplify the frequency measurement, two crystals are used. One crystal is exposed to wet sample and the other to a dry reference gas for a short period. Then sample and reference gas flows are switched so that moisture is absorbed by one while being desorbed by the other crystal. The frequency difference between the two crystals is in proportion to their mass difference and the moisture content of the gas

Silicon Oxide: Sensor can be an optical device that changes its refractive index as water is absorbed into the sensitive layer or a different impedance type in which silicon replaces the aluminum.

Aluminum Oxide: Has two metal layers that form the electrodes of a capacitor. The number of water molecules adsorbed will cause a change in the dielectric constant of the sensor. The sensor impedance correlates to the water concentration.

14.2.3 Conductivity Analyzers:

Conductive Method: Between two or more opposite electrode surfaces with defined distance and dimensions, a voltage is applied. The measuring transmitter converts the arising potential difference by means of compensation equations into conductivity.

Inductive Method: Two coils potted in synthetic material (e.g. PEEK, PFA) are flown through by the liquid. Due to the ions in the liquid, the primary coil induces a current in the secondary coil. The measuring transmitter can convert this current by means of compensation equations into conductivity.

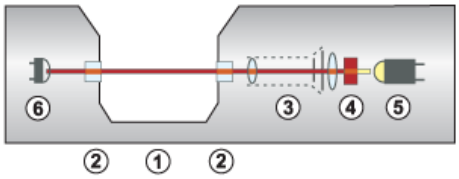
14.2.4 pH / ORP Analyzers:

pH Method: pH is determined by measuring the voltage of an electrochemical cell. The cell consists of a measuring electrode, a reference electrode, a temperature sensing element and the liquid being measured. The voltage of the cell is directly proportional to the ORP of the liquid. The cell voltage is the algebraic sum of the potentials of the measuring electrode, the reference electrode and the liquid junction. The potential of the measuring electrode depends only on the pH of the solution. The pH of the reference electrode is unaffected by pH, so it provides a stable reference voltage. The liquid junction potential is usually small and relatively constant. pH probe ends should be kept wet at all times (stored in buffer solution when not in use) and not allowed to dry out.

ORP Method: (Oxidation-Reduction Potential) Depends on the ratio of the concentrations of oxidized and reduced substances in the sample. In a typical system, the measuring electrode is an inert metal such as gold or platinum, and the reference electrode is usually silver / silver chloride electrode. An ORP measurement cell is similar to a pH cell. The major difference is the glass measuring electrode has been replaced with an ORP electrode. The cell voltage is the ORP of the sample.

14.2.5 Infrared Adsorption Analyzers (NIR / MIR / FTIR):

Infrared radiation interacts with all molecules, with a few exceptions by exciting molecular vibrations and rotations. The oscillating electric field of the IR wave interacts with the electric dipole of the molecule, and when the IR frequency matches the natural frequency of the molecule, some of the IR power is absorbed. The pattern of wavelengths, or frequencies, absorbed identifies the molecules in the sample. The strength of absorption at particular frequencies is a measure of their concentration.



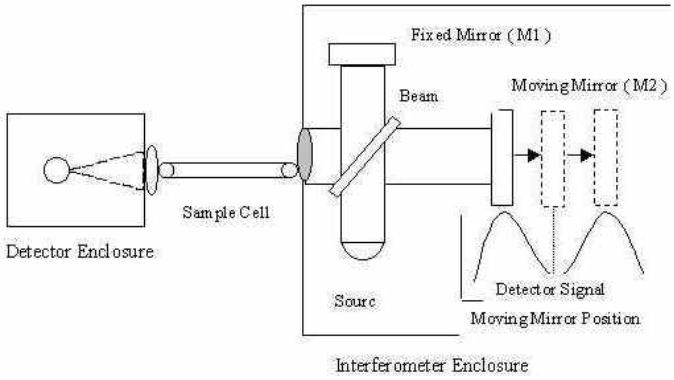
1. Optical Path Length (OPL)
2. Sapphire Windows
3. Optics module
4. NIR Filter
5. Tungsten Lamp
6. Photo Diode

Single beam dual wavelength (SBDW) infrared analyzers are able to make measurements using one source, one measurement cell and one detector. Typically, a lens is used to focus the light for a straight pass through the cell. Thus, the SBDW analyzer does not depend on internal reflections to increase energy throughput or increase effective path length. In a practical sense, effects of component aging and window contamination are minimized, since aging and contamination effects both the measurement and reference wavelengths equally. In fact using the SBDW principle an infrared analyzer can perform to specifications with up to a 50% coating on the windows. After this point, energy transmission falls to a point where noise in the data results in decreasing analytical precision.

In the dual-beam configuration the IR radiation is allowed by the chopper to pass alternately through the sample and the reference tube . The reference tube provides a true zero reference, as it is filled with nonabsorbing gases. A narrow bandpass optical filter is placed in front of the detector to limit the IR energy it receives to the wavelength which is characteristic of the component of interest. Therefore if the sample contains the component of interest, this will attenuate the magnitude of the detected signal in the absorption band of the bandpass filter. The use of the reference cell in the dual-beam configuration reduces the drift causes by power supply or temperature fluctuations. The use of collimating optics also eliminates the need for internal reflection from the interior surfaces of the tubes, thus simplifying their construction and elimination the associated drift.

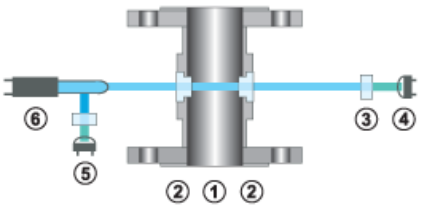
NIR wavelengths: 12,500 to 4000 cm^{-1}
MIR wavelengths: 4000 to 650 cm^{-1}

FTIR: Fourier transform infrared (FTIR) spectrophotometers are dispersive devices that are being used for on-line analysis in the near infrared region. A schematic of a typical FTIR spectrophotometer is presented in the figure below. Collimated light from the source is directed through the beam splitter. Approximately 50% of the light passes through the beam splitter to the fixed mirror (Ml). The balance of the light is reflected onto the moving mirror (M2). When these two beams are reflected off the mirror surfaces they recombine at the beam splitter to give constructive and destructive interference. This interference is dependent on the position of the moving mirror relative to the fixed mirror. Helium neon laser detector systems are used in FTIR analyzers to monitor the position and velocity of the moving mirror. This results in excellent wavelength accuracy in FTIR analyzers. The displacement of the moving mirror induces phase differences which result in an interferogram. In order to produce a spectrum of absorbance versus wavelength, the interferogram must be transformed digitally from a plot of detector response versus optical path difference. The calculation of the spectrum is carried out using a "Fast Fourier Transform" algorithm on a computer.
Near infrared FTIR analyzers measure all of the wavelengths in the spectrum simultaneously on one detector. The entire scanned spectrum is used for quantitative measurements. FTIR spectrophotometers are used for multicomponent applications that require high resolution to separate interfering components.



14.2.6 UV Absorption Analyzers:

The UV region consists of wavelengths from 200 to 400 nanometers (nm). The visible region extends from 400 to 800 nm, and the near IR (NIR) region covers 0.8 to 2.50 micrometers (j~m). Nanometer units are commonly used in the UV/VIS region, while micrometers or microns are normally used in the NIR region. The UVVIS-NIR is a relatively small part of the electromagnetic radiation spectrum, and the shorter the wavelength the more penetrating the radiation. The region where a compound absorbs radiation depends on the energy of the molecular transitions. High-energy electronic transitions are observed in the low-wavelength UV/VIS regions. Moderate-energy vibrational and rotational transitions are observed in the high-wavelength IR region.



1. Sensor Body
2. Sapphire Windows
3. UV Filter
4. Measurement Photo Diode
5. Lamp Reference Photo Diode
6. Mercury Vapor Lamp

Light from a Mercury Vapor UV lamp (6) passes through a sapphire window (2), and then through the process stream (1). The resulting light passes through a UV filter (3) and is detected by a photodiode (4). The narrow-band pass UV filter blocks all wavelengths except the specified UV wavelength. The photocurrents induced are directly proportional to the remaining light intensity at this wavelength. Light from the Mercury Vapor Lamp UV (6) also passes through a lamp reference filter (5) and lamp reference photodiode (5) installed within the lamp assembly. This compensates for any variations or intensity fluctuations of the UV lamp. The resulting photocurrents are precisely amplified, converted, and analyzed by the converter. The converter provides real time measurements and can send outputs to the process control system.

14.2.7 Gas Chromatographic Analyzers:

Gas chromatography (GC), also sometimes known as Gas-Liquid chromatography, (GLC), is a separation technique in which the mobile phase is a gas. Gas chromatography is always carried out in a column, which is typically "packed" or "capillary"

Thermal: The Thermal Conductivity Detector uses four spiral wound filament wires supported inside cavities in a metal block. The filaments are manufactured from a material whose electrical resistance varies significantly with variations in temperature. A constant DC current of up to 100 mA is applied to the filaments in an electronic bridge circuit. With pure carrier and reference gas flowing across the filaments, the heat loss is constant leading to a constant filament temperature. This consistent filament temperature produces a constant filament resistance. The currents in the electronic bridge can be balanced to produce a zero signal level as a reference. When component peaks enter the TCD with the carrier, the heat dissipated from the filaments on the measure side changes. This leads to an imbalance in the electronic bridge. The resulting electrical signal is then used to measure the quantity of the component.

FID: (Flame Ionization Detector) When sample material enters the FID, the compounds are burned by the flame. The ions produced result in a proportional current flow inside the detector. This current is amplified and converted to a voltage signal. The amplitude of the signal is proportional to the concentration of the component ionized.

FPD: (Flame Photometric Detector) The FPD provides a hydrogen rich flame where sulfur compounds are reduced to the elemental S2 species which produce a blue chemiluminescent emission in the visible light spectrum. The blue light emission is passed through a narrow band (395 nanometer) optical filter to the photomultiplier tube. The signal from the photomultiplier tube is further amplified by an electrometer and interface board. To minimize condensate buildup in the detector, the FPD block is typically maintained above 100° C. The photomultiplier tube housing is maintained at a cooler temperature to extend its service life. The intensity of the blue light emission is approximately proportional to the square of the sulfur concentration in the flame. In some applications, a small amount of methyl mercaptan (2 to 5 ppm) is continuously added to the FPD air to bias the output level upward to the linear portion of the output curve. Generally, a permeation device is used to provide a consistent addition of methyl mercaptan to the FPD air. A small sample size is also used to ensure operation in the linear portion of the curve.

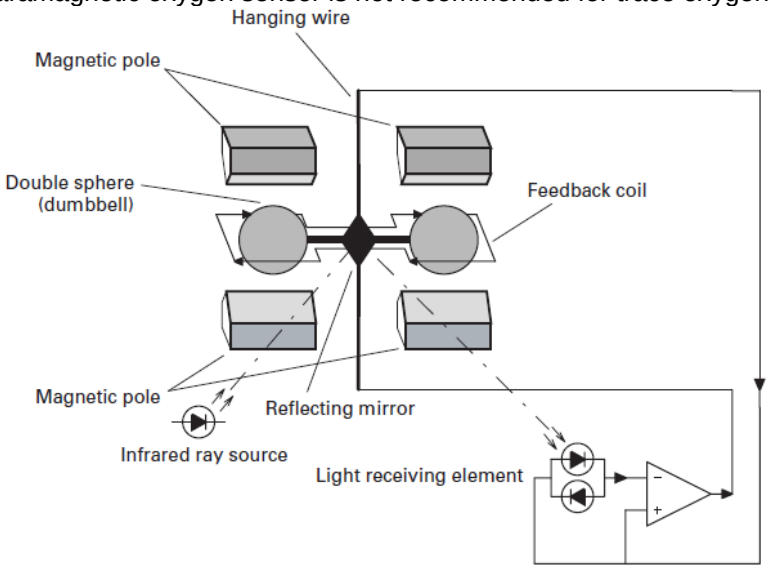
PID: (Photoionization Detector) The TID uses a modified Flame Ionization Detector body. The FID ignitor is removed and a modified Swagelock fitting is mounted on the top of the FID cap. The TID source is mounted in this fitting. The thermionic source is heated by passing a precisely controlled current through a wire inside the mineral source. The current heats the mineral source to its operating temperature causing it to generate a flow of ions between the source and the collector in the detector. The heat of the source also burns the hydrogen and air mixture in the detector.

14.2.8 Liquid Chromatographic Analyzers:

Liquid chromatography (LC) is a separation technique in which the mobile phase is a liquid. Liquid chromatography can be carried out either in a column or a plane. Present day liquid chromatography that generally utilizes very small packing particles and a relatively high pressure is referred to as high performance liquid chromatography (HPLC). In the HPLC technique, the sample is forced through a column that is packed with irregularly or spherically shaped particles or a porous monolithic layer (stationary phase) by a liquid (mobile phase) at high pressure. HPLC is historically divided into two different sub-classes based on the polarity of the mobile and stationary phases. Technique in which the stationary phase is more polar than the mobile phase (e.g. toluene as the mobile phase, silica as the stationary phase) is called normal phase liquid chromatography (NPLC) and the opposite (e.g. water-methanol mixture as the mobile phase and C18 = octadecylsilyl as the stationary phase) is called reversed phase liquid chromatography (RPLC). Ironically the "normal phase" has fewer applications and RPLC is therefore used considerably more.

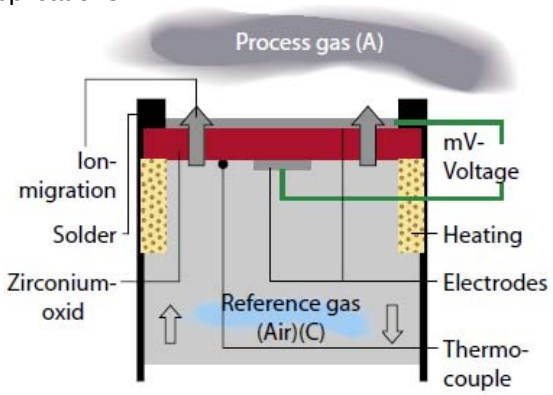
14.2.9 Oxygen Content (in Gas) Analyzers:

- **Paramagnetic Oxygen Analyzer**
  Within this category, the magnetodynamic or `dumbbell' type of design is the predominate sensor type. Oxygen has a relatively high magnetic susceptibility as compared to other gases such as nitrogen, helium, argon, etc. and displays a paramagnetic behavior. The paramagnetic oxygen sensor consists of a cylindrical shaped container inside of which is placed a small glass dumbbell. The dumbbell is filled with an inert gas such as nitrogen and suspended on a taut platinum wire within a non-uniform magnetic field. The dumbbell is designed to move freely as it is suspended from the wire. When a sample gas containing oxygen is processed through the sensor, the oxygen molecules are attracted to the stronger of the two magnetic fields. This causes a displacement of the dumbbell which results in the dumbbell rotating. A precision optical system consisting of a light source, photodiode, and amplifier circuit is used to measure the degree of rotation of the dumbbell. In some paramagnetic oxygen sensor designs, an opposing current is applied to restore the dumbbell to its normal position. The current required to maintain the dumbbell in it normal state is directly proportional to the partial pressure of oxygen and is represented electronically in percent oxygen. There are design variations associated with the various manufacturers of magnetodynamic paramagnetic oxygen analyzer types. Also, other types of sensors have been developed that use the susceptibility of oxygen to a magnetic field which include the thermomagnetic or `magnetic wind' type and the magnetopneumatic sensor. In general, paramagnetic oxygen sensors offer very good response time characteristics and use no consumable parts, making sensor life, under normal conditions, quite good. It also offers excellent precision over a range of 1% to 100% oxygen. The magnetodynamic sensor is quite delicate and is sensitive to vibration and/or position. Due to the loss in measurement sensitivity, in general, the paramagnetic oxygen sensor is not recommended for trace oxygen measurements.



- **Zirconium Oxide Oxygen Analyzer**
  The type of oxygen analyzer that uses this type of oxygen sensor is occasionally referred to as the “high temperature” electrochemical sensor and is based on the Nernst principle. Zirconium oxide sensors use a solid state electrolyte typically fabricated from zirconium oxide stabilized with yttrium oxide. The zirconium oxide probe is plated on opposing sides with platinum which serves as the sensor electrodes. For a zirconium oxide sensor to operate properly, it must be heated to approximately 650 degrees Centigrade. At this temperature, on a molecular basis, the zirconium lattice becomes porous, allowing the movement of oxygen ions from a higher concentration of oxygen to a lower one, based on the partial pressure of oxygen. To create this partial pressure differential, one electrode is usually exposed to air (20.9% oxygen) while the other electrode is exposed to the sample gas. The movement of oxygen ions across the zirconium oxide produces a voltage

between the two electrodes, the magnitude of which is based on the oxygen partial pressure differential created by the reference gas and sample gas. The zirconium oxide oxygen sensor exhibits excellent response time characteristics. Another virtue is that the same sensor can be used to measure 100% oxygen, as well as parts per billion concentrations. Due to the high temperatures of operation, the life of the sensor can be shortened by on/off operation. The coefficients of expansions associated with the materials of construction are such that the constant heating and cooling often causes “sensor fatigue”. A major limitation of the zirconium oxide oxygen analyzer is their unsuitability for trace oxygen measurements when reducing gases (hydrocarbons of any species, hydrogen, and carbon monoxide) are present in the sample gas. At operating temperatures of 650 degrees Centigrade, the reducing gases will react with the oxygen, consuming it prior to measurement thus producing a lower than actual oxygen reading. The magnitude of the error is proportional to the concentration of reducing gas. The zirconium oxide oxygen analyzer is the “defacto standard” for in-situ combustion control applications.



14.2.10 Dissolved Oxygen Analyzers:

Dissolved oxygen (DO) is the term used for the measurement of the amount of oxygen dissolved in a unit volume of water, usually presented in units of mg/L or ppm.
Most commonly used DO probe is the polarographic type. The polarographic cell has two noble metal electrodes and requires a polarizing voltage to reduce the oxygen. The DO is the sample diffuses through the membrane into the electrolyte which is an aqueous KCl solution. If there is a polarizing voltage across the electrodes, the oxygen is reduced at the cathode and the resulting current flow is directly proportional to the oxygen content of the electrolyte. The polarographic cell is affected by temperature, therefore temperature compensation is required to attain high accuracy.

14.2.11 Mass Spectrometric Analyzers:

Analytical technique for the determination of the elemental composition of a sample or molecule. It is also used for elucidating the chemical structures of molecules, such as peptides and other chemical compounds. The MS principle consists of ionizing chemical compounds to generate charged molecules or molecule fragments and measurement of their mass-to-charge ratios. In a typical MS procedure:

- a sample is loaded onto the MS instrument, and
- the components of the sample are ionized by one of a variety of methods (e.g., by impacting them with an electron beam), which results in the formation of charged particles (ions)
- directing the ions into an electric and/or magnetic field
- computation of the mass-to-charge ratio of the particles based on the details of motion of the ions as they transit through electromagnetic fields, and
- detection of the ions, which in step 4 were sorted according to m/z.

MS instruments consist of three modules: an *ion source*, which can convert gas phase sample molecules into ions (or, in the case of electrospray ionization, move ions that exist in solution into the gas phase); a *mass analyzer*, which sorts the ions by their masses by applying electromagnetic fields; and a *detector*, which measures the value of

an indicator quantity and thus provides data for calculating the abundances of each ion present. The technique has both qualitative and quantitative uses. These include identifying unknown compounds, determining the isotopic composition of elements in a molecule, and determining the structure of a compound by observing its fragmentation. Other uses include quantifying the amount of a compound in a sample or studying the fundamentals of gas phase ion chemistry (the chemistry of ions and neutrals in a vacuum). MS is now in very common use in analytical laboratories that study physical, chemical, or biological properties of a great variety of compounds.

14.2.12 Turbidity Analyzers:

Measure the cloudiness of a fluid by detecting the intensity of transmitted or reflected light. The suspended finely dispersed particles which when exposed to infrared or visible light will scatter the light. The cloudier the process (the higher its turbidity), the more scattering will occur and therefore the less light will be transmitted through a sample. The amount of light scatter is measured and compared to the amount of scatter from known mixtures. The amount of the unknown is determined from a standard curve.
Turbidity meters are available in three different forms:

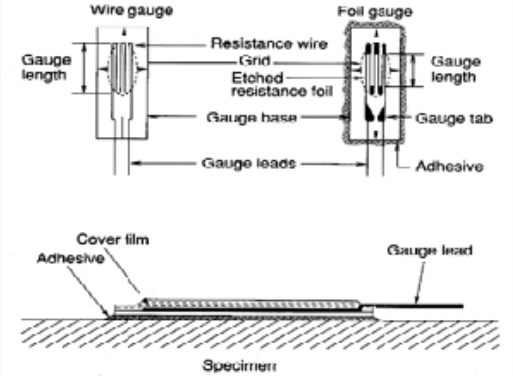
- Perpendicular (nephelometry)
- Back Scattering
- Forward Scattering

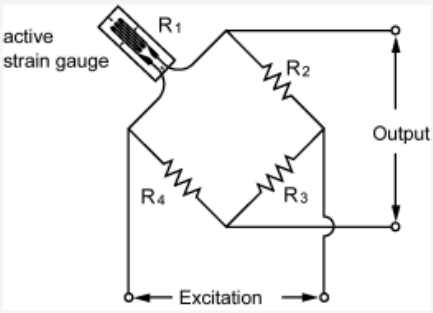
Units of Measure:

- JTU: Jackson Turbidity Unit
- NTU: Nephelometric Turbidity Units
- FTU: Formazin Turbidity Unit

14.2.13 Load Cells:

A load cell is an electronic device (transducer) that is used to convert a force into an electrical signal. This conversion is indirect and happens in two stages. Through a mechanical arrangement, the force being sensed deforms a strain gauge. The strain gauge converts the deformation (strain) to electrical signals. A load cell usually consists of four strain gauges in a Wheatstone bridge configuration. Load cells of one or two strain gauges are also available. The electrical signal output is typically in the order of a few millivolts and requires amplification by an instrumentation amplifier before it can be used. The output of the transducer is plugged into an algorithm to calculate the force applied to the transducer.
Although strain gauge load cells are the most common, there are other types of load cells as well. In industrial applications, hydraulic (or hydrostatic) is probably the second most common, and these are utilized to eliminate some problems with strain gauge load cell devices. As an example, a hydraulic load cell is immune to transient voltages (lightning) so might be a more effective device in outdoor environments.
A strain gage is constructed by bonding a fine electric resistance wire or photographically etched metallic resistance foil to an electrically insulated backing, and attaching wire leads. The strain gage is then used for strain measurement by bonding it to the surface of the specimen with a special adhesive.





15 Final Control Elements

15.1 Control Valves

15.1.1 Selection Guide

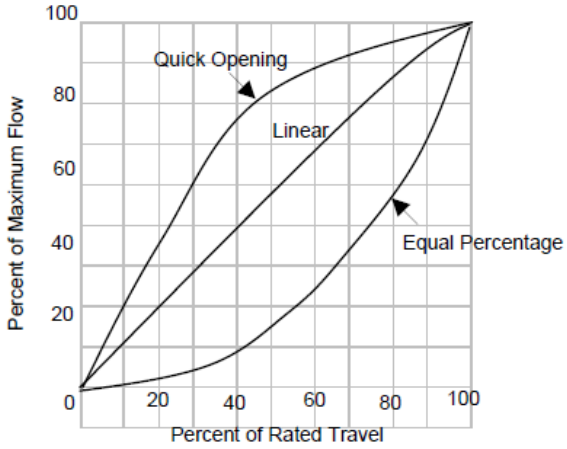
Conveying Media	Nature of Media	Valve Function	Type of Disc
Liquid	Neutral (Water, Oil, etc.)	On / Off	Gate
			Rotary Ball
			Plug
			Diaphragm
			Butterfly
			Plug Gate
		Modulating	Globe
			Butterfly
			Plug Gate
			Diaphragm
			Needle
	Corrosive (Acid, Alkaline, etc.)	On / Off	Gate
			Rotary Ball
			Plug
			Diaphragm
			Butterfly
			Plug Gate
		Modulating	Globe
			Diaphragm
			Butterfly
			Plug Gate
	Hygenic (Food, Beverages, Drugs, etc.)	On / Off	Butterfly
			Diaphragm
		Modulating	Butterfly
			Diaphragm
			Squeeze
			Pinch
	Slurry	On / Off	Rotary Ball
			Butterfly
			Diaphragm
			Plug
			Pinch
			Squeeze
		Modulating	Butterfly
			Diaphragm
			Squeeze
			Pinch
			Gate
	Fibrous Suspensions	On / Off Modulating	Gate
			Diaphragm
			Squeeze
			Pinch

Conveying Media	Nature of Media	Valve Function	Type of Disc
Gas	Neutral (Air, Steam, etc.)	On / Off	Gate
			Globe
			Rotary Ball
			Plug
			Diaphragm
		Modulating	Globe
			Needle
			Butterfly
			Diaphragm
			Gate
	Corrosive (Acid Vapors, Chlorine, etc.)	On / Off	Butterfly
			Rotary Ball
			Diaphragm
			Plug
		Modulating	Butterfly
			Globe
			Needle
			Diaphragm
	Vacuum	On / Off	Gate
			Globe
			Rotary Ball
			Butterfly
Solids	Abrasive Powder (Silica, etc.)	On / Off Modulating	Pinch
			Squeeze
	Lubricating Powder (Graphite, Talcum, etc.)	On / Off Modulating	Pinch
			Gate
			Squeeze

Control Valve Application Comparison Chart

Valve Type	Characteristic and Rangeability	Uses on slurries, Dirty solid bearing fluids	Relative Cost	Rating as Control Valve
Globe body with characterized plug or cage Sizes from needle up to 24 inches	Equal percentage or linear Max 50:1 Approx. 35:1 for needle	Very poor, can be constructed of corrosion resistant materials	High, very high in larger sizes	Excellent; any desired characteristic can be designed into this type valve
Ball valve availability up to 42 inches	Equal percentage Approx. 50:1 Ball can be characterized	Reasonably good, can be constructed of corrosion resistance materials	Medium	Excellent, if characteristic is suitable
Butterfly valve availability up to 150 inches	Equal percentage or linear Approx. 30:1 (some can characterized for quick opening)	Poor, a variety of material for construction available	Lowest cost for large size valves	Good, if characteristic is suitable
Saunders valve availability up to 20 inches	Approx. Linear 3:1 conventional 15:1 dual range	Very good, available with liner to resist corrosion	Medium	Conventional is poor; dual range is fair. Use only when ability is needed to handle dirty flow
Pinch valve availability up to 24 inches	Approx. Linear 3:1 to 15:1, depending on type	Excellent, several materials available to resist corrosion	Low	Poor to fair. Use only when ability is needed to handle dirty flow

15.1.2 Control Valve Characteristics



Linear flow characteristic shows that the flow rate is directly proportional to the valve travel (i.e. 50% of rated travel = 50% of maximum flow). This proportional relationship produces a characteristic with a constant slope so that with constant pressure drop, the valve gain will be the same for all flows.
Valve gain is the ration of an incremental change in flow rate to an incremental change in valve plug position. Gain is a function of valve size and configuration, system operating conditions and valve plug characteristic.
Control Valves with linear flow characteristic are commonly specified for liquid level control and for certain flow control applications requiring constant gain.

Equal % flow characteristic, equal increments of valve travel produce equal % changes in the existing flow. The change in the flow rate is always proportional to the flow rate just before the change in valve plug position. The change in flow rate will be relatively small at the low end and high end of plug travel.
A valve with equal % provides precise throttling control through the lower portion of the travel range and rapidly increasing capacity as the valve plug nears wide open.
Control valves with equal % flow characteristic are typically used on pressure control, on applications where a large % of the pressure drop is normally absorbed by the system itself with only a relatively small % available at the valve.

Quick Opening flow characteristic provides a maximum change in flow rate at low travels and small changes when the plug is near maximum. The curve is basically linear through the first 40% of plug travel, and then flattens out to indicate little increase in flow rate as plug travel approached wide open position.
Control Valves with quick opening flow characteristic are typically used for on/off type applications where significant flow rate must be established quickly as the valve begins to open.

15.1.3 Control Valve Plug Guiding

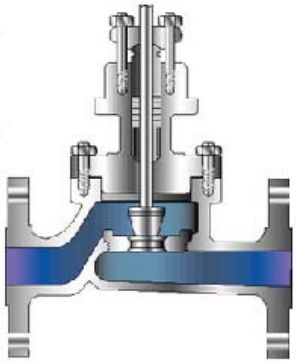
Cage Guiding: The outside diameter of the valve plug is close to the inside wall surface of the cylindrical cage throughout the travel range. This ensures correct valve plug / seat ring alignment when the valve closes. Cage guiding can provide stable control at high pressure drops, in addition cage guiding reduces vibration and mechanical noise.
The most common maintenance problem with cage guiding is galling and sticking due to the close metal-to-metal contact between cage and plug.



Top Guiding: Valve guide is aligned by a single guide bushing in the bonnet or valve body or by packing arrangement.



Stem Guiding: Valve plug is aligned with the seat ring by a guide bushing in the bonnet that acts on the valve plug stem.



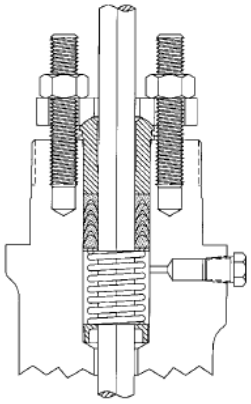
Top-and-Bottom Guiding: Valve plug is aligned by guide bushings in the bonnet and bottom flange.



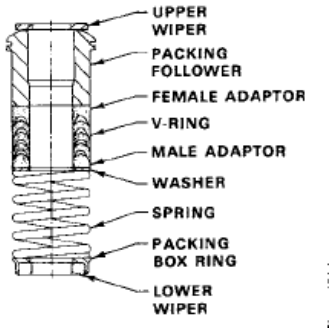
Port Guiding: Valve plug is aligned by the valve body port. This construction is typical for valves utilizing small diameter plugs with fluted skirt projections to control low flow rates.

15.1.4 Control Valve Packing

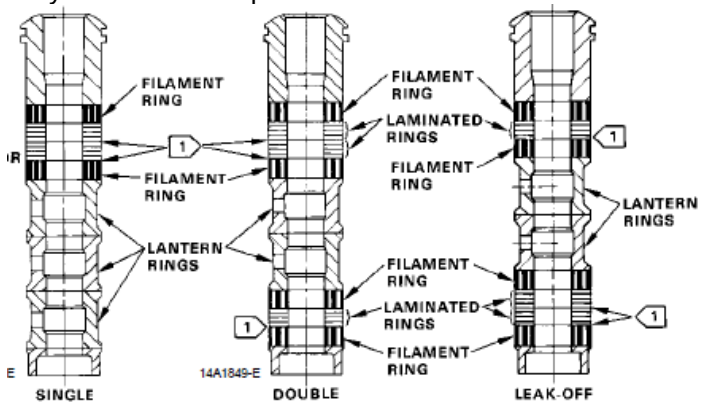
Most control valves use packing boxes with the packing retained and adjusted by a flange and stud bolts.



PTFE V-Ring (Chevron): Molded V-shaped rings that are spring loaded and self-adjusting in the packing box. Packing lubrication is not required.
Recommended temperature limits: -40 to +450°F



Laminated and Filament Graphite: Provides leak free operation, high thermal conductivity, and long service life. *However,* this packing produces high stem friction and resultant hysteresis. Packing lubrication is not required, but an extension bonnet or steel yoke should be used when packing temperature exceeds 800°F.
Recommended temperature limits: Cryogenic to 1200°F.
Primarily used when temperatures exceed that of PTFE limits.



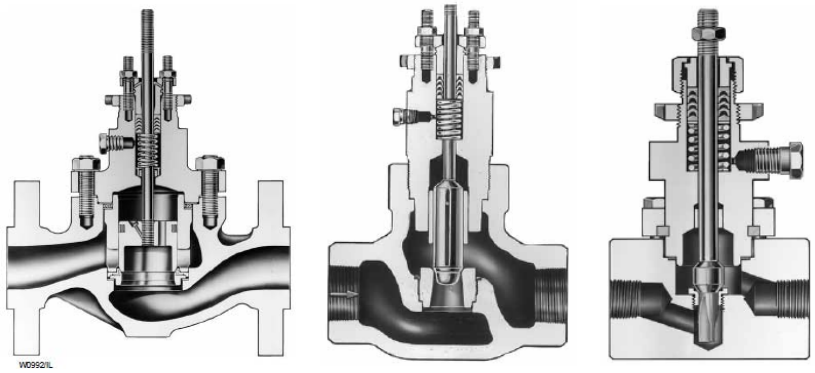
1 LOCATION OF SACRIFICIAL ZINC WASHER, IF USED.

15.1.5 Control Valve Bonnets

The bonnet normally provides a means of mounting the actuator to the body and houses the packing box. Generally rotary style valves do not have bonnets.

Bolted Flange Bonnet: Most common type of bonnet. May also be a screwed in bonnet or slip-on flange held in place with a split ring

Flanged *Screwed* *Split-Ring*



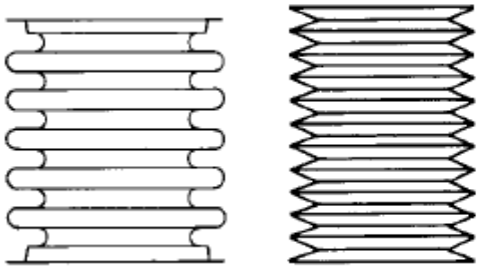
Extension Bonnet: May be used for either high or low temperature service to protect the valve stem packing from extreme process temperatures.



Bellows Seal Bonnet: Used when no leakage (< 10 x 10^{-6} cc/sec of He) along the stem can be tolerated. They are often used when the process fluid is toxic, volatile, radioactive or highly expensive.

Two type of bellows seal designs are used for control valves (mechanically formed and welded leaf). The welded-leaf design may have a shorter life expectancy than that of the mechanically formed, but it does offer a shorter package height.

Mechanical *Welded*



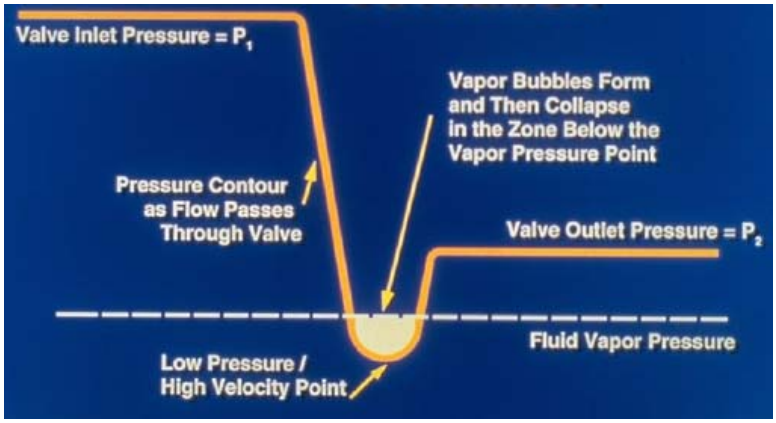
15.1.6 Control Valve Shutoff Classifications:

Class I	No requirements
Class II	0.05% of rated valve capacity
Class III	0.1% of valve rated capacity
Class IV	0.01% of valve rated capacity
Class V	5×10^{-4} ml/min of water per in of seat diameter *This is the best expected of a metal seat valve*
Class VI	The numbers range from: 0.15ml for a 1" valve to 6.75ml for an 8" valve *Soft Seat which will have temperature and pressure limitations.*

15.1.7 Control Valve Flashing / Cavitation:

Choked flow causes flashing and cavitation.

Cavitation: If the speed through the valve is high enough, the pressure in the liquid may drop to a level where the fluid may start bubble or flash. The pressure recovers sufficiently and the bubbles collapse upon themselves. Cavitation may be noisy but is usually of low intensity and low frequency. The collapsing bubbles are extremely destructive and may wear out the trim and body parts of the valve in short time.



Flashing: Is a one-stage phenomenon similar to cavitation. The difference is the downstream pressure does not recover enough to be above the fluid's vapor pressure. The vapor bubbles do not collapse and they remain in the fluid as vapor. This results in 2-phase flow downstream of the valve.

15.1.7.1 Control Valve Noise:

OSHA has set a noise-exposure limit of a weighted 90-dBA maximum over 8 hours. 85-dBA is the accepted maximum for control valve noise.

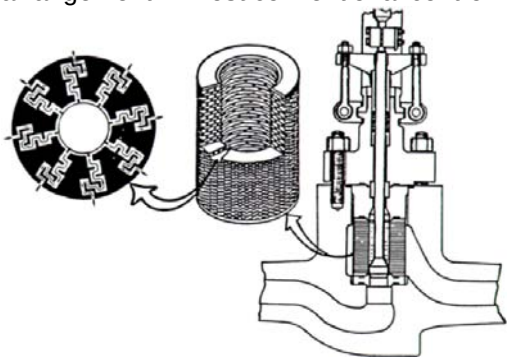
Sources of control valve noise:

- Mechanical Noise: Sound produced by this type of vibration will normally have a frequency < 1500Hz. The physical damaged incurred by the valve components is generally more of a concern than the noise emitted. A second source of mechanical vibration is a valve component vibrating at its natural (resonant) frequency (typical between 3000 & 7000 Hz.)
- Hydrodynamic Noise (Liquid Flow): Cavitation & Flashing
- Noise generated as a by product of a turbulent gas stream.

IEC noise standard recognizes three different categories of noise reducing trim.

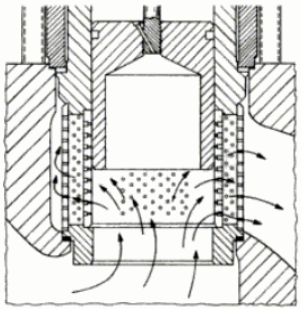
- Single stage, multiple flow passage trim (torturous path)
- Single flow path, multi-stage pressure reduction trim
- Multi-path, multi-stage trim (Whisper flow)

“Single stage” means that flowing fluid goes from upstream pressure condition at the valve inlet (P1) to the downstream pressure condition at the valve outlet (P2) in one step or stage. This is the typical arrangement in most conventional control valves.

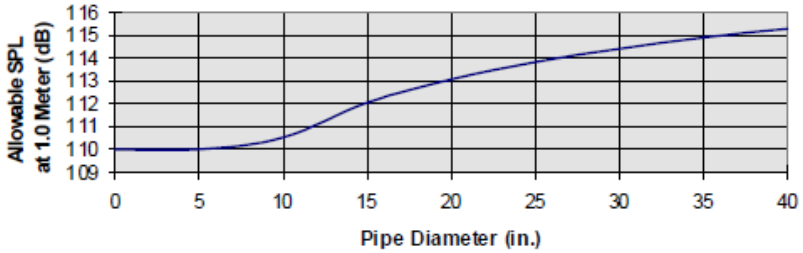


“Multiple flow passage” means that the flowing fluid, in going from the valve inlet to the valve outlet, passes through several flow openings rather than just one orifice. There are a couple of restrictive conditions on this definition, which are important to remember:

- The flow passages must be sufficiently separated in distance so there is no interaction between the jets emanating from each flow opening.
- The calculation procedures of the standard require that all of the multiple flow passages have the same hydraulic diameter. “Hydraulic diameter” is a term used to account for the fact that each flow opening might have some unusual or irregular shape other than circular. Hydraulic diameter then simply becomes the diameter of a circular hole that has the same area as the irregularly shaped flow passage. In the case of a drilled hole cage, the hydraulic diameter would simply be the diameter of each identical hole.



Recommended Max Valve Noise Levels for Structural Integrity of Piping



BEL = Log_{10} N
BEL = Log (Measured Power ÷ Reference Power)
dB = 10 Log (Measured Power ÷ Reference Power)
Gain (dB) = 20 Log (Gain Ratio)

POWER RATIO (Meas. Power)/(Ref. Power)	DECIBELS (dB)
0.01	-20 dB
0.1	-10 dB
1	0 dB
10	10 dB
100	20 dB

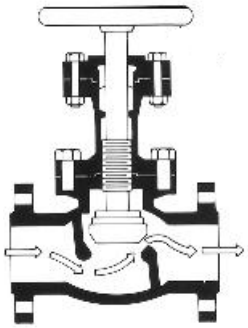
Environmental Noise	
Weakest sound heard	0dB
Whisper Quiet Library	30dB
Normal conversation (3-5')	60-70dB
Telephone dial tone	80dB
City Traffic (inside car)	85dB
Train whistle at 500', Truck Traffic	90dB
Subway train at 200'	95dB
Level at which sustained exposure may result in hearing loss	*90 - 95dB*
Power mower at 3'	107dB
Snowmobile, Motorcycle	100dB
Power saw at 3'	110dB
Sandblasting, Loud Rock Concert	115dB
Pain begins	*125dB*
Pneumatic riveter at 4'	125dB
Even short term exposure can cause permanent damage - Loudest recommended exposure WITH hearing protection	*140dB*
Jet engine at 100', Gun Blast	140dB
Death of hearing tissue	180dB
Loudest sound possible	194dB

Decibel comparison chart

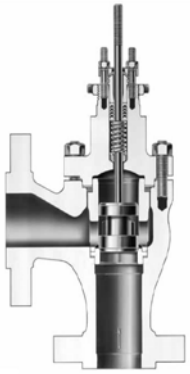
15.1.8 Control Valve Types:

15.1.8.1 Sliding Stem:

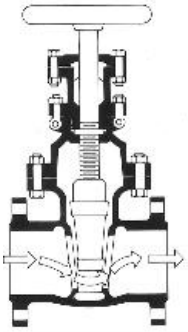
Globe: Available is single-port and double-port design.
Single-Port is the most common in use, and are generally specified for applications with stringent shutoff requirements. Generally are top-guided and they use metal-to-metal seating surfaces or soft-seating. Normal flow direction is up through the seat ring (has a self flushing effect).
Double-Port has more leakage than single port as it is almost impossible to close the two ports simultaneously. The advantage in the double-port design is the reduction of the required actuator forces.



Angle: Almost always single ported. Used where space is at a premium and the valve can also serve as an elbow. Commonly used in boiler feedwater and heater drain service

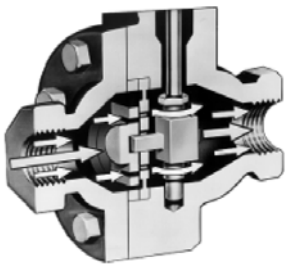


Gate: Valve that opens by lifting a round or rectangular gate/wedge out of the path of the fluid. The distinct feature of a gate valve is the sealing surfaces between the gate and seats are planar. Typically gate valves are for on/off service.



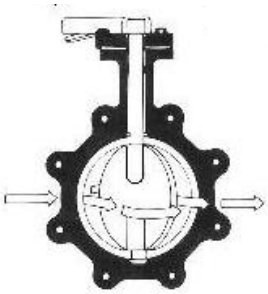
Sliding Gate: Uses a stationary plate and a movable disk, which slide against one another. Both components are slotted. When the slots are aligned, flow proportional to the flow area passes through.



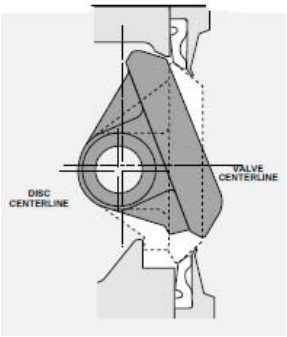


15.1.8.2 Rotary Valves:

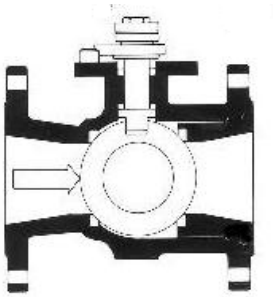
Butterfly: Typical application size is 2" through 72". May require large actuators if pressure drop across the valve is quite high. Conventional contoured disks provide throttling control for up to 60° disk rotation. Exhibit an approximate of the equal % flow characteristic.

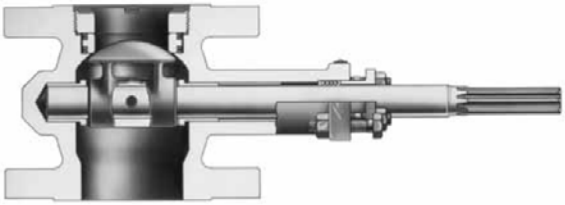


Eccentric Disk: An alternative to the single seated globe valve. Primarily used in steam service, high temperature gases and abrasive media.

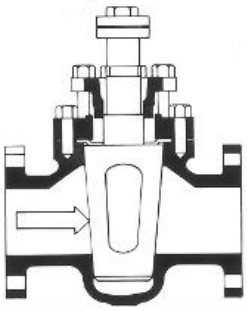


V-Ball (Segmental): Similar to a ball valve but the v-notch produces an equal % flow characteristic. Have good shutoff capability. They have one seat, generally on the upstream side rather than two as the traditional full sphere ball valve has. Their design also does not trap media when closed as the traditional full sphere ball valve does.



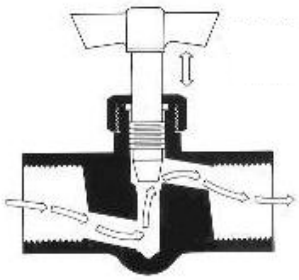


Plug Valve: Valve with cylindrical or conically-tapered "plugs" which can be rotated inside the valve body to control flow through the valve. The plugs in plug valves have one or more hollow passageways going sideways through the plug, so that fluid can flow through the plug when the valve is open. Typically used for on/off service.

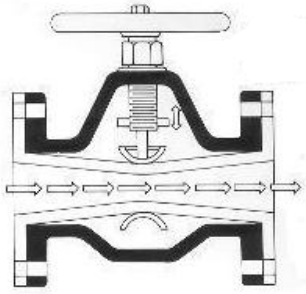


15.1.8.3 Special Purpose Valves:

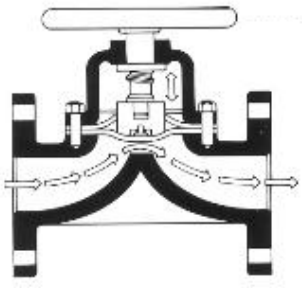
Needle Valve: Has a relatively small orifice with a long, tapered, conical seat. A needle-shaped plunger, on the end of a screw, exactly fits this seat. As the screw is turned and the plunger retracted, flow between the seat and the plunger is possible.



Pinch Valve: Employs a flexible body liner that is forced together to restrict flow.



Diaphragm Valve: Consists of a valve body with two or more ports, a diaphragm, and a "saddle" or seat upon which the diaphragm closes the valve. Their application is generally as shut-off valves in process systems within the food and beverage, pharmaceutical and biotech industries. The older generation of these valves is not suited for regulating and controlling process flows, however newer developments in this area have successfully tackled this problem.



Squeeze Valve: Similar to a needle valve, but operates similar to that of a valve on the end of a garden hose.



15.1.8.4 Actuators:

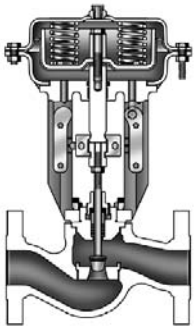
Hydraulic / Pneumatic: used on linear or quarter-turn valves. Sufficient air or fluid pressure acts on a piston to provide thrust in a linear motion for gate or globe valves. Alternatively, the thrust may be mechanically converted to rotary motion to operate a quarter-turn valve. These may be spring return or double-acting. Note that double acting actuators do not have an inherent fail safe action.

Direct Acting: (Air to Extend) Fail Open
Reverse Acting: (Air to Retract) Fail Close

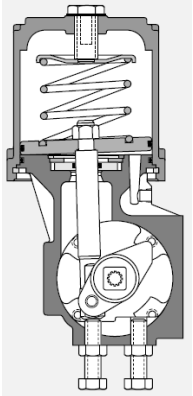
(reference section 15.1.6 for shutoff classifications table)

Actuator Types:

- **Spring & Diaphragm**: Spring opposed pneumatic cylinders. A 3-way solenoid valve is used to operate a spring return actuator. The normal position is the position of the valve with the spring extended and the solenoid de-energized. This style may be single-acting or double acting.



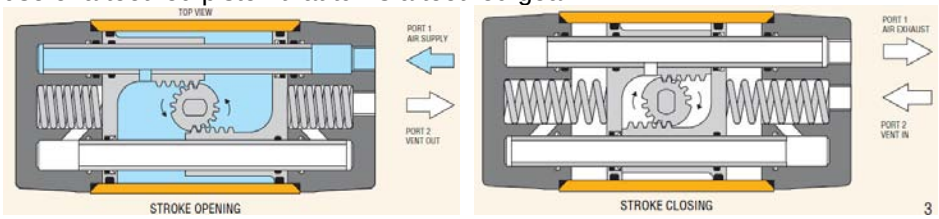
- **Piston:** The piston is covered by a diaphragm, or seal, which keeps the air in the upper portion of the cylinder, allowing air pressure to force the diaphragm downward, moving the piston underneath, which in turn moves the valve stem, which is linked to the internal parts of the actuator. Generally tolerates higher actuating pressures than that of spring & diaphragm.



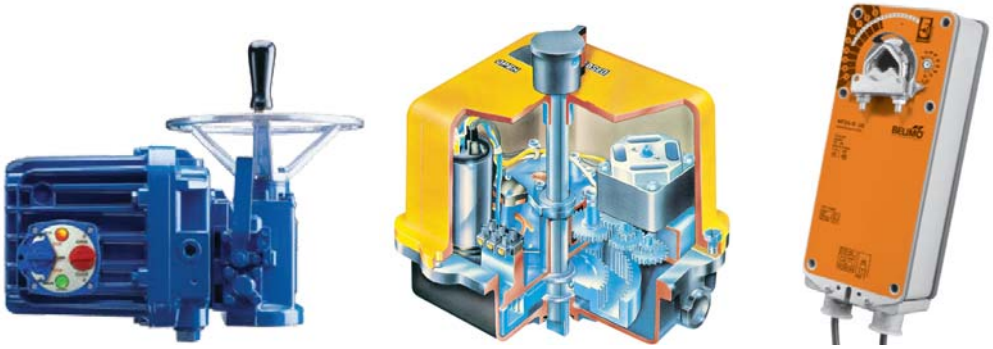
- **Vane:** Rotary actuator that directs energy in a circular motion through the use of two arm-like mechanisms, or vanes. Can be thought of as a rotary piston.



- **Rack & Pinion:** Rotary actuator that directs energy in a circular motion through the use of a toothed piston that turns a toothed gear



- **Electric:** Has a motor drive that provides torque to operate a valve. Electric actuators are frequently used on multi-turn valves such as gate or HVAC dampers. With the addition of a quarter-turn gearbox, they can be utilized on ball, plug, or other quarter-turn valves. Typically well suited for remote locations where compressed air is not available. Electric actuators inherently fail in the last position, some form of backup power would be needed to make these fail safe.



Actuator Characteristics:

TYPE SOURCE	POWER SERVICES	RANGE OF TORQUE CAPABILITIES	THRUST OR APPLICATION	VALVE TYPE	STIFFNESS	MATERIALS OF CONSTRUCTION	FAIL SAFE ACTION	WEIGHT PER UNIT FORCE	PHYSICAL ENVELOPE	INITIAL COST	MAINTENANCE COST	AVAILABLE ACCESSORIES	RELIABILITY
Spring and diaphragm	Air to 30 psig, spring return	Throttling, on-off general service	Light to moderate	Vertical direct, rotary with linkage	Low to moderate	Various elastomeric diaphragms and frame materials	Inherent due to built-in spring	high in large sizes	Require high vertical clearance	Low	Low handwheels,	Positioners, limit switches, technology microprocessors	Excellent, simple, mature technology
Pneumatic Piston	Air to 50-150 psig, opposed pistons	Throttling, on-off general service	Moderate to high	Vertical direct, rotary with linkage	Moderate to high	Available in CS,SS, corrosion-resistant materials	Require spring or reserve air cylinders	Low	Moderately compact	Low to moderate	Low to moderate	Positioners, limit switches, microprocessors	Excellent, mature technology
Electric Motor Operated	Electric motor and gear train	Limited throttling good on-off service	Very high	Best on larger remote rotary valves	Very high	CS,SS, aluminum and epoxy filled composites	Require stand-by power source	Moderate to high	Compact	High	Low to moderate remote operators positioners	Limit Switches, microprocessors, mature technology	Good depending on power source,
Electro-Hydraulic	Electric motor and pump	Good in throttling services	Moderate to very high	Vertical direct, rotary w/ linkage, handles dampers well	High	Available in various materials to suit environment	Require stand-by power source or spring	Moderate	Require high vertical clearance	Highest	Highest switches	Positioners, limit enance, multiple moving parts	Higher maint-

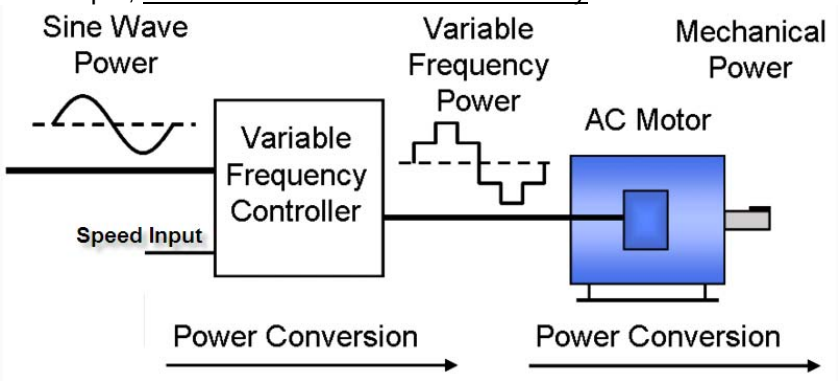
Actuator Advantages / Disadvantages:

ADVANTAGES	DISADVANTAGES
SPRING AND DIAPHRAGM	
Lowest first cost	Limited force output
Operating simplicity and reliability	Large size and weight per unit force
Throttles well without positioner	Low Stiffness
Inherent fail-safe operation	Does not handle high ambient temperatures well
Low air pressure supply	
Easily adjustable to local conditions	
Few moving parts	
Ease of maintenance	
Most mature technology	
PNEUMATIC PISTON	
High force and torque capacity	Requires spring or accessories for fail safe operation
Moderately compact silhouette	Must have positioner to function
Low weight to power ratio	Higher first cost than S&D actuator
Can stand high ambient temperatures	Requires high pressure air to operate
Fast stroking speed	
Good throttling control	
Few Moving parts	
Relatively high actuator stiffness	
Available in smart valve configuration	
Available in corrosion resistant materials	
ELECTRIC MOTOR DRIVE	
High force output	High first cost
Very high stiffness	Not inherently fail safe
Compact silhouette	Limited duty cycle of motor drive
Slow stroking speed (if required)	Multiple moving parts
Does not require air supply or I/P	
Corrosion-resistant seals and construction are available	
Infrared emote setting unit available	
Relatively mature technology	
Good on large valves in remote locations	
ELECTROHYDRAULIC	
High force output	Highest first cost
High actuator stiffness	Multiple moving parts, highest maintenance
Excellent throttling ability	Fails safe only with spring or accessories
Fast stroking speed	Least mature actuator technology
Does not require air supply or I/P	
Accepts harsh environments well	

15.2 Variable Frequency Drives / Motors:

15.2.1 Types of Variable Frequency Drives (AC):

All VFDs use their output devices (IGBTs, transistors, thyristors) only as switches to approximate a sine wave output. Due to the heating effect from the staircase design sine wave output, motors must be rated for inverter duty.

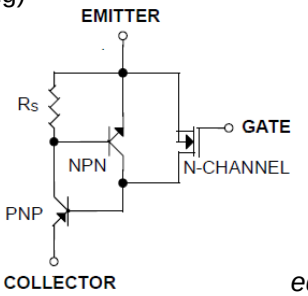


In addition steps shall be taken to remove the "reflected wave" phenomena" associated with VFDs. Addition of output filters to the VFDs will help reduce the "reflected wave" phenomena, in addition to inverted duty rated motors, motor feeder lengths should be kept as short as possible.

Maximum cable length vs drive type for 480VAC motor insulation:

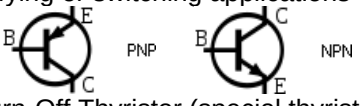
Drive Type	Semiconductor t_{rise} range (µs)	Motor Allowable Line to Line Surge Voltage		
		1000 V_{PK}	1200 V_{PK}	1600V_{PK}
IGBT	0.05 - 0.4	30-100 Ft	50-200 Ft.	Unlimited
BJT	0.5 - 2	250-600 Ft	500-1000 Ft	Unlimited
GTO	2-4	600-1000 Ft.	1000-2000 Ft.	Unlimited

IGBT: Insulated Gate Bipolar Transistor (three-terminal power semiconductor device, noted for high efficiency and fast switching)

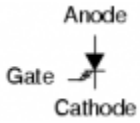


equivalent circuit diagram

BJT: Bipolar Junction Transistor (three-terminal electronic device constructed of doped semiconductor material and may be used in amplifying or switching applications



GTO: Gate Turn-Off Thyristor (special thyristor which can be turned on by a positive gate signal and can be turned off by a negative signal



Constant Torque: Requires motor to produce full load torque at zero speed.

- Crane
- Elevator
- Conveyor
- P.D. Pumps

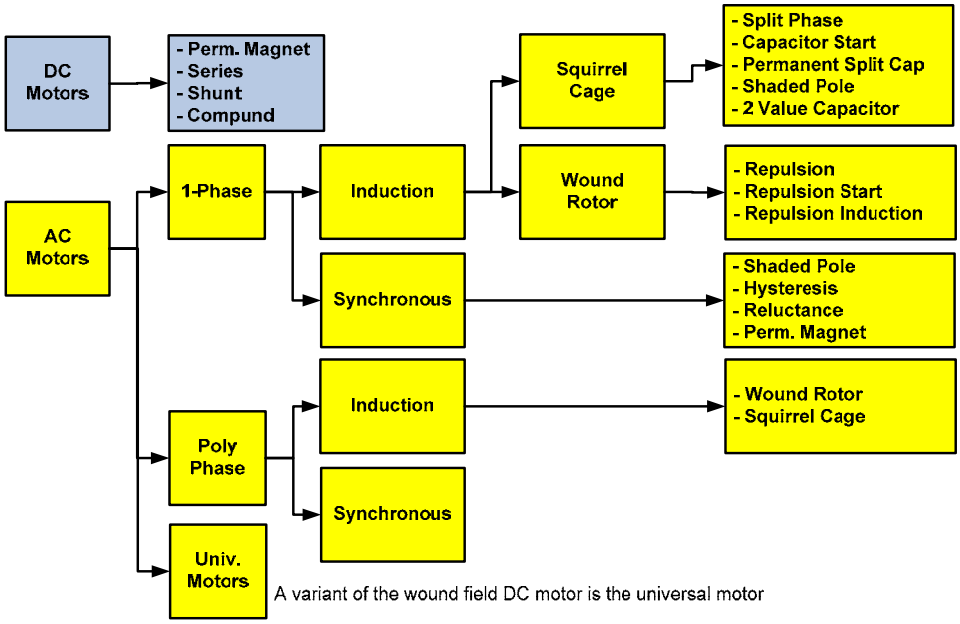
Variable Torque: Load that requires high torque at low speeds and low torque at high speeds. Horsepower remains constant as speed and torque are inversely proportional

- Centrifugal and axial pumps
- Fans and blowers
- Mixers and agitators

Constant HP: Load that has decreasing torque requirements at higher speeds.

- Grinders or Lathes
- Winding machines

15.2.2 Types of Motors:

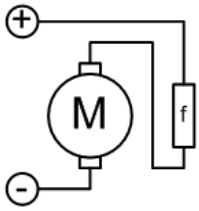


A variant of the wound field DC motor is the universal motor

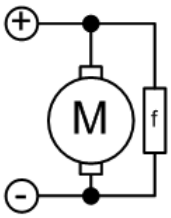
15.2.2.1 DC Motors

Brush Type DC Motors:

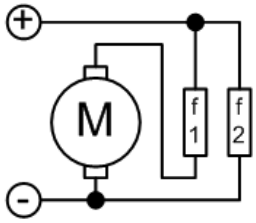
Series: The field coils and the armature in a shunt-wound motor are connected in parallel, also known as shunt, formation, causing the field current to be proportional to the load on the motor. Develops high torque at low speeds.



Shunt: In series-wound motors, the field coils and armature are connected in a series and the current flows through the field coils only. Has better speed control than does a series motor, but does not develop the low speed high torque.



Compound: A compound-wound motor is hybrid of both the shunt-wound and series-wound types and features both configurations.



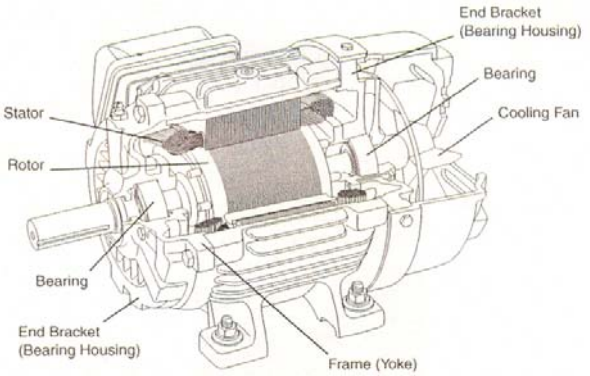
Brushless Type DC Motors:

Perm. Magnet: The field in a permanent magnet motor is created by permanent magnets as the name allows. Typically fractional horsepower motors.

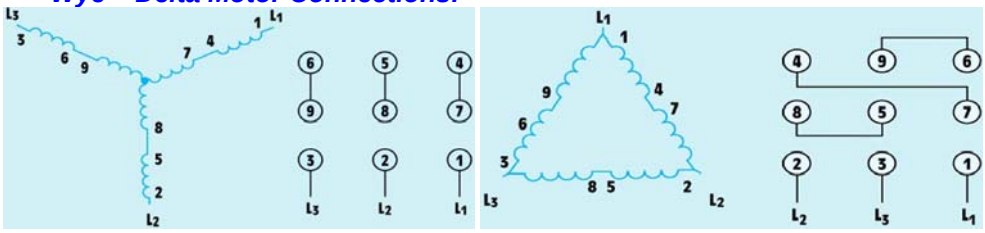
15.2.2.2 AC Induction Motors

$$RPM = \frac{120\,f}{\#\,Poles}$$

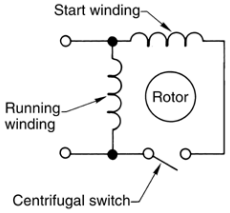
- **Squirrel Cage:** These motors are probably the simplest and most rugged of all electric motors. They consist of two basic electrical assemblies: the wound stator and the rotor assembly.



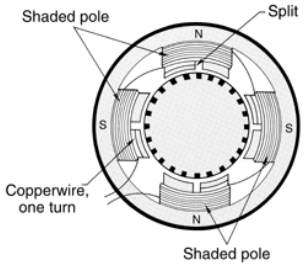
Wye – Delta Motor Connections:



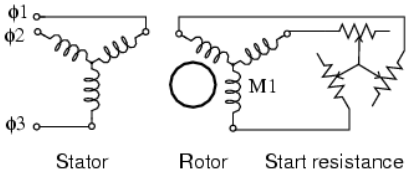
- **Split Phase:** Use both a starting and running winding. The starting winding is displaced 90 electrical degrees from the running winding. The running winding has many turns of large diameter wire wound in the bottom of the stator slots to get high reactance. Therefore, the current in the starting winding leads the current in the running winding, causing a rotating field. During startup, both windings are connected to the line, Figure 7. As the motor comes up to speed (at about 25% of full-load speed), a centrifugal switch actuated by the rotor, or an electronic switch, disconnects the starting winding. Split phase motors are considered low or moderate starting torque motors and are limited to about 1/3 hp.



- **Capacitor Start:** Are similar to split phase motors. The main difference is that a capacitor is placed in series with the auxiliary winding. This type of motor produces greater locked rotor and accelerating torque per ampere than does the split phase motor. Sizes range from fractional to 10 hp at 900 to 3600 rpm.
- **Shaded Pole:** Have a continuous copper loop wound around a small portion of each pole. The loop causes the magnetic field through the ringed portion to lag behind the field in the un-ringed portion. This produces a slightly rotating field in each pole face sufficient to turn the rotor. As the rotor accelerates, its torque increases and rated speed is reached. Shaded pole motors have low starting torque and are available only in fractional and sub-fractional horsepower sizes. Slip is about 10%, or more at rated load.



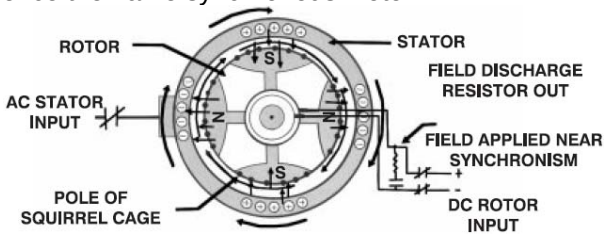
- **Split Capacitor:** Also have an auxiliary winding with a capacitor, but they remain continuously energized and aid in producing a higher power factor than other capacitor designs. This makes them well suited to variable speed applications.
- **Wound Rotor:** Motor has a stator like the squirrel cage induction motor, but a rotor with insulated windings brought out via slip rings and brushes. However, no power is applied to the slip rings. Their sole purpose is to allow resistance to be placed in series with the rotor windings while starting. This resistance is shorted out once the motor is started to make the rotor look electrically like the squirrel cage counterpart. Placing resistance in series with the rotor windings not only decreases start current, locked rotor current (LRC), but also increases the starting torque, locked rotor torque (LRT).



- **Repulsion:** An alternating-current commutator motor designed for single-phase operation. The chief distinction between the repulsion motor and the single-phase series motors is the way in which the armature receives its power. In the series motor the armature power is supplied by conduction from the line power supply. In the repulsion motor, however, armature power is supplied by induction (transformer action) from the field of the stator winding.
- **Repulsion – Start:** An alternating-current motor that starts as a repulsion motor; at a predetermined speed the commutator bars are short-circuited to give the equivalent of a squirrel-cage winding for operation as an induction motor with constant-speed characteristics.
- **Repulsion – Induction:** A repulsion motor that has a squirrel-cage winding in the rotor in addition to the repulsion-motor winding.

15.2.2.3 Synchronous Motors

AC motor distinguished by a rotor spinning with coils passing magnets at the same rate as the alternating current and resulting magnetic field which drives it. Another way of saying this is that it has zero slip under usual operating conditions. Contrast this with an induction motor, which must slip in order to produce torque. Synchronous motor is like an induction motor except the rotor is excited by a DC field. Slip rings and brushes are used to conduct current to rotor. The rotor poles connect to each other and move at the same speed hence the name synchronous motor.



15.2.2.4 TWO Speed Motors

Can classified into two different winding types:

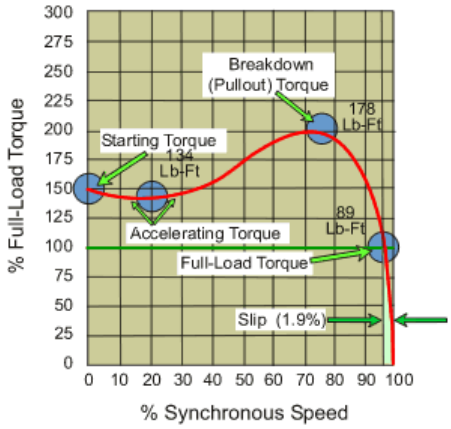
TWO SPEED, TWO WINDING: The two winding motor is made in such a manner that it is really two motors wound into one stator. One winding, when energized, gives one of the speeds. When the second winding is energized, the motor takes on the speed that is determined by the second winding. The two speed, two winding motor can be used to get virtually any combination of normal motor speeds and the two different speeds need not be related to each other by a 2:1 speed factor. Thus, a two speed motor requiring 1750 RPM and 1140 RPM would, of necessity, have to be a two winding motor.

TWO SPEED, ONE WINDING: The second type of motor is the two speed, single winding motor. In this type of motor, a 2:1 relationship between the low and high speed must exist. Two speed, single winding motors are of the design that is called consequent pole. These motors are wound for one speed but when the winding is reconnected, the number of magnetic poles within the stator is doubled and the motor speed is reduced to one-half of the original speed. The two speed, one winding motor is, by nature, more economical to manufacture than the two speed, two winding motor. This is because the same winding is used for both speeds and the slots in which the conductors are placed within the motor do not have to be nearly as large as they would have to be to accommodate two separate windings that work independently. Thus, the frame size on the two speed, single winding motor can usually be smaller than on an equivalent two winding motor

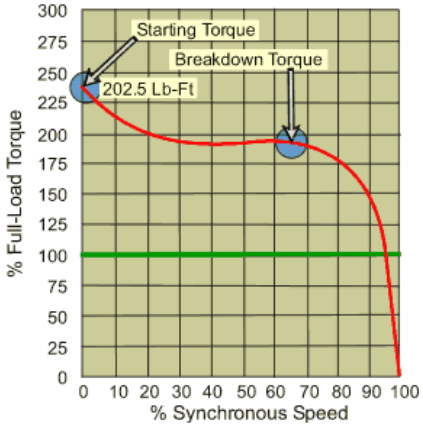
15.2.3 Motor NEMA Designations:

$$HP = \frac{T \times RPM}{5250} \quad \textbf{OR} \quad T = \frac{HP \times 5250}{RPM}$$

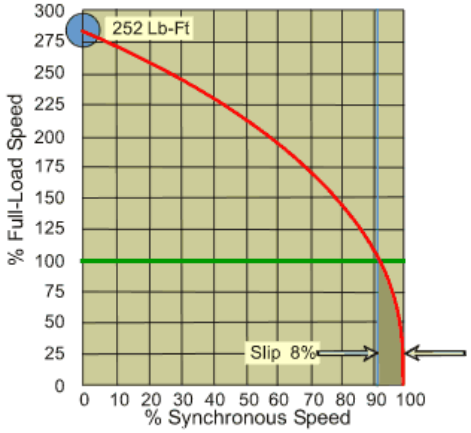
- **Design A** motors have a higher breakdown torque than Design B motors and are usually designed for a specific use. Slip is 5%, or less.
- **Design B** motors account for most of the induction motors sold. Often referred to as general purpose motors, slip is 5% or less.
- **Design C** motors have high starting torque with normal starting current and low slip. This design is normally used where breakaway loads are high at starting, but normally run at rated full load, and are not subject to high overload demands after running speed has been reached. Slip is 5% or less.
- **Design D** motors exhibit high slip (5 to 13%), very high starting torque, low starting current, and low full load speed. Because of high slip, speed can drop when fluctuating loads are encountered. This design is subdivided into several groups that vary according to slip or the shape of the speed-torque curve. These motors are usually available only on a special order basis.



Design B



Design C



Design D

15.2.4 Motor NEMA Insulation Classes:

Based Upon 20,000 hours of average insulation life:

Insulation Class	Temperature Rating
A	105° C
B	130° C
F	155° C
H	180° C

15.2.5 Motor Feeder Sizes:

****** 460VAC, 3Phase, MOTOR STARTERS, FEEDERS AND SIZES ******
(Based on 2008 NEC Tables 310.16 and 430.250)

Motor HP	Starter Size (1)	Space Factor (2)	FLA	E3 / E3+ Rating	HMCP Rating (Adj Setting)	HMCP CAM Setting	Feeder Wire Size (7)	Max. Fdr. Ampacity (3)	Fdr Cable XLP-HP OD / Area	Conduit Size Reqd. (5)	Ohms / K Feet (6)	Max Length (in feet) 5% V. D. (24V)	Max Length (in feet) 3% V. D. (14.4V)
0.5	1	1	1.1	E3 P	3A (0.69-2.5)	C	4/C #12AWG 7/C #12AWG	16	0.475" 0.1772sq in	3/4"	2.00	11000	6500
0.75	1	1	1.6	E3 A	3A (0.69-2.5)	E	4/C #12AWG 7/C #12AWG	16	0.475" 0.1772sq in	3/4"	2.00	7500	4500
1	1	1	2.1	E3 A	3A (0.69-2.5)	G	4/C #12AWG 7/C #12AWG	16	0.475" 0.1772sq in	3/4"	2.00	5700	3400
1.5	1	1	3.0	E3 A	7A (1.5-5.7)	C	4/C #12AWG 7/C #12AWG	16	0.475" 0.1772sq in	3/4"	2.00	4000	2400
2	1	1	3.4	E3 A	7A (1.5-5.7)	D	4/C #12AWG 7/C #12AWG	16	0.475" 0.1772sq in	3/4"	2.00	3500	2100
3	1	1	4.8	E3 B	15A (3.4-12.6)	B	4/C #12AWG 7/C #12AWG	16	0.475" 0.1772sq in	3/4"	2.00	2500	1500
5	1	1	7.6	E3 B	15A (3.4-12.6)	D	4/C #12AWG 7/C #12AWG	16	0.475" 0.1772sq in	3/4"	2.00	1575	950
7.5	1	1	11	E3 C	30A (6.9-25.2)	B	4/C #12AWG 7/C #12AWG	16	0.475" 0.1772sq in	3/4"	2.00	1100	650
10	1	1	14	E3 C	30A (6.9-25.2)	D	4/C #12AWG 7/C #12AWG	16	0.475" 0.1772sq in	3/4"	2.00	850	515
15	2	1	21	E3 D	50A (11.5-42.1)	C	4/C #10AWG	24	0.56" 0.2463sq in	1" 1.5"	1.20	925	550
20	2	1	27	E3 D	50A (11.5-42.1)	E	3/C #8AWG w/GRD	44	0.66" 0.3421sq in	1" 1.5	0.78	1150	675
25	2	1	34	E3 D	50A (11.5-42.1)	F	3/C #8AWG w/GRD	44	0.66" 0.3421sq in	1" 1.5	0.78	900	540
30	3	1.5	40	E3+ E	100A (23-84.5)	C	3/C #6AWG w/GRD	60	0.745" 0.4359sq in	1.5"	0.49	1225	725
40	3	1.5	52	E3+ E	100A (23-84.5)	D	3/C #6AWG w/GRD	60	0.745" 0.4359sq in	1.5"	0.49	950	560
50	3	1.5	65	E3+ E	100A (23-84.5)	F	3/C #4AWG w/GRD	76	0.89" 0.6221sq in	1.5"	0.31	1200	725
60	4	2	77	E3+ E	150A (34.6-126.7)	D	3/C #2AWG w/GRD	104	1.02" 0.8171sq in	1.5"	0.20	1550	950
75	4	2	96	E3+ F	150A (34.6-126.7)	F	3/C #1/0AWG w/GRD	136	1.24" 1.2076sq in	2"	0.12	1925	1150
100 (4)	4	2	124	E3+ G	150A (57-130.7)	D	3/C #2/0AWG w/GRD	152	1.34" 1.4103sq in	3"	0.10	1750	1050
125 (4)	5	3.5	156	E3+ G	250A (86.6-183.6)	G	3/C #4/0AWG w/GRD	208	1.578" 1.9557sq in	3"	0.063	1900	1150
150 (4)	5	3.5	180	E3+ G	250A (96.2-204)	H	3/C #4/0AWG w/GRD	208	1.578" 1.9557sq in	3"	0.06	1650	1000

(1) Starter shall be FVNR Starter Unit with Ckt. Brkr.
(2) Space factor is based upon Allen Bradley "Centerline" model MCC combination starter.
(3) The above wire sizes are based on 90 deg. C conductor termination rating, derated 80% for continuous load.
(4) Cross-the-Line starters require prior Talecris approval for these HP levels. Preferred are Allen Bradley VFDs or SMCs
(5) NEC Chapter 9 Table 4 (Art. 344 Rigid Metal Conduit) Based upon 40% fill over 2 wires. (Conduit Sizes NOT to be used 1.25" and 2.5")
Top size is for power feeder only in conduit; Bottom size is for motor feeder and control wires in same conduit.
(6) NEC Chapter 9 Table 9 AC resistance for uncoated copper wires in steel conduit.
(7) Top size shown is for motor feeder only; Bottom size shown is for motor feeder & control wires in same cable

Continuous Load Definition: A load where the maximum current is expected to continue for 3 hours or more.
Conductors that get too hot for too long can lead to a loss of temper of the metal components & a potential softening of the insulation material.

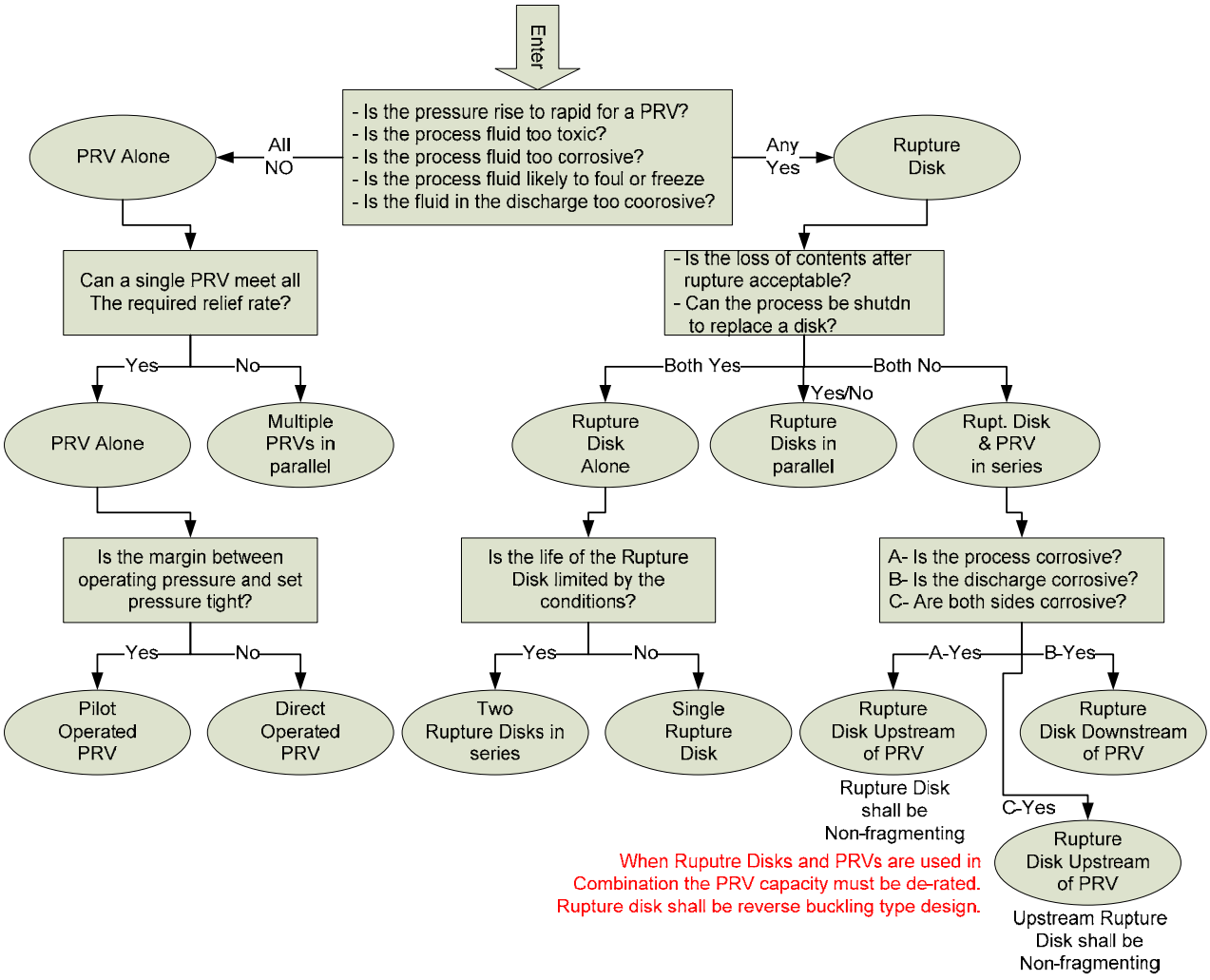
Conductors that supply a single motor used in a continuous duty application shall have an ampacity of not less than 125% of the motor's full load current (or stated another way, the conductor may only be loaded to 80% of its capacity due to a continuous load)

Control Cable 3/C #12 OD=0.435" Area = 0.1486sq in

RGS Size	40% Fill Sq. In.
3/4"	0.220sq in
1"	0.355sq in
1.5"	0.829sq in
2"	1.363sq in
3"	3.000sq in
4"	5.153sq in

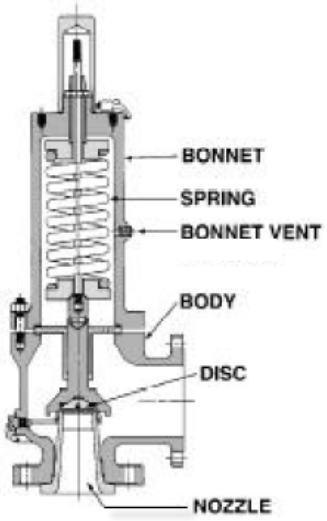
16 Relief Valves

16.1 Selection of Pressure Relief Devices

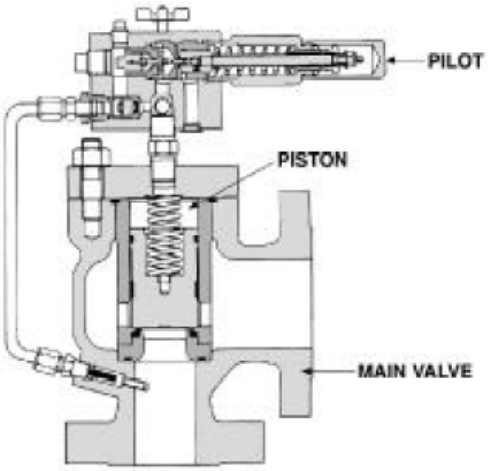


16.2 Types of Pressure Relief Devices

- **Pressure Relief Valve:** Device designed to prevent internal pressure from rising above a pre-determined maximum pressure in a pressure vessel exposed to abnormal or emergency conditions.
  - **Spring Loaded Design:** Valve consists of an inlet valve or nozzle mounted on the pressurized system, a disc held against the nozzle to prevent flow under normal operating conditions, a spring to hold the disc closed and a body/bonnet to contain the operating elements. The spring load is adjustable to vary the pressure at which the valve will open.



  - **Pilot:** Consist of a main valve with a piston or diaphragm operated disc and a pilot. Under normal operating conditions the pilot allows system pressure into the piston chamber. Since the piston area is greater than the disc seat area, the disc is held closed. When the set pressure is reached, the pilot actuates to shut off system fluid to the piston chamber and simultaneously vents the piston chamber. This causes the disc to open.



- **Safety Relief Valve:** Safety valve is a PRV actuated by static inlet pressure and characterized by rapid opening or 'pop' action. Typically used for steam and air service.

- **Emergency Relief Vent:** (Typical used on storage tanks – API rated) Emergency Pressure Vents are designed to provide emergency relief capacity beyond that furnished by the operating vent on tanks. Under normal operating conditions, the vent pallet assembly is closed providing an effective vapor seal. In the event of an emergency (fire involvement of the tank), the pallet lifts in response to the increased pressure in the tank's vapor space. Vapor is expelled, thereby protecting the tank from dangerous over-pressurization. Pallet assembly automatically closes and reseals when the pressure is reduced. Emergency vents do not provide vacuum relief. Vacuum relief must be supplied by independent operating conservation vents.



- **Conservation Vent:** Is intended for use where both pressure and vacuum relief are required. The pallets in the vent housing allow intake of air and escape of vapors as the tank normally breathes in and out. Pallets open and close to permit only that intake or outlet relief necessary to remain within permissible working pressures and avoid damage to the tank.



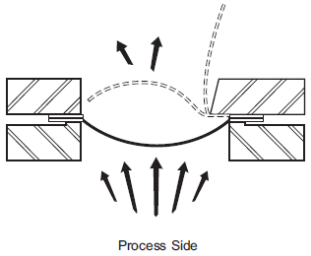
- **Flame Arrester:** Installed where it is not necessary to conserve vapors but where low flash point liquids must be protected against fire and explosion from exterior sources of ignition. The tightly spaced circular flame arrester grid plates are integral with the vent housing. Flame arresters are mounted on the end of a vent pipe from the tank (may also be integral to a conservation vent). Vapors are allowed to escape into the atmosphere and air can be drawn into the tank through the specially designed flame arrester grid assembly. If an ignition source outside the tank (unconfined deflagration) is encountered, the flame arrester provides protection for the tank's vapor space.



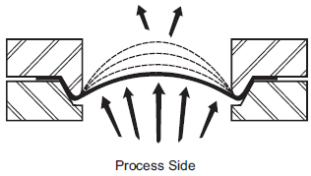
16.3 Types of Rupture Disks:

Pressure relief device that protects a vessel or system from over-pressurization. Rupture discs are a type of sacrificial part because it has a one-time-use membrane that fails at a predetermined pressure, either positive or vacuum. The membrane is usually a thin metal foil, but nearly any material can be used to suit a particular application. Rupture discs provide fast response to an increase in system pressure but once the membrane has failed it will not reseal.

- **Reverse Buckling:** By loading the Reverse Buckling disk in compression it is able to resist operating pressures up to 100% of minimum burst pressure even under pressure cycling or pulsating conditions. Designed for non fragmentation upon activation, Reverse Buckling disks are recommended for combination with pressure relief valves to isolate them from normal process conditions ensuring excellent leak tightness.



- **Forward Acting (Tension):** Applying load to the concave side, the disk is subjected to tension forces. Forward acting disks regulate burst pressures by the tensile strength of the material.



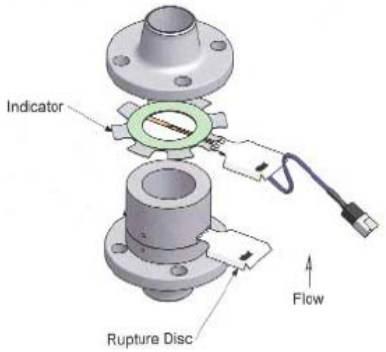
- **Accessories:**
  - **Holder (Safety Head):** Holder allows the rupture disk to be pre-assembled to insure it is properly seated before installation into the pressure system.



  - **Tell-Tale Connection:** The ASME code requires that the space between a rupture disk (bursting disc) device and a pressure relief valve be provided with a pressure gauge, tricock, free vent, or suitable telltale indicator



- **Burst Indicator:** Designed to operate in a “normally closed” electrical circuit. A membrane is used to support an electrical conducting circuit. When the pressure event (disk rupture) occurs, the flow of fluid places the membrane in tension which leads to the break of an electrical conductor. This changes the electrical status of the sensor to “normally open”.



- **Tolerances:** The burst pressure tolerance at the specified rupture disk temperature shall not exceed ±2 psi for marked pressures up to and including 40 psi and ±5 psi for marked burst pressures above 40 psi
- **Operating Ratios:** Are defined as the relationship between operating pressure and the stamped burst pressure, and are usually expressed as a % (i.e. $P_o \div P_b$) In general, good service life can be expected when operating pressures do not exceed the following:
    - 70% of stamped burst pressure for conventional pre-bulged rupture disks.
    - 80% of stamped burst pressure for composite design rupture disks.
    - 80 – 90% of stamped burst pressure for forward acting rupture disks.
    - Up to 90% of stamped burst pressure for reverse acting rupture disks.

16.4 Pressure Relief Sizing Contingencies:

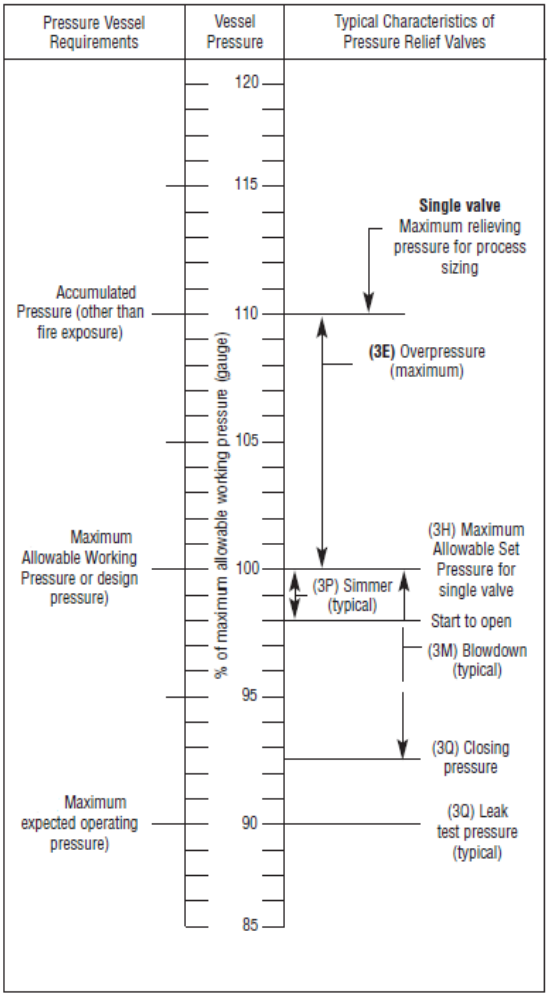
Pressure Relief Contingencies			WORST CREDIBLE SCENARIO
Designer:	Project No.:	Area:	
Process Knowledgeable Person:	P & ID No. with latest Rev. and Rev. Date:		
Date:	Relief Device Tag No.:		
Equipment to be protected:			
Possible Pressure or Vacuum Contingencies	Relief Flow Rate Lb/Hr	Orifice Area Sq In	
1 Exposure to External Fire			
2 Incorrect Valve Position			
3 Overfilling a Vessel			
4 Emptying a Vessel			
5 Thermal Expansion			
6 Coolant Failure			
7 Loss of Heat			
8 Power Failure			
9 Instrument Failure			
10 Utility Failure			
Air			
Water			
Nitrogen			
Steam			
Refrigeration			
Other (Specify)			
11 Abnormal Heat Input			
12 Abnormal Flow Input			
13 Exchanger Tube Failure			
14 Chemical Reaction			
15 Accumulation of Non-Condensables			
16 Mechanical Failures			
17 Pump Overpressure			
18 Sudden Mixing of Hot & Highly Volatile Materials			
19 Transient Pressure Surges			
20 Absorbent Flow Failure			
21 Plugging/Fouling			
22 Other (Specify)			
Comments:			

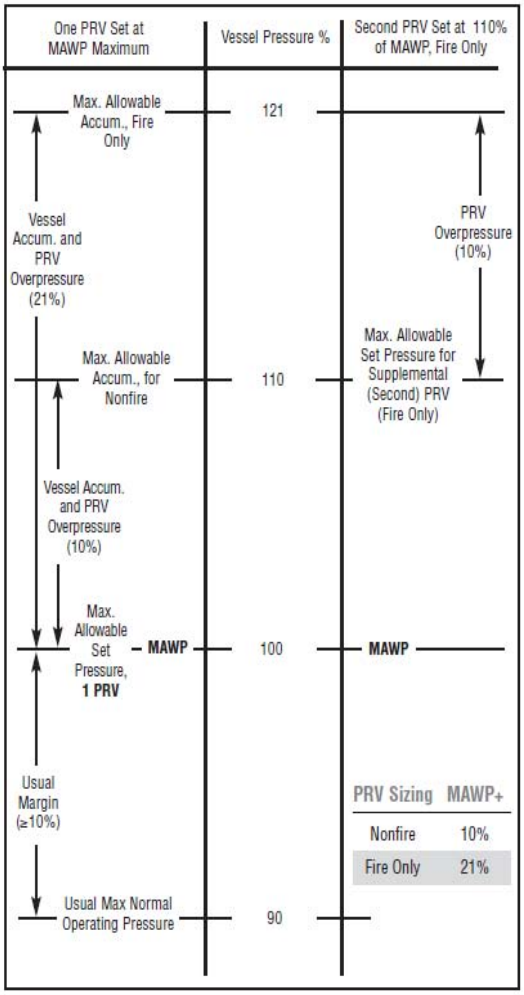
16.5 Pressure Relief Terms:

- **Back Pressure:** The pressure that exists at the outlet of a Pressure Relief Device as a result of the pressure in the discharge system. It can be either variable or constant and is the sum of the superimposed and built-up back pressure.
- **Superimposed Back Pressure:** The static pressure at the outlet of a pressure relief device at the time the device is required to operate. It is the result of pressure in the discharge system from other sources.
- **Built-Up Back Pressure:** The pressure existing at the outlet of a pressure relief device occasioned by the flow through the particular device into a discharge system, plus the effects of any other devices which may relieve simultaneously.
- **Coincident Temperature:** Normally used with rupture discs. The temperature of the flowing fluid after the disc has burst.
- **Gag:** Device which can hold a pressure relief device in the closed position. They are used to allow equipment pressure testing without removing or blinding the pressure relief valve.
- **Accumulation:** Pressure increase over the set pressure of a pressure relief valve, usually expressed as a % of the set pressure.
- **Design Pressure:** The pressure at which the vessel will normally operate at.
- **MAWP:** Maximum allowable working pressure. What the fabricated vessel is rated for.
    - Maximum vessel pressure with one working relief valve is 110% * MAWP
    - Maximum vessel pressure with two working relief valves is 116% * MAWP

Pressure relief devices for vessels within the scope of ASME Section VIII, Division 1, are used under the following conditions.

1. UG-125(c). All pressure vessels other than unfired steam boilers should be protected by a pressure-relieving device that prevents the pressure from rising more than 10% or 3 psi (whichever is greater) above the maximum allowable working pressure (MAWP), except as permitted in UG-125(c)(1) and (c)(2). See Figure 3-1.
2. UG-134(a). If a single valve is used, it must be set at a pressure not higher than the MAWP. When more than one valve is used to meet the required relieving capacity, then only one valve need be set at or below the MAWP. The additional valves can be set at higher pressures, but in no case can they be set greater than 105% of MAWP, except as permitted in UG-134(b). See Figure 3-2.
3. UG-125(c)(1). When multiple pressure-relieving devices are provided and set in accordance with UG-134(a), they must prevent the pressure from rising more than 16% or 4 psi (whichever is greater) above the MAWP. See Figure 3-2.
4. UG-125(c)(2). When an additional hazard can be created by exposure of a pressure vessel to fire or other unexpected sources of external heat, supplemental pressure relieving devices should be installed to protect against excessive pressure. Such supplemental pressure relieving devices shall be capable of preventing the pressure from rising more than 21% above the MAWP. The same pressure relieving devices can be used to satisfy the capacity requirements of (c) or (c)(1) and (c)(2), provided that the pressure setting requirements of UG-134(a) are met. See Figure 3-1.
5. UG-134(b). Protective devices permitted by UG-125(c)(2) (such as fire case) can be adjusted to operate at a pressure not greater than 110% of MAWP. However, if such a device is used to meet the requirements of both UG-125(c) and UG-125(c)(2), it should be set to operate at not more than MAWP.


Pressure Vessel Requirements
Vessel Pressure
Typical Characteristics of Pressure Relief Valves
% of maximum allowable working pressure (gauge)
120
115
110
105
100
95
90
85
Single valve
Maximum relieving pressure for process sizing
Accumulated Pressure (other than fire exposure)
(3E) Overpressure (maximum)
Maximum Allowable Working Pressure or design pressure)
(3H) Maximum Allowable Set Pressure for single valve
(3P) Simmer (typical)
Start to open
(3M) Blowdown (typical)
(3Q) Closing pressure
Maximum expected operating pressure)
(3Q) Leak test pressure (typical)

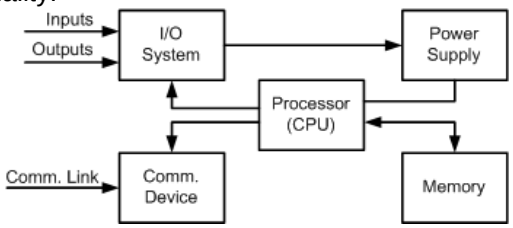

One PRV Set at MAWP Maximum
Vessel Pressure %
Second PRV Set at 110% of MAWP, Fire Only
Max. Allowable Accum., Fire Only
121
Vessel Accum. and PRV Overpressure (21%)
PRV Overpressure (10%)
Max. Allowable Accum., for Nonfire
110
Max. Allowable Set Pressure for Supplemental (Second) PRV (Fire Only)
Vessel Accum. and PRV Overpressure (10%)
Max. Allowable Set Pressure, 1 PRV
MAWP
100
MAWP
Usual Margin (≥10%)
PRV Sizing
MAWP+
Nonfire
10%
Fire Only
21%
Usual Max Normal Operating Pressure
90

17 Control System Analysis

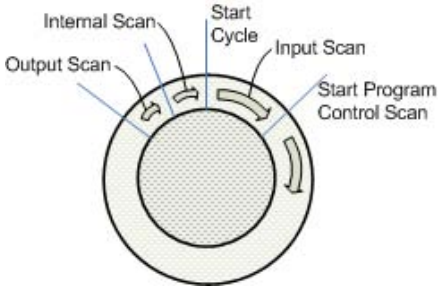
17.1 Control System Types:

17.1.1 Programmable Logic Controller (PLC):

Regardless of size, complexity or cost all PLCs share the same basic components and functionality:



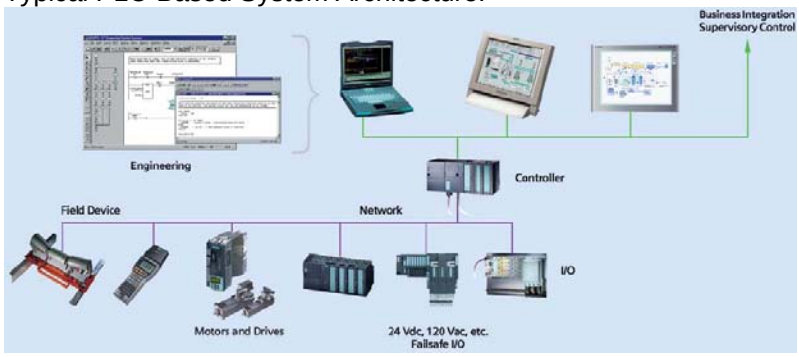
Processor (CPU): The processor consists of one or more microprocessors that perform the logic, control and memory functions of the PLC. The processor reads inputs, executes logic as determined by the application program, performs calculations and controls the outputs accordingly. The processor controls the operating cycle (scan). The operating cycle consists of a series of operations performed sequentially and repeatedly.



Input Scan: The PLC examines the external input devices for an on or off state. The status of these inputs is temporarily stored in an input image table or memory file.
Program Scan: Process scans the instructions in the control program, uses the input status from the input image file and determines if an output will or will not be energized. The resulting status is written to the output image table or memory file.
Output Scan: Based on the data in the output image table, the PLC energizes or de-energizes the associated output circuits.
Internal Scan: Processor performs housekeeping functions such as internal checks on memory, speed and operation and service any communication requests.

This operating cycle typically takes 1 – 25 milliseconds. However, the operating cycle depends upon the complexity of the control logic written by the user, a large and complex program may take as high as 250 milliseconds. These are continually repeated in a looped process.

Typical PLC-Based System Architecture:

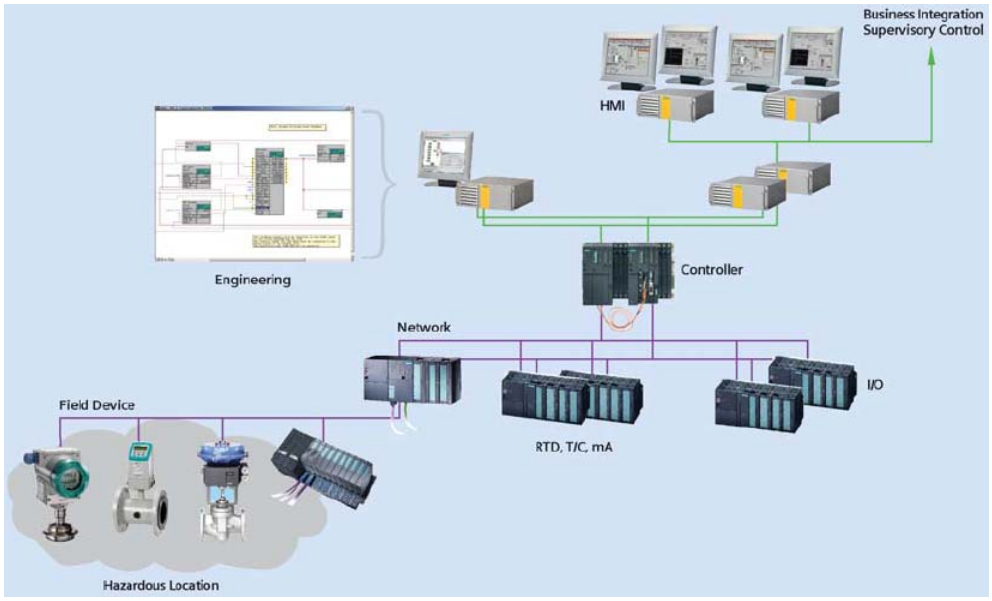


17.1.2 Distributed Control System (DCS):

The DCS is a control system which collects the data from the field and decides what to do with them. Data from the field can either be stored for future reference, used for simple process control, use in conjunction with data from another part of the plant for advanced control strategies. What must be in the DCS for it to be able to do so much?
Operator Console: These are like the monitors of our computers. They provide us with the feedback of what they are doing in the plant as well as the command we issue to the control system. These are also the places where operators issue commands to the field instruments.
Engineering Station: These are stations for engineers to configure the system and also to implement control algorithms.
History Module: This is like the harddisk of our PCs. They store the configurations of the DCS as well as the configurations of all the points in the plant. They also store the graphic files that are shown in the console and in most systems these days they are able to store some plant operating data.
Data Historian: These are usually extra pieces of software that are dedicated to store process variables, set points and output values. They are usually of higher scanning rates than that available in the history module.
Control Modules: These are like the brains of the DCS. Specially customized blocks are found here. These are customized to do control functions like PID control, ratio control, simple arithmetic and dynamic compensation. These days, advanced control features can also be found in them.
I/O: These manage the input and output of the DCS. Input and output can be digital or analogues. Digital I/Os are those like on/off, start/stop signals. Most of the process measurements and controller outputs are considered analogue. These are the points where the field instruments are hard-wired to.

All above mentioned elements are connected by using a network, nowadays very often used is Ethernet.

Unlike PLC/HMI solutions, the DCS features a single database that coordinates all configuration activities. System configuration is globally distributed in the run-time environment



Typical DCS-Based System Architecture:

17.1.3 Supervisory Control & Data Acquisition (SCADA):

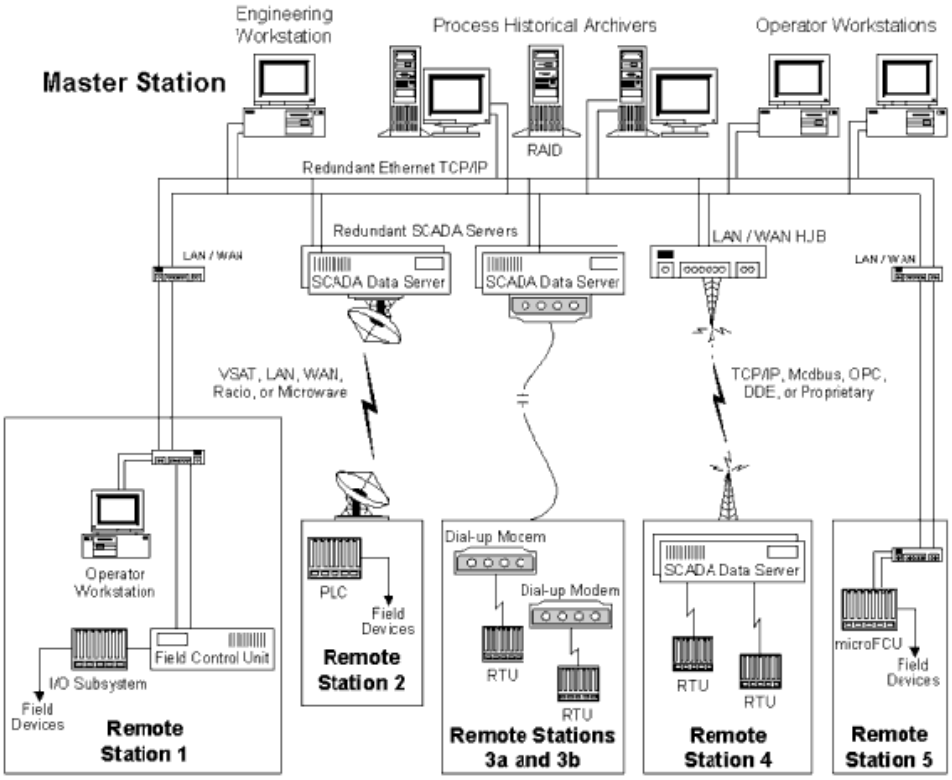
SCADA is not a specific technology, but rather a type of application. These systems encompass the transfer of data between a SCADA central host computer and a number of Remote Terminal Units (RTUs) and / or PLCs, and the central host and the operator terminals. SCADA system gathers information, transfers the information back to a central site, then updates the remote station.

A SCADA system performs four functions:

1. Data acquisition
2. Networked data communication
3. Data presentation
4. Control

These functions are performed by four kinds of SCADA components:

1. Sensors that directly interface with the managed system
2. Remote Terminal Units – These are small computerized units deployed in the field to serve as location collection points for information and delivering commands to outputs.
3. SCADA master unit – These are large computer consoles that serve as the central processor for the SCADA system. Master units provide a human interface to the system and automatically regulate the managed system in response to sensor inputs.
4. The communication network that connects the SCADA master unit to the RTUs in the field.



17.1.4 DCS vs PLC:

Seven questions to answer before choosing whether PLC or DCS control system:

1. What are you manufacturing and how?

PLC	DCS
❑ Manufacturing or assembly of specific items (aka "Things")	❑ Involves the combination and/or transformation of raw materials (aka "Stuff")
❑ Product is visible as it moves through the process	❑ Often impossible to visually see the product as it moves through the process
❑ High-speed logic control (such as motors)	❑ Regulatory/Analog (loop) control
❑ Simple Batch control	❑ Complex Batch Control

2. What is the value of the product being manufactured and the cost of downtime?

PLC	DCS
❑ Value of the individual component being manufactured is relatively low	❑ The value of a "batch" can be very high (either in raw material cost or market value)
❑ Downtime mainly results in lost production	❑ Downtime not only results in lost production, but can result in dangerous conditions
❑ Downtime does not typically damage the process equipment	❑ Downtime can result in process equipment damage (product hardens, etc.)
❑ Return to steady state production after an outage is short and relatively straightforward	❑ Return to steady state production after an unplanned outage can be long, expensive, and difficult

3. What do you view as the heart of the system?

PLC	DCS
❑ Typically, the heart of the system is the controller	❑ Typically, the heart of the system is the HMI

4. What does the operator need to be successful?

PLC	DCS
❑ The operator's primary role is to handle exceptions	❑ The operator's interaction is typically required to keep the process in its target performance range
❑ Status information (On/Off, Run/Stop) is critical information for the operator	❑ Faceplates and analog trends are critical to "see" what is happening to the process
❑ Exception-based alarming is key information for the operator	❑ Alarm management is key to safe operation of the process and for responding effectively during plant upset conditions
❑ Manufacturing might be able to run "lights-out"	❑ Failure of the HMI could force the shutdown of the process

5. What system performance is required?

PLC	DCS
❑ Fast logic scan (approx. 10ms) is required to perform motor or motion control	❑ Control loops require deterministic scan execution at a speed of 100 to 500 ms
❑ Redundancy may not be cost justified	❑ System redundancy is often required
❑ System can be taken offline to make configuration changes	❑ Online configuration changes often required
❑ Analog Control: Simple PID only	❑ Analog Control: Simple to advanced PID control up to Advanced Process Control
❑ Diagnostics to tell you when something is broken	❑ Asset Management alerts you to what might break before it does

6. What degree of customization is required?

PLC	DCS
❑ High level programming languages are available for creating custom logic	❑ Custom logic created from existing function blocks
❑ Customized routines usually required	❑ Many algorithms (i.e. PID) are complex and do not vary among applications
❑ Standard libraries considered nice features	❑ Standard application libraries are expected (function blocks and faceplates)
❑ Provisions must be available to integrate functions/ products into an integrated architecture	❑ Entire system is expected to function as a complete solution

7. What are your engineering expectations?

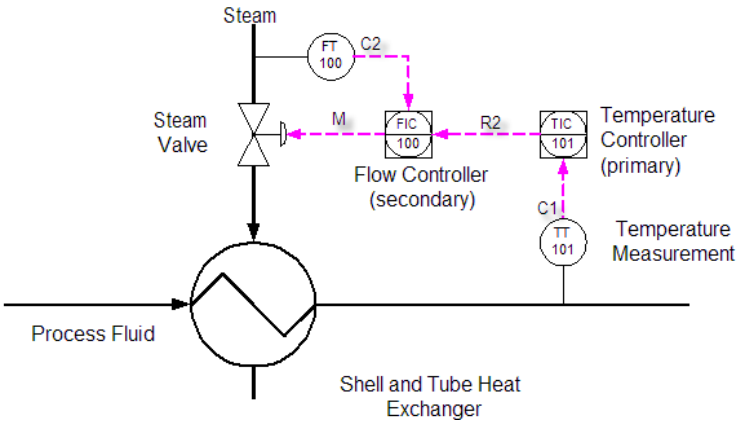
PLC	DCS
❑ Program/configure individual components, integrate later (bottom-up)	❑ Up-front design of complete system before implementation begins (top-down)
❑ Desire customizable platforms to build upon	❑ Looking for significant "out-of-the-box" functionality
❑ System designed to be flexible	❑ System designed to make it "easy" to engineer process applications
❑ Solution is generic in nature, to be applied on a wide variety of applications	❑ Use of pre-defined, pre-tested functions saves time
❑ Use ladder logic to configure application	❑ Use function block diagram to configuration application

17.2 Controller Actions:

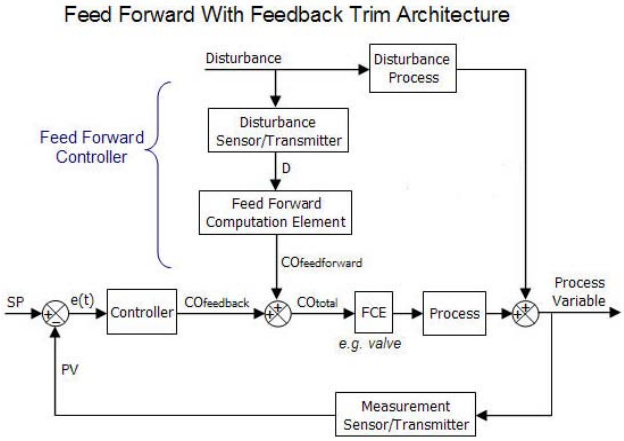
- **Direct Acting:** Increase in PV causes an increase in controller output.
- **Reverse Acting:** Increase in PV causes a decrease in controller output.
- **Process Action:** This is not to be confused with controller action. Defines the relationship between changes in the valve and changes in the measurement.
  - Increase in valve position causes an increase in the measurement. (Reverse Output = NO)
  - Increase in valve position causes a decrease in the measurement. (Reverse Output = YES)

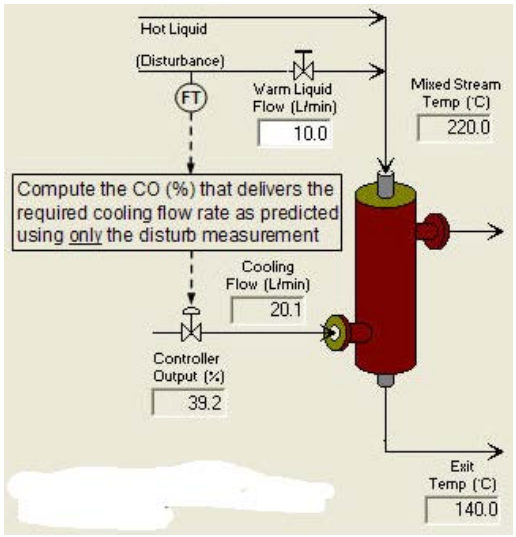
  Note: 0% always means closed to the operator, and 100% always means open to the operator.
- **PID Control:** Reference Section 18 (Loop Tuning)
- **Cascade Control:** Two controllers are used but only one process variable "M" is manipulated. The primary controller (master) maintains the primary variable "C1" at its setpoint by adjusting the setpoint "R2" of the secondary controller (slave). The secondary controller in turn, responds both to the output of the primary controller and to the secondary controlled variable "C2". The inner loop is tuned 1st, then the outer loop is tuned. There are two distinct advantages gained with cascade control:
  - Disturbances affecting the secondary variable can be corrected by the secondary controller before a pronounced influence is felt by the primary controller.
  - Closing the control loop around the secondary part of the process reduces the phase lag seen by the primary controller, resulting in increased speed of response.

  Requirements for cascade control:
  - Secondary loop process dynamics must be at least four times as fast as primary loop process dynamics.
  - Secondary loop must have influence over the primary loop.



- **Feed Forward:** The traditional PID controller takes action only when the PV has been moved from set point, SP, to produce a controller error, e(t) = SP – PV. Thus, disruption to stable operation is already in progress before a feedback controller first begins to respond. From this view, a feedback strategy simply starts too late and at best can only work to minimize the upset as events unfold. In contrast, a feed forward controller measures the disturbance, D, while it is still distant. As shown on the next page, a feed forward element receives the measured D, uses it to predict an impact on PV, and then computes preemptive control actions, CO feedforward, that counteract the predicted impact as the disturbance arrives. The goal is to maintain the process variable at set point (PV = SP) throughout the disturbance event.

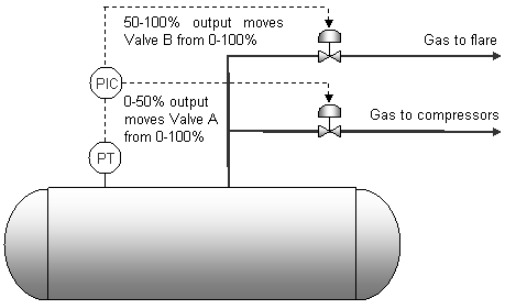




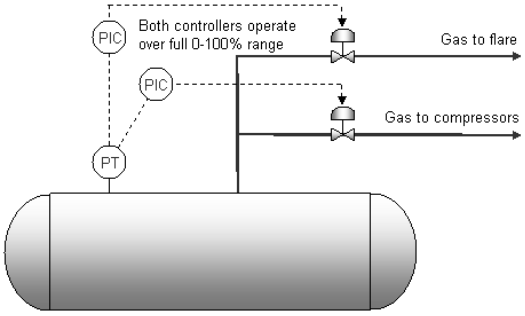
- **Split Range:** Split range control in which the output of a controller is split to two or more control valves.
  - Controller output 0% — Valve A is fully open and Valve B fully closed.
  - Controller output 25% — Valve A is 75% open and Valve B 25% open.
  - Controller output 50% — Both valves are 50% open.
  - Controller output 75% — Valve A is 25% open and Valve B 75% open.
  - Controller output 100% — Valve A is fully closed and Valve B fully open.

  OR

  - Controller output 0% — Both valves are closed.
  - Controller output 25% — Valve A is 50% open and Valve B still closed.
  - Controller output 50% — Valve A is fully open and Valve B closed.
  - Controller output 75% — Valve A is fully open and Valve B 50% open.
  - Controller output 100% — Both valves are fully open.



OR



- **Ratio:** Ratio control is used to ensure that two or more flows are kept at the same ratio even if the flows are changing.

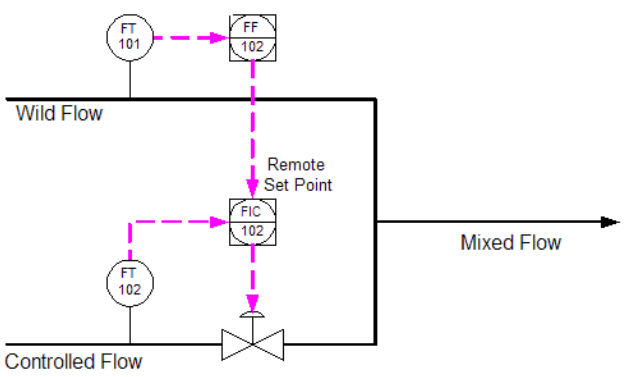
  Applications of ratio control:
  - Blending two or more flows to produce a mixture with specified composition.
  - Blending two or more flows to produce a mixture with specified physical properties.
  - Maintaining correct air and fuel mixture to combustion.

  The controlled flow is increased and decreased to keep it at the correct ratio with the wild flow.

  The "wild flow" is the flow not controlled by this loop. It may be controlled by some other control loop.

  The "controlled flow" is controlled by this loop with a setpoint equal to the measured wild flow multiplied by some value (FF-102).

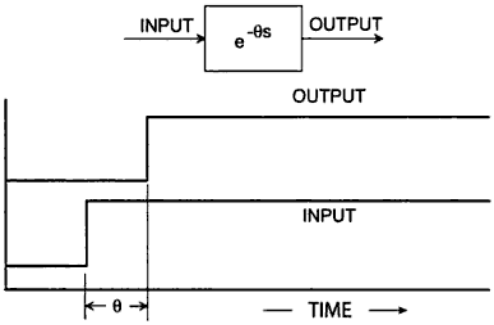
  The measured wild flow is multiplied by a value that may be fixed or may be adjustable by the operator. The result of the multiplication becomes the setpoint of the controlled flow controller.



- **Dynamic Response:**

  Dead Time: Usually associated with the physical movement of mass or energy. An example would be a well insulated flowing pipeline where the temperature is measured at two points separated by a considerable distance. The temperatures recorded from the two measuring points would be identical except they would be separated by the time required for the fluid to move from the upstream to the downstream point of measurement. This process can be described by a single parameter model that represents the dead time.

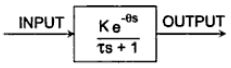
  $\frac{Ke^{-\theta s}}{\tau s+1}$ K = process gain; θ = dead time; τ = time constant

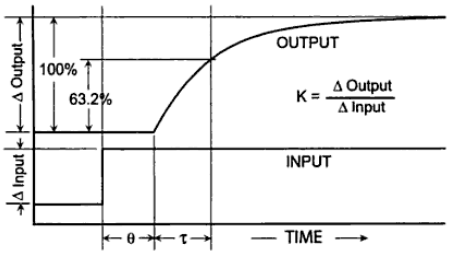


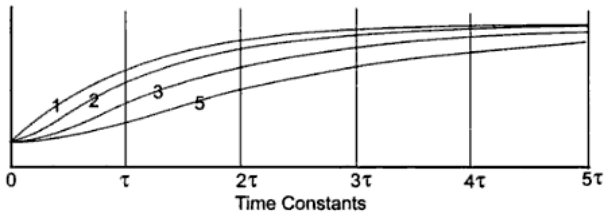
First Order Lag: A dynamic system will come to equilibrium in five time constants. The system will reach 63.2% of equilibrium in one time constant, 63.2% of the remaining amount in one more time constant, and so on.

Time since Step Input Change	Percentage of Steady-State Change
1 Time Constant	63.2%
2 Time Constants	86.5%
3 Time Constants	95.0%
4 Time Constants	98.2%
5 Time Constants	99.6%

$$\tau\frac{d}{dt}c(t)+c(t)=K\,r(t)$$





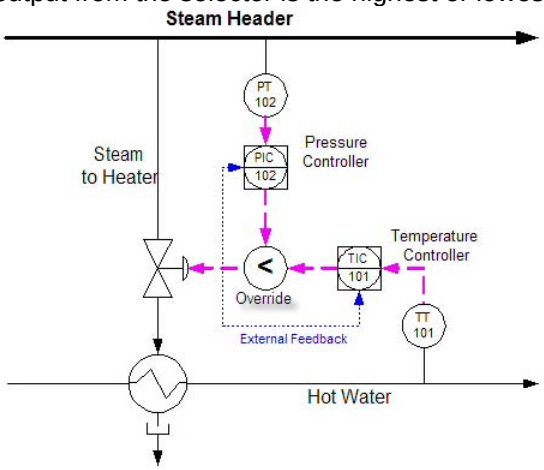


FOPDT = First Order Plus Dead Time

- **Time Constant:** In general terms, the time constant, τ, describes how fast the PV moves in response to a change in the output The time constant must be positive and it must have units of time. For controllers used on processes comprised of gases, liquids, powders, slurries and melts, τ most often has units of minutes or seconds. compute τ in five steps:
    1. Determine ΔPV, the total change that is going to occur in PV, computed as “final minus initial steady state”.
    2. Compute the value of the PV that is 63% of the total change that is going to occur, or “initial steady state PV + 0.63(ΔPV)”.
    3. Note the time when the PV passes through the 63% point of “initial steady state PV + 0.63(ΔPV)”.
    4. Subtract from it the time when the “PV starts a clear response” to the step change in the output.
    5. The passage of time from step 4 minus step 3 is the process time constant, τ.

  Summarizing in one sentence, for step test data, τ is the time that passes from when the PV shows its first response to the output step, until when the PV reaches 63% of the total ΔPV change that is going to occur.

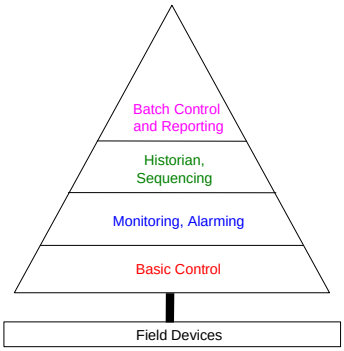
- **Override:** Override control is used to take control of an output from one loop to allow a more important loop to manipulate the output. (similar to high / low select)
  The output from two or more controllers are combined in a high or low selector. The output from the selector is the highest or lowest individual controller output.



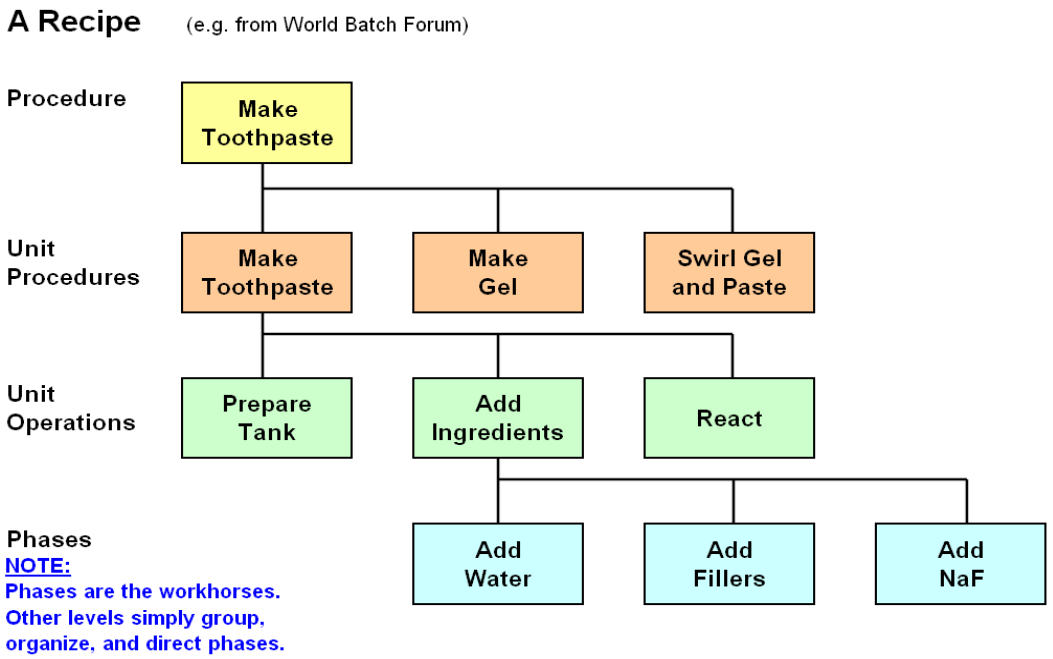
17.3 S88 Batch Control:

S88 defines hierarchical recipe management and process segmentation frameworks, which separates products from processes that make them. The standard enables reuse and flexibility of equipment and software, and provides a structure for coordinating and integrating recipe-related information across traditional ERP, MES and control domains.

17.3.1 Automation Pyramid:

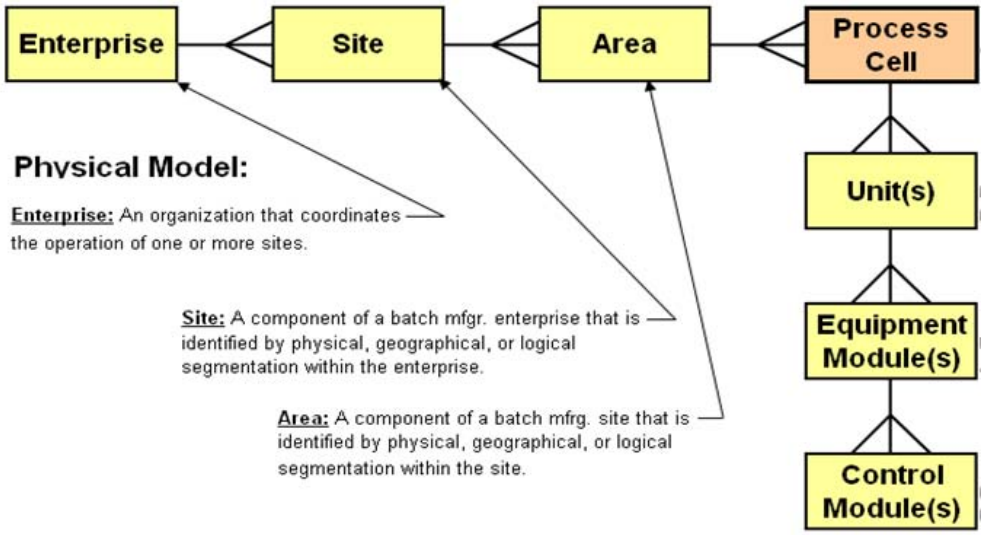


17.3.2 Procedural Model:



17.3.3 Process Cell Level:

- Contains all of the equipment, including units, required to make batches
- A process cell may be processing more than one batch at a time. However, Units only work on a single batch at a time.



17.3.4 Unit:

- Batching cannot occur without units, batching occurs *in* units.
- A Unit runs a recipe to (examples):
  - Combine ingredients
  - Perform a reaction

17.3.5 Equipment & Control Modules:

- Equipment Modules group devices for performing on or more specific processing activities (examples)
  - Clean or Dirty Timer Calculations
  - Message
  - Patch Panel
  - Temperature Control Module (TCM)
- Control Modules connect software to the process through field devices (e.g. actuators, sensors) with all the elements treated as a single entity (examples)
  - Pressure, Weight, Temperature, etc. Monitor
  - E-Stop
  - On-Off valve
- Equipment Modules run portions of a recipe; Control Modules do not.

17.3.6 Phases:

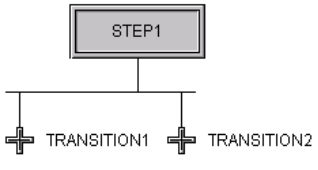
A phase is a series of steps (SFC) that cause one or more equipment- or process-oriented actions, for example, filling a tank or agitating the contents. The phase logic defines the states of the phase (running, holding, restarting, aborting, and stopping) and the logic associated with each state.

Phase Commands:

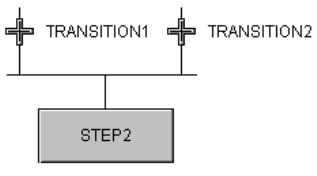
COMMAND	DESCRIPTION
Abort	Invokes the aborting logic. Abort is an emergency stop.
Hold	Used for a phase in the running or restarting state. Hold invokes the holding logic to temporarily stop the phase and proceed to the held state, from which the Restart command can be used.
Reset	Transitions the phase from an aborted, complete, or stopped state to the idle state.
Restart	Used for a phase in the held state to invoke the restarting logic.
Start	From idle, the Start command invokes the running logic.
Stop	Invokes the stopping logic.

17.3.7 Sequential Function Chart:

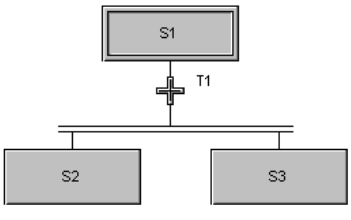
Series of steps and transitions. Steps are represented by boxes and transitions by vertical lines with crosses attached. Each step contains a set of actions that affect the process. At any given time, one or more of the steps and transitions can be active. Each time the SFC scans, the active steps and transitions are evaluated. When a transition evaluates as TRUE (for example, the transition condition is met), the steps prior to the transition are made inactive and the step(s) following the transition become active. This way, the SFC can sequence through the various control states defined by the module's diagram



Example of a Sequence Select Divergence



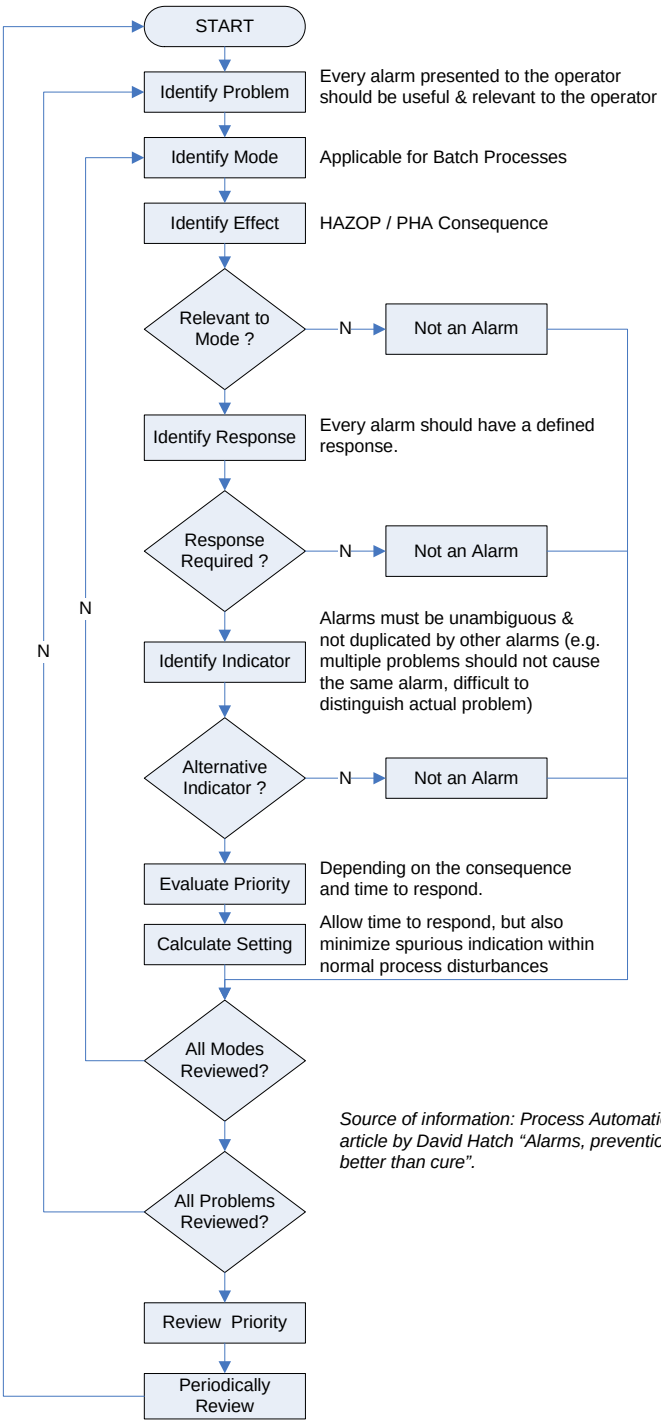
Example of a Sequence Select Convergence



Example of a Parallel Divergence

17.4 Alarm Management:

Good Guideline for Alarm Mgmt.:
EEMUA 191 suggests that <1% of alarms be critical, 5% high priority, 15% medium and the remaining 80% low priority. (EEMUA = Engineering Equipment and Materials User Association)



Source of information: Process Automation article by David Hatch "Alarms, prevention is better than cure".

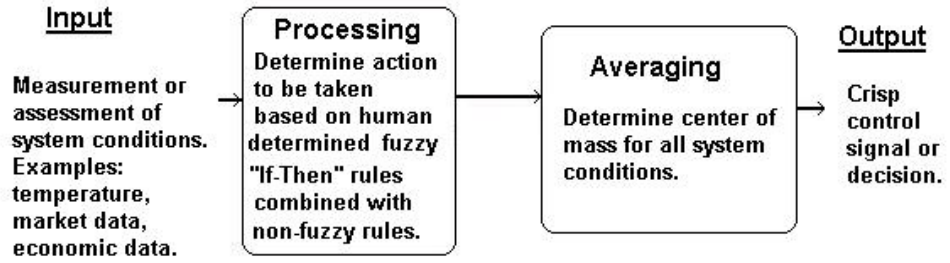
Characteristics of Good Alarm(s) Systems:

Characteristic	Reasoning
Relevant	Not spurious or of low operational value
Unique	Not duplicating another alarm
Timely	Not long before a response is needed or too late to do anything
Prioritized	Indicating the importance (criticality) that the operator respond / correct the problem
Understandable	Having a message that is clear and understandable
Diagnostic	Identifies the problem that has occurred
Advisory	Indicates the action that should be taken (corrective response)
Focusing	Draws attention to the most important issues

17.5 Fuzzy Logic:

Control system based on fuzzy logic - a mathematical system that analyzes analog input values in terms of logical variables that take on continuous values between 0 and 1, in contrast to classical or digital logic, which operates on discrete values of either 0 and 1 (true and false). Logic involved can deal with fuzzy concepts - concepts that cannot be expressed as "true" or "false" but rather as "partially true".

As the complexity of a system increases, it becomes more difficult and eventually impossible to make a precise statement about its behavior, eventually arriving at a point of complexity where the fuzzy logic method born in humans is the only way to get at the problem.



The Fuzzy Logic Control-Analysis Method

To Create Fuzzy Logic:

1. Determine the inputs.
2. Describe the cause and effect action of the system with "fuzzy rules" stated in plain English words.
3. Write a computer program to act on the inputs and determine the output, considering each input separately. The rules become "If-Then" statements in the program. (As will be seen below, where feedback loop control is involved, use of graphical triangles can help visualize and compute this input-output action.)
4. In the program, use a weighted average to merge the various actions called for by the individual inputs into one crisp output acting on the controlled system. (In the event there is only one output, then merging is not necessary, only scaling the output as needed.)

17.6 Model Predictive Control:

Model predictive control uses a mathematical model to simulate a process. This model then fits the inputs to predict the system behavior. In this way, MPC is a type of feed forward control. It uses system inputs as a basis of control. MPC is more complex than most other feed forward control types because of the way these predictions are used to optimize a process over a defined amount of time. Most feed forward control types do not take into account the process outputs much past a residence time. The MPC algorithm will compare predicted outputs to desired outputs and select signals that will minimize this difference over the time selected. This control type can see ahead into multiple time steps in the future in order to optimize the process. Normal PID type controllers use mathematical expressions based on error from a set point. The governing equation for MPC controllers are based on set points, system properties, and desired outcomes and optimization.

MPC is very specific to the process it is modeling. Unlike ratio or cascade control set ups, where it is simple to implement and change set points in various situations, MPC will model one specific process and optimize it. As previously mentioned this can either be an advantage or disadvantage. MPC is great for selecting one type of operation on one system and perfecting it to the desired conditions. This also has a downside in that the model equation will work for one and only one situation

Advantages of MPC

1. MPC can be used to handle multivariable control programs.
2. MPC can consider actuator limitations.
3. MPC can increase profits by allowing for operation close to the system constraints.
4. MPC can perform online computations quickly.
5. MPC can be used for non-minimal phase and unstable processes.
6. MPC is easy to tune.
7. MPC is able to handle structural changes.

Disadvantages of MPC

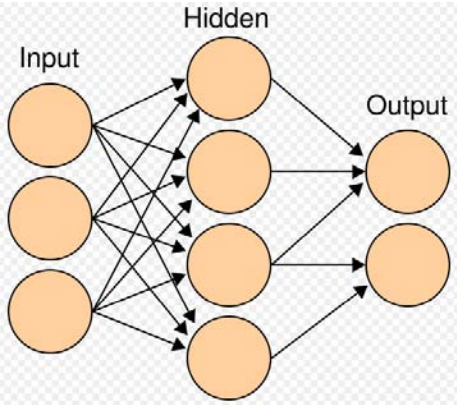
1. Several MPC models are limited to only stable, open-loop processes.
2. MPC often requires a large number of model coefficients to describe a response.
3. Some MPC models are formulated for output disturbances, and they may not handle input disturbances well.
4. Some forms of MPC use a constant output disturbance assumption. This corrects for the fact that the output predicted by the model is not exactly equal to the actual measured output. This method assumes the correction term is constant in the future, which may not yield a good performance if there is a real disturbance at the plant input.
5. If the prediction horizon is not formulated correctly, control performance will be poor even if the model is correct.
6. Some systems have a wide range of operating conditions that change frequently. Some examples of this include exothermic reactors, batch processes, and any systems where different consumers have different product specifications. An MPC linear model will not be able to handle the dynamic behavior of these processes. A nonlinear model must be used for better control performance.

17.7 Artificial Neural Networks (ANN)

Neural networks are computational models that share some properties of the brain. These networks consist of many simple units working in parallel with no central control, and learning takes place by modifying the weights between connections. The basic components of an ANN are:

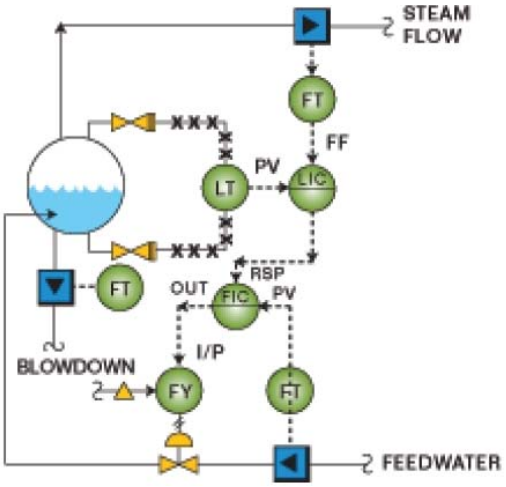
- Neurons
  - Input Neurons
  - Hidden Neurons
  - Output Neurons
- Weights
- Learning Rules

Inputs are provided to the input neurons, such as equipment parameters, and outputs are provided by the output neurons. These outputs may be a measurement of the performance of the process. The network is trained by establishing the weighted connections between input neurons and output neurons via the hidden neurons. Weights are continuously modified until the ANN is able to predict the outputs from a given set of inputs within an acceptable user defined error level. Once the network is sufficiently trained, a general model is created for the relationship between inputs and outputs. The user can then determine the impact that a specific set of process inputs has on a set of process outputs. After sufficient training, the ANN model can be used to monitor process parameters to determine, with the current parameters, if the process is moving towards an out of control situation.



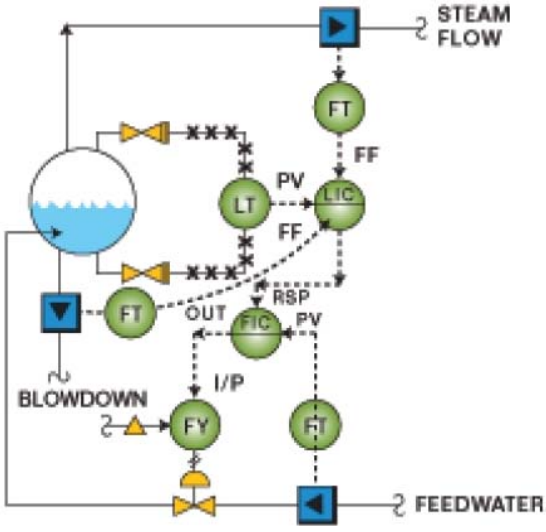
17.8 Example Boiler Control:

Three Element Control: When steam vapor is generated in the risers of the boiler, some of the volume in the tubes is displaced by steam bubbles. These steam bubbles have a smaller density than the boiler feed water. The density of these steam bubbles is also a function of temperature and pressure. As more and more of these steam bubbles are generated, the level indicator does not give a true indication of the inventory of boiler feed water in the system. As steam demand is increased, this effect becomes increasingly more pronounced, and the level in the steam drum rises rather than decreases, as one would expect. This phenomenon is known as "Shrink and Swell". Under these circumstances, a simple level controller would cut back rather than increase the amount of boiler feed water to the steam drum. Consequently, a more sophisticated control scheme is required. This control scheme is known as Three-Element Boiler Feed Water Control. With this type of control, steam rate is summed with the output of the level controller. This signal then serves as the RSP for the boiler feed water flow controller. The level controller is tuned proportional only, and the boiler feed water controller is tuned proportional-integral. With this type of control, when steam demand increases, boiler feed water increases as well.



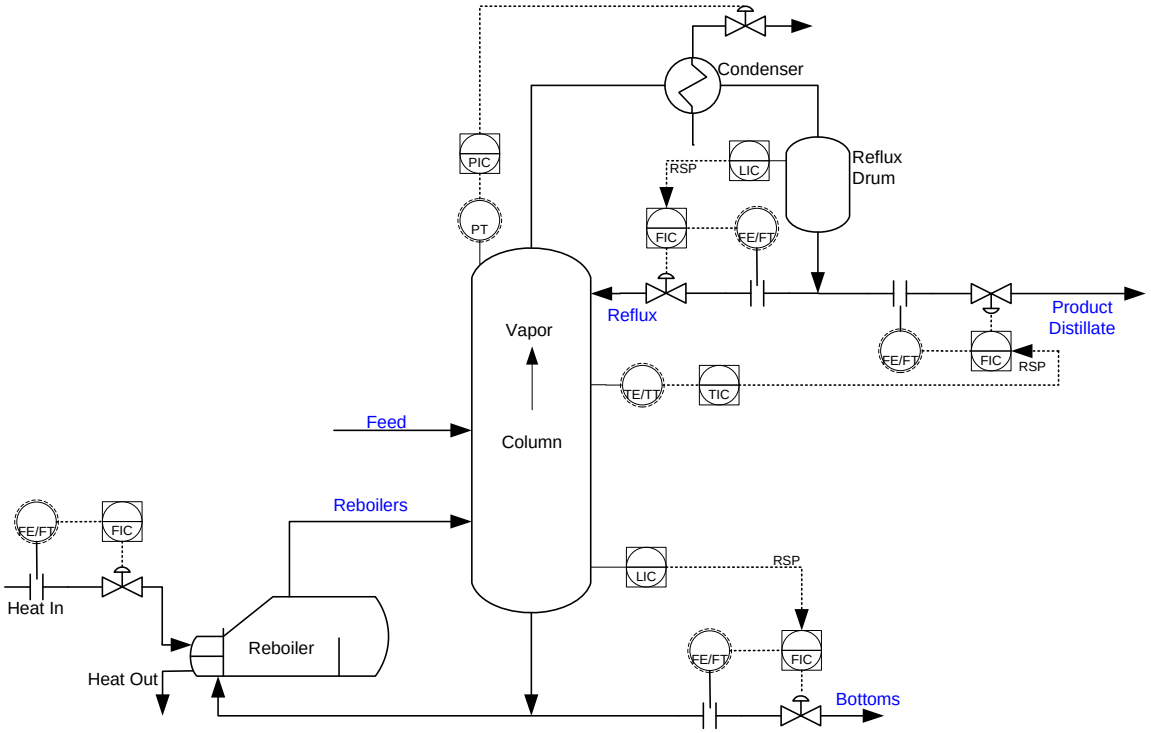
Four Element Control: Same as three element control except that blowdown is also taken into consideration. Reason for blowdown: Dissolved solids and particles entering a boiler through the make-up water will remain behind when steam is generated. During operation the concentration of solids builds up and finally a concentration level is reached where operation of the boiler becomes impossible. If solids are not purged from the boiler they can lead to

- Scale formation
- Carryover
- Corrosion
- Embrittlement



17.9 Example Distillation Column Control:

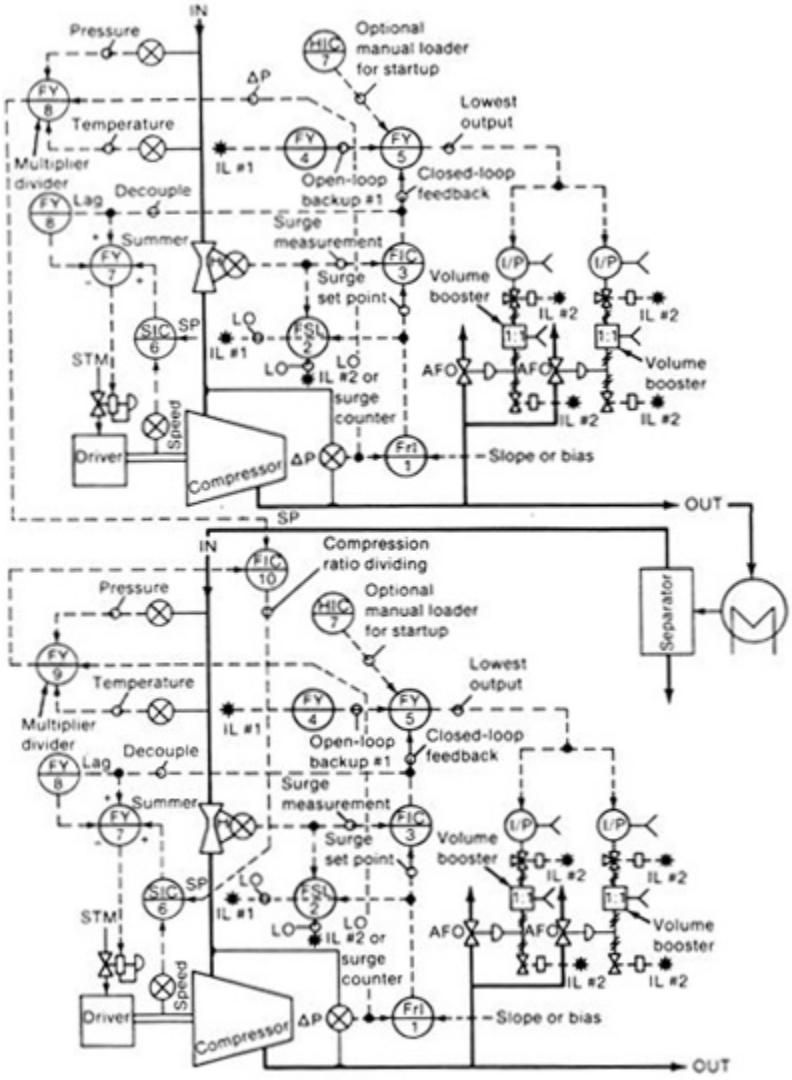
- The liquid mixture that is to be processed is known as the **feed** and this is introduced usually somewhere near the middle of the column to a tray known as the feed tray. The feed tray divides the column into a top (enriching or rectification) section and a bottom (stripping) section. The feed flows down the column where it is collected at the bottom in the reboiler.
- Heat is supplied to the reboiler to generate vapor. The source of heat input can be any suitable fluid, although in most chemical plants this is normally steam. In refineries, the heating source may be the output streams of other columns. The vapor raised in the reboiler is re-introduced into the unit at the bottom of the column. The liquid removed from the reboiler is known as the **bottoms** product or simply, bottoms (low boilers).
- The vapor moves up the column, and as it exits the top of the unit, it is cooled by a condenser. The condensed liquid is stored in a holding vessel known as the reflux drum. Some of this liquid is recycled back to the top of the column and this is called the **reflux**. The condensed liquid that is removed from the system is known as the **distillate** or top product.



17.10 Example Compressor Control:

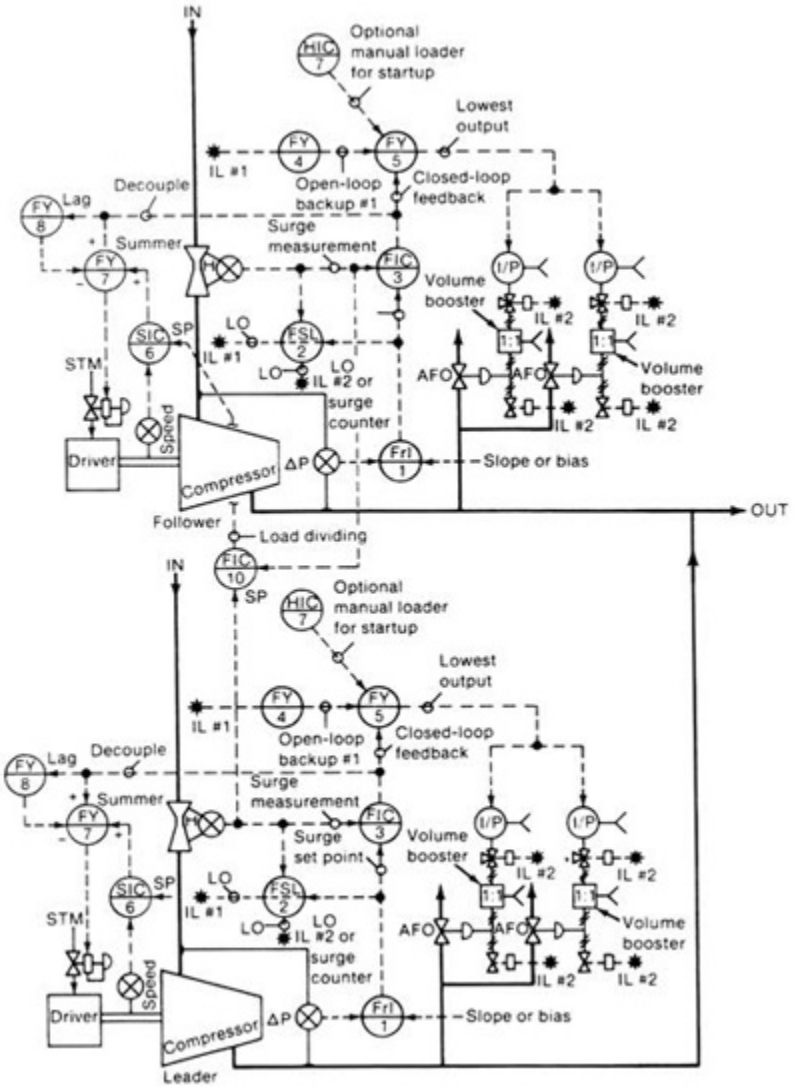
Series Operation Compressors:

Compressor stages (same driver shaft) or compressors (separate driver shafts) are installed in series to increase compression ratio. The control systems shown in the literature use a single surge control valve on the end of the train with a low-signal selector (air-fail-open surge valve) to choose the output of the surge controller on the stage or compressor that is closest to surge. Separate surge control valves are not shown in order to avoid the interaction that might result. The opening of an intermediate surge valve would increase the flow to an upstream stage or compressor but would decrease the flow to a downstream stage or compressor. However, field experience shows that the interaction is minimal. The use of a single surge valve on the end of the train delays the correction of flow for the upstream stages or compressors, especially if there are intercoolers or separators between the stages or compressors. It is best to duplicate the surge control system for each stage or compressor in series. If the throughput flow is not equal for each compressor in series (caused by separators and recycle streams), then compression ratio dividing would help prevent unnecessary opening of the surge valve



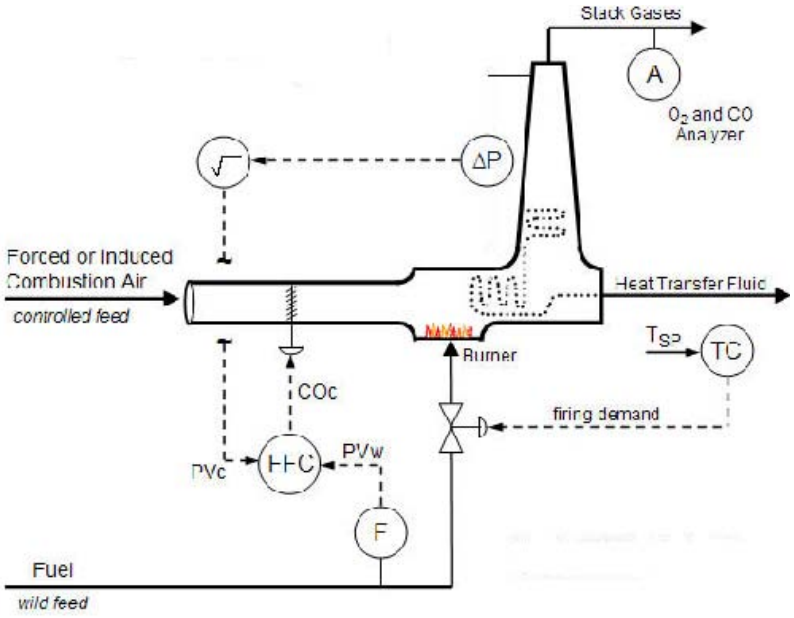
Parallel Operation Compressors:

Compressors are installed in parallel to increase the flow capacity. Most parallel compressor installations to date use a graduated set point method of distributing load among the compressors. The compressor with the lowest set point is unloaded first, while the compressor with the highest set point is unloaded last. This method of distribution is inefficient for the following reasons: (1) the lower set point compressors may be operating with surge control valves partially or fully open for load reduction; and (2) the operating pressure will be higher since the incremented set points are typically discharge pressure. The incremented set points may also upset the downstream users. A better control strategy recognizes how far each compressor operating point is from the surge curve and simultaneously loads and unloads all compressors proportionally. The leader compressor is either the largest compressor or the compressor whose operating point is closest to the surge curve for a load increase or furthest from the surge curve for a load decrease. If the bias divided by the slope of the surge set point curve is chosen to be equal for all compressors, then the ratio controller set point of each follower compressor is simply the flow of the leader compressor multiplied by the slope of the follower surge set point divided by the slope of the leader surge set point.



17.11 Example Burner Combustion Control:

A ratio control strategy can play a fundamental role in the safe and profitable operation of fired heaters, boilers, furnaces and similar fuel burning processes. This is because the air-to-fuel ratio in the combustion zone of these processes directly impacts fuel combustion efficiency and environmental emissions.
A requirement for ratio control implementation is that both the fuel feed rate and combustion air feed rate are measured and available as process variable (PV) signals



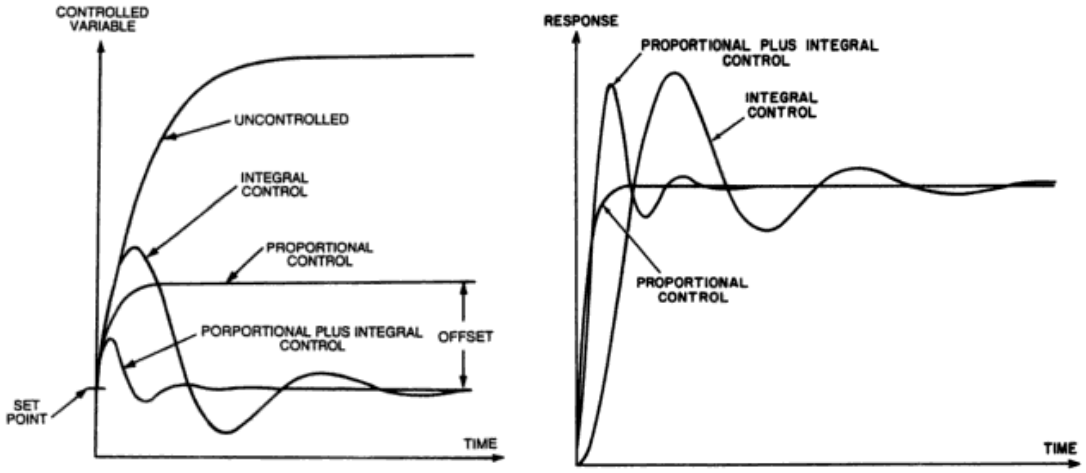
Too-small air/fuel ratio leads to incomplete combustion of our fuel. As the availability of oxygen decreases, noxious exhaust gases including carbon monoxide will form first. As the air/fuel ratio decreases further, partially burned and unburned fuel can appear in the exhaust stack, often revealing itself as smoke and soot. Carbon monoxide, partially burned and unburned fuel are all poisons whose release is regulated by the government. Incomplete combustion also means that we are wasting expensive fuel. Fuel that does not burn to provide useful heat energy, including carbon monoxide that could yield energy as it converts to carbon dioxide, literally flows up our exhaust stack as lost profit.

As the *air/fuel ratio increases above* that needed for complete combustion, the extra nitrogen and unneeded oxygen absorb heat energy, decreasing the temperature of the flame and gases in the combustion zone. As the operating temperature drops, we are less able to extract useful heat energy for our intended application.
So when the air/fuel ratio is too high, we produce a surplus of hot air. And this hot air simply carries its heat energy up and out the exhaust stack as lost profit.

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18 Loop Tuning

A proportional controller (Kp) will have the effect of reducing the rise time and will reduce, but never eliminate, the steady-state error. An integral control (Ki) will have the effect of eliminating the steady-state error, but it may make the transient response worse. A derivative control (Kd) will have the effect of increasing the stability of the system, reducing the overshoot, and improving the transient response.



18.1 Description of PID Units:

- **Proportional (Gain):** The amount added to the output based on the current error. The output moves in proportion to the error between SP and PV. The higher the gain the greater the rejection of disturbance and the greater the response to SP changes *However, any loop will cycle if the gain is increased far enough.*
  - Decreasing the dead time increases the maximum gain and the controllability.
  - Decreasing the ratio of the longest to the second longest lag also increases the controllability

  $m = K_c e + b$ m = output signal Kc = controller gain e = error

  b = live zero or bias of the output (e.g. 4mA or 3psig)

  - Proportional Gain is a multiplier (e.g. if the error is 10 and the Gain is 0.8, then the output will move 8%
  - Proportional Band: is a divider as a % (e.g. if the error is 10 and the band is 125%, then the output is (10*(100/125)) = 8%
  - Conversion between P-Gain & P-Band:

    $$P-Band = \frac{100}{P-Gain}$$

  - Reset Windup describes several situations in which the reset element of the controller continues to increase (or decrease) the output of the controller even when the change in output does not cause any change in the process measurement (controlled variable). With no resulting decrease in error, the output will continue to increase until it reaches its limit. The problem with reset windup is that, when the condition causing the windup is eliminated, the output must “wind down” for a period of time before the decreasing output has any effect on the process (more common in pneumatic controllers as electronic controllers have anti-reset windup feature built-in).

- **Integral (Reset):** The amount added to the output based on the sum of the error. The integral time is the time to repeat the change that was due to proportional action alone. Integral action causes the output to change at a rate proportional to the error. *Any loop will cycle if you reduce the integral time far enough.*

  $m = \frac{1}{T_i}\int e\ dt$ Ti = Integral time in repeats per minutes

  $P + I = m = K_c\left[e + \frac{1}{T_1}\int e\,dt\right] + b$

  - Time Constant is the time for one full repeat of the Proportional Term (e.g. if the P-Term is 8% and the time constant is 10 seconds, then the output will ramp up 8% every 10 seconds)
  - Reset Rate: is amount the output will move in one second (e.g. if the p-term is 8% and the reset rate is 0.1 repeats/sec then the output will move (0.1*8) every second, and take 10 seconds for the full repeat of the p-term 8%)
  - Integral Gain: is the same as the reset rate multiplied by the P-Gain.
  - Conversion between Time and Reset Rate:
    Reset Rate = 1 / Time Constant
    I–Gain = (1/Time Constant)*P-Gain

- **Derivative (Rate):** The amount subtracted from the output based upon the rate change of the error. This anticipatory action may be called pre-act. *If you do not use enough derivative there is no benefit at all, and there could be some harm.*

  $m = T_d\frac{d}{dt}e$ Td = Derivative time in minutes

  $P + I + D = m = K_c\left[e + \frac{1}{T_1}\int e\,dt + T_d\frac{d}{dt}e\right] + b$

  - Time Constant is the amount of time the controller will look forward.
  - Derivative Gain: is the amount of time the controller looks forward multiplied by the P-Gain.

- **Filter Time:** Time constant applied to PV before it is introduced to the controller.

  $\frac{1}{T_f S + 1}$

18.2 Description of Processes:

18.2.1 Fast Loops (Flow & Pressure)

P = Little (too much will cause cycling)
I = More
D = NOT needed

18.2.2 Slow Loops (Temperature)

P = More
I = Some (too much will cause cycling)
D = Some

18.2.3 Integrating (Level & Insulated Temperature)

P = More
I = Little (will cause cycling)
D = Must (if D is not used, loop will cycle)

18.2.4 Noisy Loops (where PV is constantly changing)

P = Low (will cause cycling)
I = Most (accumulated error)
D = Off (will cause cycling)

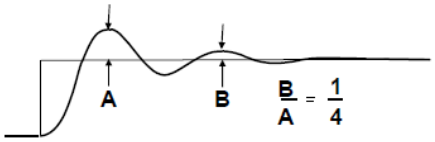
18.3 Manual Tuning:

If the system must remain online, one tuning method is to first set K_i and K_d values to zero. Increase the K_p until the output of the loop oscillates, then the K_p should be left set to be approximately half of that value for a "quarter amplitude decay" type response. Then increase K_i until any offset is correct in sufficient time for the process. However, too much K_i will cause instability. Finally, increase K_d, if required, until the loop is acceptably quick to reach its reference after a load disturbance. However, too much K_d will cause excessive response and overshoot. A fast PID loop tuning usually overshoots slightly to reach the setpoint more quickly; however, some systems cannot accept overshoot, in which case an "over-damped" closed-loop system is required, which will require a K_p setting significantly less than half that of the K_p setting causing oscillation.

Effects of Increasing parameters				
Parameter	**Rise Time**	**Overshoot**	**Settling Time**	**Steady State Error**
Kp	Decrease	Increase	Small Change	Decrease
Ki	Decrease	Increase	Increase	Eliminate
Kd	Small Change	Decrease	Decrease	None

Acceptable Response to a SP Change:

¼ Decay Ratio:



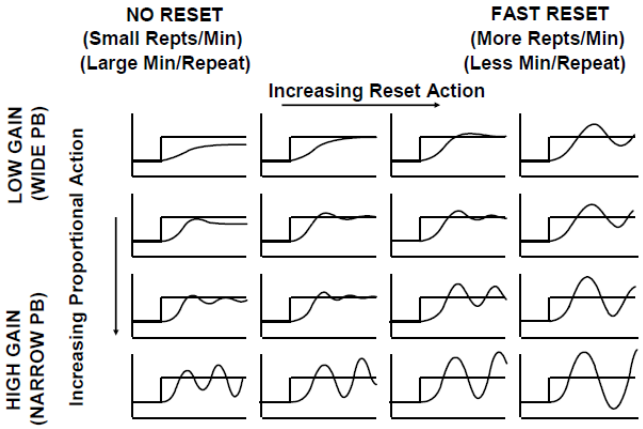
Minimize Overshoot:



18.3.1 Trial & Error Method (closed loop):

- Start With:
  - Low gain, no reset or derivative
  - Loop near normal operating point
  - Controller in Auto
- Make a small SP change
- Adjust gain until desire damping such as ¼ delay is achieved.
  - Do not worry about steady state offset at this time
  - If far from oscillation, double the gain
  - If far too oscillatory, half the gain
  - As you get closer, make smaller changes in the gain.
- Adjust reset to eliminate steady state offset
  - Test by making small SP change
  - By adding reset, you will have to reduce gain
  - If the loop oscillates, the period of oscillation should be between 1.5 and 2 times the integral time
    - If the period is longer than this, make the integral time longer (increase min/repeat or decrease repeats/min)
    - If the period meets the criteria or is shorter, decrease the gain by a factor of 1.5 to 2.
- If the loop is amenable to derivative (relatively noise free)
  - Add some derivative: try 1/8 of the integral time (minutes / repeat)
  - In order to make the loop more stable, you can:
    - Increase the gain by about 30%
    - Decrease the integral time to about 2/3 of its former value
- Now test the process again and trim any of the adjustments
- Test with a load change if possible.

18.4 Tuning Map – Gain & Reset:



18.5 Open Loop Testing:

- Ensure process is running fairly smoothly, if not, it would be hard differentiate between what the step change caused or what might have been happening with the process anyway.
- Place controller in manual, should be bumpless transfer from auto
- Step change the controller output
- Approximate the observed response by a simplified process model (trend of some sort)
- From the model parameters use “somebody’s” correlation to determine controller tuning parameters (e.g. Z-N)

18.5.1 Potential Problems with Open Loop Tuning:

- Sensitive to parameter estimation error (especially dead time)
- Simplified form of process model may not match the actual process
- Controller not in normal operating mode
- Limitations on step size may make it difficult to interpret the response – especially in the presence of noise
- Closed loop response may not be acceptably damped for a SP change.

18.6 Closed Loop Testing:

The basic idea in the closed loop method of tuning is to get the loop to cycle without getting the process into trouble.

- Ensure process is running fairly smoothly, if not, it would be hard differentiate between what the step change caused or what might have been happening with the process anyway.
- Ensure controller is in automatic
- Gain only, no reset or derivative
- Induce sustained oscillation by gradually increasing controller gain
- Note the ultimate period Pu and the ultimate gain Kcu
- From the model parameters use “somebody’s” correlation to determine controller tuning parameters (e.g. Z-N)

18.6.1 Potential Problems with Closed Loop Tuning:

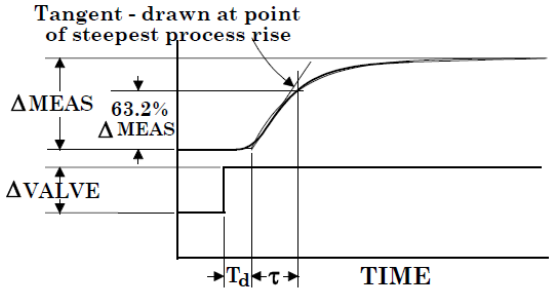
- May not be able to drive the process into oscillating condition
- May require several tests – longer testing time, than open loop method
- Can not guarantee how much the PV – nor the controller output will swing.

18.6.2 Potential Problems with Closed Loop Tuning:

- Controller is operating in its normal mode, automatic
- No artificial form of process model imposed
- Minimal uncertainty in the data

18.7 Z-N Tuning:

18.7.1 Open Loop Method:



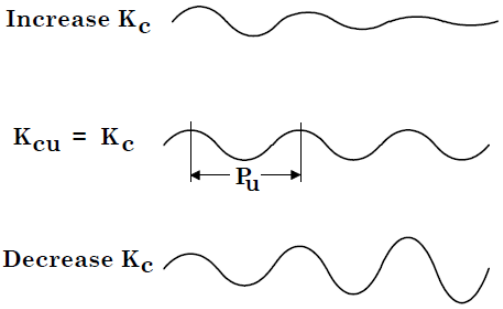
K_p = PROCESS GAIN = $\frac{\Delta MEAS}{\Delta VALVE}$

T_d = DEAD TIME

τ = TIME CONSTANT

	Tuning Parameters		
	P	**PI**	**PID**
Kc	$\frac{\tau}{K_p T_d}$	$\frac{0.9\tau}{K_p T_d}$	$\frac{1.2\tau}{K_p T_d}$
T_I	NA	$3.33T_d$	$2.0T_d$
T_D	NA	NA	$0.5T_d$

18.7.2 Closed Loop Method:



	Tuning Parameters		
	P	**PI**	**PID**
Kc	$0.5K_U$	$0.45K_U$	$0.6K_U$
T_I	NA	$0.83P_U$	$0.5P_U$
T_D	NA	NA	$0.125P_U$

18.8 Tuning Rules of Thumb:

The following rules of thumb are intended to give ballpark figure controller settings:

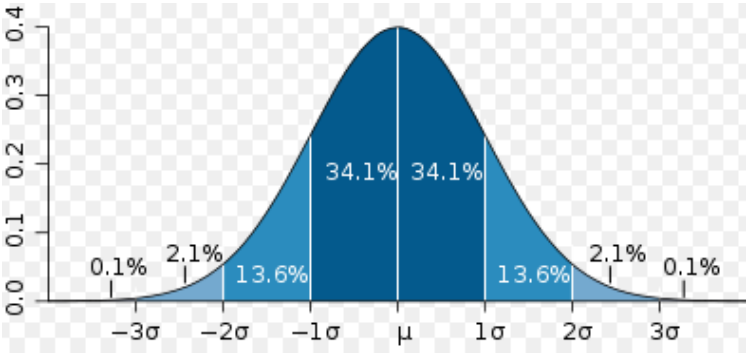
Loop Type	PB (%)	I (mins)	D (mins)
Liquid Level	< 100	10	None
Temperature	20 – 60	2-15	I / 4
Flow	150	0.1	None
Liquid Pressure	50 – 500	0.005 – 0.5	None
Gas Pressure	1 – 50	0.1 – 50	0.02 – 0.1
Chromatograph	100 - 2000	10 -120	0.1 - 20

18.9 Statistics:

Degrees of Freedom (df):

Number of values in the final calculation of a statistic that are free to vary

df = n – 1 where n = number of samples.



Standard Deviation (σ):

Example:

Consider a population consisting of the following values:

2, 4, 4, 4, 5, 5, 7, 9

There are eight data points in total, with a mean (or average) value of 5:

$$\frac{2+4+4+4+5+5+7+9}{8} = \frac{40}{8} = 5$$

To calculate the population standard deviation, we compute the difference of each data point from the mean, and square the result:

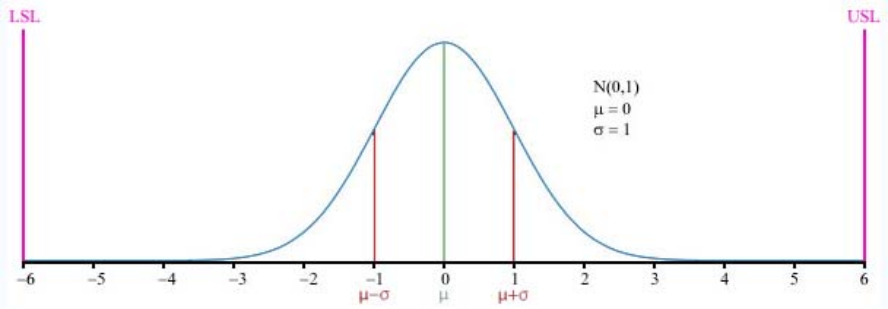
$(2-5)^2 = (-3)^2 = 9$
$(4-5)^2 = (-1)^2 = 1$
$(4-5)^2 = (-1)^2 = 1$
$(4-5)^2 = (-1)^2 = 1$
$(5-5)^2 = (-0)^2 = 0$
$(5-5)^2 = (-0)^2 = 0$
$(7-5)^2 = (-2)^2 = 2$
$(9-5)^2 = (-4)^2 = 16$

Next we average these values and take the square root, which gives the standard deviation:

$$\sqrt{\frac{9+1+1+1+0+0+4+16}{8}} = \sqrt{4} = 2$$

Six Sigma:

Comes from the notion that if one has six standard deviations between the process mean and the nearest specification limit, as shown in the graphic, there will be practically no items that fail to meet specifications.



Role of the 1.5 sigma shift:
Experience has shown that in the long term, processes usually do not perform as well as they do in the short. As a result, the number of sigmas that will fit between the process mean and the nearest specification limit is likely to drop over time, compared to an initial short-term study. To account for this real-life increase in process variation over time, an empirically-based 1.5 sigma shift is introduced into the calculation. According to this idea, a process that fits six sigmas between the process mean and the nearest specification limit in a short-term study will in the long term only fit 4.5 sigmas – either because the process mean will move over time, or because the long-term standard deviation of the process will be greater than that observed in the short term, or both.
Hence the widely accepted definition of a six sigma process is one that produces 3.4 defective parts per million opportunities (DPMO). This is based on the fact that a process that is normally distributed will have 3.4 parts per million beyond a point that is 4.5 standard deviations above or below the mean (one-sided capability study). So the 3.4 DPMO of a "Six Sigma" process in fact corresponds to 4.5 sigmas, namely 6 sigmas minus the 1.5 sigma shift introduced to account for long-term variation. This is designed to prevent underestimation of the defect levels likely to be encountered in real-life operation.

Sigma levels:
The table below gives long-term DPMO values corresponding to various short-term Sigma levels.
Note that these figures assume that the process mean will shift by 1.5 sigma towards the side with the critical specification limit. In other words, they assume that after the initial study determining the short-term sigma level, the long-term Cpk value will turn out to be 0.5 less than the short-term Cpk value. So, for example, the DPMO figure given for 1 sigma assumes that the long-term process mean will be 0.5 sigma beyond the specification limit (Cpk = –0.17), rather than 1 sigma within it, as it was in the short-term study (Cpk = 0.33). Note that the defect percentages only indicate defects exceeding the specification limit that the process mean is nearest to. Defects beyond the far specification limit are not included in the percentages.

ΣLevel	DPMO	% Defective	% Yield	Short Term C_{PK}	Long Term C_{PK}
1	691,462	69%	31%	0.33	–0.17
2	308,538	31%	69%	0.67	0.17
3	66,807	6.7%	93.3%	1.00	0.5
4	6,210	0.62%	99.38%	1.33	0.83
5	233	0.023%	99.977%	1.67	1.17
6	3.4	0.00034%	99.99966%	2.00	1.5

18.10 Damping Ratio:

Damping Ratio:

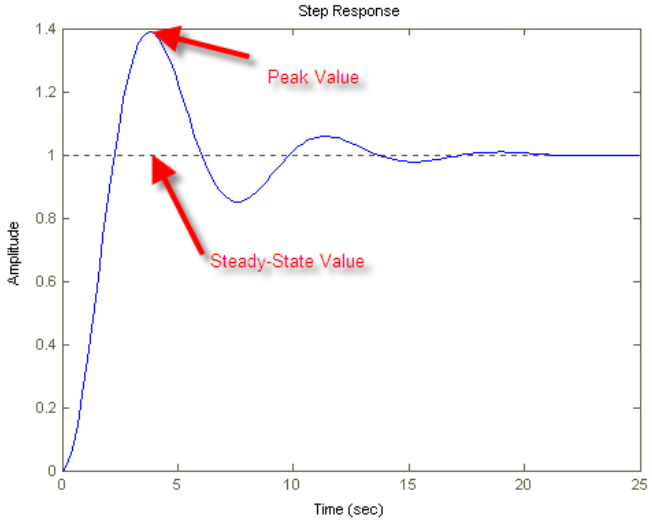
$$\zeta = \frac{-\ln\left(\frac{\%OS}{100}\right)}{\sqrt{\pi^2 + \ln^2\left(\frac{\%OS}{100}\right)}}$$

%OS = % Overshoot

Overshoot:
[(Peak Value of Unit Step Response – State Value of Unit Response)÷ State Value of Unit Response]

DC Gain: The DC Gain is the ration of the steady state step response to the magnitude of a step input:

DC Gain = Steady State Output ÷ Step Magnitude



18.11 Nyquist Stability Criterion:

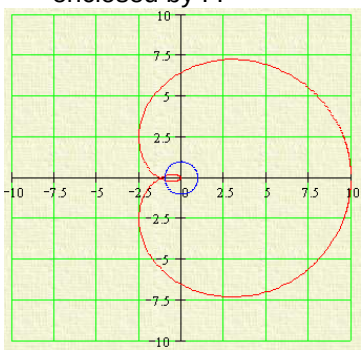
Unique method for determining stability of a closed loop system (Nyquist plot allows us also to predict the stability and performance of a closed-loop system by observing its open-loop behavior)

$N = Z - P$

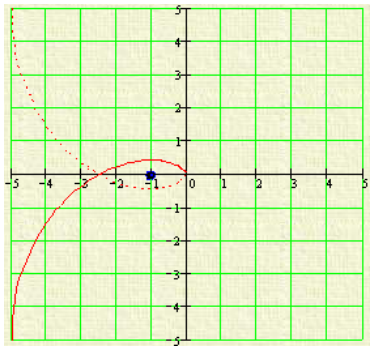
N is the number of encirclements of the (-1, 0) point.
Z is the number of zeros of the characteristic equation.
P is the number of poles of the open-loop characteristic equation.

With this equation stated, we can now state the Nyquist Stability Criterion:

- A feedback control system is stable, if and only if the contour $\Gamma_{F(s)}$ in the F(s) plane does not encircle the (-1, 0) point when P is 0.
- A feedback control system is stable, if and only if the contour $\Gamma_{F(s)}$ in the F(s) plane encircles the (-1, 0) point a number of times equal to the number of poles of F(s) enclosed by Γ.



Stable



Un-Stable

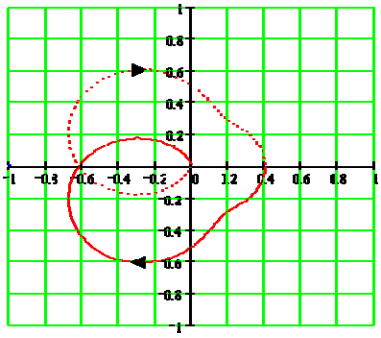
In other words, if P is zero then N must equal zero. Otherwise, N must equal P. Essentially, we are saying that Z must always equal zero, because Z is the number of zeros of the characteristic equation (and therefore the number of poles of the closed-loop transfer function) that are in the right-half of the s plane.

A closed loop system is stable if the unit circle crossing is at a lower frequency than the –180° crossing.

A closed loop system is unstable if the unit circle crossing is at a higher frequency than the –180° crossing.

Here is a Nyquist plot for a typical system.

- The arrows on the plot indicate the direction of increasing frequency.
- The solid plot is for positive frequencies, and the dotted plot is for negative frequencies.
- The plot starts at zero frequency at a gain of about 0.4
- The plot intersects the negative real axis at about -0.62



To calculate the largest gain available:

- Determine the intersection with the negative axis (e.g. say -0.62)
- Determine the factor by which the plot can be re-sized so that it would go through -1
    - Here you would be able to increase the plot size by a factor of 1÷0.62 = 1.61

        ∴ The largest gain this system can have is $K_p = 1.61$

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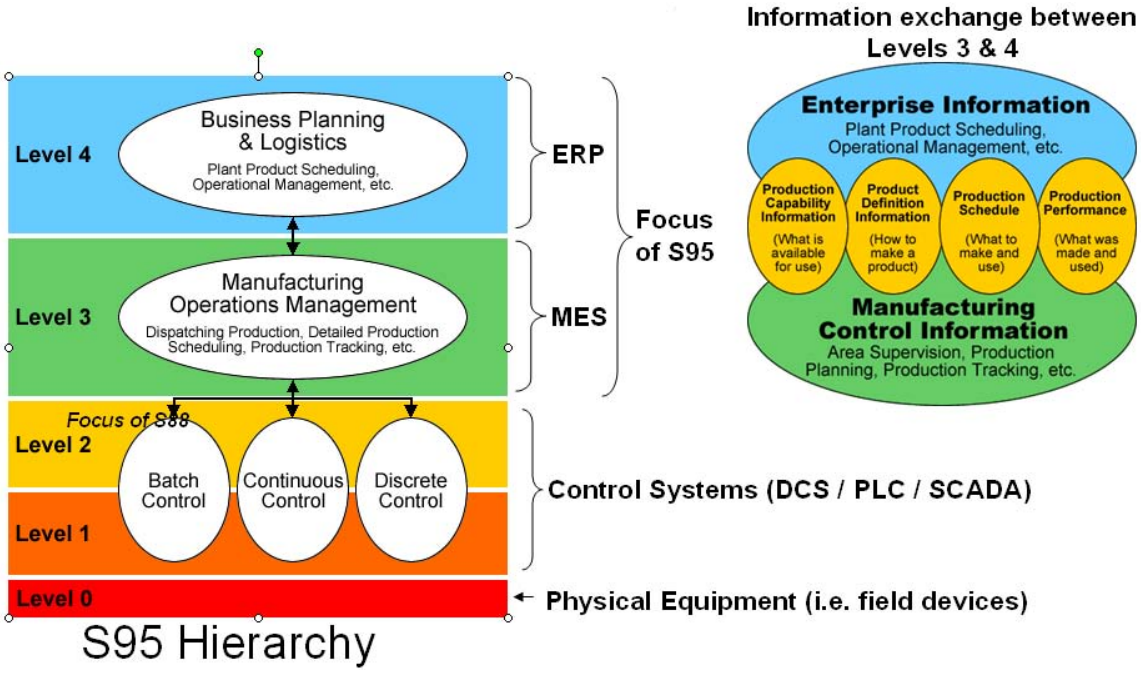
19 S95 (MES)

Premise of MES (ANSI/ISA – 95.00):

- Provide open information exchange across manufacturing production and business planning systems.
- Provide integrated real-time manufacturing applications.

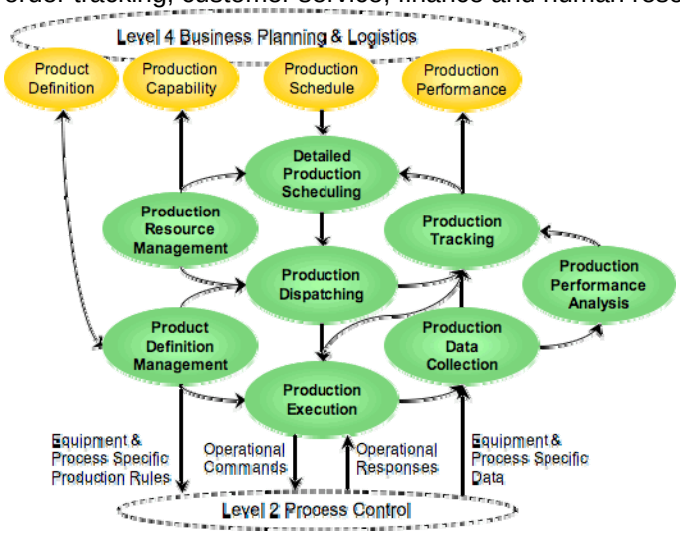
MES = Manufacturing Execution System: Has great potential, by integrating with real time production data, process plant managers can more easily schedule production, manage raw materials and optimize equipment use.

However, this is not a simple task, as it requires investment in additional HW, SW and application engineering (due to various different control systems). Essentially users have been required to implement an additional system layer on top of existing systems in order to support the information routing and numerous interfaces required across functional areas.



S95 Hierarchy

ERP = Enterprise Resource System: Business management system that integrates all facets of the business, including planning, manufacturing, sales, and marketing. ERP methodology helps business managers in business activities such as inventory control, order tracking, customer service, finance and human resources.



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20 Enclosure Ratings

20.1 NEMA

NEMA	IP Equiv	NEMA Definition
1	IP10	Enclosures constructed for indoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment and to provide a degree of protection against falling dirt.
2	IP11	Enclosures constructed for indoor used to provide a degree of protection to personnel against incidental contact with the enclosed equipment, to provide a degree of protection against falling dirt, and to provide a degree of protection against dripping and light splashing of liquids.
3	IP54	Enclosures constructed for either indoor or outdoor used to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to proved a degree of protection against falling dirt, rain, sleet, snow, and windblown dust; and that will undamaged by external formation of ice on the enclosure.
3R	IP14	Enclosures constructed for either indoor or outdoor used to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt, rain, sleet, and snow; and that will be undamaged by external formation of ice on the enclosure.
3S	IP54	Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt, rain, sleet, snow, and windblown dust; and in which the external mechanism(s) remain operable when ice laden
4	IP56	Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt, rain, sleet, snow, windblown dust, splashing water, and hose-directed water, and corrosion; and that will be undamaged by the external formation of ice on the enclosure.
4X	IP56	Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt, rain, sleet, snow, windblown dust, splashing water, hose-directed water, and corrosion; and that will be undamaged by thee external formation of ice on the enclosure.
5	IP52	Enclosures constructed for indoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt; against settling airborne dust, lint, fibers, and flyings; and to provide a degree of protection against dripping and light splashing of liquids.
6 / 6P	IP67	Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt; against hose-directed water and the entry of water during occasional temporary submersion at a limited depth; and that will be undamaged by the external formation of ice on the enclosure.
12 / 12K	IP52	Enclosures constructed (without knockouts) for indoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt; against circulating dust, lint, fibers, and flying; and against dripping and light splashing of liquids
13	IP54	Enclosures constructed for indoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt; against circulating dust, lint, fibers, and flyings; and against the spraying, splashing, and seepage of water, oil, and noncorrosive coolants.

20.2 IP

FIRST DIGIT		
Level	**Object Size Protection**	**Effective Against**
0	NA	No protection against contact and ingress of objects
1	> 50mm	Any large surface of the body, such as the back of a hand, but no protection against deliberate contact with a body part
2	> 12.5mm	Fingers or similar objects
3	> 2.5mm	Tools, thick wires, etc
4	> 1 mm	Most wires, screws, etc
6	Dust Protected	Ingress of dust is not entirely prevented, but it must not enter in sufficient quantity to interfere with the satisfactory operation of the equipment; complete protection against contact
6	Dust Tight	No ingress of dust; complete protection against contact
FIRST DIGIT		
Level	**Protect Against**	**Details**
0	not protected	NA
1	dripping water	Dripping water (vertically falling drops) shall have no harmful effect.
2	dripping water when tilted up to 15°	Vertically dripping water shall have no harmful effect when the enclosure is tilted at an angle up to 15° from its normal position
3	spraying water	Water falling as a spray at any angle up to 60° from the vertical shall have no harmful effect
4	splashing water	Water splashing against the enclosure from any direction shall have no harmful effect
5	water jets	Water projected by a nozzle against enclosure from any direction shall have no harmful effects
6	powerful water jets	Water projected in powerful jets against the enclosure from any direction shall have no harmful effects
7	immersion up to 1m	Ingress of water in harmful quantity shall not be possible when the enclosure is immersed in water under defined conditions of pressure and time (up to 1m of submersion).
8	immersion beyond 1m	The equipment is suitable for continuous immersion in water under conditions which shall be specified by the manufacturer. NOTE: Normally, this will mean that the equipment is hermetically sealed. However, with certain types of equipment, it can mean that water can enter but only in such a manner that produces no harmful effects

21 Hazardous Areas:

	Flammable Material Present Continuously	Flammable Material Present Intermittently	Flammable Material Present Abnormally
IEC / EU	Zone 0	Zone 1	Zone 2
US NEC 505	Zone 0	Zone 1	Zone 2
US NEC 500	Division 1		Division 2
CA CEC Section 18	Zone 0	Zone 1	Zone 2
CEC Annex J	Division 1		Division 2

IEC classification per IEC 60079-10
EU classification per EN 60079-10
US classification per ANSI/NFPA 70 National Electrical Code® (NEC®) Article 500 or Article 505
CA classification per CSAC22.1 Canadian Electrical Code (CEC) Section 18 or Annex J

Common protection techniques in hazardous areas are:

- **Dust-ignitionproof equipment** - excludes dust, will not permit arcs, sparks or heat to ignite exterior dust in the enclosure
- **Explosionproof equipment** - withstands an internal explosion without igniting the external atmosphere
- **Hermetically sealed equipment** - sealed against the external atmosphere to prevent the entry of hazardous gases or vapors
- **Intrinsically safe systems** - electrical circuits designed not to release sufficient energy to ignite or explosive the atmosphere
- **Nonicendive circuits and components** - designed to prevent any arc or thermal effect
- **Oil immersed equipment** - arc producing components are immersed in oil, an arc will never reach the surface of the oil
- **Purged and pressurized systems** - spaces and equipment are pressurized above the external atmosphere, hazardous gases and vapors are not able to enter from the enclosure

Appropriate Protection Techniques

Protection Technique	Hazardous Area			
	Class I		Class II	
	Div I	Div II	Div I	Div II
Dust-ignitionproof equipment	X	X	✓	✓
Explosionproof equipment	✓	✓	✓	✓
Hermetically sealed equipment	X	✓	✓	✓
Intrinsically safe systems	✓	✓	✓	✓
Nonicendive circuits and components	X	✓	✓	✓
Oil immersed equipment	X	✓	✓	✓
Purged and pressurized systems	✓	✓	✓	✓

21.1 NEC Classes (500)

- **Class I:** Atmospheric hazards are divided into four groups, A, B, C, D, but also into two divisions.

Groups:

- **Class A**: Acetylene
- **Class B:** Flammable gas. flammable liquid produced vapors or combustible liquid produced vapors mixed with air in sufficient quantities that may burn or explode, having a Maximum Safe Experimental Gap (MESG) value ≤ 0.45mm or a minimum igniting current ratio (MIC ratio) of ≤ 0.40.
- **Class C:** Flammable gas. flammable liquid produced vapors or combustible liquid produced vapors mixed with air in sufficient quantities that may burn or explode,

having a Maximum Safe Experimental Gap (MESG) value ≥ 0.45mm and ≤ 75mm or a minimum igniting current ratio (MIC ratio) of ≥ 0.40 and ≤ 0.80.

- **Class D:** Flammable gas. flammable liquid produced vapors or combustible liquid produced vapors mixed with air in sufficient quantities that may burn or explode, having a Maximum Safe Experimental Gap (MESG) value > 0.75mm or a minimum igniting current ratio (MIC ratio) of > 0.80.

Divisions:

- **Division I:** Covers locations where flammable gas. flammable liquid produced vapors or combustible liquid produced vapors are or may exist under normal operating conditions, under frequent repair or maintenance operations or where breakdown or faulty operation of process equipment might also cause simultaneous failure of electrical equipment.
- **Division II:** Flammable gas. flammable liquid produced vapors or combustible liquid produced vapors are handled, processed, or used, but in which the liquids, vapors or gases will normally be confined within a closed container or a closed system from which they can escape under accidental rupture or breakdown, or in case of abnormal operation. **OR** in which Flammable gas. flammable liquid produced vapors or combustible liquid produced vapors are prevented by positive mechanical ventilation (purge) and which may become hazardous due to the failure of the positive mechanical ventilation equipment. **OR** in areas adjacent to Division I locations, into which gases might occasionally flow, would also be division II, unless such flow is prevented by adequate positive pressure ventilation from a source of clean air, and effective safeguards against ventilation failure are provided.

- **Class II:** Atmospheric hazards cover three groups of combustible dusts.

- **Group E:** Metallic Dusts including a aluminum, magnesium and their commercial alloys or other combustible dusts whose particle size, abrasiveness and conductivity present similar hazards in the use of electrical equipment.
- **Group F:** Carbonaceous Dusts that have more than 8% total entrapped volatiles. (Coal, carbon black, charcoal and coke dust are examples of carbonaceous dusts)
- **Group G:** Combustible Dusts not included in Group E or F, including flour, grain, wood, plastic and chemicals.

Where an area is Division I or Division II depends on the quantity of dust present, except that for Group E there is only Division I.

- **Class III:** Atmospheric hazards cover locations where combustible fibers/flyings are present but not likely to be in suspension in air in quantities sufficient to produce ignitable mixtures. Division I is where they are manufactured and Division II is where they are stored.

21.2 NEC Zones (505)

- **Zone 0:** A place in which an explosive atmosphere consisting of a mixture with air of flammable substances in the form of gas, vapor or mist is presently continuously or for long periods of time.
- **Zone1:** A place in which an explosive atmosphere consisting of a mixture with air or flammable substances in the form of gas, vapor or mist is likely to occur in normal operation.
- **Zone 2:** A place in which an explosive atmosphere consisting of a mixture with air of flammable substances in the form of gas, vapor or mist is not likely to occur in normal operation but, if it does occur, will persist for a short period only.
- **Zone 20:** a place in which an explosive atmosphere in the form of a cloud of combustible dust in air is present continuously, or for long periods or frequently.
- **Zone 21:** A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is likely to occur in normal operation occasionally.
- **Zone 22:** A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is not likely to occur in normal operation but, if it does occur, will persist for a short period only.

	Zone	Duration of the occurrence of an explosive atmosphere	Equipment category
Gases, vapours, mists	0	continuously, for a long period, frequently	1G
	1	occasionally	2G
	2	rarely and for a short period	3G
Dusts	20	continuously, for a long period, frequently	1D
	21	occasionally	2D
	22	rarely and for a short period	3D

Difference Between Flash-Point and Auto-Ignition Temperature:

- **Flash-Point:** The flash-point of a flammable liquid is the lowest temperature at which the surface of the liquid emits sufficient vapor to be ignited by a small flame.
- **Auto-Ignition:** Temperature where ignition will take place, even without an external ignition source such as a spark or flame. Apparatus for use in a hazardous area must not have a surface temperature > the ignition temperature. Apparatus is therefore marked with a maximum surface temperature or “T” rating.

21.3 FM Approvals

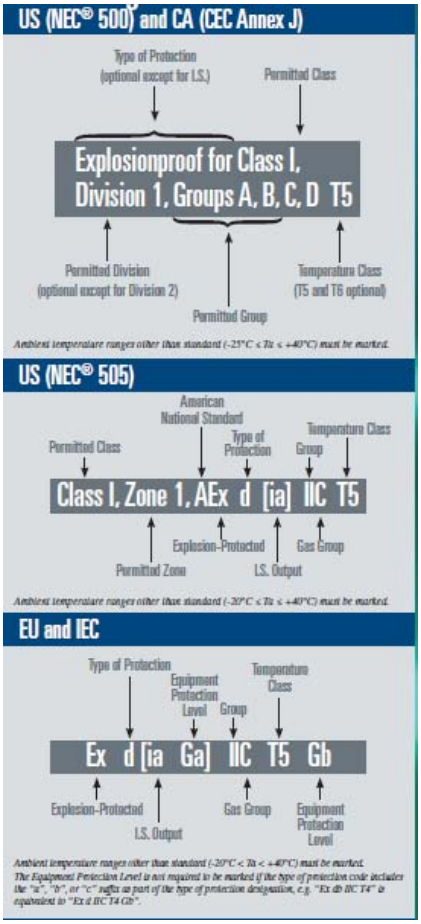
21.3.1 Protection Concepts

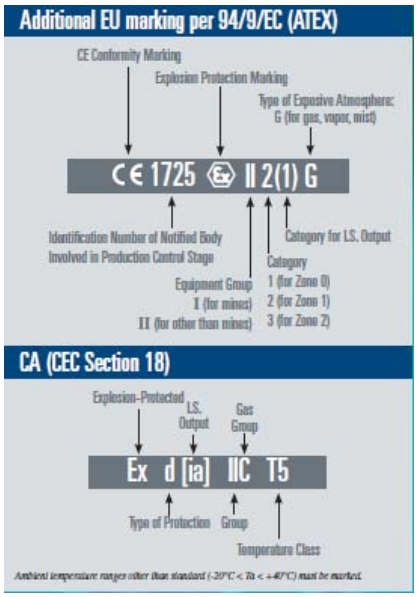
Type of Protection	Code	Market	Application	Standard	Protection Principle
General Requirements		US	Class I, Division 1 & 2	FM 3600	
		CA	Class I, Division 1 & 2	CSA C22.2 No. 0	
	AEx	US	Class I, Division 1 & 2	ISA 60079-0	
	Ex	CA	Class I, Division 1 & 2	CSA E60079-0	
	Ex	EU	Category 1G, 2G, & 3G	EN 60079-0	
	Ex	IEC	EPL Ga, Gb, Gc	IEC 60079-0	
Increased Safety	AEx e	US	Class I, Zone 1	ISA 60079-7	
	Ex e	CA	Class I, Zone 1	CSA E60079-7	
	Ex e *(or Ex eb)*	EU	Category 2G	EN E60079-7	
	Ex e *(or Ex eb)*	IEC	EPL Gb	IEC E60079-7	No arcs, sparks or hot surfaces
Non-Incendive	(NI)	US	Class I, Division 2	FM 3611	
	(NI)	CA	Class I, Division 2	C22.2 No. 213	
Non-Sparking	AEx nA	US	Class I, Zone 2	ISA 60079-15	
	Ex nA	CA	Class I, Zone 2	CSA E60079-15	
	Ex nA *(or Ex nAc)*	EU	Category 3G	EN 60079-15	
	Ex nA *(or Ex nAc)*	IEC	EPL Gc	IEC 60079-15	
Explosionproof	(XP)	US	Class I, Division 1	FM 3615	
	(XP)	CA	Class I, Division 1	C22.2 No. 30	
Flameproof	AEx d	US	Class I Zone 1	ISA 60079-1	
	Ex d	CA	Class I, Zone 1	CSA E60079-1	
	Ex d *(or Ex db)*	EU	Category 2G	EN 60079-1	Contain the explosion and extinguish the flame
	Ex d *(or Ex db)*	IEC	EPL Gb	IEC 60079-1	
Powder-Filled	AEx q	US	Class I, Zone 1	ISA 60079-5	
	Ex q	CA	Class I, Zone 1	CSA E60079-5	
	Ex q *(or Ex qb)*	EU	Category 2G	EN 60079-5	
	Ex q *(or Ex qb)*	IEC	EPL Gb	IEC 60079-5	
Enclosed Break	AEx nC	US	Class I, Zone 2	ISA 60079-15	
	Ex nC	CA	Class I, Zone 2	CSA E60079-15	
	Ex nC *(or Ex nCc)*	EU	Category 3G	EN 60079-15	
	Ex nC *(or Ex nCc)*	IEC	EPL Gc	IEC 60079-15	
Intrinsic Safety	(IS)	US	Class I, Division 1	FM 3610	
	(IS)	CA	Class I, Division 1	C22.2 No. 157	
	AEx ia	US	Class I, Zone 0	FM 3610	
	AEx ib	US	Class I, Zone 1	FM 3610	
	Ex ia	CA	Class I, Zone 0	CSA E60079-11	
	Ex ib	CA	Class I, Zone 1	CSA E60079-11	
	Ex ia	EU	Category 1G	EN E60079-11	
	Ex ib	EU	Category 2G	EN E60079-11	
	Ex ic	EU	Category 3G	EN E60079-11	Limit energy of sparks and surface temperature
	Ex ia	IEC	EPL Ga	IEC E60079-11	
	Ex ib	IEC	EPL Gb	IEC E60079-11	
	Ex ic	IEC	EPL Gc	IEC E60079-11	

Note 1: For associated intrinsically safe apparatus suitable for installation in a hazardous location, the symbols for the type of protection "ia" or "ib" are enclosed within square brackets, e.g. AEx d [ia] IIC T4.
Note 2: For associated intrinsically safe apparatus not suitable for installation in a hazardous location, both the symbol "Ex" or "AEx" and the symbol for the type of protection "ia" or "ib" are enclosed within the same square brackets, e.g. [AEx ia] IIC; in this case, a temperature class is not included.

Limited Energy	AEx nC	US	Class I, Zone 2	ISA 60079-15	
	Ex nL	CA	Class I, Zone 2	CSA E60079-15	
	Ex nL *(or Ex nLc)*	EU	Category 3G	EN 60079-15	
	Ex nL *(or Ex nLc)*	IEC	EPL Gc	IEC 60079-15	
Pressurized	Type X	US	Class I, Division 1	FM 3620 (NFPA 496)	Keep flammable gas out
	Type X	CA	Class I, Division 1	NFPA 496	
	Type Y	US	Class I, Division 1	FM 3620 (NFPA 496)	
	Type Y	CA	Class I, Division 1	NFPA 496	
	Type Z	US	Class I, Division 2	FM 3620 (NFPA 496)	
	Type Z	CA	Class I, Division 2	NFPA 496	
	AEx px	US	Class I, Zone 1	ISA 60079-2	
	Ex px	CA	Class I, Zone 1	CSA E60079-2	
	Ex px *(or Ex pxb)*	EU	Category 2G	EN 60079-2	
	Ex px *(or Ex pxb)*	IEC	EPL Gb	IEC 60079-2	
	AEx py	US	Class I, Zone 1	ISA 60079-2	
	Ex py	CA	Class I, Zone 1	CSA E60079-2	
	Ex py *(or Ex pyb)*	EU	Category 2G	EN 60079-2	
	Ex py *(or Ex pyb)*	IEC	EPL Gb	IEC 60079-2	
	AEx pz	US	Class I, Zone 2	ISA 60079-2	
	Ex pz	CA	Class I, Zone 2	CSA E60079-2	
	Ex pz *(or Ex pzc)*	EU	Category 3G	EN 60079-2	
	Ex pz *(or Ex pzc)*	IEC	EPL Gc	IEC 60079-2	
Restricted Breathing	AEx nR	US	Class I, Zone 2	ISA 60079-15	
	Ex nR	CA	Class I, Zone 2	CSA E60079-15	
	Ex nR *(or Ex nRc)*	EU	Category 3G	EN 60079-15	
	Ex nR *(or Ex nRc)*	IEC	EPL Gc	IEC 60079-15	
Encapsulation	AEx ma	US	Class I, Zone 0	ISA 60079-18	
	Ex ma	EU	Category 1G	EN 60079-18	
	Ex ma	IEC	EPL Ga	IEC 60079-18	
	AEx m	US	Class I, Zone 1	ISA 60079-18	
	Ex m	CA	Class I, Zone 1	CSA E60079-18	
	AEx mb	US	Class I, Zone 1	ISA 60079-18	
	Ex mb	EU	Category 2G	EN 60079-18	
	Ex mb	IEC	EPL Gb	IEC 60079-18	
Oil Immersion	AEx o	US	Class I, Zone 1	ISA 60079-6	
	Ex o	CA	Class I, Zone 1	CSA E60079-6	
	Ex o *(or Ex ob)*	EU	Category 2G	EN 60079-6	
	Ex o *(or Ex ob)*	IEC	EPL Gb	IEC 60079-6	

21.3.2 Ex Markings





21.3.3 Temperature Classifications

Marking	US (NEC® 505) CA (CEC) Section 18) EU IEC	US (NEC® 500) CA (CEC Annex J)
450°C	T1	T1
300°C	T2	T2
280°C		T2A
260°C		T2B
230°C		T2C
215°C		T2D
200°C	T3	T3
180°C		T3A
165°C		T3B
160°C		T3C
135°C	T4	T4
120°C		T4A
100°C	T5	T5
85°C	T6	T6

21.4 Purged & Pressurized Systems

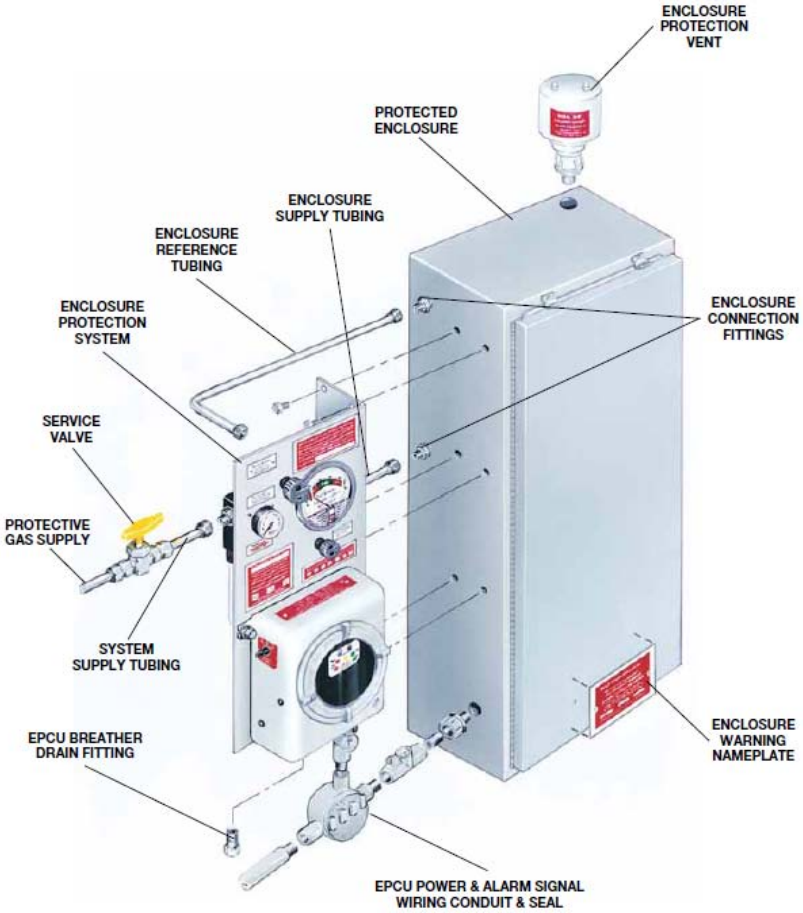
21.4.1 Type X Purge

Type X pressurization reduces the hazard classification from Division 1 to non-classified
Type X purging requires that if the enclosure pressure is lost, the supply is automatically disconnected on loss of purge pressure and a re-purge is required before the supply is restored.

- Class I applications- the Type X purge system operates by forcing air or an inert gas through the enclosure until all of the hazardous gas is removed (purging). After purging, constant pressure is maintained within the enclosure and equipment inside the enclosure is energized. If enclosure pressure drops below its minimum, enclosure power is cut and a purging sequence begins.

- Class II applications- the enclosure must be free of all combustible material. It is pressurized before powering the equipment inside. There is no purging of the Class II, Type X system.

- Vents are required for most Type X purge enclosures.

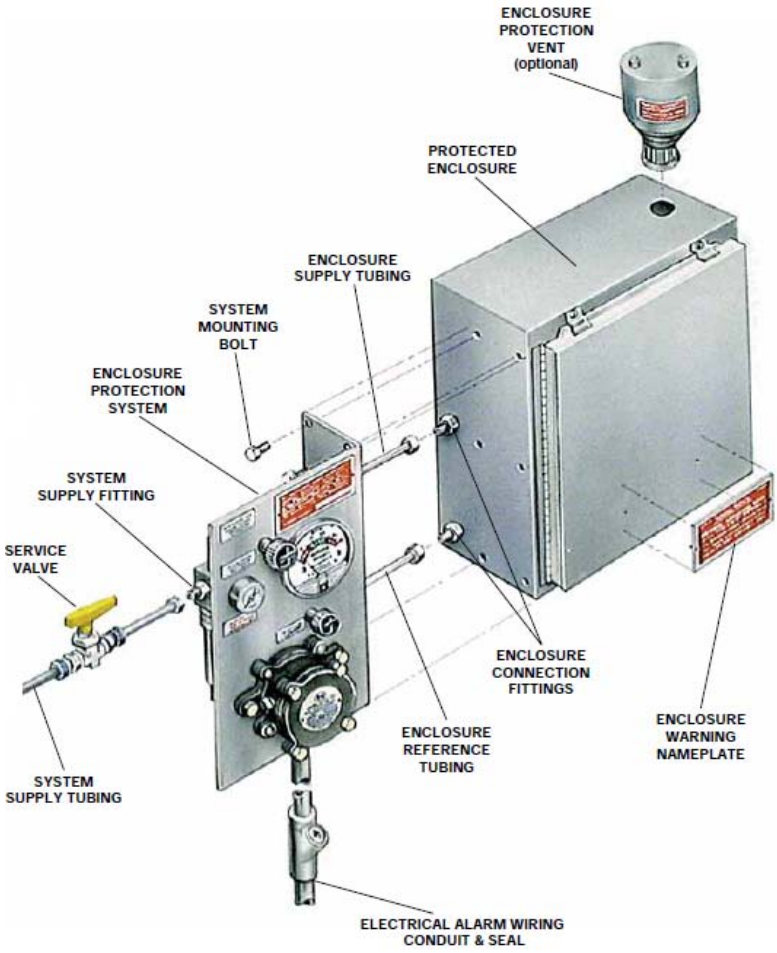


21.4.2 Type Y Purge

Type Y pressurization reduces the hazard classification from Division 1 to Division 2
Type Y purging does not require supply disconnection on loss of pressure but the equipment in the purged enclosure must be suitable for division 2

- All Equipment inside the enclosure must be Division 2 rated. For Class I applications this system operates by forcing air or an inert gas through the enclosure for a specified time until all of the hazardous gas has been removed (purging). After purging, a constant pressure with the enclosure is maintained and the equipment with the enclosure can be energized. Loss of safe pressure requires immediate attention if power is not discontinued to the enclosure.

- For Class II applications no purging is performed but the enclosure must be cleaned of all combustible material, then pressurized before energizing the equipment with the enclosure. Loss of safe pressure requires immediate attention if power is not discontinued to the enclosure.



21.4.3 Type Z Purge

Type Z pressurization reduces the hazard classification from Division 2 to non-classified. Type Z purging, because of the lower level of risk in division 2, requires only an indication of loss of purged pressure.
Z-Purge has the same components as that of a Y-Purge, except instead of reducing from Div I to Div II, it reduces from Div II to non-hazardous.

21.5 Wiring Methods

21.5.1 Class I, Division I

The following wiring methods are permitted in Class I, Division I locations:

- Threaded rigid metal conduit or threaded intermediate metal conduit (IMC).
  - Exception: Plastic / fiberglass conduit used in encased in concrete with a minimum envelope cover of 2” and the last 24” of conduit to the emergence point is either rigid metal or IMC.
- Type MI cable with termination fittings listed for the location (MI = made from copper conductors inside a copper sheath, insulated by inorganic magnesium oxide powder). Must installed with sufficient support to avoid tensile strength on the termination fittings.
- In industrial establishments with restricted public access, where the conditions of maintenance and supervision ensure that only qualified persons service the

installation. Type MC-HL cable listed for use in Class1, Zone 1 or Division I locations with a gas / vaportight continuous corrugated metallic sheath, an overall jacket of suitable polymeric material, a separate equipment grounding conductor(s) and provided with termination fittings listed for the application. (MC-HL = Metal Clad: IAC, CLX).

- In industrial establishments with restricted public access, where the conditions of maintenance and supervision ensure that only qualified persons service the installation. Type ITC-HL cable listed for use in Class1, Zone 1 or Division I locations with a gas / vaportight continuous corrugated metallic sheath, an overall jacket of suitable polymeric material, a separate equipment grounding conductor(s) and provided with termination fittings listed for the application. (ITC-HL = Metal Clad Instrument Tray Cable: IAC, CLX).

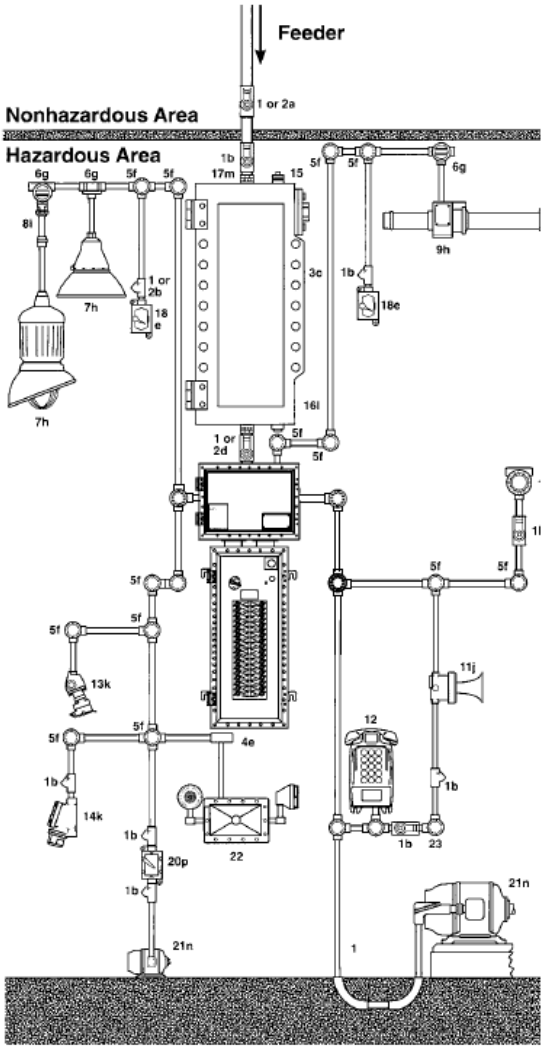
21.5.2 Class I, Division II

The following wiring methods are permitted in Class I, Division I locations:

- All wiring methods permitted for Class I, Division I (section 17.6.1)
- Threaded rigid metal conduit or threaded intermediate metal conduit (IMC).
- Enclosed gasketed busways or wireways.
- Type PLTC (Power Limited Tray Cable). Must installed with sufficient support to avoid tensile strength on the termination fittings.
- Type ITC Cable (Instrumentation Tray Cable). Must installed with sufficient support to avoid tensile strength on the termination fittings.
- Type MI, MC, MV or TC cable with termination fittings, or in cable tray systems. Must installed with sufficient support to avoid tensile strength on the termination fittings. (MI = Mineral Insulated; MC = Metal Clad; MV = Medium Voltage; TC = Tray Cable)

21.5.3 Installation Details

21.5.3.1 Class I, Division I Lighting:



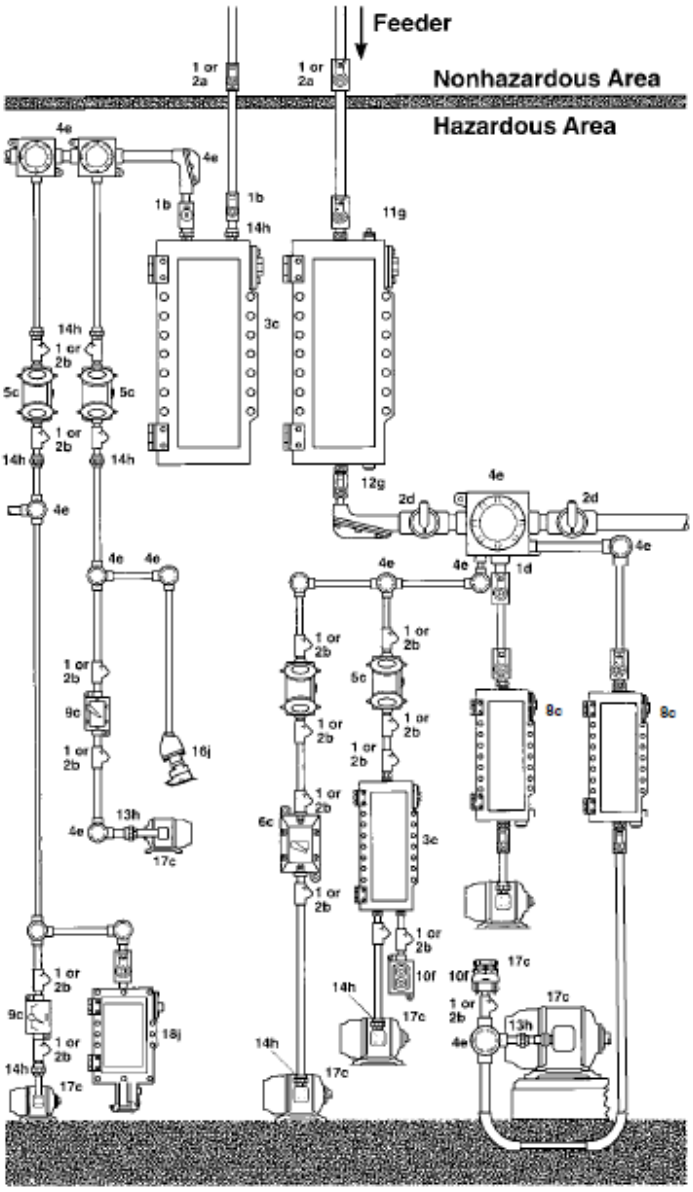
Key to Numerals

1 Sealing fitting. EYS for horizontal or vertical.
2 Sealing fitting. EZS for vertical or horizontal conduits.
3 Circuit breaker. Type EBM.
4 Panelboard. EXDC/EPL. Branch circuits are factory sealed. No seals required in mains or branches unless 2" or over in size.
5 Junction box. Series GUA, GUB, EAJ, EAB have threaded covers. Series CPS has ground flat surface covers.
6 Fixture hanger. EFHC, GUAC, or EFH.
7 Lighting Fixture. EV Series incandescent and EVM Series. H.I.D.
8 Flexible fixture support. ECHF.
9 Fluorescent fixture. EVFT.
11 Signal. ETH horns and sirens. ESR bells, Flex•Tone™ signals.
12 ETW explosionproof telephone.
13 Plug receptacle. CES delayed action.
14 Plug receptacle. FSQ. Interlocked with switch.
15 Breather. ECD.
16 Drain. ECD.
17 Union. UNY.
18 Switch. Series EFS.
19 Instrument enclosure. EIH.
20 Manual line starter. EMN.
21 Motors. Explosionproof.
22 Emergency lighting system. ELPS.
23 ETC Power Relay.

***National Electrical Code* References**

a Sec. 501.15(A)(4). Seal required where conduit passes from hazardous to nonhazardous area.
b Sec. 501.15(A)(1)(1). Seals required within 18 inches of all arcing devices.
c Sec. 408.36. Circuit breaker protection required ahead of panelboard.
d Sec. 501.15(A)(1)(2). Seals required if conduit is 2 inches or larger.
e Sec. 501.115(A). All arcing devices must be explosionproof.
f Sec. 501.10(A). All boxes must be explosionproof and threaded for rigid or IMC conduit.
g Sec. 501.130(A)(4). All boxes and fittings for support of luminaires must be approved for Class I locations.
h Sec. 501.130(A)(1). All lighting fixtures, fixed or portable, must be explosionproof.
i Sec. 501.130(A)(3). Pendant fixture stems must be threaded rigid or IMC conduit. Conduit stems if over 12 inches must have flexible connector, or must be braced.
j Sec. 501.150(A). All signal and alarm equipment irrespective of voltage must be approved for Class I, Division 1 locations.
k Sec. 501.145. Receptacles and plugs must be explosionproof and provide grounding connectionsfor portable equipment.
l Sec. 501.15(F)(I). Breathers and drains needed in all humid locations.
m Sec. 501.10(A). All joints and fittings must be explosionproof.
n Sec. 501.125(A). Motor must be suitable for Class I.
p Art. 430. Motor overcurrent protection.

21.5.3.2 Class I, Division I Power:



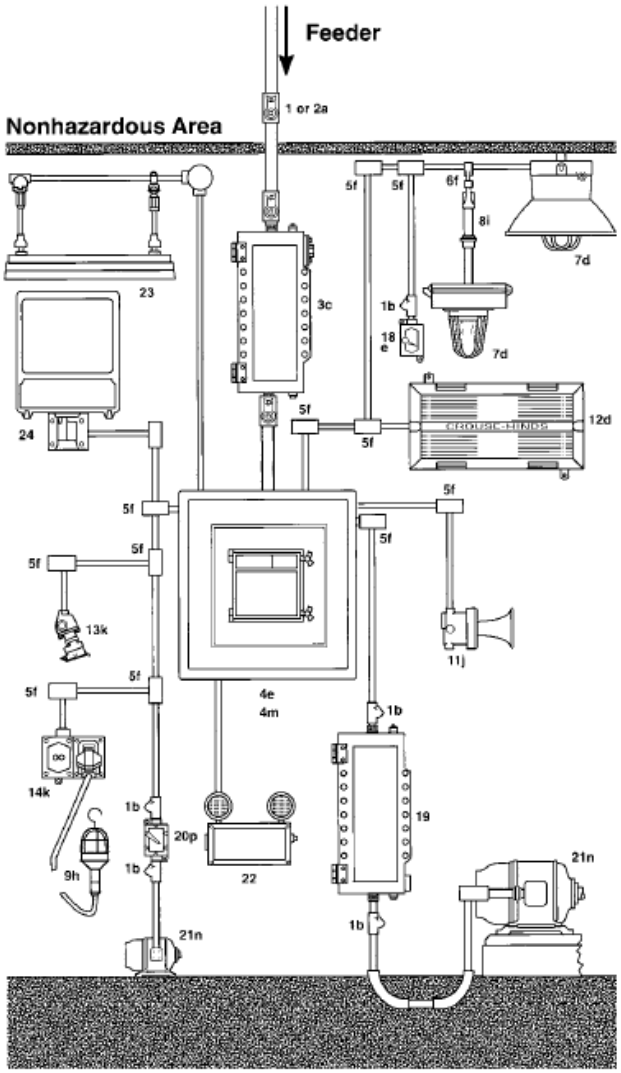
Key to Numerals

1 Sealing fitting. EYS for horizontal or vertical.
2 Sealing fitting. EZS for vertical or horizontal conduits.
3 Circuit breaker EBMB.
4 Junction box Series GUA, GUB, EAB, and EAJ have threaded covers. Series CPS and Type LBH have ground flat surface covers.
5 Circuit breaker FLB.
6 Manual line starter EMN.
7 Magnetic line starter EBMS.
8 Combination circuit breaker and line starter EPC.
9 Switch or motor starter. Series EFS, EDS, or EMN.
10 Pushbutton station. Series EFS or OAC.
11 Breather. ECD.
12 Drain. ECD.
13 Union. UNF.
14 Union. UNY.
15 Flexible coupling. EC.
16 Plug receptacle. CES. Factory sealed.
17 Motor for hazardous location.
18 Plug receptacle. EBBR Interlocked Arktite® receptacle with circuit breaker.

***National Electrical Code* References**

a Sec. 501.15(A)(4). Seals required where conduits pass from hazardous to nonhazardous area.
b Sec. 501.15(A)(1). Seals required within 18 inches of all arcing devices.
c Art. 430 should be studied for detailed requirements for conductors, motor feeders, motor feeder and motor branch circuit protection, motor overcurrent protection, motor controllers, and motor disconnecting means.
d Sec. 501.15(A)(2). Seals required if conduit is 2 inches or larger.
e Sec. 501.10(A). All boxes must be explosionproof and threaded for rigid or IMC conduit.
f Sec. 501.115(A). Pushbutton stations must be explosionproof.
g Sec. 501.15(F)(1). Breathers and drains needed in all humid locations.
h Sec. 501.40(A). All joints and fittings must be explosionproof.
i Sec. 501.10(A). Flexible connections must be explosionproof.
j Sec. 501.145. Receptacles and plugs must be explosionproof, and provide grounding connections for portable devices.

21.5.3.3 Class I, Division II Power & Lighting:



Key to Numerals

1 Sealing fitting. EYS for horizontal or vertical.
2 Sealing fitting. EZS for vertical or horizontal conduits.
3 Circuit breaker. Type EPC.
4 Panelboard. D2PB, N2PB. Branch circuits are factory sealed.
5 Junction box or conduit fitting. NJB, Condulet®.
6 Fixture hanger. AHG, GS, UNJ.
7 Lighting fixture. VMV, DMV, and LMV (CHAMP®).
8 Flexible fixture support. ECHF.
9 Handlamp. EVH.
11 Signal. ETH horns and sirens. ESR bells, Flex•Tone™, and W2H.
12 Compact fluorescent lighting fixture. FVS.
13 Plug receptacle. CES delayed action.
14 Plug receptacle. ENR or CPS delayed action with GFS-1 ground fault circuit interrupter.
15 Breather. ECD.
16 Drain. ECD.
17 Union. UNY.
18 Switch. Series EFS.
19 Magnetic line starter. EBMS.
20 Manual line starter. EMN.
21 Motors. Suitable for Class I, Division 2 locations.
22 Emergency lighting system. N2LPS.
23 Fluorescent fixture. FVN.
24 Floodlight. FMV.

***National Electrical Code* References**

a Sec. 501.15(B)(2). Seal required where conduit passes from hazardous to non-hazardous area.
b Sec. 501.15(B)(1). Seals required within 18 inches of all arcing devices.
c Sec. 408.36. Circuit breaker protection required ahead of panelboard.
d Sec. 501.130(B)(2). All fixed lighting fixtures shall be enclosed and gasketed and not exceed ignition temperature of the gas.
e Sec. 501.115(B)(1). Most arcing devices must be explosionproof.
f Sec. 501.10(B). All boxes must be threaded for rigid or IMC conduit.
h Sec. 501.130(B)(1). All portable lighting fixtures must be explosion proof.
i Sec. 501.130(B)(3). Pendant fixture stems must be threaded rigid conduit or IMC. Rigid stems if over 12 inches must have flexible connector, or must be braced.
j Sec. 501.150(B). All signaling equipment must be approved for Class I location.
k Sec. 501.145. Receptacles and plugs must be explosionproof and provide grounding connections for portable equipment.
l Sec. 501.15(F)(1). Breathers and drains needed in all humid locations.
m Sec. 501.10(B). Not all joints and fittings are required to be explosionproof.
n Sec. 501.125(B). Motor shall be suitable for Division 2.
p Art. 430. Motor overcurrent protection.

21.6 Hazardous Substances Used in Industry

TABLE I

Class I* Group	Substance	Auto-Ignition Temp.*		Flash Point**		Flammable Limits** Percent by Volume		Vapor Density** (Air Equals 1.0)
		°F	°C	°F	°C	Lower	Upper	
C	Acetaldehyde	347	175	-38	-39	4.0	60	1.5
D	Acetic Acid	867	464	103	39	4.0	19.9 @ 200°F	2.1
D	Acetic Anhydride	600	316	120	49	2.7	10.3	3.5
D	Acetone	869	465	-4	-20	2.5	13	2.0
D	Acetone Cyanohydrin	1270	688	165	74	2.2	12.0	2.9
D	Acetonitrile	975	524	42	6	3.0	16.0	1.4
A	Acetylene	581	305	gas	gas	2.5	100	0.9
B(C)	Acrolein (inhibited)[2]	455	235	-15	-26	2.8	31.0	1.9
D	Acrylic Acid	820	438	122	50	2.4	8.0	2.5
D	Acrylonitrile	898	481	32	0	3.0	17	1.8
D	Adiponitrile	—	—	200	93	—	—	—
C	Allyl Alcohol	713	378	70	21	2.5	18.0	2.0
D	Allyl Chloride	905	485	-25	-32	2.9	11.1	2.6
B(C)	Allyl Glycidyl Ether[1]	—	—	—	—	—	—	—
D	Ammonia[2]	928	498	gas	gas	15	28	0.6
D	n-Amyl Acetate	680	360	60	16	1.1	7.5	4.5
D	sec-Amyl Acetate	—	—	89	32	—	—	4.5
D	Aniline	1139	615	158	70	1.3	11	3.2
D	Benzene	928	498	12	-11	1.3	7.9	2.8
D	Benzyl Chloride	1085	585	153	67	1.1	—	4.4
B(D)	1,3-Butadiene[1]	788	420	gas	gas	2.0	12.0	1.9
D	Butane	550	288	-76	-60	1.6	8.4	2.0
D	1-Butanol	650	343	98	37	1.4	11.2	2.6
D	2-Butanol	761	405	75	24	1.7 @ 212°F	9.8 @ 212°F	2.6
D	n-Butyl Acetate	790	421	72	22	1.7	7.6	4.0
D	iso-Butyl Acetate	790	421	—	—	—	—	—
D	sec-Butyl Acetate	—	—	88	31	1.7	9.8	4.0
D	t-Butyl Acetate	—	—	—	—	—	—	—
D	n-Butyl Acrylate (inhibited)	559	293	118	48	1.5	9.9	4.4
C	n-Butyl Formal	—	—	—	—	—	—	—
B(C)	n-Butyl Glycidyl Ether[1]	—	—	—	—	—	—	—
C	Butyl Mercaptan	—	—	35	2	—	—	3.1
D	t-Butyl Toluene	—	—	—	—	—	—	—
D	Butylamine	594	312	10	-12	1.7	9.8	2.5
D	Butylene	725	385	gas	gas	1.6	10.0	1.9
C	n-Butyraldehyde	425	218	-8	-22	1.9	12.5	2.5
D	n-Butyric Acid	830	443	161	72	2.0	10.0	3.0
-[3]	Carbon Disulfide	194	90	-22	-30	1.3	50.0	2.6
C	Carbon Monoxide	1128	609	gas	gas	12.5	74.0	1.0
C	Chloroacetaldehyde	—	—	—	—	—	—	—
D	Chlorobenzene	1099	593	82	28	1.3	9.6	3.9
C	1-Chloro-1-Nitropropane	—	—	144	62	—	—	4.3
D	Chloroprene	—	—	-4	-20	4.0	20.0	3.0
D	Cresol	1038-1110	559-599	178-187	81-86	1.1-1.4	—	—
C	Crotonaldehyde	450	232	55	13	2.1	15.5	2.4
D	Cumene	795	424	96	36	0.9	6.5	4.1
D	Cyclohexane	473	245	-4	-20	1.3	8.0	2.9
D	Cyclohexanol	572	300	154	68	—	—	3.5
D	Cyclohexanone	473	245	111	44	1.1 @ 212°F	9.4	3.4
D	Cyclohexene	471	244	<20	<-7	—	—	2.8
D	Cyclopropane	938	503	gas	gas	2.4	10.4	1.5
D	p-Cymene	817	436	117	47	0.7 @ 212°F	5.6	4.6
C	n-Decaldehyde	—	—	—	—	—	—	—
D	n-Decanol	550	288	180	82	—	—	5.5
D	Decene	455	235	<131	<55	—	—	4.84
D	Diacetone Alcohol	1118	603	148	64	1.8	6.9	4.0
D	o-Dichlorobenzene	1198	647	151	66	2.2	9.2	5.1
D	1,1-Dichloroethane	820	438	22	-6	5.6	—	—
D	1,2-Dichloroethylene	860	460	36	2	5.6	12.8	3.4
C	1,1-Dichloro-1-Nitroethane	—	—	168	76	—	—	5.0
D	1,3-Dichloropropene	—	—	95	35	5.3	14.5	3.8
C	Dicyclopentadiene	937	503	90	32	—	—	—
D	Diethyl Benzene	743-842	395-450	133-135	56-57	—	—	4.6
C	Diethyl Ether	320	160	-49	-45	1.9	36.0	2.6
C	Diethylamine	594	312	-9	-23	1.8	10.1	2.5
C	Diethylaminoethanol	—	—	—	—	—	—	—
C	Diethylene Glycol Monobutyl Ether	442	228	172	78	0.85	24.6	5.6
C	Diethylene Glycol Monomethyl Ether	465	241	205	96	—	—	—
D	Di-isobutyl Ketone	745	396	120	49	0.8 @ 200°F	7.1 @ 200°F	4.9
D	Di-isobutylene	736	391	23	-5	0.8	4.8	3.9
C	Di-isopropylamine	600	316	30	-1	1.1	7.1	3.5
C	N-N-Dimethyl Aniline	700	371	145	63	—	—	4.2
D	Dimethyl Formamide	833	455	136	58	2.2 @ 212°F	15.2	2.5
D	Dimethyl Sulfate	370	188	182	83	—	—	4.4
C	Dimethylamine	752	400	gas	gas	2.8	14.4	1.6
C	1,4-Dioxane	356	180	54	12	2.0	22	3.0
D	Dipentene	458	237	113	45	0.7 @ 302°F	6.1 @ 302°F	4.7
C	Di-n-propylamine	570	299	63	17	—	—	3.5
C	Dipropylene Glycol Methyl Ether	—	—	186	86	—	—	5.11
D	Dodecene	491	255	—	—	—	—	—

TABLE I

Class I* Group	Substance	Auto-Ignition Temp.*		Flash Point**		Flammable Limits** Percent by Volume		Vapor Density** (Air Equals 1.0)
		°F	°C	°F	°C	Lower	Upper	
C	Epichlorohydrin	772	411	88	31	3.8	21.0	3.2
D	Ethane	882	472	gas	gas	3.0	12.5	1.0
D	Ethanol	685	363	55	13	3.3	19	1.6
D	Ethyl Acetate	800	427	24	-4	2.0	11.5	3.0
D	Ethyl Acrylate (inhibited)	702	372	50	10	1.4	14	3.5
D	Ethyl sec-Amyl Ketone	—	—	—	—	—	—	—
D	Ethyl Benzene	810	432	70	21	0.8	6.7	3.7
D	Ethyl Butanol	—	—	—	—	—	—	—
D	Ethyl Butyl Ketone	—	—	115	46	—	—	4.0
D	Ethyl Chloride	966	519	-58	-50	3.8	15.4	2.2
D	Ethyl Formate	851	455	-4	-20	2.8	16.0	2.6
D	2-Ethyl Hexanol	448	231	164	73	0.88	9.7	4.5
D	2-Ethyl Hexyl Acrylate	485	252	180	82	—	—	—
C	Ethyl Mercaptan	572	300	<0	<-18	2.8	18.0	2.1
C	n-Ethyl Morpholine	—	—	—	—	—	—	—
C	2-Ethyl-3-Propyl Acrolein	—	—	155	68	—	—	4.4
D	Ethyl Silicate	—	—	125	52	—	—	7.2
D	Ethylamine	725	385	<0	<-18	3.5	14.0	1.6
C	Ethylene	842	450	gas	gas	2.7	36.0	1.0
D	Ethylene Chlorohydrin	797	425	140	60	4.9	15.9	2.8
D	Ethylene Dichloride	775	413	56	13	6.2	16	3.4
C	Ethylene Glycol Monobutyl Ether	460	238	143	62	1.1 @ 200°F	12.7 @ 275°F	4.1
C	Ethylene Glycol Monobutyl Ether Acetate	645	340	160	71	0.88 @ 200°F	8.54 @ 275°F	—
C	Ethylene Glycol Monoethyl Ether	455	235	110	43	1.7 @ 200°F	15.6 @ 200°F	3.0
C	Ethylene Glycol Monoethyl Ether Acetate	715	379	124	52	1.7	—	4.72
D	Ethylene Glycol Monomethyl ether	545	285	102	39	1.8 @ STP	14 @ STP	2.6
B(C)	Ethylene Oxide[1]	804	429	-20	-28	3.0	100	1.5
D	Ethylenediamine	725	385	104	40	2.5	12.0	2.1
C	Ethylenimine	608	320	12	-11	3.3	54.8	1.5
C	2-Ethylhexaldehyde	375	191	112	44	0.85 @ 200°F	7.2 @ 275°F	4.4
B	Formaldehyde (Gas)	795	429	gas	gas	7.0	73	1.0
D	Formic Acid (90%)	813	434	122	50	18	57	1.6
B	Fuel and Combustible ProcessGas (containing more than 30 percent H_2 by volume)	—	—	—	—	—	—	—
D	Fuel Oils	410-765	210-407	100-336	38-169	0.7	5	—
C	Furfural	600	316	140	60	2.1	19.3	3.3
C	Furfuryl Alcohol	915	490	167	75	1.8	16.3	3.4
D	Gasoline	536-880	280-471	-36 to -50	-38 to -46	1.2-1.5	7.1-7.6	3-4
D	Heptane	399	204	25	-4	1.05	6.7	3.5
D	Heptene	500	260	<32	<0	—	—	3.39
D	Hexane	437	225	-7	-22	1.1	7.5	3.0
D	Hexanol	—	—	145	63	—	—	3.5
D	2-Hexanone	795	424	77	25	—	8	3.5
D	Hexenes	473	245	<20	<-7	—	—	3.0
D	sec-Hexyl Acetate	—	—	—	—	—	—	—
C	Hydrazine	74-518	23-270	100	38	2.9	9.8	1.1
B	Hydrogen	968	520	gas	gas	4.0	75	0.1
C	Hydrogen Cyanide	1000	538	0	-18	5.6	40.0	0.9
C	Hydrogen Selenide	—	—	—	—	—	—	—
C	Hydrogen Sulfide	500	260	gas	gas	4.0	44.0	1.2
D	Isoamyl Acetate	680	360	77	25	1.0 @ 212°F	7.5	4.5
D	Isoamyl Alcohol	662	350	109	43	1.2	9.0 @ 212°F	3.0
D	Isobutyl Acrylate	800	427	86	30	—	—	4.42
C	Isobutyraldehyde	385	196	-1	-18	1.6	10.6	2.5
C	Isodecaldehyde	—	—	185	85	—	—	5.4
C	Iso-octyl Alcohol	—	—	180	82	—	—	—
C	Iso-octyl Aldehyde	387	197	—	—	—	—	—
D	Isophorone	860	460	184	84	0.8	3.8	—
D	Isoprene	428	220	-65	-54	1.5	8.9	2.4
D	Isopropyl Acetate	860	460	35	2	1.8 @ 100°F	8	3.5
D	Isopropyl Ether	830	443	-18	-28	1.4	7.9	3.5
C	Isopropyl Glycidyl Ether	—	—	—	—	—	—	—
D	Isopropylamine	756	402	-35	-37	—	—	2.0
D	Kerosene	410	210	110-162	43-72	0.7	5	—
D	Liquefied Petroleum Gas	761-842	405-450	—	—	—	—	—
	Manufactured Gas (see Fuel and Combustible Process Gas)	—	—	—	—	—	—	—
D	Mesityl Oxide	652	344	87	31	1.4	7.2	3.4
D	Methane	999	537	gas	gas	5.0	15.0	0.6
D	Methanol	725	385	52	11	6.0	36	1.1
D	Methyl Acetate	850	454	14	-10	3.1	16	2.8
D	Methyl Acrylate	875	468	27	-3	2.8	25	3.0
D	Methyl Amyl Alcohol	—	—	106	41	1.0	5.5	—
D	Methyl n-Amyl Ketone	740	393	102	39	1.1 @ 151°F	7.9 @ 250°F	3.9
C	Methyl Ether	662	350	gas	gas	3.4	27.0	1.6
D	Methyl Ethyl Ketone	759	404	16	-9	1.7 @ 200°F	11.4 @ 200°F	2.5
D	2-Methyl-5-Ethyl Pyridine	—	—	155	68	1.1	6.6	4.2
C	Methyl Formal	460	238	—	—	—	—	—
D	Methyl Formate	840	449	-2	-19	4.5	23	2.1
D	Methyl Isocyanate	994	534	19	-7	5.3	26	1.97
C	Methyl Mercaptan	—	—	—	—	3.9	21.8	1.7

TABLE I

Class I* Group	Substance	Auto-Ignition Temp.* °F	Auto-Ignition Temp.* °C	Flash Point** °F	Flash Point** °C	Flammable Limits** Percent by Volume Lower	Flammable Limits** Percent by Volume Upper	Vapor Density** (Air Equals 1.0)
D	Methyl Methacrylate	792	422	50	10	1.7	8.2	3.6
D	2-Methyl-1-Propanol	780	416	82	28	1.7 @ 123°F	10.6 @ 202°F	2.6
D	2-Methyl-2-Propanol	892	478	52	11	2.4	8.0	2.6
D	alpha-Methyl Styrene	1066	574	129	54	1.9	6.1	—
C	Methylacetylene	—	—	gas	gas	1.7	—	1.4
C	Methylacetylene-Propadiene (stabilized)	—	—	—	—	—	—	—
D	Methylamine	806	430	gas	gas	4.9	20.7	1.0
D	Methylcyclohexane	482	250	25	-4	1.2	6.7	3.4
D	Methylcyclohexanol	565	296	149	65	—	—	3.9
D	o-Methylcyclohexanone	—	—	118	48	—	—	3.9
D	Monoethanolamine	770	410	185	85	—	—	2.1
D	Monoisopropanolamine	705	374	171	77	—	—	2.6
C	Monomethyl Aniline	900	482	185	85	—	—	3.7
C	Monomethyl Hydrazine	382	194	17	-8	2.5	92	1.6
C	Morpholine	590	310	98	37	1.4	11.2	3.0
D	Naphtha (Coal Tar)	531	277	107	42	—	—	—
D	Naphtha (Petroleum)[4]	550	288	<0	<-18	1.1	5.9	2.5
D	Nitrobenzene	900	482	190	88	1.8 @ 200°F	—	4.3
C	Nitroethane	778	414	82	28	3.4	—	2.6
C	Nitromethane	785	418	95	35	7.3	—	2.1
C	1-Nitropropane	789	421	96	36	2.2	—	3.1
C	2-Nitropropane	802	428	75	24	2.6	11.0	3.1
D	Nonane	401	205	88	31	0.8	2.9	4.4
D	Nonene	—	—	78	26	—	—	4.35
D	Nonyl Alcohol	—	—	165	74	0.8 @ 212°F	6.1 @ 212°F	5.0
D	Octane	403	206	56	13	1.0	6.5	3.9
D	Octene	446	230	70	21	—	—	3.9
D	n-Octyl Alcohol	—	—	178	81	—	—	4.5
D	Pentane	470	243	<-40	<-40	1.5	7.8	2.5
D	1-Pentanol	572	300	91	33	1.2	10.0 @ 212°F	3.0
D	2-Pentanone	846	452	45	7	1.5	8.2	3.0
D	1-Pentene	527	275	0	-18	1.5	8.7	2.4
D	Phenylhydrazine	—	—	190	88	—	—	—
D	Propane	842	450	gas	gas	2.1	9.5	1.6
D	1-Propanol	775	413	74	23	2.2	13.7	2.1
D	2-Propanol	750	399	53	12	2.0	12.7 @ 200°F	2.1
D	Propiolactone	—	—	165	74	2.9	—	2.5
C	Propionaldehyde	405	207	-22	-30	2.6	17	2.0
D	Propionic Acid	870	466	126	52	2.9	12.1	2.5
D	Propionic Anhydride	545	285	145	63	1.3	9.5	4.5
D	n-Propyl Acetate	842	450	55	13	1.7 @ 100°F	8	3.5
C	n-Propyl Ether	419	215	70	21	1.3	7.0	3.53
B	Propyl Nitrate	347	175	68	20	2	100	—
D	Propylene	851	455	gas	gas	2.0	11.1	1.5
D	Propylene Dichloride	1035	557	60	16	3.4	14.5	3.9
B(C)	Propylene Oxide[1]	840	449	-35	-37	2.3	36	2.0
D	Pyridine	900	482	68	20	1.8	12.4	2.7
D	Styrene	914	490	88	31	0.9	6.8	3.6
C	Tetrahydrofuran	610	321	6	-14	2.0	11.8	2.5
D	Tetrahydronaphthalene	725	385	160	71	0.8 @ 212°F	5.0 @ 302°F	4.6
C	Tetramethyl Lead	—	—	100	38	—	—	6.5
D	Toluene	896	480	40	4	1.1	7.1	3.1
D	Tridecene	—	—	—	—	—	—	—
C	Triethylamine	480**	249**	16	-9	1.2	8.0	3.5
D	Triethylbenzene	—	—	181	83	—	—	5.6
D	Tripropylamine	—	—	105	41	—	—	4.9
D	Turpentine	488	253	95	35	0.8	—	—
D	Undecene	—	—	—	—	—	—	—
C	Unsymmetrical Dimethyl Hydrazine (UDMH)	480	249	5	-15	2	95	2.0
C	Valeraldehyde	432	222	54	12	—	—	3.0
D	Vinyl Acetate	756	402	18	-8	2.6	13.4	3.0
D	Vinyl Chloride	882	472	-108.4	-78	3.6	33.0	2.2
D	Vinyl Toluene	921	494	127	53	0.8	11.0	4.1
D	Vinylidene Chloride	1058	570	-19	-28	6.5	15.5	3.4
D	Xylenes	867-984	464-529	81-90	27-32	1.0-1.1	7.0	3.7

[1] If equipment is isolated by sealing all conduit 1/2 in. or larger, in accordance with Section 501.15(A) of NFPA 70, *National Electrical Code*, equipment for the group classification shown in parentheses is permitted.

[2] For classification of areas involving Ammonia, see *Safety Code for Mechanical Refrigeration*, ANSI/ASHRAE 15, and *Safety Requirements for the Storage and Handling of Anhydrous Ammonia*, ANSI/CGA G2.1.

[3] Certain chemicals may have characteristics that require safeguards beyond those required for any of the above groups. Carbon disulfide is one of these chemicals because of its low autoignition temperature and the small joint clearance to arrest its flame propagation.

[4] Petroleum Naphtha is a saturated hydrocarbon mixture whose boiling range is 20° to 135°C. It is also known as benzine, ligroin, petroleum ether, and naphtha.

*Data from NFPA 497 - 2004, *Recommended Practice for the Classification of Flammable Liquids, Gases, Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas.*

**Data from NFPA 325M-1991, *Fire Hazard Properties of Flammable Liquids, Gases, and Volatile Solids.*

TABLE II	Minimum Cloud or Layer Ignition Temp.[1]		
Class II, Group G	°F		°C
Aluminum, atomized collector fines	1022	Cl	550
Aluminum, A422 flake	608		320
Aluminum — cobalt alloy (60-40)	1058		570
Aluminum — copper alloy (50-50)	1526		830
Aluminum — lithium alloy (15% Li)	752		400
Aluminum — magnesium alloy (Dowmetal)	806	Cl	430
Aluminum — nickel alloy (58-42)	1004		540
Aluminum — silicon alloy (12% Si)	1238	NL	670
Boron, commercial-amorphous (85% B)	752		400
Calcium Silicide	1004		540
Chromium, (97%) electrolytic, milled	752		400
Ferromanganese, medium carbon	554		290
Ferrosilicon (88%, 9% Fe)	1472		800
Ferrotitanium (19% Ti, 74.1% Fe, 0.06% C)	698	Cl	370
Iron, 98%, H_2 reduced	554		290
Iron, 99%, Carbonyl	590		310
Magnesium, Grade B, milled	806		430
Manganese	464		240
Silicon, 96%, milled	1436	Cl	780
Tantalum	572		300
Thorium, 1.2%, O_2	518	Cl	270
Tin, 96%, atomized (2% Pb)	806		430
Titanium, 99%	626	Cl	330
Titanium Hydride, (95% Ti, 3.8% H_2)	896	Cl	480
Vanadium, 86.4%	914		490
Zirconium Hydride, (93.6% Zr, 2.1% H_2)	518		270
Class II, Group F			
CARBONACEOUS DUSTS			
Asphalt, (Blown Petroleum Resin)	950	Cl	510
Charcoal	356		180
Coal, Kentucky Bituminous	356		180
Coal, Pittsburgh Experimental	338		170
Coal, Wyoming	—		—
Gilsonite	932		500
Lignite, California	356		180
Pitch, Coal Tar	1310	NL	710
Pitch, Petroleum	1166	NL	630
Shale, Oil	—		—
Class II, Group G			
AGRICULTURAL DUSTS			
Alfalfa Meal	392		200
Almond Shell	392		200
Apricot Pit	446		203
Cellulose	500		260
Cherry Pit	428		220
Cinnamon	446		230
Citrus Peel	518		270
Cocoa Bean Shell	698		370
Cocoa, natural, 19% fat	464		240
Coconut Shell	428		220
Corn	482		250
Corncob Grit	464		240
Corn Dextrine	698		370
Cornstarch, commercial	626		330
Cornstarch, modified	392		200
Cork	410		210
Cottonseed Meal	392		200
Cube Root, South Amer.	446		230
Flax Shive	446		230
Garlic, dehydrated	680	NL	360
Guar Seed	932	NL	500
Gum, Arabic	500		260
Gum, Karaya	464		240
Gum, Manila (copal)	680	Cl	360
Gum, Tragacanth	500		260
Hemp Hurd	428		220

TABLE II	Minimum Cloud or Layer Ignition Temp.[1]		
Class II, Group G	°F		°C
Lycopodium	590		310
Malt Barley	482		250
Milk, Skimmed	392		200
Pea Flour	500		260
Peach Pit Shell	410		210
Peanut Hull	410		210
Peat, Sphagnum	464		240
Pecan Nut Shell	410		210
Pectin	392		200
Potato Starch, Dextrinated	824	NL	440
Pyrethrum	410		210
Rauwolfia Vomitoria Root	446		230
Rice	428		220
Rice Bran	914	NL	490
Rice Hull	428		220
Safflower Meal	410		210
Soy Flour	374		190
Soy Protein	500		260
Sucrose	662	Cl	350
Sugar, Powdered	698	Cl	370
Tung, Kernels, Oil-Free	464		240
Walnut Shell, Black	428		220
Wheat	428		220
Wheat Flour	680		360
Wheat Gluten, gum	968	NL	520
Wheat Starch	716	NL	380
Wheat Straw	428		220
Woodbark, Ground	482		250
Wood Flour	500		260
Yeast, Torula	500		260
CHEMICALS			
Acetoacetanilide	824	M	440
Acetoacet-p-phenetidide	1040	NL	506
Adipic Acid	1022	M	550
Anthranilic Acid	1076	M	580
Aryl-nitrosomethylamide	914	NL	490
Azelaic Acid	1130	M	610
2,2-Azo-bis-butyronitrile	662		350
Benzoic Acid	824	M	440
Benzotriazole	824	M	440
Bisphenol-A	1058	M	570
Chloroacetoacetanilide	1184	M	640
Diallyl Phthalate	896	M	480
Dicumyl Peroxide (suspended on $CaCO_3$), 40-60	356		180
Dicyclopentadiene Dioxide	788	NL	420
Dihydroacetic Acid	806	NL	430
Dimethyl Isophthalate	1076	M	580
Dimethyl Terephthalate	1058	M	570
3,5 - Dinitrobenzoic Acid	860	NL	460
Dinitrotoluamide	932	NL	500
Diphenyl	1166	M	630
Ditertiary Butyl Paracresol	878	NL	470
Ethyl Hydroxyethyl Cellulose	734	NL	390
Fumaric Acid	968	M	520
Hexamethylene Tetramine	770	S	410
Hydroxyethyl Cellulose	770	NL	410
Isotoic Anhydride	1292	NL	700
Methionine	680		360
Nitrosoamine	518	NL	270
Para-oxy-benzaldehyde	716	Cl	380
Paraphenylene Diamine	1148	M	620
Paratertiary Butyl Benzoic Acid	1040	M	560
Pentaerythritol	752	M	400
Phenylbetanaphthylamine	1256	NL	680
Phthalic Anydride	1202	M	650
Phthalimide	1166	M	630
Salicylanilide	1130	M	610

TABLE II	Minimum Cloud or Layer Ignition Temp.[1]		
Class II, Group G	°F		°C
Sorbic Acid	860		460
Stearic Acid, Aluminum Salt	572		300
Stearic Acid, Zinc Salt	950	M	510
Sulfur	428		220
Terephthalic Acid	1256	NL	680
DRUGS			
2-Acetylamino-5-nitrothiazole	842		450
2-Amino-5-nitrothiazole	860		460
Aspirin	1220	M	660
Gulasonic Acid, Diacetone	788	NL	420
Mannitol	860	M	460
Nitropyridone	806	M	430
l-Sorbose	698	M	370
Vitamin B1, mononitrate	680	NL	360
Vitamin C (Ascorbic Acid)	536		280
DYES, PIGMENTS, INTERMEDIATES			
Beta-naphthalene-azo-Dimethylaniline	347	347	175
Green Base Harmon Dye	347	347	175
Red Dye Intermediate	347	347	175
Violet 200 Dye	347	347	175
PESTICIDES			
Benzethonium Chloride	716	CI	380
Bis(2-Hydroxy-5-chlorophenyl) methane	1058	NL	570
Crag No. 974	590	CI	310
Dieldrin (20%)	1022	NL	550
2, 6-Ditertiary-butyl-paracresol	788	NL	420
Dithane	356		180
Ferbam	302		150
Manganese Vancide	2448		120
Sevin	284		140
∝ ∝ - Trithiobis (N,N-Dimethylthio-formamide)	446		230
THERMOPLASTIC RESINS AND MOLDING COMPOUNDS			
Acetal Resins			
Acetal, Linear (Polyformaldehyde)	824	NL	440
Acrylic Resins			
Acrylamide Polymer	464		240
Acrylonitrile Polymer	860		460
Acrylonitrile-Vinyl Pyridine Copolymer	464		240
Acrylonitrile-Vinyl Chloride-Vinylidene Chloride Copolymer (70-20-10)	410		210
Methyl Methacrylate Polymer	824	NL	440
Methyl Methacrylate-Ethyl Acrylate Copolymer	896	NL	408
Methyl Methacrylate-Ethyl Acrylate-Styrene Copolymer	824	NL	440
Methyl Methacrylate-Styrene-Butadiene-Acrylonitrile Copolymer	896	NL	480
Methacrylic Acid Polymer	554		290
Cellulosic Resins			
Cellulose Acetate	644		340
Cellulose Triacetate	806	NL	430
Cellulose Acetate Butyrate	698	NL	370
Cellulose Propionate	860	NL	460
Ethyl Cellulose	608	CI	320
Methyl Cellulose	644		340

[1]Normally, the minimum ignition temperature of a layer of a specific dust is lower than the minimum ignition temperature of a cloud of that dust. Since this is not universally true, the lower of the two minimum ignition temperatures is listed. If no symbol appears between the two temperature columns, then the layer ignition temperature is shown. "CI" means the cloud ignition temperature is shown. "NL" means that no layer ignition temperature is available and the cloud ignition temperature is shown. "M" signifies that the dust layer melts before it ignites; the cloud ignition temperature is shown. "S" signifies that the dust layer sublimes before it ignites; the cloud ignition temperature is shown.

[2]Certain metal dusts may have characteristics that require safe-guards beyond those required for atmospheres containing the dusts of aluminum, magnesium, and their commercial alloys. For example, zirconium, thorium, and uranium dusts have extremely low ignition temperatures (as low as 20°C) and minimum ignition energies lower than any material classified in any of the Class I or Class II groups.

Data from NFPA 499 - 2004, *Recommended Practice for the Classification of Combustible Dusts and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas.*

TABLE II	Minimum Cloud or Layer Ignition Temp.[1]		
Class II, Group G	°F		°C
Carboxymethyl Cellulose	554		290
Hydroxyethyl Cellulose	644		340
Chlorinated Polyether Resins			
Chlorinated Polyether Alcohol	806		460
Nylon (Polyamide) Resins			
Nylon Polymer (Polyhexa-methylene Adipamide)	806		430
Polycarbonate Resins			
Polycarbonate	1310	NL	710
Polyethylene Resins			
Polyethylene, High Pressure Process	716		380
Polyethylene, Low Pressure Process	788	NL	420
Polyethylene Wax	752	NL	400
Polymethylene Resins			
Carboxypolymethylene	968	NL	520
Polypropylene Resins			
Polypropylene (No Antioxidant)	788	NL	420
Rayon Resins			
Rayon (Viscose) Flock	482		250
Styrene Resins			
Polystyrene Molding Cmpd.	1040	NL	560
Polystyrene Latex	932		500
Styrene-Acrylonitrile (70-30)	932	NL	500
Styrene-Butadiene Latex (>75% Styrene; Alum Coagulated)	824	NL	440
Vinyl Resins			
Polyvinyl Acetate	1022	NL	550
Polyvinyl Acetate/Alcohol	824		440
Polyvinyl Butyral	734	NL	390
Vinyl Chloride-Acrylonitrile Copolymer	878		470
Polyvinyl Chloride-Dioctyl Phthalate Mixture	608	NL	320
Vinyl Toluene-Acrylonitrile Butadiene Copolymer	936	NL	530
THERMOSETTING RESINS AND MOLDING COMPOUNDS			
Allyl Resins			
Allyl Alcohol Derivative (CR-39)	932	NL	500
Amino Resins			
Urea Formaldehyde Molding Compound	860	NL	460
Urea Formaldehyde-Phenol Formaldehyde Molding Compound (Wood Flour Filler)	464		240
Epoxy Resins			
Epoxy	1004	NL	540
Epoxy - Bisphenol A	950	NL	510
Phenol Furfural	590		310
Phenolic Resins			
Phenol Formaldehyde	1076	NL	580
Phenol Formaldehyde Molding Cmpd. (Wood Flour Filler)	932	NL	500
Phenol Formaldehyde, Polyalkylene-Polyamine Modified	554		290
Polyester Resins			
Polyethylene Terephthalate	1076	NL	580
Styrene Modified Polyester-Glass Fiber Mixture	680		360
Polyurethane Resins			
Polyurethane Foam, No Fire Retardant	824		440
Polyurethane Foam, Fire Retardant	734		390
SPECIAL RESINS AND MOLDING COMPOUNDS			
Alkyl Ketone Dimer Sizing Compound	320		160
Cashew Oil, Phenolic, Hard	356		180
Chlorinated Phenol	1058	NL	570
Coumarone-Indene, Hard	968	NL	520
Ethylene Oxide Polymer	662	NL	350
Ethylene-Maleic Anhydride Copolymer	1004	NL	540
Lignin, Hydrolized, Wood-Type, Fines	842	NL	450
Petrin Acrylate Monomer	428	NL	220
Petroleum Resin (Blown Asphalt)	932		500
Rosin, DK	734	NL	390
Rubber, Crude, Hard	662	NL	350
Rubber, Synthetic, Hard (33% S)	608	NL	320
Shellac	752	NL	400
Sodium Resinate	428		220
Styrene — Maleic Anhydride Copolymer	878	CI	470

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22 Safety Instrumented Systems (SIS)

- Safety Integrity Level (SIL) is defined as a relative level of risk-reduction provided by a safety function, or to specify a target level of risk reduction. In simple terms, SIL is a measurement of performance required for a Safety Instrumented Function (SIF).
- Four SILs are defined, with SIL4 being the most dependable and SIL1 being the least. A SIL is determined based on a number of quantitative factors in combination with qualitative factors such as development process and safety life cycle management.

SIL	IEC 61508	ANSI S84.01	PFD	Availabilty Required	1/PFD
4	YES	NO	10^{-5} to 10^{-4}	> 99.99%	100,000 to 10,000
3	YES	YES	10^{-4} to 10^{-3}	99.90 to 99.99%	10,000 to 1,000
2	YES	YES	10^{-3} to 10^{-2}	99.90 to 99.99%	1,000 to 100
1	YES	YES	10^{-2} to 10^{-1}	99.90 to 99.99%	100 to 10

PFD (Probability of Failure on Demand) and RRF (Risk Reduction Factor = 1/PFD)

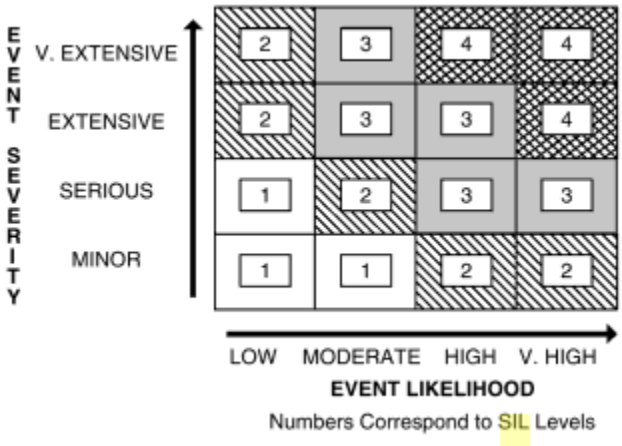
Qualitative View of SIL

SIL	Generalized View
4	Catastrophic Community Impact
3	Employee and Community Impact
2	Major Property and Production Protection. Possible Injury to employee
1	Minor Property and Production Protection

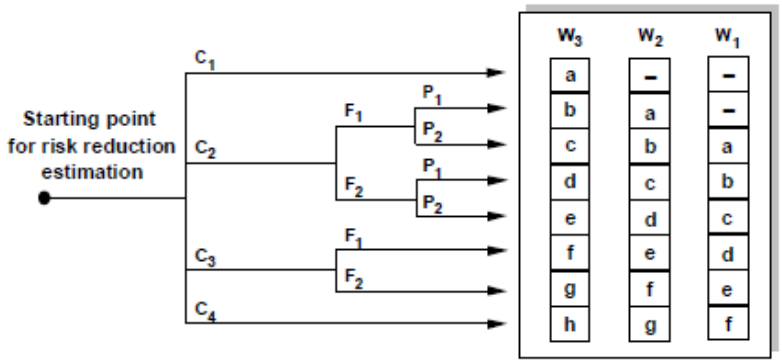
Consequence only decision table

SIL	Generalized View
4	Potential for fatalities in the community
3	Potential for multiple fatalities
2	Potential for major serious injuries or one fatality
1	Potential for minor injuries

Hazard or Risk Analysis Matrix



Hazard or Risk Analysis Graph



a, b, c, d, e, f, g, h represent the necessary minimum risk reduction. The link between the necessary minimum risk reduction and the safety integrity level is shown in the table.

C = Consequence risk parameter

F = Frequency and exposure time risk parameter

P = Possibility of avoiding hazard risk parameter

W = Probability of the unwanted occurrence

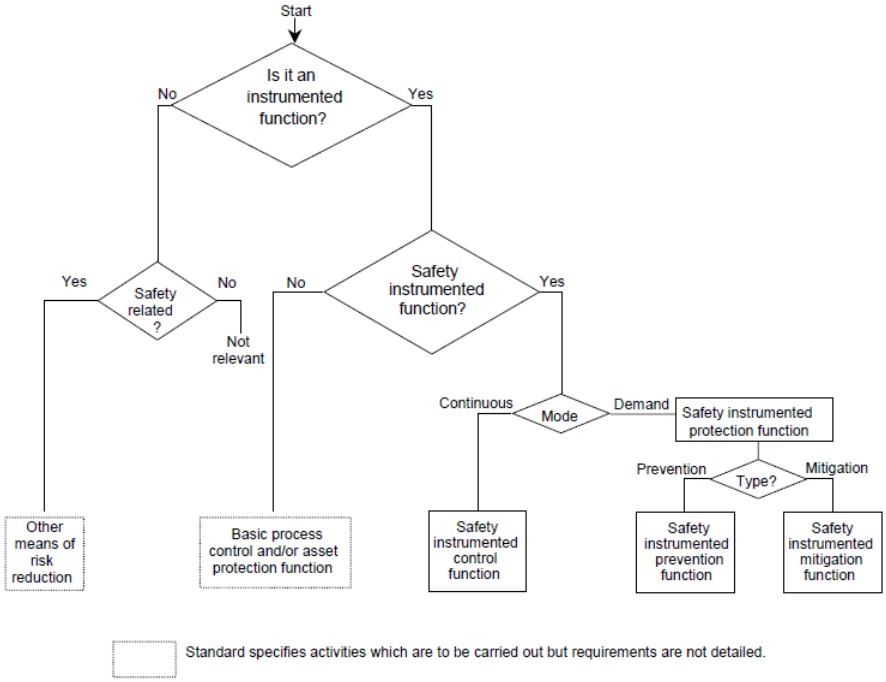
a, b, c ... h = Estimates of the required risk reduction for the SRSs

Necessary minimum risk reduction	Safety integrity level
–	No safety requirements
a	No special safety requirements
b, c	1
d	2
e, f	3
g	4
h	An E/E/PE SRS is not sufficient

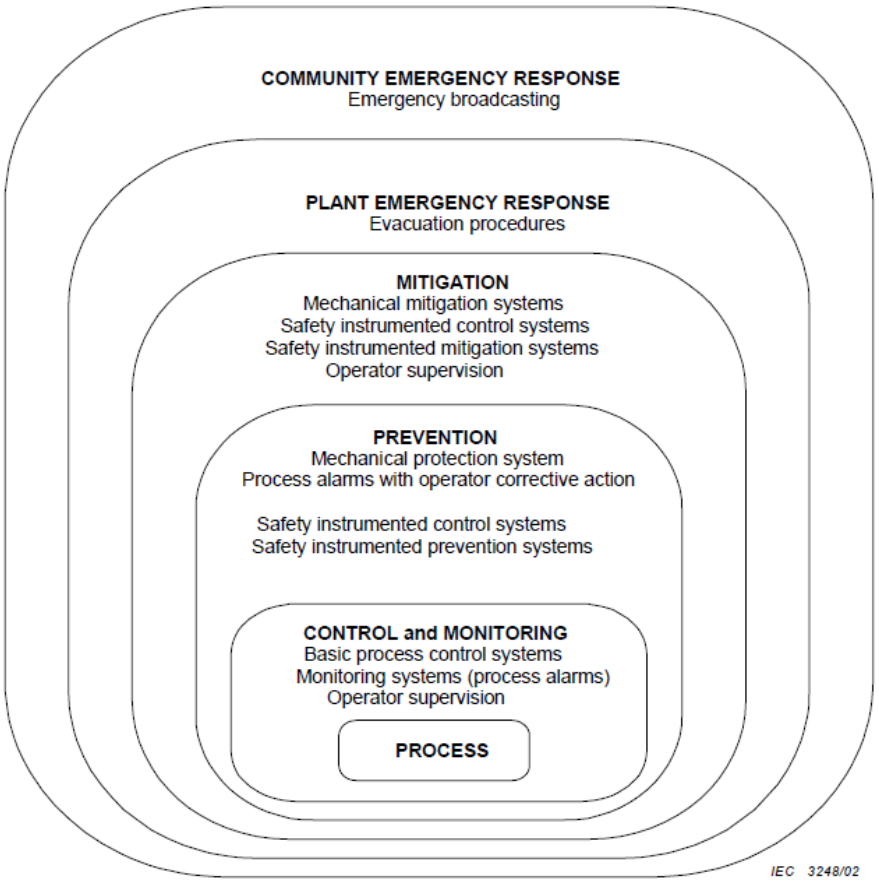
Risk parameter		Classification
Consequence (C)	C_1 C_2 C_3 C_4	Minor injury Serious permanent injury to one or more persons; death to one person Death to several people Very many people killed
Frequency of, and exposure time in, the hazardous zone (F)	F_1 F_2	Rare to more often exposure in the hazardous zone Frequent to permanent exposure in the hazardous zone
Possibility of avoiding the hazardous event (P)	P_1 P_2	Possible under certain conditions Almost impossible
Probability of the unwanted occurrence (W)	W_1 W_2 W_3	A very slight probability that the unwanted occurrences will come to pass and only a few unwanted occurrences are likely A slight probability that the unwanted occurrences will come to pass and few unwanted occurrences are likely A relatively high probability that the unwanted occurrences will come to pass and frequent unwanted occurrences are likely

General System Recommendations			
SIL	Sensors	Logic	Final Elements
1	Simplex switches or transmitters	Relays Solid state systems General purpose PLCs	Simplex “dumb”
2	Redundant switches / xmtrs Xmtrs with comparison Simplex safety xmtrs	Relays Fail safe or fully tested solid state systems Certified safety PLCs	Redundant “dumb” Simplex “smart” (e.g. partial stroking valves)
3	Redundant xmtrs	Certified safety PLCs	Redundant “smart” (e.g. partial stroking valves)

Relationship between Safety Instrumented Function and other Functions



Typical Risk Reduction Methods found in process plants



22.1 Determining PFD (Probability of Failure on Demand):

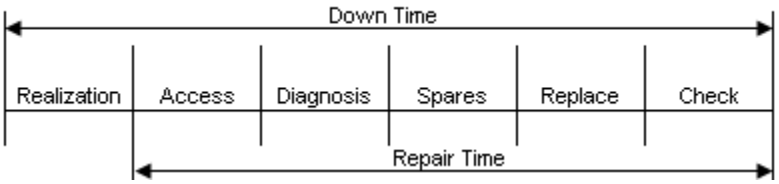
Failure Rate = # of Failures / Unit of Time (expressed in hours)
Constant Failure Rate assumed for normal life of device
MTTF = 1 / Failure Rate

Basic Reliability Formulas:

Configuration	$MTTF_{sp}$	PFD
1oo1	$\frac{1}{\lambda_s}$	$\lambda_{du} = \frac{TI}{2}$
1oo2	$\frac{1}{2\lambda_s}$	$\left[(\lambda_{du})^2 \times \frac{TI^2}{3} \right]$
2oo2	$\left[\frac{1}{2(\lambda_s)^2 \times MTTR} \right]$	$(\lambda_{du})^2 \times TI$
2oo3	$\left[\frac{1}{6(\lambda_s)^2 \times MTTR} \right]$	$(\lambda_{du})^2 \times TI^2$

λ = Failure Rate
MTTR = Mean Time to Repair
TI = Test Interval
λ_s = Safe Failure = $MTTF_s$ / ($MTTF_s$ + MTTR)
λ_{du} = Dangerous Undetected Failure = $MTTF_d$ / ($MTTF_d$ + TI / 2 + MTTR)

Downtime vs Repair Time:



Fault-free system	Degradation 1	Degradation 2	Degradation 3	System structure
2oo4	1oo2	Shutdown		Safety related and fault tolerant
1oo3	Shutdown			Safety related
2oo3	1oo2	1oo1*	Shutdown	Safety related and fault tolerant *with time restriction
2oo3	1oo2	Shutdown		Safety related and fault tolerant
1oo2D	1oo1D	Shutdown		Safety related and fault tolerant with time restriction
1oo2	Shutdown			Safety related
2oo2	1oo1	Shutdown		Safety related and fault tolerant
1oo1	Shutdown			Safety related

Part 4 of the IEC 61508 gives the defintion of
MooN : M out of N channel architecture (for example 1oo2 is 1 out of 2 architecture, where either of the two channels can perform the safety function)
MooND : M out of N channel architecture with diagnostic

23 Codes Standards & Regulations

ISA Standards Listing: Instrumentation, Systems, and Automation Society, formerly the Instrument Society of America, a professional organization for instrument engineers

- ISA2 - Manometer Tables
- ISA5 - Documentation of Measurement and Control Instruments and Systems
- ISA7 - Instrument Air Standards Committee
- ISA12 - Electrical Equipment for Hazardous Locations
- ISA18 - Instrument Signals and Alarms
- ISA20 - Instrument Specification Forms
- ISATAG31 - Electrical Apparatus for Explosive Atmospheres
- ISA37 - Measurement Transducers
- ISA42 - Nomenclature for Instrument Tube Fittings
- ISA50 - Signal Compatibility of Electrical Instruments (includes ISA-50.02 Fieldbus Standard for Use in Industrial Control Systems)
- ISA52 - Environments for Standards Laboratories
- ISA55 - Hardware Testing of Digital Process Computers
- ISA60 - Control Centers
- ISATAG65 - Industrial-Process Measurement and Control
- ISATAG66 - Safety of Measuring Control and Associated Equipment
- ISA67 - Nuclear Power Plant Standards
- ISA71 - Environmental Conditions for Process Measurement and Control
- ISA74 - Continuous Weighing Instrumentation
- ISA75 - Control Valve Standards
- ISA76 - Composition Analyzers
- ISA77 - Fossil Power Plant Standards
- ISA82 - Electrical and Electronic Instrumentation
- ISA84 - Electrical/Electronic/Programmable Electronic Systems (E/E/P)
- ISATAG85 - Measuring Equipment for Basic Electrical Quantities
- ISA88 - Batch Control Systems (includes the batch processing standards ISA-88.01 Models and Terminology, ISA-88.00.02 Data Structures and Guidelines for Languages, ISA-88.00.03 General and Site Recipe Models and Representation, and ISA-88-00.04 Batch Production Records.
- ISA90 - Instrumentation and Computer Grounding
- ISA91 - Criticality Ranking for Instrumentation
- ISA92 - Performance Requirements for Industrial Air Measurement Instrumentation Related to Health and Safety
- ISA93 - Sealing Technologies Committee
- ISA95 - Enterprise/Control Integration Committee
- ISA96 - Valve Actuator Committee
- ISA97 - In-Line Sensors Committee
- ISA98 - Personnel Certification Standards Committee
- ISA99 - Manufacturing and Control Systems Security
- ISA100 - Wireless Systems for Automation
- ISA101 - Human-Machine Interface
- ISA102 - High-Power Research & Development Electrical Systems
- ISA103 - Field Device Tools (FDT)
- ISA104 - Electronic Device Description Language (EDDL)

ASME = American Society of Mechanical Engineers

- Boiler & Pressure Vessel Code
- Pressure Relief Devices

NFPA = National Fire Protection Agency

- NFPA 70 - National Electrical Code
- NFPA 70E - Standard for Electrical Safety in the Workplace
- NFPA 77 – Static Electricity
- NFPA 79 – Industrial Machinery
- NFPA 496 - Standard for Purged and Pressurized Enclosures for Electrical Equipment

- NFPA 497 - Recommended Practice for the Classification of Flammable Liquids, Gases, or Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas
- NFPA 780 – Lightning Protection

IEC = International Electrotechnical Commission: not-for-profit, non-governmental international standards organization that prepares and publishes International Standards for all electrical, electronic and related technologies

- IEC 61508 - Functional safety of electrical/electronic/programmable electronic safety-related systems
- IEC 61511 - Functional safety - safety instrumented systems for the process industry sector

CSA = Canadian Standards Association is a not-for-profit association composed of representatives from government, industry, and consumer groups

- C22.1 – Canadian National Electrical Code

UL = Underwriters Laboratories is a U.S. privately owned and operated, independent, third party product safety testing and certification organization

FM = Factory Mutual FM Approvals offers worldwide certification and testing services of industrial and commercial property loss prevention products

CE = "Conformité Européenne" **CE marking** (also known as **CE mark**) is a mandatory conformity mark on many products placed on the single market in the European Economic Area (EEA). The CE marking certifies that a product has met EU consumer safety, health or environmental requirements.

- No 3rd party testing: The approach to conformity enables manufacturers to use what is called as "SELF DECLARATION" where the manufacturer himself declares conformity by signing the "Declaration of Conformity (DOC)" and then affixes the CE Mark on his product by following a seven step procedure.

ANSI = American National Standards Institute is a private non-profit organization that oversees the development of voluntary consensus standards for products, services, processes, systems, and personnel in the United States. The organization also coordinates U.S. standards with international standards so that American products can be used worldwide.

IEEE = Institute of Electrical & Electronic Engineers is an international non-profit, professional organization for the advancement of technology related to electricity.

AICHE = American Institute of Chemical Engineers

OSHA = Occupational Safety and Health Administration Its mission is to prevent work-related injuries, illnesses, and deaths by issuing and enforcing rules (called standards) for workplace safety and health. The agency is headed by a Deputy Assistant Secretary of Labor. Standards are covered under 29CFR.

- PART 1910 Occupational Safety and Health Standards

  Subpart S -- Electrical

  GENERAL

  1910.301 Introduction.

  DESIGN SAFETY STANDARDS FOR ELECTRICAL SYSTEMS

  1910.302 Electric utilization systems.
  1910.303 General requirements.
  1910.304 Wiring design and protection.
  1910.305 Wiring methods, components, and equipment for general use.
  1910.306 Specific purpose equipment and installations.
  1910.307 Hazardous (classified) locations.
  1910.308 Special systems.
  1910.309 - 1910.330 [Reserved]

  SAFETY-RELATED WORK PRACTICES

  1910.331 Scope.
  1910.332 Training.
  1910.333 Selection and use of work practices.
  1910.334 Use of equipment.
  1910.335 Safeguards for personnel protection.
  1910.336 - 1910.360 [Reserved]

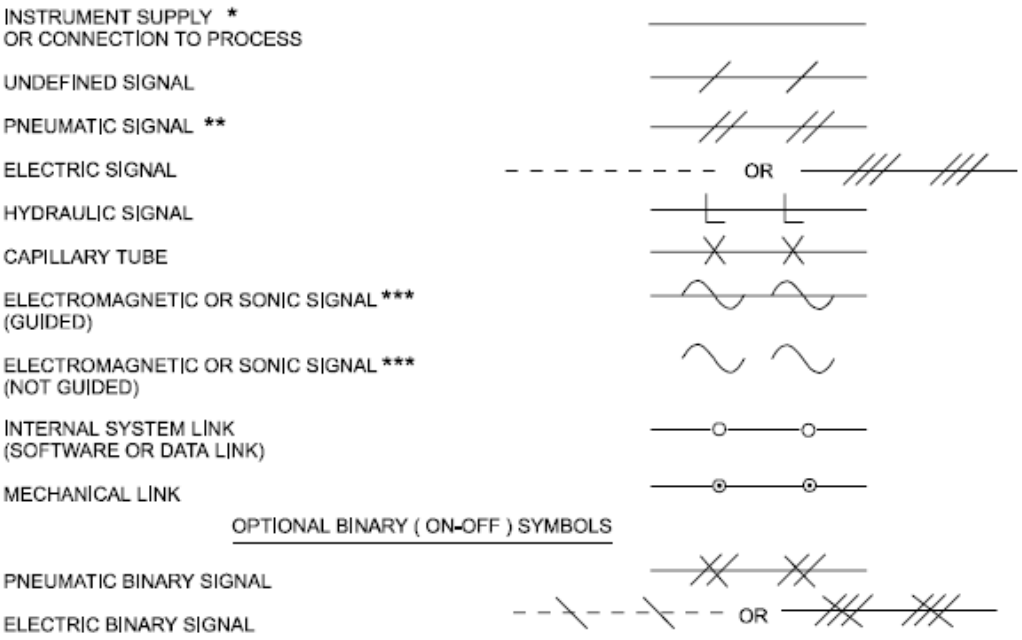
24 System Documentation

24.1 ISA:

24.1.1 Identification Letters

	FIRST-LETTER (4)		SUCCEEDING-LETTERS (3)		
	MEASURED OR INITIATING VARIABLE	MODIFIER	READOUT OR PASSIVE FUNCTION	OUTPUT FUNCTION	MODIFIER
A	Analysis (5,19)		Alarm		
B	Burner, Combustion		User's Choice (1)	User's Choice (1)	User's Choice (1)
C	User's Choice (1)			Control (13)	
D	User's Choice (1)	Differential (4)			
E	Voltage		Sensor (Primary Element)		
F	Flow Rate	Ratio (Fraction) (4)			
G	User's Choice (1)		Glass, Viewing Device (9)		
H	Hand				High (7, 15, 16)
I	Current (Electrical)		Indicate (10)		
J	Power	Scan (7)			
K	Time, Time Schedule	Time Rate of Change (4, 21)		Control Station (22)	
L	Level		Light (11)		Low (7, 15, 16)
M	User's Choice (1)	Momentary (4)			Middle, Intermediate (7,15)
N	User's Choice (1)		User's Choice (1)	User's Choice (1)	User's Choice (1)
O	User's Choice (1)		Orifice, Restriction		
P	Pressure, Vacuum		Point (Test) Connection		
Q	Quantity	Integrate, Totalize (4)			
R	Radiation		Record (17)		
S	Speed, Frequency	Safety (8)		Switch (13)	
T	Temperature			Transmit (18)	
U	Multivariable (6)		Multifunction (12)	Multifunction (12)	Multifunction (12)
V	Vibration, Mechanical Analysis (19)			Valve, Damper, Louver (13)	
W	Weight, Force		Well		
X	Unclassified (2)	X Axis	Unclassified (2)	Unclassified (2)	Unclassified (2)
Y	Event, State or Presence (20)	Y Axis		Relay, Compute, Convert (13, 14, 18)	
Z	Position, Dimension	Z Axis		Driver, Actuator, Unclassified Final Control Element	

24.1.2 Instrument Line Symbols



NOTE: 'OR' means user's choice. Consistency is recommended.

* The following abbreviations are suggested to denote the types of power supply. These designations may also be applied to purge fluid supplies.

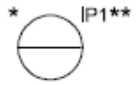
AS - Air Supply
IA - Instrument Air
PA - Plant Air
} Options
ES - Electric Supply
GS - Gas Supply
HS - Hydraulic Supply
NS - Nitrogen Supply
SS - Steam Supply
WS - Water Supply

The supply level may be added to the instrument supply line, e.g., AS-100, a 100-psig air supply; ES-24DC, a 24-volt direct current power supply.

** The pneumatic signal symbol applies to a signal using any gas as the signal medium. If a gas other than air is used, the gas may be identified by a note on the signal symbol or otherwise.

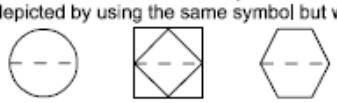
*** Electromagnetic phenomena include heat, radio waves, nuclear radiation, and light.

24.1.3 Instrument & Function Symbols

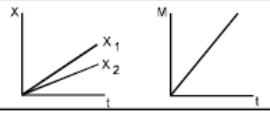
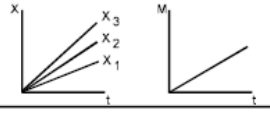
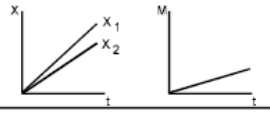
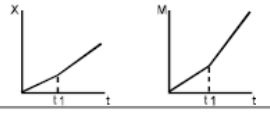
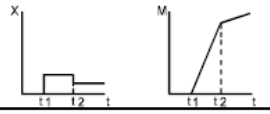
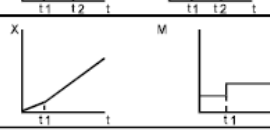
	PRIMARY LOCATION ***NORMALLY ACCESSIBLE TO OPERATOR	FIELD MOUNTED	AUXILIARY LOCATION ***NORMALLY ACCESSIBLE TO OPERATOR
DISCRETE INSTRUMENTS	1 	2 	3 
SHARED DISPLAY, SHARED CONTROL	4 	5 	6 
COMPUTER FUNCTION	7 	8 	9 
PROGRAMMABLE LOGIC CONTROL	10 	11 	12 

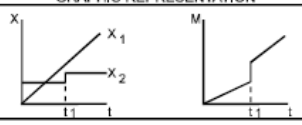
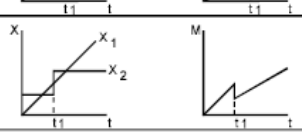
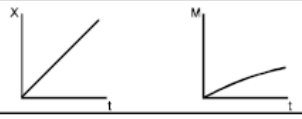
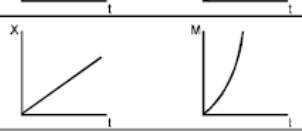
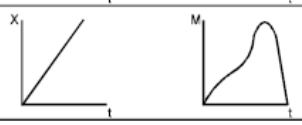
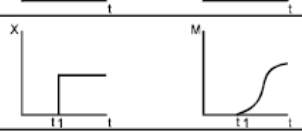
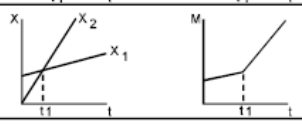
* Symbol size may vary according to the user's needs and the type of document. A suggested square and circle size for large diagrams is shown above. Consistency is recommended.

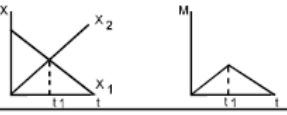
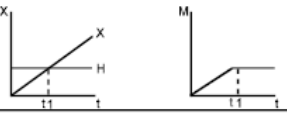
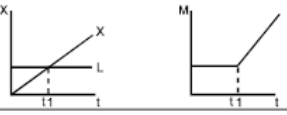
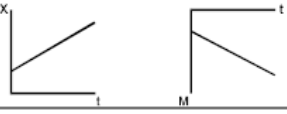
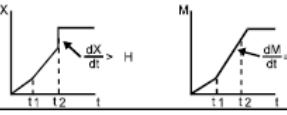
** Abbreviations of the user's choice such as IP1 (Instrument Panel #1), IC2 (Instrument Console #2), CC3 (Computer Console #3), etc., may be used when it is necessary to specify instrument or function location.

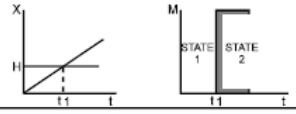
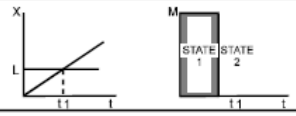
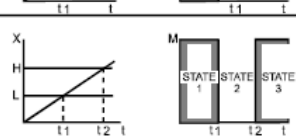
*** Normally inaccessible or behind-the-panel devices or functions may be depicted by using the same symbol but with dashed horizontal bars, i.e. 

24.1.4 Function Blocks – Function Designations

NO	FUNCTION	SYMBOL	MATH EQUATION	GRAPHIC REPRESENTATION	DEFINITION
1	SUMMING	Σ	$M = X_1 + X_2 + \ldots + X_n$		THE OUTPUT EQUALS THE ALGEBRAIC SUM OF THE INPUTS. (THE INPUTS MAY BE LABELED WITH POSITIVE OR NEGATIVE SIGNS).
2	AVERAGING	$\Sigma/_n$	$M = \frac{X_1 + X_2 + \ldots + X_n}{n}$		THE OUTPUT EQUALS THE ALGEBRAIC SUM OF THE INPUTS DIVIDED BY THE NUMBER OF INPUTS.
3	DIFFERENCE	Δ	$M = X_1 - X_2$		THE OUTPUT EQUALS THE ALGEBRAIC DIFFERENCE OF THE TWO INPUTS.
4	PROPORTIONAL	K 1:1 2:1	$M = KX$		THE OUTPUT IS DIRECTLY PROPORTIONAL TO THE INPUT. IN THE CASE OF A VOLUME BOOSTER, "K" MAY BE REPLACED BY 1:1 FOR INTEGER GAINS, 2:1, 3:1, ETC., MAY BE SUBSTITUTED FOR K.
5	INTEGRAL	$\int$	$M = \frac{1}{T_I} \int X dt$		THE OUTPUT VARIES IN ACCORDANCE WITH BOTH MAGNITUDE AND DURATION OF THE INPUT, THE OUTPUT IS PROPORTIONAL TO THE TIME INTEGRAL OF THE INPUT.
6	DERIVATIVE	$d/_{dt}$	$M = T_D \frac{dX}{dt}$		THE OUTPUT IS PROPORTIONAL TO THE RATE OF CHANGE (DERIVATIVE) OF THE INPUT.

NO	FUNCTION	SYMBOL	MATH EQUATION	GRAPHIC REPRESENTATION	DEFINITION
7	MULTIPLYING	X	$M = X_1 X_2$		THE OUTPUT EQUALS THE PRODUCT OF THE TWO INPUTS.
8	DIVIDING	÷	$M = \frac{X_1}{X_2}$		THE OUTPUT EQUALS THE QUOTIENT OF THE TWO INPUTS.
9	ROOT EXTRACTION	$\sqrt[n]{}$	$M = \sqrt[n]{X}$		THE OUTPUT EQUALS THE ROOT (I.E., CUBE ROOT, FOURTH ROOT, 3/2 ROOT, ETC.) OF THE INPUT. IF n IS OMITTED, A SQUARE ROOT IS ASSUMED.
10	EXPONENTIAL	X^n	$M = X^n$		THE OUTPUT EQUALS THE INPUT RAISED TO A POWER (I.E., SECOND, THIRD, FOURTH, ETC.).
11	NONLINEAR OR UNSPECIFIED FUNCTION	f (X)	M = f (X)		THE OUTPUT EQUALS SOME NONLINEAR OR UNSPECIFIED FUNCTION OF THE INPUT.
12	TIME FUNCTION	f (t)	M = Xf (t) M = f (t)		THE OUTPUT EQUALS THE INPUT TIMES SOME FUNCTION OF TIME OR EQUALS SOME FUNCTION OF TIME ALONE.
13	HIGH SELECTING	>	$M = \begin{cases} X_1 \text{FOR } X_1 \geq X_2 \\ X_2 \text{FOR } X_1 \leq X_2 \end{cases}$		THE OUTPUT IS EQUAL TO THE GREATER OF THE INPUTS.

NO	FUNCTION	SYMBOL	MATH EQUATION	GRAPHIC REPRESENTATION	DEFINITION
14	LOW SELECTING	<	$M = \begin{cases} X_1 \text{ FOR } X_1 \le X_2 \\ X_2 \text{ FOR } X_1 \ge X_2 \end{cases}$		THE OUTPUT IS EQUAL TO THE LESSER OF THE INPUTS.
15	HIGH LIMITING	>	$M = \begin{cases} X \text{ FOR } X \le H \\ H \text{ FOR } X \ge H \end{cases}$		THE OUTPUT EQUALS THE INPUT OR THE HIGH LIMIT VALUE WHICHEVER IS LOWER.
16	LOW LIMITING	<	$M = \begin{cases} X \text{ FOR } X \ge L \\ L \text{ FOR } X \le L \end{cases}$		THE OUTPUT EQUALS THE INPUT OR THE LOW LIMIT VALUE WHICHEVER IS HIGHER.
17	REVERSE PROPORTIONAL	−K	$M = -KX$		THE OUTPUT IS REVERSELY PROPORTIONAL TO THE INPUT.
18	VELOCITY LIMITER	V	$\frac{dM}{dt} = \frac{dX}{dt} \begin{cases} \frac{dX}{dt} \le H \text{ AND} \\ M = X \end{cases}$ $\frac{dM}{dt} = H \begin{cases} \frac{dX}{dt} \ge H \text{ OR} \\ M \ne X \end{cases}$		THE OUTPUT EQUALS THE INPUT AS LONG AS THE RATE OF CHANGE OF THE INPUT DOES NOT EXCEED A LIMIT VALUE, THE OUTPUT WILL CHANGE AT THE RATE ESTABLISHED BY THIS LIMIT UNTIL THE OUTPUT AGAIN EQUALS THE INPUT.
19	BIAS	+ − ±	$M = X \pm b$		THE OUTPUT EQUALS THE INPUT PLUS (OR MINUS) SOME ARBITRARY VALUE (BIAS).
20	CONVERT	*/*	OUTPUT = f (INPUT)	NONE	THE FORM OF THE OUTPUT SIGNAL IS DIFFERENT FROM THAT OF THE INPUT. *E - VOLTAGE, I - CURRENT, P - PNEUMATIC, A - ANALOG, B - BINARY *H - HYDRAULIC, O - ELECTROMAGNETIC, SONIC, R - RESISTANCE (ELECT.), D - DIGITAL

NO	FUNCTION	SYMBOL	MATH EQUATION	GRAPHIC REPRESENTATION	DEFINITION
21	SIGNAL MONITOR	** H	STATE 1 $X \le H$ STATE 2 (ENERGIZED OR ALARM STATE) $X > H$		THE OUTPUT HAS DISCRETE STATES WHICH ARE DEPENDENT ON THE VALUE OF THE INPUT. WHEN THE INPUT EXCEEDS (OR BECOMES LESS THAN) AN ARBITRARY LIMIT VALUE THE OUTPUT CHANGES STATE.
		** L	STATE 1 (ENERGIZED OR ALARM STATE) $X < L$ STATE 2 $X \ge L$		
		** HL	STATE 1 (FIRST OUTPUT M_1 ENERGIZED OR ALARM STATE) $X < L$ STATE 2 (BOTH OUTPUTS INACTIVE OR DE-ENERGIZED) $L \le X \le H$ STATE 3 (SECOND OUTPUT M_2 ENERGIZED OR ALARM STATE) $X > H$		

THE VARIABLES USED IN THE TABLE ARE:

- b - ANALOG BIAS VALUE.
- $\frac{d}{dt}$ - DERIVATIVE WITH RESPECT TO TIME.
- H - AN ARBITRARY ANALOG HIGH LIMIT VALUE.
- $\frac{1}{T_1}$ - INTEGRATING RATE.
- L - AN ARBITRARY ANALOG LOW LIMIT VALUE.
- M - ANALOG OUTPUT VARIABLE.
- n - NUMBER OF ANALOG INPUTS OR VALUE OF EXPONENT.
- t - TIME
- T_D - DERIVATIVE TIME.
- X - ANALOG INPUT VARIABLE.
- $X_1, X_2, X_3, \ldots\ldots, X_n$ - ANALOG INPUT VARIABLE (1 TO N IN NUMBER).
- * - TABLE 1 LETTER DESIGNATORS.

NOTE: THE SQUARE MAY BE USED AS A FLAG

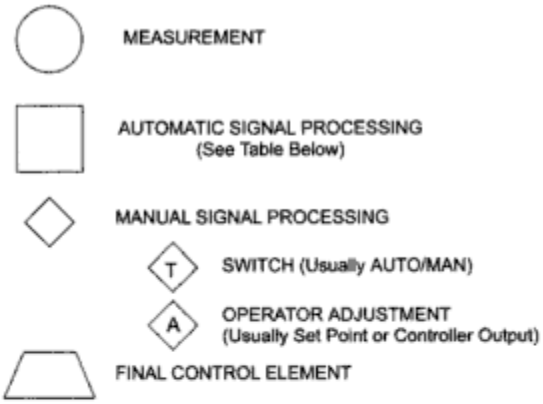
I-O	ON-OFF
REV	REVERSE ACTION

THIS TABLE HAS BEEN MODIFIED FROM SAMA PMC 22-11-1981 WITH PERMISSION OF THE COPYRIGHT HOLDER, SCIENTIFIC APPARATUS MAKERS ASSOCIATION. COPIES OF PMC 22-11-1981 ARE AVAILABLE FROM ISA, 67 ALEXANDER DR., RESEARCH TRIANGLE PARK, NC 27709; TELEPHONE (919) 549-8411.

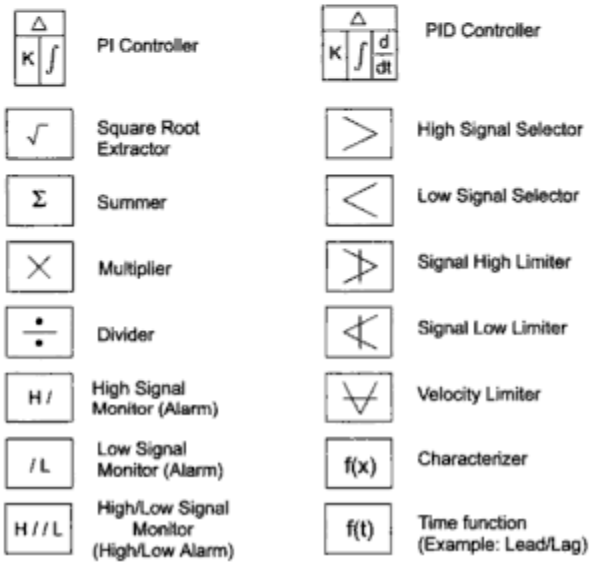
24.2 SAMA

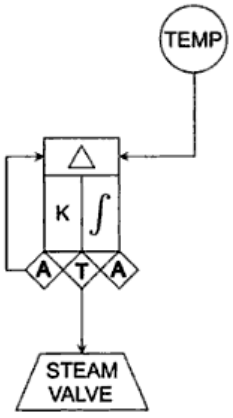
Scientific Apparatus Makers Association

GENERAL SYMBOLS

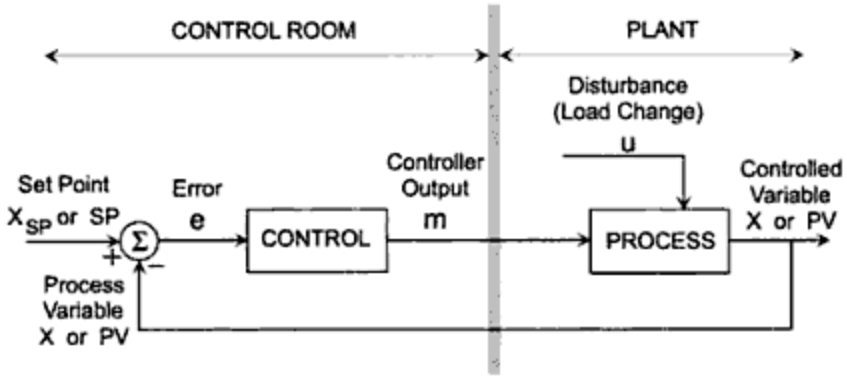


REPRESENTATIVE AUTOMATIC SIGNAL PROCESSING SYMBOLS

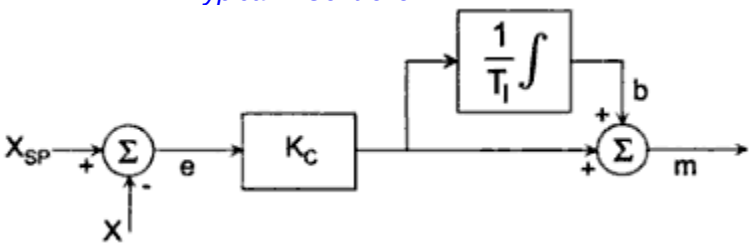




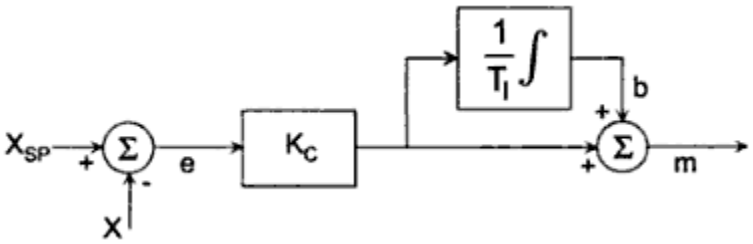
24.3 Block Diagram:



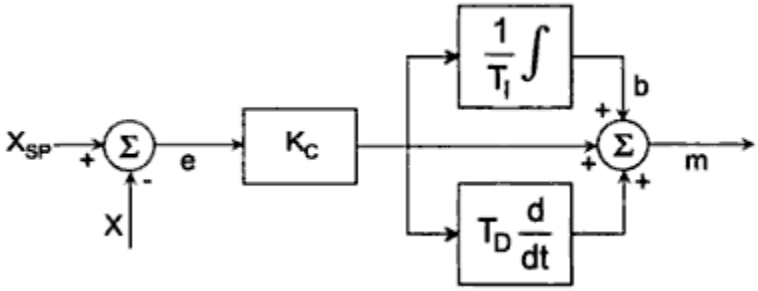
Typical P Controller



Typical PI Controller



Typical PID Controller



Block Diagrams:

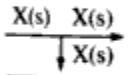
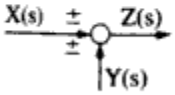
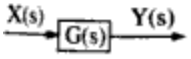
Two main symbols are a circle, which indicates a summation of two signals, and a rectangle, which indicates multiplication of a signal by a constant K or by a transfer function G(s). In control analysis the most frequent block diagram manipulations involve feedback loops of some sort. Rules for reducing such systems into a signle transfer function are identified below:

$$\text{Output} = \frac{(\text{Input})\left(\begin{array}{c}\text{Product of blocks in the forward}\\ \text{path between input signal}\\ \text{and output signal}\end{array}\right)}{1 + \left(\begin{array}{c}\text{Product of blocks in the}\\ \text{control loop}\end{array}\right)}$$

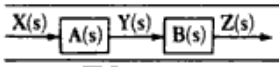
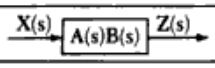
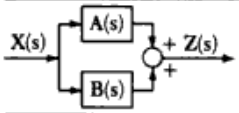
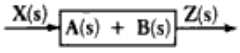
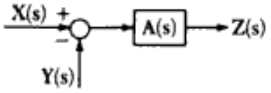
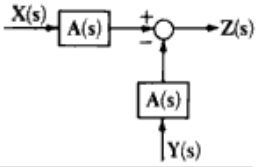
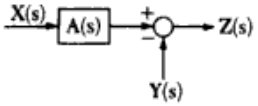
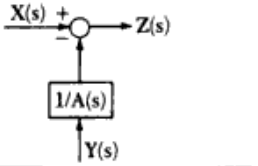
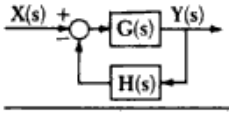
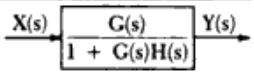
For Systems with more than one input:

$$\text{Output} = \sum_{i=1}^{N} \frac{(\text{Input-i})\left(\begin{array}{c}\text{Product of blocks in the}\\ \text{forward path between input-i}\\ \text{and the output signal}\end{array}\right)}{1 + \left(\begin{array}{c}\text{Product of blocks in the}\\ \text{control loop}\end{array}\right)}$$

Block Diagram Symbols:

Symbol	*Interpretation*
	Input or output signal; arrow gives direction.
	Branch point. Division of a signal to give two or more paths without modification.
	Summing point. $Z(s) = \pm X(s) \pm Y(s)$
	System element. $Y(s) = G(s)X(s)$

Manipulations of Block Diagrams:

Original Form	*Converted Form*
	
	
	
	
	

25 Miscellaneous Tables / Information

25.1 Wet Bulb / Dry Bulb

Dry Bulb Temperature - *Tdb*

The Dry Bulb temperature, usually referred to as air temperature, is the air property that is most common used. When people refer to the temperature of the air, they are normally referring to its dry bulb temperature.

The Dry Bulb Temperature refers basically to the ambient air temperature. It is called "Dry Bulb" because the air temperature is indicated by a thermometer not affected by the moisture of the air.

Dry-bulb temperature - *Tdb*, can be measured using a normal thermometer freely exposed to the air but shielded from radiation and moisture. The temperature is usually given in degrees Celsius (°C) or degrees Fahrenheit (°F). The SI unit is Kelvin (K). Zero Kelvin equals to -273°C.

The dry-bulb temperature is an indicator of heat content and is shown along the bottom axis of the psychometric chart. Constant dry bulb temperatures appear as vertical lines in the psychometric chart.

Wet Bulb Temperature - *Twb*

The Wet Bulb temperature is the temperature of adiabatic saturation. This is the temperature indicated by a moistened thermometer bulb exposed to the air flow.

Wet Bulb temperature can be measured by using a thermometer with the bulb wrapped in wet muslin. The adiabatic evaporation of water from the thermometer and the cooling effect is indicated by a "wet bulb temperature" lower than the "dry bulb temperature" in the air.

The rate of evaporation from the wet bandage on the bulb, and the temperature difference between the dry bulb and wet bulb, depends on the humidity of the air. The evaporation is reduced when the air contains more water vapor.

The wet bulb temperature is always lower than the dry bulb temperature but will be identical with 100% relative humidity (the air is at the saturation line).

Combining the dry bulb and wet bulb temperature in a psychometric diagram or Mollier chart, gives the state of the humid air. Lines of constant wet bulb temperatures run diagonally from the upper left to the lower right in the Psychometric Chart.

Dew Point Temperature - *Tdp*

The Dew Point is the temperature at which water vapor starts to condense out of the air, the temperature at which air becomes completely saturated. Above this temperature the moisture will stay in the air.

If the dew-point temperature is close to the air temperature, the relative humidity is high, and if the dew point is well below the air temperature, the relative humidity is low.

If moisture condensates on a cold bottle from the refrigerator, the dew-point temperature of the air is above the temperature in the refrigerator.

The Dew Point temperature can be measured by filling a metal can with water and ice cubes. Stir by a thermometer and watch the outside of the can. When the vapor in the air starts to condensate on the outside of the can, the temperature on the thermometer is pretty close to the dew point of the actual air.

The Dew Point is given by the saturation line in the psychometric chart.

Entropy's: measured in BTU/lbm-°R, represents the unavailability of energy (°R=Rankine temperature scale where 0°R = absolute zero and 460°R = 0°F). The second law of thermodynamics states that when heat is transferred from high temperature to low temperature regions, some of the heat will be rejected and not converted into mechanical work. Entropy is a measure of how much heat must be rejected to a lower temperature receiver at a given pressure and temperature.

Enthalpy 'h': *(inherent heat)* measured in British thermal units per pound (mass), or BTU/lbm, represents the total energy content of steam. It expresses the internal energy and flow work, or the total potential energy and kinetic energy contained within a substance. The advantage of *enthalpy* is that we can express in one term all of the energy in a substance which is due to its pressure and temperature. Enthalpy values are used to represent the energy level of steam entering a turbine, a value useful for determining turbine efficiency. By superheating steam, we can add enthalpy to steam without raising the pressure of the steam. For example, steam at 620 psig and 850°F can do more work in a turbine than steam that is 620 psig and 650°F

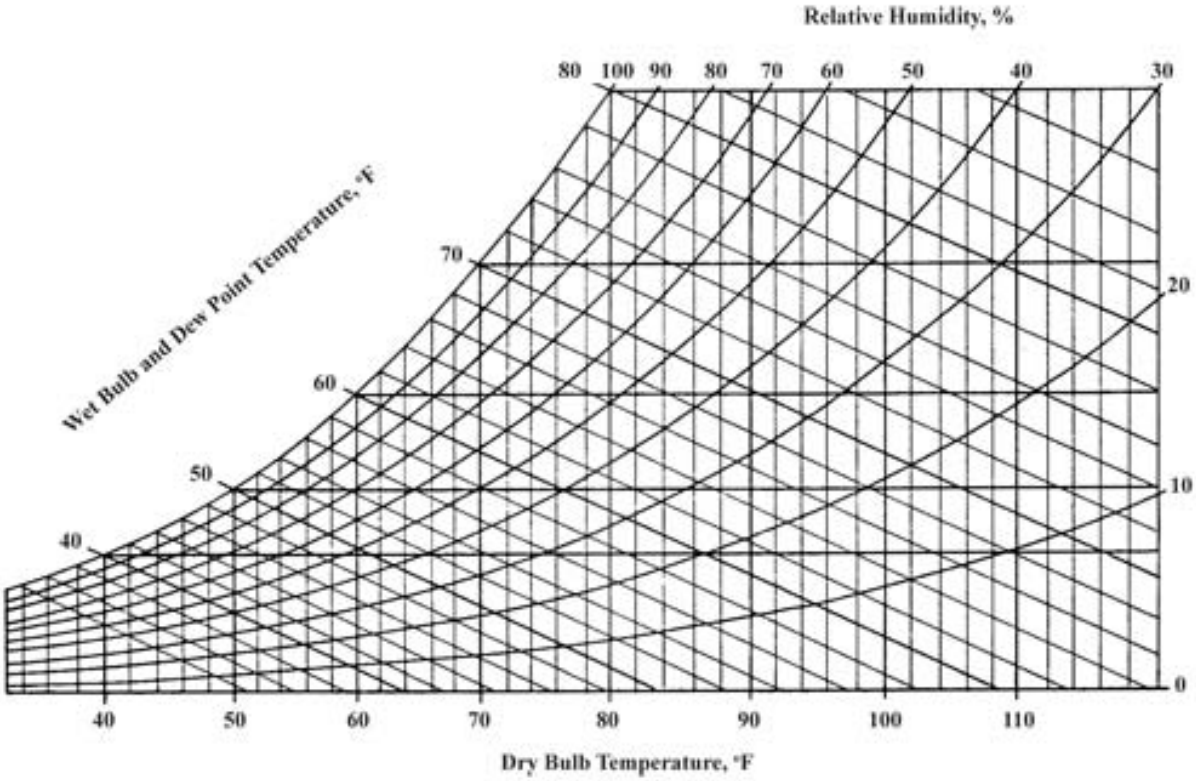
Adiabatic process: is a state change where there is no transfer of heat to or from the system during the process. Because heat transfer is relatively slow, any rapidly performed process can approach being adiabatic. Compression and expansion of working fluids are frequently achieved adiabatically with pumps and turbines.

Isothermal process: is a state change in which no temperature change occurs. Note that heat transfer can occur without causing a change in temperature of the working fluid. In the DFT (Deaerating Feed Tank), auxiliary exhaust heats incoming condensate, then condenses to liquid and falls to the bottom of the tank. Throughout this process, the temperature of the auxiliary exhaust remains constant at 246-249°F.

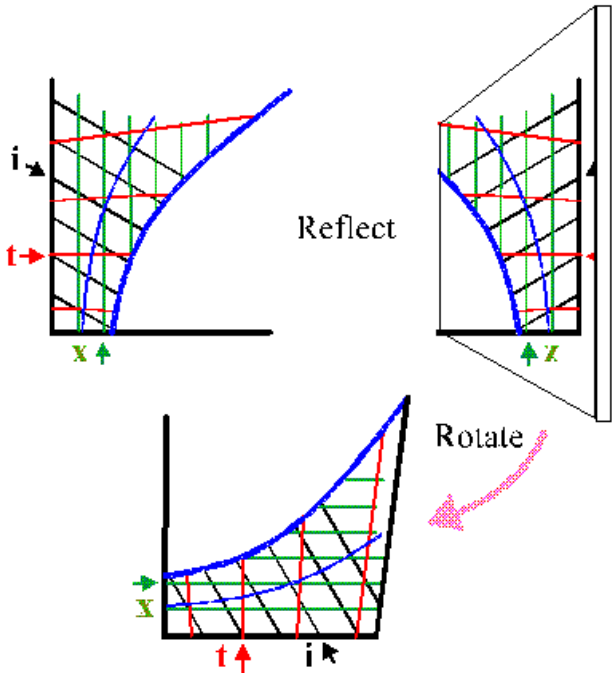
Heat Addition and Temperature:

- When heat is added to a material, one of two things will occur: the material will change temperature or the material will change state. When a substance is below the temperature at a given pressure required to change state, the addition of *sensible heat* will raise the temperature of the substance. Sensible heat applied to a pot of water will raise its temperature until it boils. Once the substance reaches the necessary temperature at a given pressure to change state, the addition of *latent heat* causes the substance to change state. Adding latent heat to the boiling water does not get the water any hotter, but changes the liquid (water) into a gas (steam).
- One can state that a certain amount of heat is required to raise the temperature of a substance 1°F. This energy is called the *specific heat capacity*. The specific heat capacity of a substance depends upon the volume and pressure of the material. For water, the specific heat capacity is 1 BTU/lbm-°F and remains constant. This means that if we add 1 BTU of heat to 1 lbm of water, the temperature will rise 1°F.

25.2 Psychometric Chart

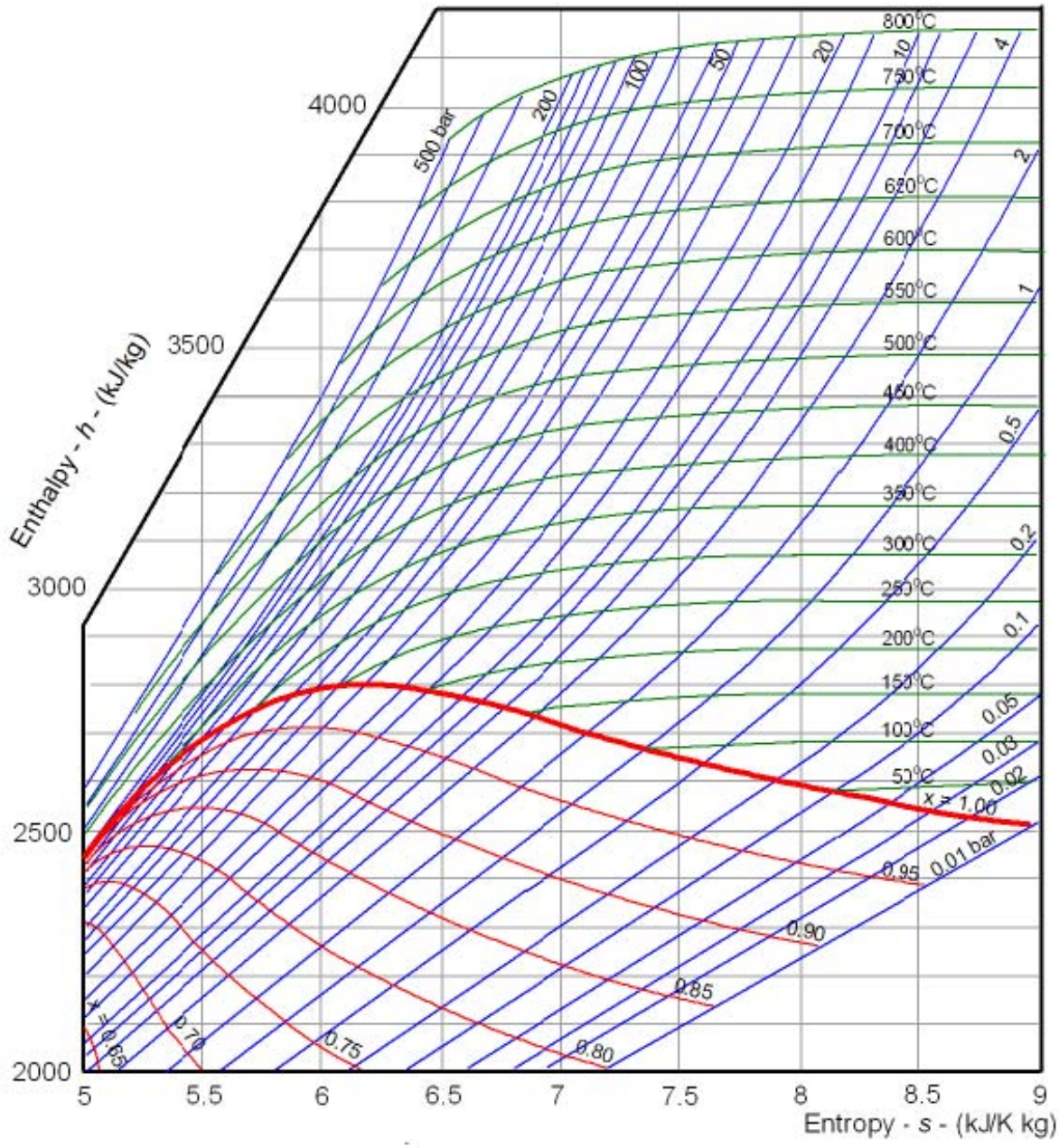


The Psychometric Chart is the same as the Mollier diagram, first reflected in a vertical mirror and then rotated through 90 degrees:



25.3 Mollier Steam Diagram

The diagram below can be used to determine enthalpy versus entropy of water and steam. The Mollier diagram is useful when analyzing the performance of adiabatic steady-flow processes, such as flow in nozzles, diffusers, turbines and compressors.



25.3.1 How To Read Mollier Diagram

Example:

Superheated Steam at 700psia and 680°F is expanded at constant entropy to 140psia.

- Locate point1 at the intersection of the 700psia and 680°F line – then read 'h' (h = 1333 BTU/lbm).
- Follow the entropy line downward vertically to the 140psia line and read 'h' (h = 1178 BTU/lbm)

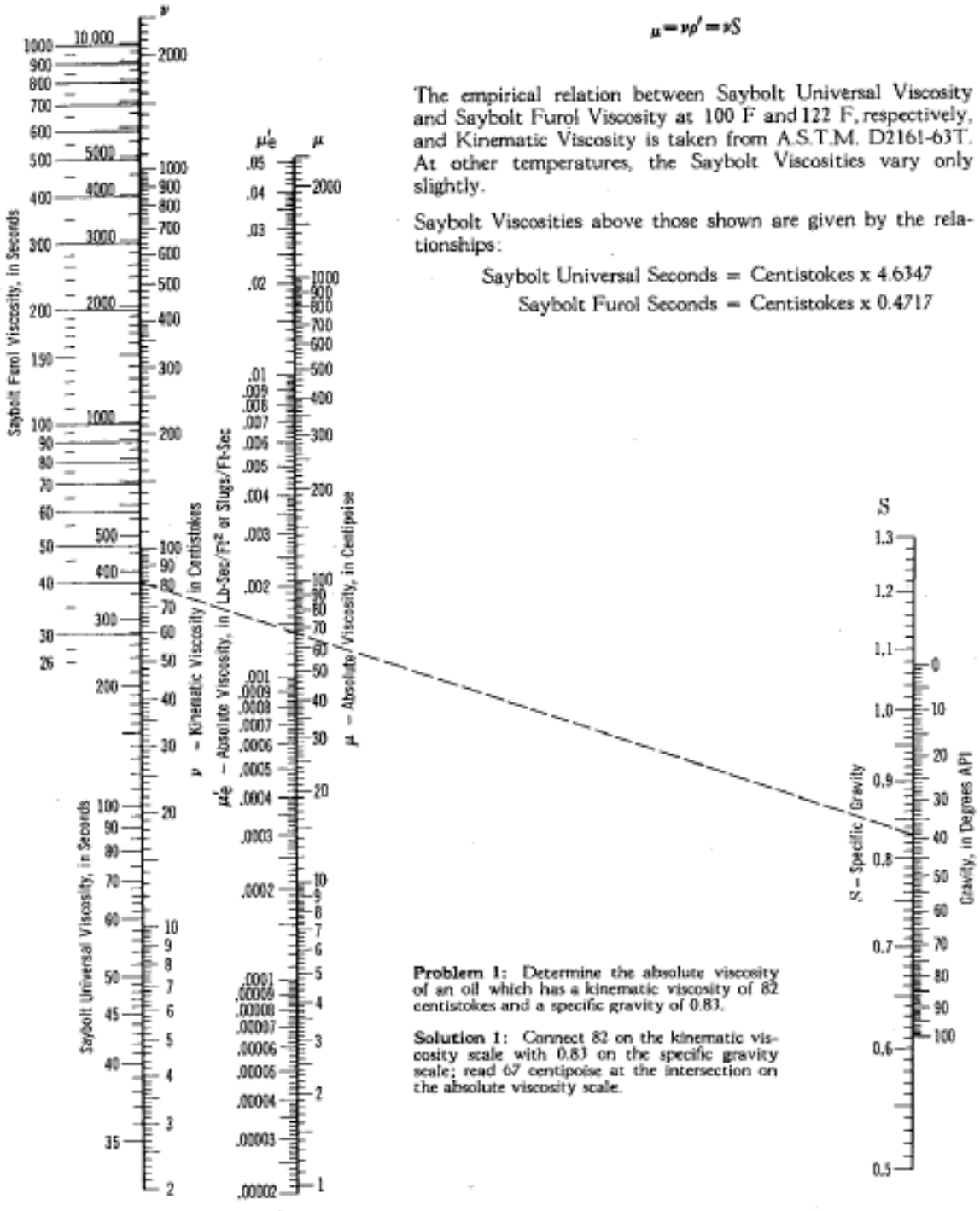
∴ h = 1178 – 1333 = –155 BTU/lbm

25.3.2 Properties of Saturated Steam:

Pressure PSIG	Saturated Temp. °F	Specific Volume Cu. Ft. per lb.		Heat Content BTU per lb.		Latent Heat of Vaporization (h_{fg}) BTU per lb.
		Saturated Liquid (v_f)	Saturated Vapor (v_g)	Saturated Liquid (h_f)	Saturated Vapor (h_g)	
0	212	0.0167	26.8	180	1150	970
1	215	0.0167	24.3	183	1151	967
2	218	0.0167	23.0	186	1153	965
3	222	0.0168	21.8	190	1154	963
4	224	0.0168	20.7	193	1155	961
5	227	0.0168	19.8	195	1156	959
6	230	0.0168	18.9	198	1157	958
7	232	0.0169	18.1	200	1158	956
8	235	0.0169	17.4	203	1158	955
9	237	0.0169	16.7	205	1159	953
10	239	0.0169	16.1	208	1160	952
11	242	0.0169	15.6	210	1161	950
12	244	0.0170	15.0	212	1161	949
13	246	0.0170	14.5	214	1162	947
14	248	0.0170	14.0	216	1163	946
15	250	0.0170	13.6	218	1164	945
16	252	0.0170	13.2	220	1164	943
17	254	0.0170	12.8	222	1165	942
18	255	0.0170	12.5	224	1165	941
19	257	0.0171	12.1	226	1166	940
20	259	0.0171	11.1	227	1166	939
25	267	0.0171	10.4	236	1169	933
30	274	0.0172	9.4	243	1171	926
35	281	0.0173	8.5	250	1173	923
40	287	0.0173	7.74	256	1175	919
45	292	0.0174	7.14	262	1177	914
50	298	0.0174	6.62	267	1178	911
55	302	0.0175	6.17	272	1179	907
60	307	0.0175	5.79	277	1181	903
65	312	0.0176	5.45	282	1182	900
70	316	0.0176	5.14	286	1183	897
75	320	0.0176	4.87	290	1184	893
80	324	0.0177	4.64	294	1185	890
85	327	0.0177	4.42	298	1186	888
90	331	0.0178	4.24	301	1189	887
95	334	0.0178	4.03	305	1190	884
100	338	0.0178	3.88	308	1190	882
105	341	0.0179	3.72	312	1189	877
110	343	0.0179	3.62	314	1191	877
115	347	0.0180	3.44	318	1191	872
120	350	0.0180	3.34	321	1193	872
125	353	0.0180	3.21	324	1193	867
130	355	0.0180	3.12	327	1194	867
135	358	0.0181	3.02	329	1194	864
140	361	0.0181	2.92	332	1195	862
145	363	0.0181	2.84	335	1196	860
150	366	0.0182	2.75	337	1196	858
155	368	0.0182	2.67	340	1196	854
160	370	0.0182	2.60	342	1196	854
165	373	0.0183	2.53	345	1197	852
170	375	0.0183	2.47	347	1197	850
175	378	0.0183	2.40	350	1198	848
180	380	0.0184	2.34	352	1198	846
185	382	0.0184	2.29	355	1199	844
190	384	0.0184	2.23	357	1199	842
195	386	0.0184	2.18	359	1199	840
200	388	0.0185	2.14	361	1199	838
210	392	0.0185	2.05	365	1200	835
220	396	0.0186	1.96	369	1200	831
230	399	0.0186	1.88	373	1201	828
240	403	0.0187	1.81	377	1201	824
250	406	0.0187	1.75	380	1201	821
260	410	0.0188	1.68	384	1201	817
270	413	0.0188	1.63	387	1202	814
280	416	0.0189	1.57	391	1202	811
290	419	0.0190	1.52	394	1202	807
300	421	0.0190	1.47	397	1202	805

25.4 Viscosity Nomograph:

Equivalents of Kinematic, Saybolt Universal, Saybolt Furol, and Absolute Viscosity



$$\mu = \nu\rho' = \nu S$$

The empirical relation between Saybolt Universal Viscosity and Saybolt Furol Viscosity at 100 F and 122 F, respectively, and Kinematic Viscosity is taken from A.S.T.M. D2161-63T. At other temperatures, the Saybolt Viscosities vary only slightly.

Saybolt Viscosities above those shown are given by the relationships:

Saybolt Universal Seconds = Centistokes x 4.6347
Saybolt Furol Seconds = Centistokes x 0.4717

Problem 1: Determine the absolute viscosity of an oil which has a kinematic viscosity of 82 centistokes and a specific gravity of 0.83.

Solution 1: Connect 82 on the kinematic viscosity scale with 0.83 on the specific gravity scale; read 67 centipoise at the intersection on the absolute viscosity scale.

25.5 RTD Resistance Table

25.5.1 100Ω Platinum in °C

°C	a 0.00385	a 0.00392	°C	a 0.00385	a 0.00392	°C	a 0.00385	a 0.00392	°C	a 0.00385	a 0.00392
-200	18.52	17.08	-140	43.88	42.87	-80	68.33	67.76	-20	92.16	92.02
-199	18.95	17.52	-139	44.29	43.29	-79	68.73	68.17	-19	92.55	92.42
-198	19.38	17.96	-138	44.70	43.72	-78	69.13	68.57	-18	92.95	92.82
-197	19.82	18.40	-137	45.12	44.14	-77	69.53	68.98	-17	93.34	93.22
-196	20.25	18.83	-136	45.53	44.56	-76	69.93	69.39	-16	93.73	93.62
-195	20.68	19.27	-135	45.94	44.98	-75	70.33	69.80	-15	94.12	94.02
-194	21.11	19.71	-134	46.36	45.40	-74	70.73	70.21	-14	94.52	94.42
-193	21.54	20.15	-133	46.77	45.82	-73	71.13	70.61	-13	94.91	94.82
-192	21.97	20.58	-132	47.18	46.24	-72	71.53	71.02	-12	95.30	95.22
-191	22.40	21.02	-131	47.59	46.66	-71	71.93	71.43	-11	95.69	95.62
-190	22.83	21.46	-130	48.00	47.07	-70	72.33	71.84	-10	96.09	96.02
-189	23.25	21.89	-129	48.42	47.49	-69	72.73	72.24	-9	96.48	96.41
-188	23.68	22.33	-128	48.83	47.91	-68	73.13	72.65	-8	96.87	96.81
-187	24.11	22.76	-127	49.24	48.33	-67	73.53	73.06	-7	97.26	97.21
-186	24.54	23.20	-126	49.65	48.75	-66	73.93	73.47	-6	97.65	97.61
-185	24.97	23.63	-125	50.06	49.17	-65	74.33	73.87	-5	98.04	98.01
-184	25.39	24.07	-124	50.47	49.58	-64	74.73	74.28	-4	98.44	98.41
-183	25.82	24.50	-123	50.88	50.00	-63	75.13	74.68	-3	98.83	98.81
-182	26.24	24.94	-122	51.29	50.42	-62	75.53	75.09	-2	99.22	99.20
-181	26.67	25.37	-121	51.70	50.84	-61	75.93	75.50	-1	99.61	99.60
-180	27.10	25.80	-120	52.11	51.25	-60	76.33	75.90	0	100.00	100.00
-179	27.52	26.23	-119	52.52	51.67	-59	76.73	76.31	1	100.39	100.40
-178	27.95	26.67	-118	52.93	52.09	-58	77.12	76.71	2	100.78	100.80
-177	28.37	27.10	-117	53.34	52.50	-57	77.52	77.12	3	101.17	101.19
-176	28.80	27.53	-116	53.75	52.92	-56	77.92	77.52	4	101.56	101.59
-175	29.22	27.96	-115	54.15	53.33	-55	78.32	77.93	5	101.95	101.99
-174	29.64	28.39	-114	54.56	53.75	-54	78.72	78.33	6	102.34	102.39
-173	30.07	28.82	-113	54.97	54.16	-53	79.11	78.74	7	102.73	102.78
-172	30.49	29.25	-112	55.38	54.58	-52	79.51	79.14	8	103.12	103.18
-171	30.91	29.68	-111	55.79	54.99	-51	79.91	79.55	9	103.51	103.58
-170	31.33	30.11	-110	56.19	55.41	-50	80.31	79.95	10	103.90	103.97
-169	31.76	30.54	-109	56.60	55.82	-49	80.70	80.36	11	104.29	104.37
-168	32.18	30.97	-108	57.01	56.24	-48	81.10	80.76	12	104.68	104.77
-167	32.60	31.40	-107	57.41	56.65	-47	81.50	81.16	13	105.07	105.16
-166	33.02	31.83	-106	57.82	57.06	-46	81.89	81.57	14	105.46	105.56
-165	33.44	32.26	-105	58.23	57.48	-45	82.29	81.97	15	105.85	105.95
-164	33.86	32.69	-104	58.63	57.89	-44	82.69	82.38	16	106.24	106.35
-163	34.28	33.11	-103	59.04	58.30	-43	83.08	82.78	17	106.63	106.75
-162	34.70	33.54	-102	59.44	58.72	-42	83.48	83.18	18	107.02	107.14
-161	35.12	33.97	-101	59.85	59.13	-41	83.87	83.58	19	107.40	107.54
-160	35.54	34.39	-100	60.26	59.54	-40	84.27	83.99	20	107.79	107.93
-159	35.96	34.82	- 99	60.66	59.96	-39	84.67	84.39	21	108.18	108.33
-158	36.38	35.25	- 98	61.07	60.37	-38	85.06	84.79	22	108.57	108.72
-157	36.80	35.67	- 97	61.47	60.78	-37	85.46	85.20	23	108.96	109.12
-156	37.22	36.10	- 96	61.88	61.19	-36	85.85	85.60	24	109.35	109.52
-155	37.64	36.52	- 95	62.28	61.60	-35	86.25	86.00	25	109.73	109.91
-154	38.05	36.95	- 94	62.68	62.01	-34	86.64	86.40	26	110.12	110.30
-153	38.47	37.37	- 93	63.09	62.43	-33	87.04	86.79	27	110.51	110.70
-152	38.89	37.80	- 92	63.49	62.84	-32	87.43	87.21	28	110.90	111.09
-151	39.31	38.22	- 91	63.90	63.25	-31	87.83	87.61	29	111.29	111.49
-150	39.72	38.65	- 90	64.30	63.66	-30	88.22	88.01	30	111.67	111.88
-149	40.14	39.07	- 89	64.70	64.07	-29	88.62	88.41	31	112.06	112.28
-148	40.56	39.49	- 88	65.11	64.48	-28	89.01	88.81	32	112.45	112.67
-147	40.97	39.92	- 87	65.51	64.89	-27	89.40	89.21	33	112.83	113.07
-146	41.39	40.34	- 86	65.91	65.30	-26	89.80	89.61	34	113.22	113.46
-145	41.80	40.76	- 85	66.31	65.71	-25	90.19	90.02	35	113.61	113.85
-144	42.22	41.19	- 84	66.72	66.12	-24	90.59	90.42	36	114.00	114.25
-143	42.63	41.61	- 83	67.12	66.53	-23	90.98	90.82	37	114.38	114.64
-142	43.05	42.03	- 82	67.52	66.94	-22	91.37	91.22	38	114.77	115.03
-141	43.46	42.45	- 81	67.92	67.35	-21	91.77	91.62	39	115.15	115.43

°C	a 0.00385	a 0.00392	°C	a 0.00385	a 0.00392	°C	a 0.00385	a 0.00392	°C	a 0.00385	a 0.00392
40	115.54	115.82	100	138.51	139.20	160	161.05	162.16	220	183.19	184.69
41	115.93	116.21	101	138.88	139.59	161	161.43	162.54	221	183.55	185.06
42	116.31	116.61	102	139.26	139.97	162	161.80	162.91	222	183.92	185.43
43	116.70	117.00	103	139.64	140.36	163	162.17	163.29	223	184.28	185.81
44	117.08	117.39	104	140.02	140.74	164	162.54	163.67	224	184.65	186.18
45	117.47	117.79	105	140.40	141.13	165	162.91	164.05	225	185.01	186.55
46	117.86	118.18	106	140.78	141.51	166	163.29	164.43	226	185.38	186.92
47	118.24	118.57	107	141.16	141.90	167	163.66	164.81	227	185.74	187.29
48	118.63	118.96	108	141.54	142.29	168	164.03	165.19	228	186.11	187.66
49	119.01	119.35	109	141.91	142.67	169	164.40	165.56	229	186.47	188.03
50	119.40	119.75	110	142.29	143.06	170	164.77	165.94	230	186.84	188.41
51	119.78	120.14	111	142.67	143.44	171	165.14	166.32	231	187.20	188.78
52	120.17	120.53	112	143.05	143.83	172	165.51	166.70	232	187.56	189.15
53	120.55	120.92	113	143.43	144.21	173	165.89	167.07	233	187.93	189.52
54	120.94	121.31	114	143.80	144.59	174	166.26	167.45	234	188.29	189.89
55	121.32	121.71	115	144.18	144.98	175	166.63	167.83	235	188.66	190.26
56	121.71	122.10	116	144.56	145.36	176	167.00	168.21	236	189.02	190.63
57	122.09	122.49	117	144.94	145.75	177	167.37	168.58	237	189.38	191.00
58	122.47	122.88	118	145.31	146.13	178	167.74	168.96	238	189.75	191.37
59	122.86	123.27	119	145.69	146.52	179	168.11	169.34	239	190.11	191.74
60	123.24	123.66	120	146.07	146.90	180	168.48	169.71	240	190.47	192.11
61	123.63	124.05	121	146.44	147.28	181	168.85	170.09	241	190.84	192.48
62	124.01	124.44	122	146.82	147.67	182	169.22	170.47	242	191.20	192.85
63	124.39	124.83	123	147.20	148.05	183	169.59	170.84	243	191.56	193.22
64	124.78	125.22	124	147.58	148.43	184	169.96	171.22	244	191.92	193.59
65	125.16	125.61	125	147.95	148.82	185	170.33	171.60	245	192.29	193.96
66	125.54	126.00	126	148.33	149.20	186	170.70	171.97	246	192.65	194.32
67	125.93	126.39	127	148.70	149.58	187	171.07	172.35	247	193.01	194.69
68	126.31	126.78	128	149.08	149.97	188	171.44	172.73	248	193.37	195.06
69	126.69	127.17	129	149.46	150.35	189	171.80	173.10	249	193.74	195.43
70	127.08	127.56	130	149.83	150.73	190	172.17	173.48	250	194.10	195.80
71	127.46	127.95	131	150.21	151.11	191	172.54	173.85	251	194.46	196.17
72	127.84	128.34	132	150.58	151.50	192	172.91	174.23	252	194.82	196.54
73	128.22	128.73	133	150.96	151.88	193	173.28	174.60	253	195.18	196.90
74	128.61	129.12	134	151.33	152.26	194	173.65	174.98	254	195.55	197.27
75	128.99	129.51	135	151.71	152.64	195	174.02	175.35	255	195.91	197.64
76	129.37	129.90	136	152.08	153.02	196	174.38	175.73	256	196.27	198.01
77	129.75	130.29	137	152.46	153.41	197	174.75	176.10	257	196.63	198.38
78	130.13	130.68	138	152.83	153.79	198	175.12	176.48	258	196.99	198.74
79	130.52	131.07	139	153.21	154.17	199	175.49	176.85	259	197.35	199.11
80	130.90	131.45	140	153.58	154.55	200	175.86	177.23	260	197.71	199.48
81	131.28	131.84	141	153.96	154.93	201	176.22	177.60	261	198.07	199.85
82	131.66	132.23	142	154.33	155.31	202	176.59	177.97	262	198.43	200.21
83	132.04	132.62	143	154.71	155.70	203	176.96	178.35	263	198.79	200.58
84	132.42	133.01	144	155.08	156.08	204	177.33	178.72	264	199.15	200.95
85	132.80	133.39	145	155.46	156.46	205	177.69	179.10	265	199.51	201.31
86	133.18	133.78	146	155.83	156.84	206	178.06	179.47	266	199.87	201.68
87	133.57	134.17	147	156.20	157.22	207	178.43	179.84	267	200.23	202.05
88	133.95	134.56	148	156.58	157.60	208	178.79	180.22	268	200.59	202.41
89	134.33	134.95	149	156.95	157.98	209	179.16	180.59	269	200.95	202.78
90	134.71	135.33	150	157.33	158.36	210	179.53	180.96	270	201.31	203.15
91	135.09	135.72	151	157.70	158.74	211	179.89	181.34	271	201.67	203.51
92	135.47	136.11	152	158.07	159.12	212	180.26	181.71	272	202.03	203.88
93	135.85	136.49	153	158.45	159.50	213	180.63	182.08	273	202.39	204.24
94	136.23	136.88	154	158.82	159.88	214	180.99	182.46	274	202.75	204.61
95	136.61	137.27	155	159.19	160.26	215	181.36	182.83	275	203.11	204.98
96	136.99	137.65	156	159.56	160.64	216	181.72	183.20	276	203.47	205.34
97	137.37	138.04	157	159.94	161.02	217	182.09	183.57	277	203.80	205.82
98	137.75	138.43	158	160.31	161.40	218	182.46	183.95	278	204.19	206.07
99	138.13	138.81	159	160.68	161.78	219	182.82	184.32	279	204.55	206.44

°C	a 0.00385	a 0.00392	°C	a 0.00385	a 0.00392	°C	a 0.00385	a 0.00392	°C	a 0.00385	a 0.00392
280	204.90	206.80	340	226.21	228.49	400	247.09	249.76	460	267.56	270.60
281	205.26	207.17	341	226.56	228.85	401	247.44	250.11	461	267.90	270.95
282	205.62	207.53	342	226.91	229.21	402	247.78	250.46	462	268.24	271.29
283	205.98	207.90	343	227.26	229.56	403	248.13	250.81	463	268.57	271.63
284	206.34	208.26	344	227.61	229.92	404	248.47	251.16	464	268.91	271.98
285	206.70	208.63	345	227.96	230.28	405	248.81	251.51	465	269.25	272.32
286	207.05	208.99	346	228.31	230.64	406	249.16	251.86	466	269.59	272.66
287	207.41	209.35	347	228.66	230.99	407	249.50	252.21	467	269.92	273.01
288	207.77	209.72	348	229.02	231.35	408	249.85	252.56	468	270.26	273.35
289	208.13	210.08	349	229.37	231.71	409	250.19	252.91	469	270.60	273.69
290	208.48	210.45	350	229.72	232.06	410	250.53	253.26	470	270.93	274.03
291	208.84	210.81	351	230.07	232.42	411	250.88	253.61	471	271.27	274.38
292	209.20	211.17	352	230.42	232.78	412	251.22	253.96	472	271.61	274.72
293	209.56	211.54	353	230.77	233.13	413	251.56	254.31	473	271.94	275.06
294	209.91	211.90	354	231.12	233.49	414	251.91	254.66	474	272.28	275.40
295	210.29	212.26	355	231.47	233.85	415	252.25	255.01	475	272.61	275.75
296	210.63	212.63	356	231.82	234.20	416	252.59	255.36	476	272.95	276.09
297	210.98	212.99	357	232.17	234.56	417	252.93	255.71	477	273.29	276.43
298	211.34	213.35	358	232.52	234.92	418	253.28	256.06	478	273.62	276.77
299	211.70	213.72	359	232.87	235.27	419	253.62	256.40	479	273.96	277.11
300	212.05	214.08	360	233.21	235.63	420	253.96	256.75	480	274.29	277.46
301	212.41	214.44	361	233.56	235.98	421	254.30	257.10	481	274.63	277.80
302	212.76	214.80	362	233.91	236.34	422	254.65	257.45	482	274.96	278.14
303	213.12	215.17	363	234.26	236.69	423	254.99	257.80	483	275.30	278.48
304	213.48	215.53	364	234.61	237.05	424	255.33	258.15	484	275.63	278.82
305	213.83	215.89	365	234.96	237.40	425	255.67	258.49	485	275.97	279.16
306	214.19	216.25	366	235.31	237.76	426	256.01	258.84	486	276.30	279.50
307	214.54	216.61	367	235.66	238.11	427	256.36	259.19	487	276.64	279.84
308	214.90	216.98	368	236.00	238.47	428	256.70	259.54	488	276.97	280.18
309	215.25	217.34	369	236.35	238.82	429	257.04	259.88	489	277.31	280.52
310	215.61	217.70	370	236.70	239.18	430	257.38	260.23	490	277.64	280.87
311	215.96	218.06	371	237.05	239.53	431	257.72	260.58	491	277.98	281.21
312	216.32	218.42	372	237.40	239.89	432	258.06	260.93	492	278.31	281.55
313	216.67	218.78	373	237.75	240.24	433	258.40	261.27	493	278.64	281.89
314	217.03	219.14	374	238.09	240.59	434	258.74	261.62	494	278.98	282.23
315	217.38	219.51	375	238.44	240.95	435	259.08	261.97	495	279.31	282.57
316	217.74	219.87	376	238.79	241.30	436	259.42	262.31	496	279.64	282.91
317	218.09	220.23	377	239.14	241.66	437	259.76	262.66	497	279.98	283.24
318	218.44	220.59	378	239.48	242.01	438	260.10	263.01	498	280.31	283.58
319	218.80	220.95	379	239.83	242.36	439	260.44	263.35	499	280.64	283.92
320	219.15	221.31	380	240.18	242.72	440	260.78	263.70	500	280.98	284.26
321	219.51	221.67	381	240.52	243.07	441	261.12	264.05	501	281.31	284.60
322	219.86	222.03	382	240.87	243.42	442	261.46	264.39	502	281.64	284.94
323	220.21	222.39	383	241.22	243.78	443	261.80	264.74	503	281.98	285.28
324	220.57	222.75	384	241.56	244.13	444	262.14	265.08	504	282.31	285.62
325	220.92	223.11	385	241.91	244.48	445	262.48	265.43	505	282.64	285.96
326	221.27	223.47	386	242.26	244.83	446	262.82	265.78	506	282.97	286.30
327	221.63	223.83	387	242.60	245.19	447	263.16	266.12	507	283.31	286.63
328	221.98	224.19	388	242.95	245.54	448	263.50	266.47	508	283.64	286.97
329	222.33	224.55	389	243.29	245.89	449	263.84	266.81	509	283.97	287.31
330	222.69	224.91	390	243.64	246.24	450	264.18	267.16	510	284.30	287.65
331	223.04	225.26	391	243.99	246.59	451	264.52	267.50	511	284.63	287.99
332	223.39	225.62	392	244.33	246.95	452	264.86	267.85	512	284.97	288.32
333	223.74	225.98	393	244.68	247.30	453	265.20	268.19	513	285.30	288.66
334	224.10	226.34	394	245.02	247.65	454	265.53	268.54	514	285.63	289.00
335	224.45	226.70	395	245.37	248.00	455	265.87	268.88	515	285.96	289.34
336	224.80	227.06	396	245.71	248.35	456	266.21	269.23	516	286.29	289.67
337	225.15	227.42	397	246.06	248.70	457	266.55	269.57	517	286.62	290.01
338	225.50	227.78	398	246.40	249.06	458	266.89	269.91	518	286.95	290.35
339	225.85	228.13	399	246.75	249.41	459	267.22	270.26	519	287.29	290.69

°C	a 0.00385	a 0.00392	°C	a 0.00385	a 0.00392	°C	a 0.00385	a 0.00392	
520	287.62	291.02	580	307.25	311.02	640	326.48	330.60	
521	287.95	291.36	581	307.58	311.35	641	326.79	330.92	
522	288.28	291.70	582	307.90	311.68	642	327.11	331.24	
523	288.61	292.03	583	308.23	312.01	643	327.43	331.57	
524	288.94	292.37	584	308.55	312.34	644	327.74	331.89	
525	289.27	292.71	585	308.87	312.67	645	328.06	332.21	
526	289.60	293.04	586	309.20	313.00	646	328.38	332.53	
527	289.93	293.38	587	309.52	313.33	647	328.69	332.85	
528	290.26	293.71	588	309.84	313.66	648	329.01	333.18	
529	290.59	294.05	589	310.16	313.99	649	329.32	333.50	
530	290.92	294.39	590	310.49	314.31	650	329.64	333.82	
531	291.25	294.72	591	310.81	314.64	651	329.96	334.14	
532	291.58	295.06	592	311.13	314.97	652	330.27	334.46	
533	291.91	295.39	593	311.45	315.30	653	330.59	334.78	
534	292.24	295.73	594	311.78	315.63	654	330.90	335.11	
535	292.56	296.06	595	312.10	315.96	655	331.22	335.43	
536	292.89	296.40	596	312.42	316.28	656	331.53	335.75	
537	293.22	296.73	597	312.74	316.61	657	331.85	336.07	
538	293.55	297.07	598	313.06	316.94	658	332.16	336.39	
539	293.88	297.40	599	313.39	317.27	659	332.48	336.71	
540	294.21	297.74	600	313.71	317.59	660	332.79	337.03	
541	294.54	298.07	601	314.03	317.92	661	333.11	337.35	
542	294.87	298.40	602	314.35	318.25				
543	295.19	298.74	603	314.67	318.58				
544	295.52	299.07	604	314.99	318.90				
545	295.85	299.41	605	315.31	319.23				
546	296.18	299.74	606	315.64	319.56				
547	296.50	300.07	607	315.96	319.88				
548	296.83	300.41	608	316.28	320.21				
549	297.16	300.74	609	316.60	320.54				
550	297.49	301.08	610	316.92	320.86				
551	297.81	301.41	611	317.24	321.19				
552	298.14	301.74	612	317.56	321.51				
553	298.47	302.07	613	317.88	321.84				
554	298.80	302.41	614	318.20	322.17				
555	299.12	302.74	615	318.52	322.49				
556	299.45	303.07	616	318.84	322.82				
557	299.78	303.41	617	319.16	323.14				
558	300.10	303.74	618	319.48	323.47				
559	300.43	304.07	619	319.80	323.79				
560	300.75	304.40	620	320.12	324.12				
561	301.08	304.73	621	320.43	324.44				
562	301.41	305.07	622	320.75	324.77				
563	301.73	305.40	623	321.07	325.09				
564	302.06	305.73	624	321.39	325.42				
565	302.38	306.06	625	321.71	325.74				
566	302.71	306.39	626	322.03	326.07				
567	303.03	306.72	627	322.35	326.39				
568	303.36	307.06	628	322.67	326.72				
569	303.69	307.39	629	322.98	327.04				
570	304.01	307.72	630	323.30	327.36				
571	304.34	308.05	631	323.62	327.69				
572	304.66	308.38	632	323.94	328.01				
573	304.98	308.71	633	324.26	328.34				
574	305.31	309.04	634	324.57	328.66				
575	305.63	309.37	635	324.89	328.98				
576	305.96	309.70	636	325.21	329.31				
577	306.28	310.03	637	325.53	329.63				
578	306.61	310.36	638	325.84	329.95				
579	306.93	310.69	639	326.16	330.28				

25.5.2 10Ω Copper RTD in °F

°F	0	1	2	3	4	5	6	7	8	9	10	°F
						Resistance in Ohms						
-320	1.242	1.219	1.196	1.173	1.150	1.127	1.104	1.081	1.058			-320
-310	1.472	1.449	1.426	1.403	1.380	1.357	1.334	1.311	1.288	1.265	1.242	-310
-300	1.701	1.678	1.655	1.632	1.609	1.587	1.564	1.541	1.518	1.495	1.472	-300
-290	1.930	1.907	1.884	1.861	1.839	1.816	1.793	1.770	1.747	1.724	1.701	-290
-280	2.158	2.136	2.113	2.090	2.067	2.044	2.021	1.999	1.976	1.953	1.930	-280
-270	2.386	2.364	2.341	2.318	2.295	2.272	2.250	2.227	2.204	2.181	2.158	-270
-260	2.614	2.591	2.568	2.546	2.523	2.500	2.477	2.455	2.432	2.409	2.386	-260
-250	2.841	2.818	2.795	2.773	2.750	2.727	2.705	2.682	2.659	2.637	2.614	-250
-240	3.067	3.045	3.022	2.999	2.977	2.954	2.931	2.909	2.886	2.863	2.841	-240
-230	3.293	3.271	3.248	3.226	3.203	3.180	3.158	3.135	3.113	3.090	3.067	-230
-220	3.519	3.496	3.474	3.451	3.429	3.406	3.384	3.361	3.338	3.316	3.293	-220
-210	3.744	3.721	3.699	3.676	3.654	3.631	3.609	3.586	3.564	3.541	3.519	-210
-200	3.968	3.946	3.923	3.901	3.879	3.856	3.834	3.811	3.789	3.766	3.744	-200
-190	4.192	4.170	4.148	4.125	4.103	4.080	4.058	4.036	4.013	3.991	3.968	-190
-180	4.416	4.394	4.371	4.349	4.327	4.304	4.282	4.259	4.237	4.215	4.192	-180
-170	4.639	4.617	4.594	4.572	4.550	4.527	4.505	4.483	4.461	4.438	4.416	-170
-160	4.862	4.839	4.817	4.795	4.773	4.750	4.728	4.706	4.684	4.661	4.639	-160
-150	5.084	5.061	5.039	5.017	4.995	4.973	4.950	4.928	4.906	4.884	4.862	-150
-140	5.305	5.283	5.261	5.239	5.217	5.195	5.172	5.150	5.128	5.106	5.084	-140
-130	5.526	5.504	5.482	5.460	5.438	5.416	5.394	5.372	5.350	5.327	5.305	-130
-120	5.747	5.725	5.703	5.681	5.659	5.637	5.615	5.593	5.571	5.548	5.526	-120
-110	5.967	5.945	5.923	5.901	5.879	5.857	5.835	5.813	5.791	5.769	5.747	-110
-100	6.187	6.165	6.143	6.121	6.099	6.077	6.055	6.033	6.011	5.989	5.967	-100
-90	6.406	6.384	6.362	6.340	6.318	6.296	6.275	6.253	6.231	6.209	6.187	-90
-80	6.625	6.603	6.581	6.559	6.537	6.515	6.494	6.472	6.450	6.428	6.406	-80
-70	6.843	6.821	6.799	6.777	6.756	6.734	6.712	6.690	6.668	6.647	6.625	-70
-60	7.061	7.039	7.017	6.995	6.974	6.952	6.930	6.908	6.886	6.865	6.843	-60
-50	7.276	7.254	7.233	7.211	7.190	7.168	7.147	7.126	7.104	7.082	7.061	-50
-40	7.490	7.469	7.447	7.426	7.404	7.383	7.362	7.340	7.319	7.297	7.276	-40
-30	7.705	7.683	7.662	7.640	7.619	7.598	7.576	7.555	7.533	7.512	7.490	-30
-20	7.919	7.898	7.876	7.855	7.834	7.812	7.791	7.769	7.748	7.726	7.705	-20
-10	8.134	8.112	8.091	8.070	8.048	8.027	8.005	7.984	7.962	7.941	7.919	-10
0	8.348	8.327	8.306	8.284	8.263	8.241	8.220	8.198	8.177	8.155	8.134	0
0	8.348	8.370	8.391	8.413	8.434	8.456	8.477	8.499	8.520	8.542	8.563	0
10	8.563	8.584	8.606	8.627	8.649	8.670	8.692	8.713	8.735	8.756	8.778	10
20	8.778	8.799	8.820	8.842	8.863	8.885	8.906	8.928	8.949	8.971	8.992	20
30	8.992	9.014	9.035	9.056	9.078	9.099	9.121	9.142	9.164	9.185	9.207	30
40	9.207	9.228	9.250	9.271	9.292	9.314	9.335	9.357	9.378	9.400	9.421	40
50	9.421	9.443	9.464	9.486	9.507	9.528	9.550	9.571	9.593	9.614	9.636	50
60	9.636	9.657	9.679	9.700	9.722	9.743	9.764	9.786	9.807	9.829	9.850	60
70	9.850	9.872	9.893	9.915	9.936	9.958	9.979	10.000	10.022	10.043	10.065	70
80	10.065	10.086	10.108	10.129	10.151	10.172	10.194	10.215	10.236	10.258	10.279	80
90	10.279	10.301	10.322	10.344	10.365	10.387	10.408	10.430	10.451	10.472	10.494	90
100	10.494	10.515	10.537	10.558	10.580	10.601	10.623	10.644	10.666	10.687	10.708	100
110	10.708	10.730	10.751	10.773	10.794	10.816	10.837	10.859	10.880	10.902	10.923	110
120	10.923	10.944	10.966	10.987	11.009	11.030	11.052	11.073	11.095	11.116	11.138	120
130	11.138	11.159	11.180	11.202	11.223	11.245	11.266	11.288	11.309	11.331	11.352	130
140	11.352	11.374	11.395	11.416	11.438	11.459	11.481	11.502	11.524	11.545	11.567	140
150	11.567	11.588	11.610	11.631	11.652	11.674	11.695	11.717	11.738	11.760	11.781	150
160	11.781	11.803	11.824	11.846	11.867	11.888	11.910	11.931	11.953	11.974	11.996	160
170	11.996	12.017	12.039	12.060	12.082	12.103	12.124	12.146	12.167	12.189	12.210	170
180	12.210	12.232	12.253	12.275	12.296	12.318	12.339	12.360	12.382	12.403	12.425	180
190	12.425	12.446	12.468	12.489	12.511	12.532	12.554	12.575	12.596	12.618	12.639	190
200	12.639	12.661	12.682	12.704	12.725	12.747	12.768	12.790	12.811	12.832	12.854	200
210	12.854	12.875	12.897	12.918	12.940	12.961	12.983	13.004	13.026	13.047	13.068	210
220	13.068	13.090	13.111	13.133	13.154	13.176	13.197	13.219	13.240	13.262	13.283	220
230	13.283	13.304	13.326	13.347	13.369	13.390	13.412	13.433	13.455	13.476	13.498	230
240	13.498	13.519	13.540	13.562	13.583	13.605	13.626	13.648	13.669	13.691	13.712	240
250	13.712	13.734	13.755	13.776	13.798	13.819	13.841	13.862	13.884	13.905	13.927	250
260	13.927	13.948	13.970	13.991	14.012	14.034	14.055	14.077	14.098	14.120	14.141	260
270	14.141	14.163	14.184	14.206	14.227	14.248	14.270	14.291	14.313	14.334	14.356	270
280	14.356	14.377	14.399	14.420	14.442	14.463	14.484	14.506	14.527	14.549	14.570	280
290	14.570	14.592	14.613	14.635	14.656	14.678	14.699	14.720	14.742	14.763	14.785	290

300	14.785	14.806	14.828	14.849	14.871	14.893	14.914	14.936	14.958	14.979	15.001	300
310	15.001	15.022	15.044	15.066	15.087	15.109	15.131	15.152	15.174	15.196	15.217	310
320	15.217	15.239	15.260	15.282	15.304	15.325	15.347	15.369	15.390	15.412	15.434	320
330	15.434	15.455	15.477	15.499	15.520	15.542	15.563	15.585	15.607	15.628	15.650	330
340	15.650	15.672	15.693	15.715	15.737	15.758	15.780	15.802	15.823	15.845	15.866	340
350	15.866	15.888	15.910	15.931	15.953	15.975	15.996	16.018	16.040	16.061	16.083	350
360	16.083	16.105	16.126	16.148	16.170	16.191	16.213	16.234	16.256	16.278	16.299	360
370	16.299	16.321	16.343	16.364	16.386	16.408	16.429	16.451	16.473	16.494	16.516	370
380	16.516	16.538	16.559	16.581	16.603	16.624	16.646	16.667	16.689	16.711	16.732	380
390	16.732	16.754	16.776	16.797	16.819	16.841	16.862	16.884	16.906	16.927	16.949	390
400	16.949	16.971	16.992	17.014	17.036	17.057	17.079	17.101	17.122	17.144	17.166	400
410	17.166	17.187	17.209	17.231	17.252	17.274	17.296	17.317	17.339	17.360	17.382	410
420	17.382	17.404	17.425	17.447	17.469	17.490	17.512	17.534	17.555	17.577	17.599	420
430	17.599	17.620	17.642	17.664	17.685	17.707	17.729	17.750	17.772	17.794	17.815	430
440	17.815	17.837	17.859	17.880	17.902	17.924	17.945	17.967	17.989	18.010	18.032	440
450	18.032	18.054	18.075	18.097	18.119	18.140	18.162	18.184	18.205	18.227	18.249	450
460	18.249	18.270	18.292	18.314	18.335	18.357	18.379	18.400	18.422	18.444	18.465	460
470	18.465	18.487	18.509	18.530	18.552	18.574	18.595	18.617	18.639	18.661	18.682	470
480	18.682	18.704	18.726	18.747	18.769	18.791	18.812	18.834	18.856	18.877	18.899	480
490	18.899	18.921	18.942	18.964	18.986	19.007	19.029	19.051	19.072	19.094	19.116	490
500	19.116											500

25.5.3 120Ω Nickel RTD in °F

°F	0	1	2	3	4	5	6	7	8	9	10	°F
						Resistance in Ohms						
-110	67.32	66.96	66.60									-110
-100	70.93	70.57	70.21	69.85	69.49	69.13	68.77	68.41	68.04	67.68	67.32	-100
-90	74.55	74.19	73.83	73.46	73.10	72.74	72.38	72.02	71.66	71.30	70.93	-90
-80	78.17	77.81	77.45	77.08	76.72	76.36	76.00	75.64	75.27	74.91	74.55	-80
-70	81.80	81.43	81.07	80.71	80.35	79.98	79.62	79.26	78.89	78.53	78.17	-70
-60	85.44	85.07	84.71	84.34	83.98	83.62	83.25	82.89	82.52	82.16	81.80	-60
-50	89.09	88.72	88.36	87.99	87.62	87.26	86.89	86.53	86.16	85.80	85.44	-50
-40	92.76	92.39	92.02	91.65	91.29	90.92	90.55	90.19	89.82	89.45	89.09	-40
-30	96.44	96.07	95.70	95.33	94.97	94.60	94.23	93.86	93.49	93.12	92.76	-30
-20	100.15	99.78	99.41	99.04	98.67	98.30	97.92	97.55	97.18	96.81	96.44	-20
-10	103.89	103.52	103.14	102.77	102.39	102.02	101.65	101.27	100.90	100.53	100.15	-10
0	107.66	107.28	106.91	106.53	106.15	105.77	105.40	105.02	104.64	104.27	103.89	0
0	107.66	108.04	108.42	108.80	109.18	109.56	109.94	110.32	110.70	111.09	111.47	0
10	111.47	111.85	112.23	112.62	113.00	113.39	113.77	114.16	114.54	114.93	115.32	10
20	115.32	115.71	116.09	116.48	116.87	117.26	117.65	118.04	118.43	118.82	119.21	20
30	119.21	119.61	120.00	120.39	120.79	121.18	121.58	121.97	122.37	122.77	123.16	30
40	123.16	123.56	123.96	124.36	124.76	125.16	125.56	125.96	126.36	126.76	127.17	40
50	127.17	127.57	127.98	128.38	128.79	129.19	129.60	130.00	130.41	130.82	131.23	50
60	131.23	131.64	132.05	132.46	132.87	133.28	133.69	134.11	134.52	134.93	135.35	60
70	135.35	135.76	136.18	136.59	137.01	137.43	137.85	138.26	138.68	139.10	139.52	70
80	139.52	139.95	140.37	140.79	141.21	141.64	142.06	142.48	142.91	143.34	143.76	80
90	143.76	144.19	144.62	145.05	145.48	145.90	146.34	146.77	147.20	147.63	148.06	90
100	148.06	148.50	148.93	149.37	149.80	150.24	150.67	151.11	151.55	151.99	152.43	100
110	152.43	152.87	153.31	153.75	154.19	154.63	155.07	155.52	155.96	156.41	156.85	110
120	156.85	157.30	157.75	158.19	158.64	159.09	159.54	159.99	160.44	160.89	161.34	120
130	161.34	161.80	162.25	162.70	163.16	163.61	164.07	164.53	164.98	165.44	165.90	130
140	165.90	166.36	166.82	167.28	167.74	168.20	168.67	169.13	169.59	170.06	170.52	140
150	170.52	170.99	171.46	171.92	172.39	172.86	173.33	173.80	174.27	174.74	175.21	150
160	175.21	175.68	176.16	176.63	177.10	177.58	178.06	178.53	179.01	179.49	179.97	160
170	179.97	180.44	180.92	181.40	181.89	182.37	182.85	183.33	183.82	184.30	184.78	170
180	184.78	185.27	185.76	186.24	186.73	187.22	187.71	188.20	188.69	189.18	189.67	180
190	189.67	190.16	190.65	191.15	191.64	192.13	192.63	193.12	193.62	194.12	194.62	190
200	194.62	195.11	195.61	196.11	196.61	197.11	197.62	198.12	198.62	199.12	199.63	200
210	199.63	200.13	200.64	201.15	201.65	202.16	202.67	203.18	203.69	204.20	204.71	210
220	204.71	205.22	205.73	206.24	206.76	207.27	207.79	208.30	208.82	209.34	209.85	220
230	209.85	210.37	210.89	211.41	211.93	212.45	212.97	213.50	214.02	214.54	215.07	230
240	215.07	215.59	216.12	216.65	217.17	217.70	218.23	218.76	219.29	219.82	220.35	240
250	220.35	220.88	221.42	221.95	222.48	223.02	223.56	224.09	224.63	225.17	225.70	250
260	225.70	226.24	226.78	227.32	227.87	228.41	228.95	229.49	230.04	230.58	231.13	260
270	231.13	231.67	232.22	232.77	233.32	233.86	234.41	234.96	235.52	236.07	236.62	270
280	236.62	237.17	237.73	238.28	238.84	239.39	239.95	240.51	241.06	241.62	242.18	280
290	242.18	242.74	243.30	243.86	244.43	244.99	245.55	246.12	246.68	247.25	247.82	290
300	247.82	248.38	248.95	249.52	250.09	250.66	251.23	251.80	252.37	252.94	253.52	300
310	253.52	254.09	254.67	255.24	255.82	256.40	256.98	257.55	258.13	258.71	259.30	310
320	259.30	259.88	260.46	261.04	261.63	262.21	262.80	263.39	263.97	264.56	265.15	320
330	265.15	265.74	266.33	266.92	267.52	268.11	268.70	269.30	269.89	270.49	271.09	330
340	271.09	271.69	272.29	272.89	273.49	274.09	274.69	275.30	275.90	276.51	277.11	340

°F	0	1	2	3	4	5	6	7	8	9	10	°F
					Resistance in Ohms							
350	277.11	277.72	278.33	278.94	279.55	280.16	280.77	281.38	282.00	282.61	283.23	350
360	283.23	283.84	284.46	285.08	285.70	286.32	286.94	287.56	288.19	288.81	289.44	360
370	289.44	290.06	290.69	291.32	291.95	292.58	293.21	293.84	294.48	295.11	295.75	370
380	295.75	296.38	297.02	297.66	298.30	298.94	299.58	300.22	300.87	301.51	302.16	380
390	302.16	302.81	303.45	304.10	304.75	305.41	306.06	306.71	307.37	308.02	308.68	390
400	308.68	309.34	310.00	310.66	311.32	311.98	312.64	313.31	313.97	314.64	315.31	400
410	315.31	315.98	316.65	317.32	317.99	318.67	319.34	320.02	320.70	321.38	322.06	410
420	322.06	322.74	323.42	324.10	324.79	325.47	326.16	326.85	327.54	328.23	328.92	420
430	328.92	329.61	330.30	331.00	331.70	332.39	333.09	333.79	334.49	335.19	335.90	430
440	335.90	336.60	337.31	338.02	338.72	339.43	340.14	340.86	341.57	342.28	343.00	440
450	343.00	343.71	344.43	345.15	345.87	346.59	347.32	348.04	348.76	349.49	350.22	450
460	350.22	350.95	351.68	352.41	353.14	353.87	354.61	355.34	356.08	356.82	357.56	460
470	357.56	358.30	359.04	359.79	360.53	361.28	362.02	362.77	363.52	364.27	365.02	470
480	365.02	365.78	366.53	367.29	368.04	368.80	369.56	370.32	371.08	371.84	372.61	480
490	372.61	373.37	374.14	374.91	375.67	376.44	377.21	377.99	378.76	379.53	380.31	490
500	380.31											500

25.5.4 120Ω Nickel-Iron (Balco) RTD in °F

°F	0	1	2	3	4	5	6	7	8	9	10	°F
					Resistance in Ohms							
-320	248.80	248.36	247.92	247.48	247.05	246.62	246.19	245.77	245.34			-320
-310	253.42	252.94	252.47	252.00	251.53	251.07	250.61	250.16	249.70	249.25	248.80	-310
-300	258.35	257.85	257.34	256.84	256.34	255.85	255.35	254.87	254.38	253.90	253.42	-300
-290	263.61	263.07	262.53	262.00	261.47	260.94	260.42	259.90	259.38	258.86	258.35	-290
-280	269.18	268.61	268.04	267.48	266.91	266.36	265.80	265.25	264.70	264.15	263.61	-280
-270	275.08	274.47	273.87	273.28	272.68	272.09	271.50	270.92	270.34	269.76	269.18	-270
-260	281.29	280.66	280.03	279.40	278.77	278.15	277.53	276.91	276.30	275.69	275.08	-260
-250	287.83	287.16	286.50	285.84	285.18	284.52	283.87	283.22	282.58	281.93	281.29	-250
-240	294.69	293.99	293.29	292.60	291.91	291.22	290.53	289.85	289.18	288.50	287.83	-240
-230	301.86	301.13	300.40	299.68	298.95	298.23	297.52	296.81	296.10	295.39	294.69	-230
-220	309.36	308.59	307.83	307.08	306.32	305.57	304.82	304.08	303.34	302.60	301.86	-220
-210	317.18	316.38	315.59	314.80	314.01	313.23	312.45	311.67	310.90	310.13	309.36	-210
-200	325.31	324.48	323.66	322.84	322.02	321.20	320.39	319.58	318.78	317.98	317.18	-200
-190	333.77	332.91	332.05	331.20	330.35	329.50	328.66	327.82	326.98	326.14	325.31	-190
-180	342.55	341.66	340.77	339.88	339.00	338.12	337.24	336.37	335.50	334.63	333.77	-180
-170	351.65	350.72	349.80	348.88	347.97	347.06	346.15	345.24	344.34	343.44	342.55	-170
-160	361.06	360.11	359.16	358.21	357.26	356.32	355.37	354.44	353.50	352.57	351.65	-160
-150	370.80	369.81	368.83	367.85	366.87	365.89	364.92	363.95	362.99	362.02	361.06	-150
-140	380.86	379.84	378.82	377.81	376.80	375.79	374.79	373.79	372.79	371.79	370.80	-140
-130	391.24	390.19	389.14	388.09	387.05	386.01	384.97	383.94	382.91	381.89	380.86	-130
-120	401.94	400.86	399.77	398.70	397.62	396.55	395.48	394.42	393.35	392.30	391.24	-120
-110	412.96	411.84	410.73	409.62	408.51	407.41	406.31	405.21	404.12	403.03	401.94	-110
-100	424.30	423.15	422.01	420.86	419.72	418.59	417.46	416.33	415.20	414.08	412.96	-100
-90	435.96	434.78	433.60	432.43	431.26	430.09	428.92	427.76	426.61	425.45	424.30	-90
-80	447.94	446.73	445.52	444.31	443.11	441.91	440.71	439.52	438.33	437.14	435.96	-80
-70	460.24	459.00	457.75	456.52	455.28	454.05	452.82	451.60	450.37	449.16	447.94	-70
-60	472.86	471.58	470.31	469.04	467.77	466.51	465.25	463.99	462.74	461.49	460.24	-60
-50	485.80	484.49	483.19	481.89	480.59	479.29	478.00	476.71	475.42	474.14	472.86	-50
-40	499.06	497.72	496.39	495.05	493.72	492.39	491.07	489.75	488.43	487.11	485.80	-40
-30	512.65	511.27	509.90	508.54	507.17	505.81	504.46	503.10	501.75	500.41	499.06	-30
-20	526.55	525.14	523.74	522.34	520.95	519.56	518.17	516.78	515.40	514.02	512.65	-20
-10	540.77	539.33	537.90	536.47	535.04	533.62	532.20	530.78	529.37	527.96	526.55	-10
0	555.31	553.84	552.38	550.92	549.46	548.00	546.55	545.10	543.65	542.21	540.77	0
0	555.31	556.78	558.26	559.74	561.22	562.70	564.19	565.68	567.18	568.67	570.17	0
10	570.17	571.68	573.19	574.70	576.21	577.73	579.25	580.77	582.30	583.82	585.36	10
20	585.36	586.89	588.43	589.97	591.52	593.07	594.62	596.18	597.73	599.30	600.86	20
30	600.86	602.43	604.00	605.54	607.09	608.64	610.19	611.74	613.30	614.85	616.41	30
40	616.41	617.98	619.54	621.11	622.67	624.24	625.82	627.39	628.97	630.55	632.13	40
50	632.13	633.71	635.30	636.89	638.48	640.07	641.66	643.26	644.86	646.46	648.06	50
60	648.06	649.67	651.28	652.89	654.50	656.11	657.73	659.35	660.97	662.59	664.22	60
70	664.22	665.85	667.48	669.11	670.74	672.38	674.02	675.66	677.30	678.95	680.59	70
80	680.59	682.24	683.89	685.55	687.20	688.86	690.52	692.19	693.85	695.52	697.19	80
90	697.19	698.86	700.53	702.21	703.89	705.57	707.25	708.93	710.62	712.31	714.00	90
100	714.00	715.69	717.39	719.09	720.79	722.49	724.19	725.90	727.61	729.32	731.03	100
110	731.03	732.75	734.47	736.19	737.91	739.63	741.36	743.09	744.82	746.55	748.29	110
120	748.29	750.03	751.77	753.51	755.25	757.00	758.75	760.50	762.25	764.00	765.76	120
130	765.76	767.52	769.28	771.05	772.81	774.58	776.35	778.12	779.90	781.67	783.45	130
140	783.45	785.24	787.02	788.80	790.59	792.38	794.18	795.97	797.77	799.57	801.37	140

°F	0	1	2	3	4	5	6	7	8	9	10	°F
						Resistance in Ohms						
150	801.37	803.17	804.98	806.78	808.59	810.41	812.22	814.04	815.86	817.68	819.50	150
160	819.50	821.32	823.15	824.98	826.81	828.65	830.48	832.32	834.16	836.01	837.85	160
170	837.85	839.70	841.55	843.40	845.25	847.11	848.97	850.83	852.69	854.56	856.42	170
180	856.42	858.29	860.16	862.04	863.91	865.79	867.67	869.55	871.44	873.33	875.21	180
190	875.21	877.11	879.00	880.90	882.79	884.69	886.60	888.50	890.41	892.32	894.23	190
200	894.23	896.14	898.05	899.97	901.89	903.81	905.74	907.67	909.59	911.52	913.46	200
210	913.46	915.39	917.33	919.27	921.21	923.16	925.10	927.05	929.00	930.95	932.91	210
220	932.91	934.87	936.82	938.79	940.75	942.72	944.68	946.65	948.63	950.60	952.58	220
230	952.58	954.56	956.54	958.52	960.51	962.50	964.49	966.48	968.47	970.47	972.47	230
240	972.47	974.47	976.47	978.48	980.49	982.50	984.51	986.52	988.54	990.56	992.58	240
250	992.58	994.60	996.63	998.66	1000.7	1002.7	1004.8	1006.8	1008.8	1010.9	1012.9	250
260	1012.9	1015.0	1017.0	1019.1	1021.1	1023.2	1025.2	1027.3	1029.3	1031.4	1033.5	260
270	1033.5	1035.5	1037.6	1039.7	1041.7	1043.8	1045.9	1048.0	1050.1	1052.1	1054.2	270
280	1054.2	1056.3	1058.4	1060.5	1062.6	1064.7	1066.8	1068.9	1071.0	1073.1	1075.2	280
290	1075.2	1077.3	1079.4	1081.6	1083.7	1085.8	1087.9	1090.0	1092.2	1094.3	1096.4	290
300	1096.4	1098.6	1100.7	1102.8	1105.0	1107.1	1109.3	1111.4	1113.6	1115.7	1117.9	300
310	1117.9	1120.0	1122.2	1124.3	1126.5	1128.7	1130.8	1133.0	1135.2	1137.3	1139.5	310
320	1139.5	1141.7	1143.9	1146.0	1148.2	1150.4	1152.6	1154.8	1157.0	1159.2	1161.4	320
330	1161.4	1163.6	1165.8	1168.0	1170.2	1172.4	1174.6	1176.8	1179.0	1181.2	1183.5	330
340	1183.5	1185.7	1187.9	1190.1	1192.4	1194.6	1196.8	1199.1	1201.3	1203.5	1205.8	340
350	1205.8	1208.0	1210.3	1212.5	1214.8	1217.0	1219.3	1221.5	1223.8	1226.0	1228.3	350
360	1228.3	1230.6	1232.8	1235.1	1237.4	1239.6	1241.9	1244.2	1246.5	1248.8	1251.0	360
370	1251.0	1253.3	1255.6	1257.9	1260.2	1262.5	1264.8	1267.1	1269.4	1271.7	1274.0	370
380	1274.0	1276.3	1278.6	1280.9	1283.3	1285.6	1287.9	1290.2	1292.5	1294.9	1297.2	380
390	1297.2	1299.5	1301.9	1304.2	1306.5	1308.9	1311.2	1313.6	1315.9	1318.3	1320.6	390
400	1320.6											400

25.6 Copper Resistance Table:

The resistance, *R*, of a length of wire is described by the expression:

$$R = \frac{\rho L}{A}$$

ρ = resistivity of the material composing the wire,
L = length of the wire, and
A = area of the conducting cross section of the wire.

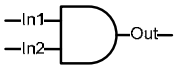
Table 8 Conductor Properties

Size (AWG or kcmil)	Area		Conductors							Direct-Current Resistance at 75°C (167°F)					
			Stranding			Overall				Copper				Aluminum	
				Diameter		Diameter		Area		Uncoated		Coated			
	mm²	Circular mils	Quantity	mm	in.	mm	in.	mm²	in.²	ohm/km	ohm/kFT	ohm/km	ohm/kFT	ohm/km	ohm/kFT
18	0.823	1620	1	—	—	1.02	0.040	0.823	0.001	25.5	7.77	26.5	8.08	42.0	12.8
18	0.823	1620	7	0.39	0.015	1.16	0.046	1.06	0.002	26.1	7.95	27.7	8.45	42.8	13.1
16	1.31	2580	1	—	—	1.29	0.051	1.31	0.002	16.0	4.89	16.7	5.08	26.4	8.05
16	1.31	2580	7	0.49	0.019	1.46	0.058	1.68	0.003	16.4	4.99	17.3	5.29	26.9	8.21
14	2.08	4110	1	—	—	1.63	0.064	2.08	0.003	10.1	3.07	10.4	3.19	16.6	5.06
14	2.08	4110	7	0.62	0.024	1.85	0.073	2.68	0.004	10.3	3.14	10.7	3.26	16.9	5.17
12	3.31	6530	1	—	—	2.05	0.081	3.31	0.005	6.34	1.93	6.57	2.01	10.45	3.18
12	3.31	6530	7	0.78	0.030	2.32	0.092	4.25	0.006	6.50	1.98	6.73	2.05	10.69	3.25
10	5.261	10380	1	—	—	2.588	0.102	5.26	0.008	3.984	1.21	4.148	1.26	6.561	2.00
10	5.261	10380	7	0.98	0.038	2.95	0.116	6.76	0.011	4.070	1.24	4.226	1.29	6.679	2.04
8	8.367	16510	1	—	—	3.264	0.128	8.37	0.013	2.506	0.764	2.579	0.786	4.125	1.26
8	8.367	16510	7	1.23	0.049	3.71	0.146	10.76	0.017	2.551	0.778	2.653	0.809	4.204	1.28
6	13.30	26240	7	1.56	0.061	4.67	0.184	17.09	0.027	1.608	0.491	1.671	0.510	2.652	0.808
4	21.15	41740	7	1.96	0.077	5.89	0.232	27.19	0.042	1.010	0.308	1.053	0.321	1.666	0.508
3	26.67	52620	7	2.20	0.087	6.60	0.260	34.28	0.053	0.802	0.245	0.833	0.254	1.320	0.403
2	33.62	66360	7	2.47	0.097	7.42	0.292	43.23	0.067	0.634	0.194	0.661	0.201	1.045	0.319
1	42.41	83690	19	1.69	0.066	8.43	0.332	55.80	0.087	0.505	0.154	0.524	0.160	0.829	0.253
1/0	53.49	105600	19	1.89	0.074	9.45	0.372	70.41	0.109	0.399	0.122	0.415	0.127	0.660	0.201
2/0	67.43	133100	19	2.13	0.084	10.62	0.418	88.74	0.137	0.3170	0.0967	0.329	0.101	0.523	0.159
3/0	85.01	167800	19	2.39	0.094	11.94	0.470	111.9	0.173	0.2512	0.0766	0.2610	0.0797	0.413	0.126
4/0	107.2	211600	19	2.68	0.106	13.41	0.528	141.1	0.219	0.1996	0.0608	0.2050	0.0626	0.328	0.100
250	127	—	37	2.09	0.082	14.61	0.575	168	0.260	0.1687	0.0515	0.1753	0.0535	0.2778	0.0847
300	152	—	37	2.29	0.090	16.00	0.630	201	0.312	0.1409	0.0429	0.1463	0.0446	0.2318	0.0707
350	177	—	37	2.47	0.097	17.30	0.681	235	0.364	0.1205	0.0367	0.1252	0.0382	0.1984	0.0605
400	203	—	37	2.64	0.104	18.49	0.728	268	0.416	0.1053	0.0321	0.1084	0.0331	0.1737	0.0529
500	253	—	37	2.95	0.116	20.65	0.813	336	0.519	0.0845	0.0258	0.0869	0.0265	0.1391	0.0424
600	304	—	61	2.52	0.099	22.68	0.893	404	0.626	0.0704	0.0214	0.0732	0.0223	0.1159	0.0353
700	355	—	61	2.72	0.107	24.49	0.964	471	0.730	0.0603	0.0184	0.0622	0.0189	0.0994	0.0303
750	380	—	61	2.82	0.111	25.35	0.998	505	0.782	0.0563	0.0171	0.0579	0.0176	0.0927	0.0282
800	405	—	61	2.91	0.114	26.16	1.030	538	0.834	0.0528	0.0161	0.0544	0.0166	0.0868	0.0265
900	456	—	61	3.09	0.122	27.79	1.094	606	0.940	0.0470	0.0143	0.0481	0.0147	0.0770	0.0235
1000	507	—	61	3.25	0.128	29.26	1.152	673	1.042	0.0423	0.0129	0.0434	0.0132	0.0695	0.0212
1250	633	—	91	2.98	0.117	32.74	1.289	842	1.305	0.0338	0.0103	0.0347	0.0106	0.0554	0.0169
1500	760	—	91	3.26	0.128	35.86	1.412	1011	1.566	0.02814	0.00858	0.02814	0.00883	0.0464	0.0141
1750	887	—	127	2.98	0.117	38.76	1.526	1180	1.829	0.02410	0.00735	0.02410	0.00756	0.0397	0.0121
2000	1013	—	127	3.19	0.126	41.45	1.632	1349	2.092	0.02109	0.00643	0.02109	0.00662	0.0348	0.0106

Table 9 Alternating-Current Resistance and Reactance for 600-Volt Cables, 3-Phase, 60 Hz, 75°C (167°F) — Three Single Conductors in Conduit

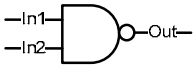
Size (AWG or kcmil)	Ohms to Neutral per Kilometer / Ohms to Neutral per 1000 Feet														Size (AWG or kcmil)
	X_L (Reactance) for All Wires		Alternating-Current Resistance for Uncoated Copper Wires			Alternating-Current Resistance for Aluminum Wires			Effective Z at 0.85 *PF* for Uncoated Copper Wires			Effective Z at 0.85 *PF* for Aluminum Wires			
	PVC, Aluminum Conduits	Steel Conduit	PVC Conduit	Aluminum Conduit	Steel Conduit	PVC Conduit	Aluminum Conduit	Steel Conduit	PVC Conduit	Aluminum Conduit	Steel Conduit	PVC Conduit	Aluminum Conduit	Steel Conduit	
14	0.190 0.058	0.240 0.073	10.2 3.1	10.2 3.1	10.2 3.1	— —	— —	— —	8.9 2.7	8.9 2.7	8.9 2.7	— —	— —	— —	14
12	0.177 0.054	0.223 0.068	6.6 2.0	6.6 2.0	6.6 2.0	10.5 3.2	10.5 3.2	10.5 3.2	5.6 1.7	5.6 1.7	5.6 1.7	9.2 2.8	9.2 2.8	9.2 2.8	12
10	0.164 0.050	0.207 0.063	3.9 1.2	3.9 1.2	3.9 1.2	6.6 2.0	6.6 2.0	6.6 2.0	3.6 1.1	3.6 1.1	3.6 1.1	5.9 1.8	5.9 1.8	5.9 1.8	10
8	0.171 0.052	0.213 0.065	2.56 0.78	2.56 0.78	2.56 0.78	4.3 1.3	4.3 1.3	4.3 1.3	2.26 0.69	2.26 0.69	2.30 0.70	3.6 1.1	3.6 1.1	3.6 1.1	8
6	0.167 0.051	0.210 0.064	1.61 0.49	1.61 0.49	1.61 0.49	2.66 0.81	2.66 0.81	2.66 0.81	1.44 0.44	1.48 0.45	1.48 0.45	2.33 0.71	2.36 0.72	2.36 0.72	6
4	0.157 0.048	0.197 0.060	1.02 0.31	1.02 0.31	1.02 0.31	1.67 0.51	1.67 0.51	1.67 0.51	0.95 0.29	0.95 0.29	0.98 0.30	1.51 0.46	1.51 0.46	1.51 0.46	4
3	0.154 0.047	0.194 0.059	0.82 0.25	0.82 0.25	0.82 0.25	1.31 0.40	1.35 .0.41	1.31 0.40	0.75 0.23	0.79 0.24	0.79 0.24	1.21 0.37	1.21 0.37	1.21 0.37	3
2	0.148 0.045	0.187 0.057	0.62 0.19	0.66 0.20	0.66 0.20	1.05 0.32	1.05 0.32	1.05 0.32	0.62 0.19	0.62 0.19	0.66 0.20	0.98 0.30	0.98 0.30	0.98 0.30	2
1	0.151 0.046	0.187 0.057	0.49 0.15	0.52 0.16	0.52 0.16	0.82 0.25	0.85 0.26	0.82 0.25	0.52 0.16	0.52 0.16	0.52 0.16	0.79 0.24	0.79 0.24	0.82 0.25	1
1/0	0.144 0.044	0.180 0.055	0.39 0.12	0.43 0.13	0.39 0.12	0.66 0.20	0.69 0.21	0.66 0.20	0.43 0.13	0.43 0.13	0.43 0.13	0.62 0.19	0.66 0.20	0.66 0.20	1/0
2/0	0.141 0.043	0.177 0.054	0.33 0.10	0.33 0.10	0.33 0.10	0.52 0.16	0.52 0.16	0.52 0.16	0.36 0.11	0.36 0.11	0.36 0.11	0.52 0.16	0.52 0.16	0.52 0.16	2/0
3/0	0.138 0.042	0.171 0.052	0.253 0.077	0.269 0.082	0.259 0.079	0.43 0.13	0.43 0.13	0.43 0.13	0.289 0.088	0.302 0.092	0.308 0.094	0.43 0.13	0.43 0.13	0.46 0.14	3/0
4/0	0.135 0.041	0.167 0.051	0.203 0.062	0.220 0.067	0.207 0.063	0.33 0.10	0.36 0.11	0.33 0.10	0.243 0.074	0.256 0.078	0.262 0.080	0.36 0.11	0.36 0.11	0.36 0.11	4/0
250	0.135 0.041	0.171 0.052	0.171 0.052	0.187 0.057	0.177 0.054	0.279 0.085	0.295 0.090	0.282 0.086	0.217 0.066	0.230 0.070	0.240 0.073	0.308 0.094	0.322 0.098	0.33 0.10	250
300	0.135 0.041	0.167 0.051	0.144 0.044	0.161 0.049	0.148 0.045	0.233 0.071	0.249 0.076	0.236 0.072	0.194 0.059	0.207 0.063	0.213 0.065	0.269 0.082	0.282 0.086	0.289 0.088	300
350	0.131 0.040	0.164 0.050	0.125 0.038	0.141 0.043	0.128 0.039	0.200 0.061	0.217 0.066	0.207 0.063	0.174 0.053	0.190 0.058	0.197 0.060	0.240 0.073	0.253 0.077	0.262 0.080	350
400	0.131 0.040	0.161 0.049	0.108 0.033	0.125 0.038	0.115 0.035	0.177 0.054	0.194 0.059	0.180 0.055	0.161 0.049	0.174 0.053	0.184 0.056	0.217 0.066	0.233 0.071	0.240 0.073	400
500	0.128 0.039	0.157 0.048	0.089 0.027	0.105 0.032	0.095 0.029	0.141 0.043	0.157 0.048	0.148 0.045	0.141 0.043	0.157 0.048	0.164 0.050	0.187 0.057	0.200 0.061	0.210 0.064	500
600	0.128 0.039	0.157 0.048	0.075 0.023	0.092 0.028	0.082 0.025	0.118 0.036	0.135 0.041	0.125 0.038	0.131 0.040	0.144 0.044	0.154 0.047	0.167 0.051	0.180 0.055	0.190 0.058	600
750	0.125 0.038	0.157 0.048	0.062 0.019	0.079 0.024	0.069 0.021	0.095 0.029	0.112 0.034	0.102 0.031	0.118 0.036	0.131 0.040	0.141 0.043	0.148 0.045	0.161 0.049	0.171 0.052	750

25.7 Boolean Logic Operations:



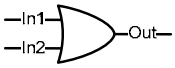
AND Gate: All inputs must be true for output to be true

In1	In2	Out
0	0	0
0	1	0
1	0	0
1	1	1



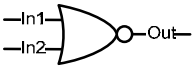
NAND Gate: All inputs must be false for output to be true

In1	In2	Out
0	0	1
0	1	1
1	0	1
1	1	0



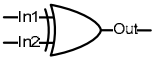
OR Gate: Any input can be true for output to be true

In1	In2	Out
0	0	0
0	1	1
1	0	1
1	1	1



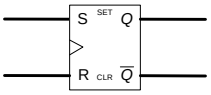
NOR Gate: If any input is true the output will be false

In1	In2	Out
0	0	0
0	1	1
1	0	1
1	1	1



XOR Gate: All the inputs must be different for the output to be true

In1	In2	Out
0	0	1
0	1	0
1	0	1
1	1	0



S-R Flip-Flop: Latch Circuit

S	R	Q	Qnot
0	0	Keep output state	Keep output state
0	1	0	1
1	0	1	0
1	1	Unstable condition	Unstable condition

25.8 Instrument Air Quality:

- **Dewpoint:**

The pressure dewpoint as measured at the dryer outlet shall be at least 10°C (18°F) below the minimum temperature to which any part of the instrument air system will be exposed. The pressure dewpoint shall not exceed 4°C (39°F) at line pressure.

- **Particle Size:**

A maximum 40µmeter particle size in the instrument air system is acceptable for the majority of pneumatic devices.

- **Lubricant Content:**

The lubricant content should be as close to zero as possible, and under no circumstances shall it exceed one (1) ppm w/w or v/v. Any lubricant in the compressed air system shall be evaluated for compatibility with the end of use pneumatic devices.

- **Contaminates:**

Instrument air shall be free of corrosive contaminates and hazardous gases which could be drawn into the instrument air supply. The air intake should be monitored for contaminates (e.g. paint vapor, chemical vapors, engine exhaust)

Traditionally, regenerative desiccant dryers for instrument air systems are sized to provide -40°C (-40°F) dew point air at pressure. However, in extremely cold climates, instrument air applications may require dew points as low as -73°C (-100°F) at operating pressure.

25.9 Derivative Tables:

Derivatives of simple functions

$$c' = 0$$
$$x' = 1$$
$$(cx)' = c$$
$$|x|' = \frac{x}{|x|} = \operatorname{sgn} x, \qquad x \neq 0$$
$$(x^c)' = cx^{c-1} \qquad \text{where both } x^c \text{ and } cx^{c-1} \text{ are defined}$$
$$\left(\frac{1}{x}\right)' = \left(x^{-1}\right)' = -x^{-2} = -\frac{1}{x^2}$$
$$\left(\frac{1}{x^c}\right)' = \left(x^{-c}\right)' = -cx^{-(c+1)} = -\frac{c}{x^{c+1}}$$
$$\left(\sqrt{x}\right)' = \left(x^{\frac{1}{2}}\right)' = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}, \qquad x > 0$$

Derivatives of exponential and logarithmic functions

$$(c^x)' = c^x \ln c, \qquad c > 0$$

note that the equation above is true for all c, but the derivative yields a complex number.

$$(e^x)' = e^x$$
$$(\log_c x)' = \frac{1}{x \ln c}, \qquad c > 0, c \neq 1$$

the equation above is also true for all c but yields a complex number.

$$(\ln x)' = \frac{1}{x}, \qquad x \neq 0$$
$$(\ln |x|)' = \frac{1}{x}$$
$$(x^x)' = x^x(1 + \ln x)$$

$$(\sin x)' = \cos x$$
$$(\cos x)' = -\sin x$$
$$(\tan x)' = \sec^2 x = \frac{1}{\cos^2 x}$$
$$(\sec x)' = \sec x \tan x$$
$$(\csc x)' = -\csc x \cot x$$
$$(\cot x)' = -\csc^2 x = \frac{-1}{\sin^2 x}$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$
$$(\arccos x)' = \frac{-1}{\sqrt{1-x^2}}$$
$$(\arctan x)' = \frac{1}{1+x^2}$$
$$(\operatorname{arcsec} x)' = \frac{1}{|x|\sqrt{x^2-1}}$$
$$(\operatorname{arccsc} x)' = \frac{-1}{|x|\sqrt{x^2-1}}$$
$$(\operatorname{arccot} x)' = \frac{-1}{1+x^2}$$

25.10 Integral Tables:

Rational functions

more integrals: List of integrals of rational functions

$$\int a\,dx = ax + C$$
$$\int x^a\,dx = \frac{x^{a+1}}{a+1} + C$$
$$\int \frac{1}{x}\,dx = \ln|x| + C$$

Exponential functions

more integrals: List of integrals of exponential functions

$$\int e^x\,dx = e^x + C$$
$$\int a^x\,dx = \frac{a^x}{\ln a} + C$$

Logarithms

more integrals: List of integrals of logarithmic functions

$$\int \ln x\,dx = x\ln x - x + C$$
$$\int \log_a x\,dx = x\log_a x - \frac{x}{\ln a} + C$$

Special functions

$$\int \operatorname{Ci}(x)dx = x\operatorname{Ci}(x) - \sin x$$
$$\int \operatorname{Si}(x)dx = x\operatorname{Si}(x) + \cos x$$
$$\int \operatorname{Ei}(x)dx = x\operatorname{Ei}(x) - e^x$$
$$\int \operatorname{li}(x)dx = x\operatorname{li}(x) - \operatorname{Ei}(2\ln x)$$
$$\int \frac{\operatorname{li}(x)}{x}\,dx = \ln x\operatorname{li}(x) - x$$
$$\int \operatorname{erf}(x)\,dx = \frac{e^{-x^2}}{\sqrt{\pi}} + x\operatorname{erf}(x)$$

Trigonometric functions

more integrals: List of integrals of trigonometric functions

$$\int \sin x\,dx = -\cos x + C$$

$$\int \cos x\,dx = \sin x + C$$

$$\int \tan x\,dx = -\ln|\cos x| + C = \ln|\sec x| + C$$

$$\int \cot x\,dx = \ln|\sin x| + C$$

$$\int \sec x\,dx = \ln|\sec x + \tan x| + C$$

$$\int \csc x\,dx = -\ln|\csc x + \cot x| + C$$

$$\int \sec^2 x\,dx = \tan x + C$$

$$\int \csc^2 x\,dx = -\cot x + C$$

$$\int \sec x\,\tan x\,dx = \sec x + C$$

$$\int \csc x\,\cot x\,dx = -\csc x + C$$

$$\int \sin^2 x\,dx = \frac{1}{2}\left(x - \frac{\sin 2x}{2}\right) + C = \frac{1}{2}(x - \sin x\cos x) + C$$

$$\int \cos^2 x\,dx = \frac{1}{2}\left(x + \frac{\sin 2x}{2}\right) + C = \frac{1}{2}(x + \sin x\cos x) + C$$

$$\int \sec^3 x\,dx = \frac{1}{2}\sec x\tan x + \frac{1}{2}\ln|\sec x + \tan x| + C$$

(see integral of secant cubed)

$$\int \sin^n x\,dx = -\frac{\sin^{n-1} x\cos x}{n} + \frac{n-1}{n}\int \sin^{n-2} x\,dx$$

$$\int \cos^n x\,dx = \frac{\cos^{n-1} x\sin x}{n} + \frac{n-1}{n}\int \cos^{n-2} x\,dx$$

Inverse trigonometric functions

more integrals: List of integrals of inverse trigonometric functions

$$\int \arcsin x\,dx = x\,\arcsin x + \sqrt{1-x^2} + C$$

$$\int \arccos x\,dx = x\,\arccos x - \sqrt{1-x^2} + C$$

$$\int \arctan x\,dx = x\,\arctan x - \frac{1}{2}\ln\left|1+x^2\right| + C$$

$$\int \operatorname{arcsec} x\,dx = x\,\operatorname{arcsec} x + \frac{\sqrt{x^2-1}\ln\left(x+\sqrt{x^2-1}\right)}{x\sqrt{1-\frac{1}{x^2}}} + C$$

$$\int \operatorname{arccsc} x\,dx = x\,\operatorname{arccsc} x + \frac{\sqrt{x^2-1}\ln\left(x+\sqrt{x^2-1}\right)}{x\sqrt{1-\frac{1}{x^2}}} + C$$

$$\int \operatorname{arccot} x\,dx = x\,\operatorname{arccot} x + \frac{1}{2}\ln\left|1+x^2\right| + C$$

Hyperbolic functions

more integrals: List of integrals of hyperbolic functions

$$\int \sinh x \, dx = \cosh x + C$$
$$\int \cosh x \, dx = \sinh x + C$$
$$\int \tanh x \, dx = \ln|\cosh x| + C$$
$$\int \operatorname{cosech} x \, dx = \ln\left|\tanh\frac{x}{2}\right| + C$$
$$\int \operatorname{sech} x \, dx = \arctan(\sinh x) + C$$
$$\int \coth x \, dx = \ln|\sinh x| + C$$
$$\int \operatorname{sech}^2 x \, dx = \tanh x + C$$

Inverse hyperbolic functions

more integrals: List of integrals of inverse hyperbolic functions

$$\int \operatorname{arsinh} x \, dx = x \operatorname{arsinh} x - \sqrt{x^2+1} + C$$
$$\int \operatorname{arcosh} x \, dx = x \operatorname{arcosh} x - \sqrt{x^2-1} + C$$
$$\int \operatorname{artanh} x \, dx = x \operatorname{artanh} x + \frac{1}{2}\ln(1-x^2) + C$$
$$\int \operatorname{arcsch} x \, dx = x \operatorname{arcsch} x + \ln\left[x\left(\sqrt{1+\frac{1}{x^2}}+1\right)\right] + C$$
$$\int \operatorname{arsech} x \, dx = x \operatorname{arsech} x - \arctan\left(\frac{x}{x-1}\sqrt{\frac{1-x}{1+x}}\right) + C$$
$$\int \operatorname{arcoth} x \, dx = x \operatorname{arcoth} x + \frac{1}{2}\ln(x^2-1) + C$$

Composed functions

$$\int \cos ax \, e^{bx} \, dx = \frac{e^{bx}}{a^2+b^2}(a\sin ax + b\cos ax) + C$$
$$\int \sin ax \, e^{bx} \, dx = \frac{e^{bx}}{a^2+b^2}(b\sin ax - a\cos ax) + C$$
$$\int \cos ax \, \cosh bx \, dx = \frac{1}{a^2+b^2}(a\sin ax \, \cosh bx + b\cos ax \, \sinh bx) + C$$
$$\int \sin ax \, \cosh bx \, dx = \frac{1}{a^2+b^2}(b\sin ax \, \sinh bx - a\cos ax \, \cosh bx) + C$$

Absolute value functions

$$\int |(ax+b)^n| \, dx = \frac{(ax+b)^{n+2}}{a(n+1)|ax+b|} + C \ [\, n \text{ is odd, and } n \neq -1 \,]$$
$$\int |\sin ax| \, dx = \frac{-1}{a}|\sin ax|\cot ax + C$$
$$\int |\cos ax| \, dx = \frac{1}{a}|\cos ax|\tan ax + C$$
$$\int |\tan ax| \, dx = \frac{\tan(ax)[-\ln|\cos ax|]}{a|\tan ax|} + C$$
$$\int |\csc ax| \, dx = \frac{-\ln|\csc ax + \cot ax|\sin ax}{a|\sin ax|} + C$$
$$\int |\sec ax| \, dx = \frac{\ln|\sec ax + \tan ax|\cos ax}{a|\cos ax|} + C$$
$$\int |\cot ax| \, dx = \frac{\tan(ax)[\ln|\sin ax|]}{a|\tan ax|} + C$$

25.11 Laplace Tables:

The following Table of Laplace Transforms is very useful when solving problems in science and engineering that require Laplace transform:

Time Function $f(t)$ $f(t) = \mathcal{L}^{-1}\{F(s)\}$	**Laplace Transform of $f(t)$** $F(s) = \mathcal{L}\{f(t)\}$
1	$\frac{1}{s} \quad s > 0$
t (unit-ramp function)	$\frac{1}{s^2} \quad s > 0$
t^n (n, a positive integer)	$\frac{n!}{s^{n+1}} \quad s > 0$
e^{at}	$\frac{1}{s-a} \quad s > a$
$\sin \omega t$	$\frac{\omega}{s^2+\omega^2} \quad s > 0$
$\cos \omega t$	$\frac{s}{s^2+\omega^2} \quad s > 0$
$t^n g(t)$, for $n = 1, 2, \ldots$	$(-1)^n \frac{d^n G(s)}{ds^n}$
$t \sin \omega t$	$\frac{2\omega s}{(s^2+\omega^2)^2} \quad s > \lvert\omega\rvert$
$t \cos \omega t$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2} \quad s > \lvert\omega\rvert$
$g(at)$	$\frac{1}{a} G(\frac{s}{a})$ Scale property
$e^{at} g(t)$	$G(s-a)$ Shift property
$e^{at} t^n$, for $n = 1, 2, \ldots$	$\frac{n!}{(s-a)^{n+1}} \quad s > a$
te^{-t}	$\frac{1}{(s+1)^2} \quad s > -1$
$1 - e^{-t/T}$	$\frac{1}{s(1+Ts)} \quad s > -1/T$
$e^{at} \sin \omega t$	$\frac{\omega}{(s-a)^2+\omega^2} \quad s > a$
$e^{at} \cos \omega t$	$\frac{s-a}{(s-a)^2+\omega^2} \quad s > a$
$u(t)$	$\frac{1}{s} \quad s > 0$
$u(t-a)$	$\frac{e^{-as}}{s} \quad s > 0$
$u(t-a)g(t-a)$	$e^{-as}G(s)$ Time-displacement theorem
$g'(t)$	$sG(s) - g(0)$
$g''(t)$	$s^2 \cdot G(s) - s \cdot g(0) - g'(0)$
$g^{(n)}(t)$	$s^n \cdot G(s) - s^{n-1} \cdot g(0) - s^{n-2} \cdot g'(0) - \ldots - g^{(n-1)}(0)$
$\int_0^t g(t)dt$	$\frac{G(s)}{s}$
$\int g(t)dt$	$\frac{G(s)}{s} + \frac{1}{s}\left\{\int g(t)dt\right\}_{t=0}$

26 Bibliography (References Used)

- Liptak: Instrument Engineers Handbook (3rd Edition)
- Liptak: Process Measurement & Analysis
- Liptak: Process Software and Digital Networks
- Sherman: Analytical Instrumentation
- Omega Engineering
- Cameron Hydraulic Book
- Crane Flow of Fluids Tech Paper 410
- Spink: Flow Meter Engineering
- Miller: Flow Measurement Engineering Handbook
- Ritz: Control Valve Actuator Options
- Journal of Industrial technology (ANN)
- Fisher Control Valve Handbook
- Borden/Friedmann: Control Valves
- Crosby Engineering Handbook TP-V300
- Crouse-Hinds 2008 Code Digest
- 2008 NEC Code Handbook
- Controller Tuning & Control Loop Performance – David St. Clair
- Various ISA Standards
- Various Vendor Websites
- Wikipedia